National
Qualifications
2026

X840/76/12

Human Biology
Paper 1 — Multiple choice

TUESDAY, 28 APRIL
9:00 AM – 9:40 AM

Total marks — 25

Attempt ALL questions.

You may use a calculator.

Instructions for the completion of Paper 1 are given on *page 02* of your answer booklet X840/76/02.

Record your answers on the answer grid on *page 03* of your answer booklet.

Space for rough work is provided at the end of this booklet.

You must leave your answer booklet on your desk; if you do not, you could lose all the marks for this paper.

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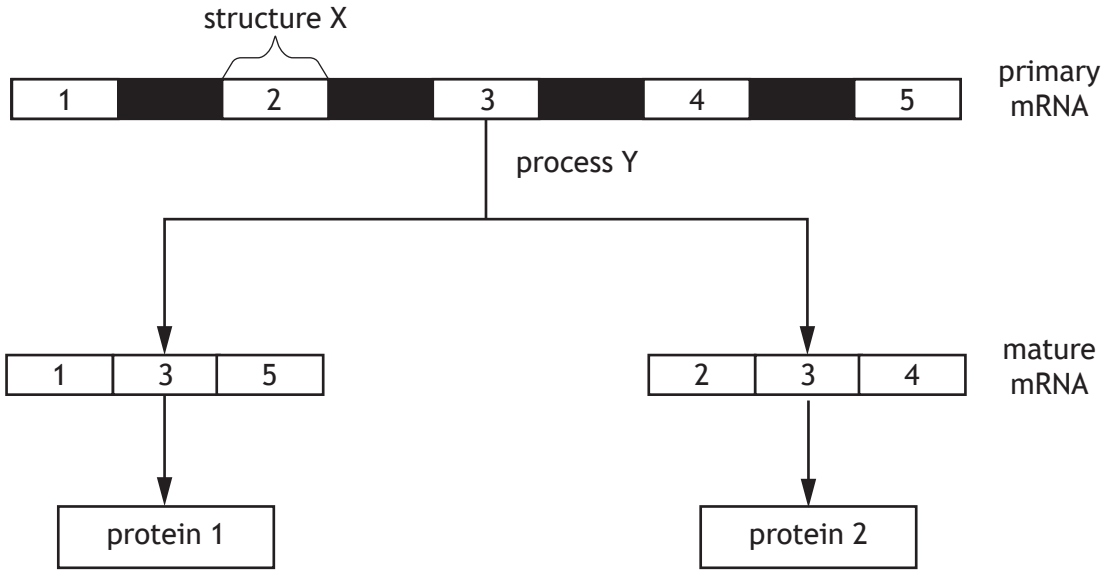
X8407612

A/PB
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Total marks — 25

Attempt ALL questions

1. The diagram shows a stage during the synthesis of two different proteins.



Identify structure X and process Y.

	Structure X	Process Y
A	exon	translation
B	intron	alternative splicing
C	exon	alternative splicing
D	intron	translation

2. One cycle of PCR takes 2 minutes.

If the original sample contained one molecule of DNA, how many DNA molecules will be present after 20 minutes?

A 10

B 512

C 1024

D 2048

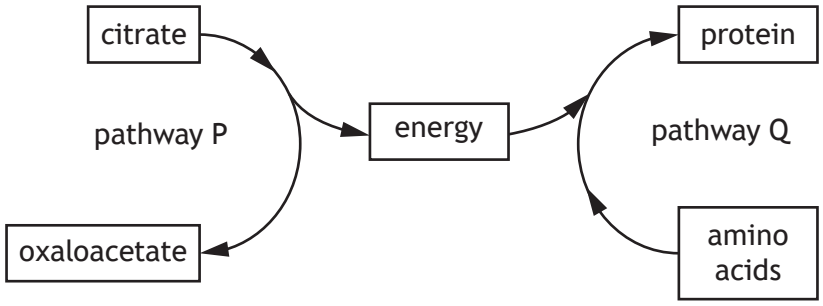
3. The following are descriptions of three chromosome structure mutations:

1. A section of a chromosome is reversed.
2. A section of a chromosome is added from its homologous partner.
3. A section of a chromosome is added to a chromosome that is not its homologous partner.

Which row in the table matches the description with the type of chromosome structure mutation?

	Chromosome structure mutation		
	Translocation	**Inversion**	**Duplication**
A	3	1	2
B	2	1	3
C	1	3	2
D	3	2	1

4. The diagram shows two linked metabolic pathways.



Which row in the table identifies the type of metabolic pathway and whether ATP is produced or required?

	Pathway	**Type of metabolic pathway**	**ATP**
A	P	anabolic	produced
B	P	catabolic	required
C	Q	anabolic	required
D	Q	catabolic	produced

[Turn over

5. An investigation was carried out into the effect of substrate concentration on enzyme activity.

Catalase is an enzyme that breaks down hydrogen peroxide to water and oxygen.

Test tubes were set up containing 5 cm^3 of hydrogen peroxide at different concentrations.

Catalase was added to each test tube and the volume of oxygen produced was measured.

Which row in the table identifies variables in this investigation?

	Independent variable	**Dependent variable**	**Controlled variable**
A	concentration of hydrogen peroxide	enzyme activity	volume of oxygen
B	concentration of hydrogen peroxide	enzyme activity	volume of hydrogen peroxide
C	enzyme activity	concentration of hydrogen peroxide	volume of hydrogen peroxide
D	enzyme activity	concentration of hydrogen peroxide	volume of oxygen

6. An investigation into the effect of substrate concentration on enzyme activity showed that enzyme activity levelled off at high substrate concentrations.

The explanation for this observation is that

A all the active sites are occupied

B no active sites are occupied

C the affinity of the active site has increased

D the affinity of the active site has decreased.

7. Three drops of iodine solution were added to a test tube containing 5 cm^3 of starch suspension, turning the starch suspension blue/black. When 1 cm^3 of enzyme amylase was added, the blue/black colour disappeared. It was concluded that the starch was broken down by the amylase.

To show that the starch was broken down by the amylase, a control test tube would contain

A 5 cm^3 of distilled water, 3 drops of amylase and 1 cm^3 of iodine solution

B 5 cm^3 of distilled water, 3 drops of iodine solution and 1 cm^3 of amylase

C 5 cm^3 of starch suspension, 3 drops of distilled water and 1 cm^3 of amylase

D 5 cm^3 of starch suspension, 3 drops of iodine solution and 1 cm^3 of distilled water.

8. The following list describes some effects of male hormones:

1. Stimulates sperm production.
2. Activates the prostate gland.
3. Inhibits the seminal vesicles.
4. Stimulates the interstitial cells.

Which of these are effects of testosterone?

A 1 only

B 1 and 2 only

C 1, 2 and 3 only

D 1, 2, 3 and 4

9. The onset of puberty is controlled by a hormone.

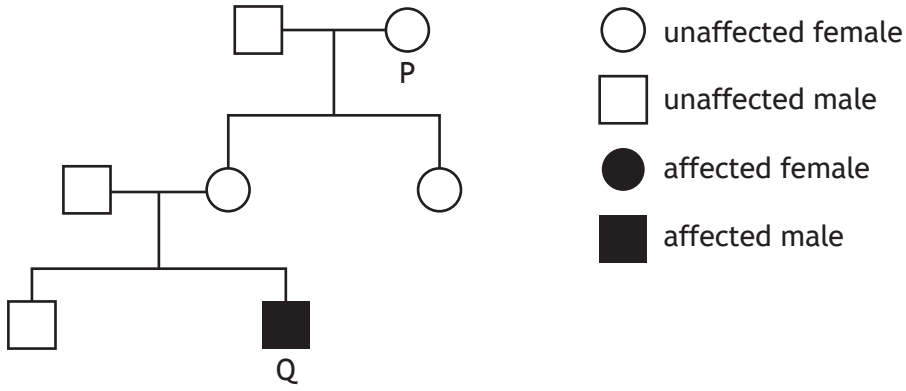
Which row in the table identifies this hormone and matches it with its site of production and target tissue?

	Hormone	**Site of production**	**Target tissue**
A	releaser hormone	pituitary gland	hypothalamus
B	FSH	hypothalamus	pituitary gland
C	releaser hormone	hypothalamus	pituitary gland
D	FSH	pituitary gland	hypothalamus

[Turn over

10. Haemophilia is a sex-linked recessive genetic disorder.

The diagram shows the pattern of inheritance of haemophilia in a family over three generations.



The allele for haemophilia is represented by 'h'.

Which row in the table shows the genotypes of individuals P and Q?

	Genotype of P	**Genotype of Q**
A	X^hX^h	X^HY
B	X^HX^h	X^HY
C	X^hX^h	X^hY
D	X^HX^h	X^hY

11. Compared to amniocentesis, chorionic villus sampling (CVS) can be carried out

A earlier in pregnancy and has a higher risk of miscarriage

B earlier in pregnancy and has a lower risk of miscarriage

C later in pregnancy and has a higher risk of miscarriage

D later in pregnancy and has a lower risk of miscarriage.

12. The table contains information about the rate of blood flow to parts of an individual's body at rest and during exercise.

	Rate of blood flow to body part (cm^3/min)				
	Heart muscle	**Skin**	**Small intestine**	**Kidneys**	**Brain**
At rest	250	500	700	1000	750
During exercise	750	1500	150	550	750

Which of the following statements describes the effect of exercise on blood flow?

A The rate of blood flow to all body parts changes.

B The decrease in blood flow to the small intestine is equal to the decrease in blood flow to the kidneys.

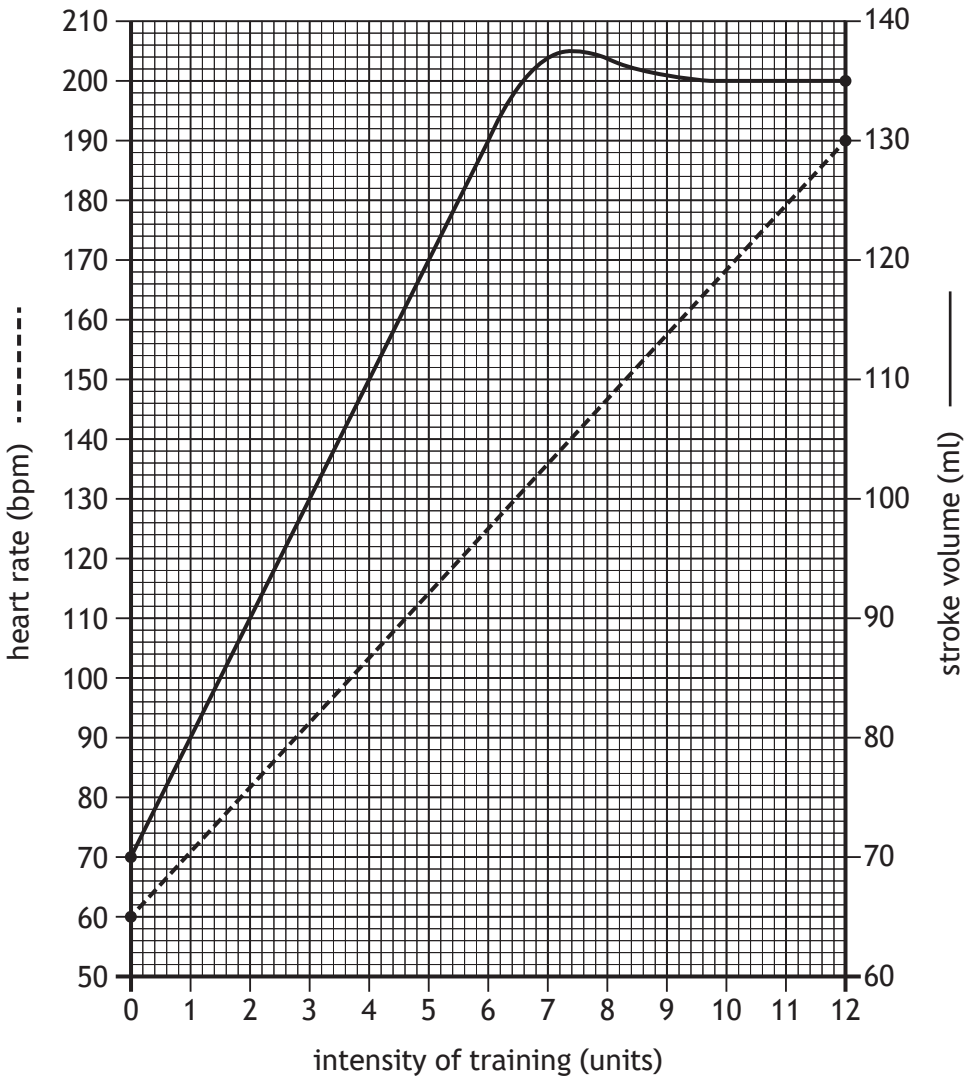
C The increase in blood flow to the skin is two times greater than the increase in blood flow to the heart muscle.

D The rate of blood flow to the heart muscle increases by 500 times.

[Turn over

13. The heart rate (HR) and stroke volume (SV) of an individual was measured throughout a training session.

The graph shows the results.



Calculate the increase in cardiac output as the intensity of training increased from 0–12 units.

A 21.45 l/min

B 21 450 l/min

C 25 650 l/min

D 510.7 l/min

14. A study was carried out to investigate the effect of a drug used to decrease blood pressure.

The following protocols were followed:

1. 50% of the participants were female.
2. Participant selection was randomised.
3. 300 000 participants completed the study.
4. The study included 20 repeated trials.

Identify the protocols that would make the study reliable.

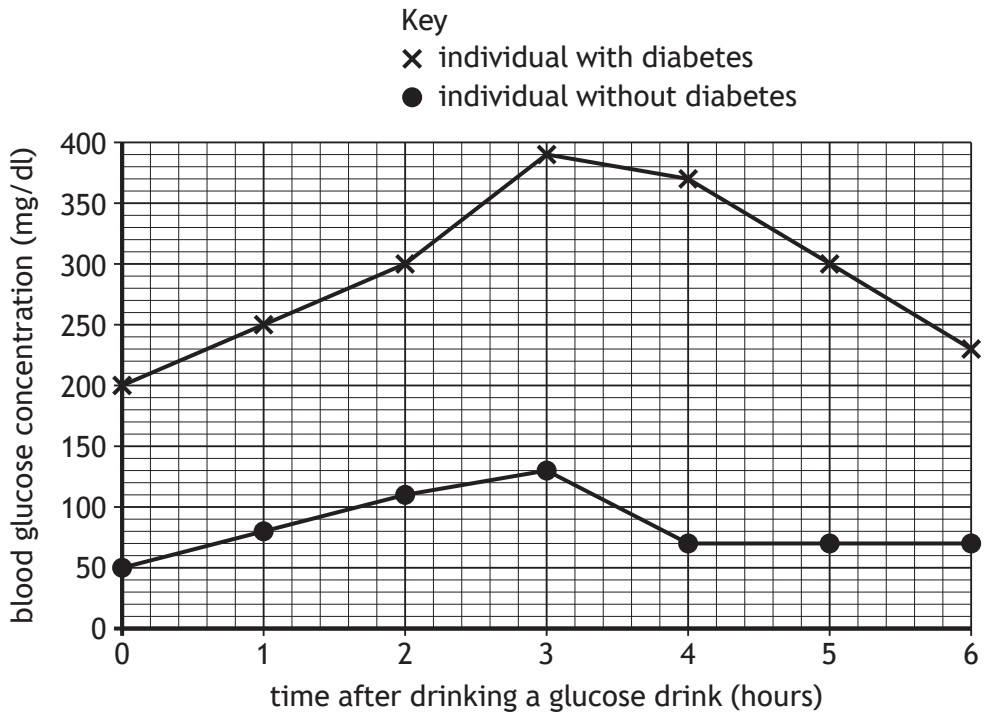
A 1 and 2

B 1 and 4

C 2 and 4

D 3 and 4

15. The graph shows the blood glucose concentrations of an individual with diabetes and an individual without diabetes, after a glucose drink.



Calculate the percentage increase in blood glucose concentration for the individual with diabetes three hours after the glucose drink.

A 49%

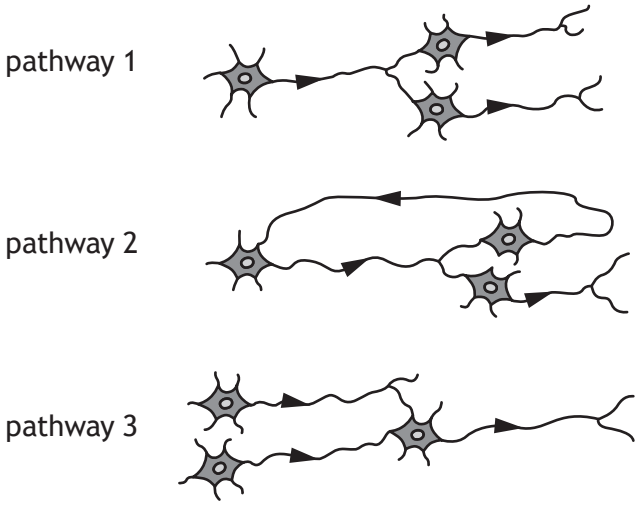
B 95%

C 160%

D 200%

[Turn over

16. The diagram shows three neural pathways. The arrowheads show the direction of impulse.



Which row in the table identifies the types of neural pathways shown?

	Neural pathway		
	Converging	**Diverging**	**Reverberating**
A	1	3	2
B	3	1	2
C	2	1	3
D	3	2	1

17. There is localisation of brain functions in the cerebral cortex.
Which of the following are functions controlled by the cerebral cortex?

A Language processing and breathing rate
B Heart rate and conscious thought
C Language processing and recall of memory
D Recall of memory and breathing rate

18. Identify the structure that transfers information between the cerebral hemispheres and the hemisphere that processes information from the right visual field.

	Structure	Hemisphere
A	corpus luteum	left
B	corpus callosum	left
C	corpus callosum	right
D	corpus luteum	right

19. Which of the following stimulate neurons involved in the reduction of the intensity of pain?

A Dopamine

B Endorphins

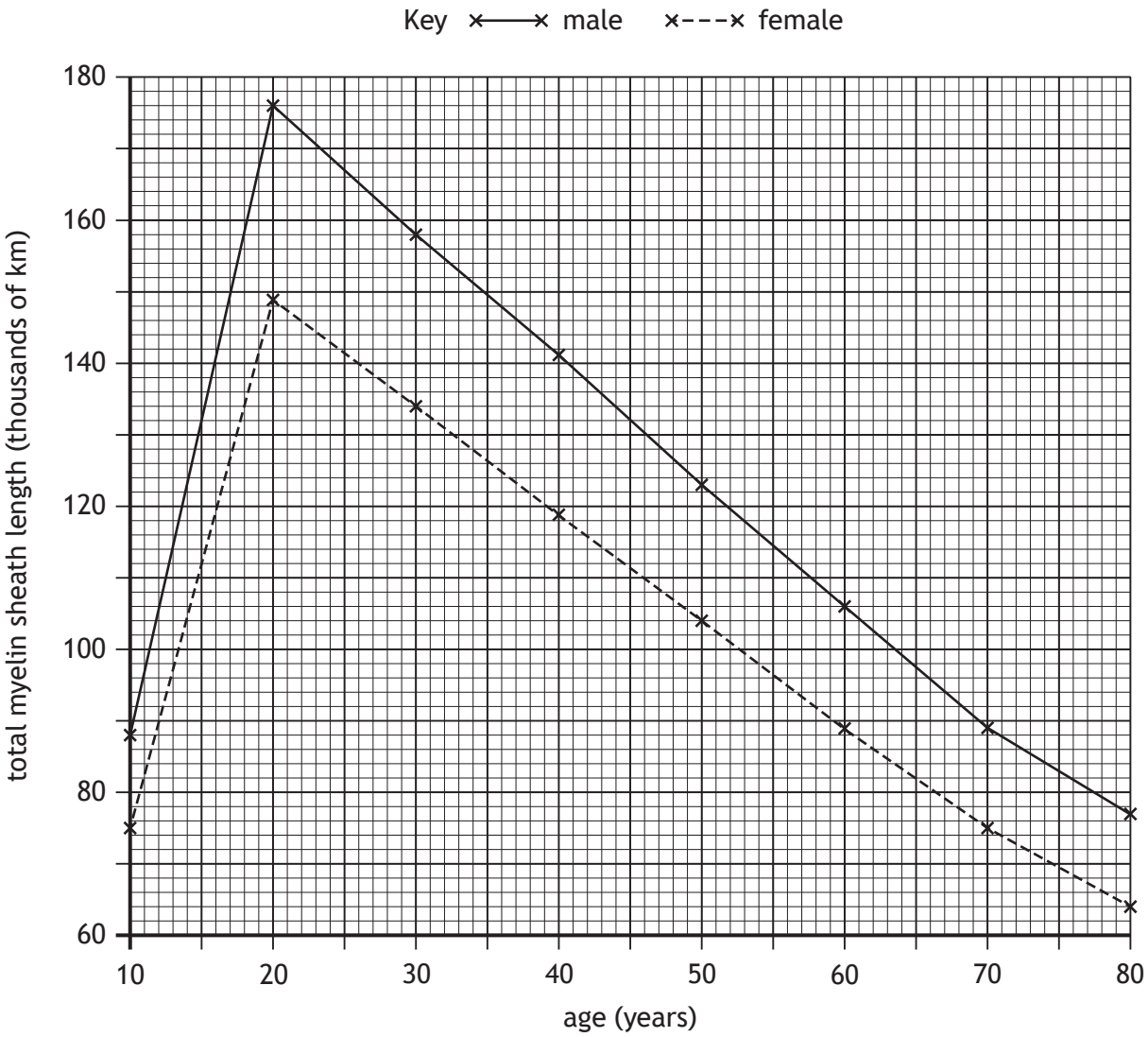
C Noradrenaline

D Acetylcholine

[Turn over

20. Myelination occurs from birth to adolescence.

The graph shows the changes in total myelin sheath lengths for different ages in males and females. Fast impulse conduction occurs when the total myelin sheath length is greater than 100 000 km.



Which row in the table is correct for the data shown?

	Sex	Age (years)	Speed of impulse conduction	Response to stimuli
A	male	20	fast	slow
B	female	20	slow	fast
C	male	80	fast	fast
D	female	80	slow	slow

21. Individuals of various ages were shown 12 different items and then asked to recall as many as they could.

The graph shows how the average number of items recalled changed with age.



Which of the following statements is **not** supported by the data in the graph?

A The average number of items recalled was greatest at 25 years of age.

B The average percentage recall at age 60 is 33%.

C The average number of items recalled was the same at ages 10 and 55.

D The average percentage recall at age 25 is 66%.

[Turn over

22. The table shows the results of an experiment to demonstrate the serial position effect.

Position of item in the list	Percentage recall
1	93
2	81
3	64
4	54
5	42
6	27
7	31
8	51
9	54
10	72

Which row in the table shows the reason for the percentage recall of item 6 and the type of memory involved?

	Reason	Memory involved
A	displacement	long term
B	displacement	short term
C	decay	long term
D	decay	short term

23. An increase in sensitivity of neurotransmitter receptors causes

A drug tolerance due to repeated exposure to agonists

B drug addiction due to repeated exposure to agonists

C drug tolerance due to repeated exposure to antagonists

D drug addiction due to repeated exposure to antagonists.

24. The following events take place at the site of tissue damage:

P Increased blood flow

Q Phagocytes accumulate

R Mast cells release histamine

S Cytokines are released

The order of these events is

A P, Q, S, R

B R, Q, P, S

C R, P, Q, S

D S, Q, R, P.

25. Two groups of subjects volunteered for a clinical trial of a new vaccine.

One group was given the new vaccine and the second group was given a placebo.

The subjects were allocated to the two groups in a randomised way, taking account of age and gender, in order to

A reduce bias

B reduce experimental error

C establish significant differences

D ensure the comparison was reliable.

[END OF QUESTION PAPER]

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National Qualifications 2026

Mark

X840/76/01

Human Biology Paper 2

TUESDAY, 28 APRIL

10:10 AM – 12:30 PM

X8407601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day | Month | Year

Scottish candidate number

Total marks — 95

Attempt ALL questions.

You may use a calculator.

Question 14 contains a choice.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. Score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Do not remove any exam materials. You must leave this booklet on your desk; if you do not, you could lose all the marks for this paper.

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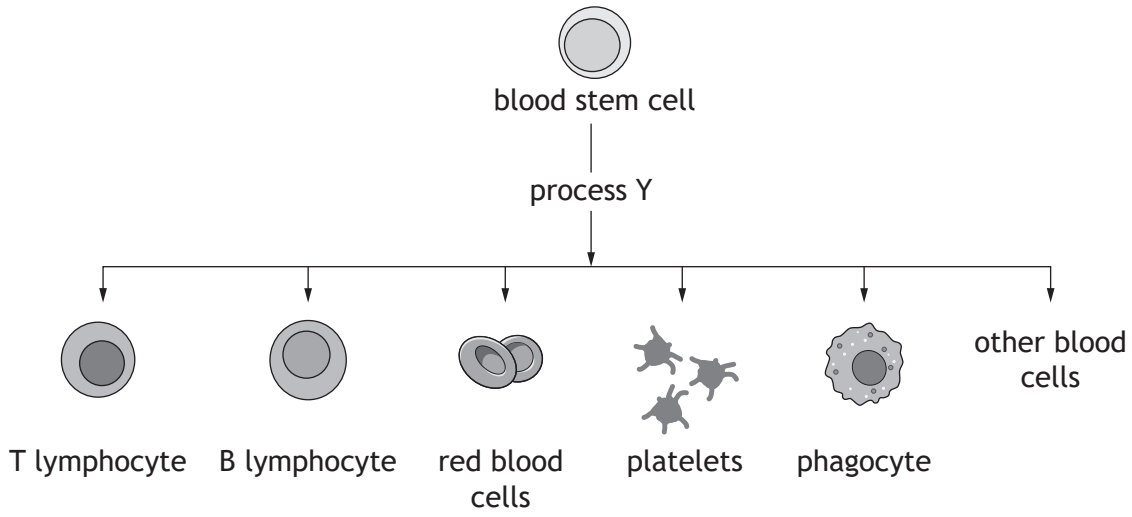
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Total marks — 95

Attempt ALL questions

Question 14 contains a choice

1. (a) The diagram shows the range of cells produced from blood stem cells found within the bone marrow.



(i) Name process Y. **1**

(ii) Using information from the diagram, explain why blood stem cells are described as multipotent. **1**

(iii) Describe the process that results in phagocytes producing different proteins to lymphocytes. **1**

(b) Leukaemia is a type of cancer that affects some blood cells.

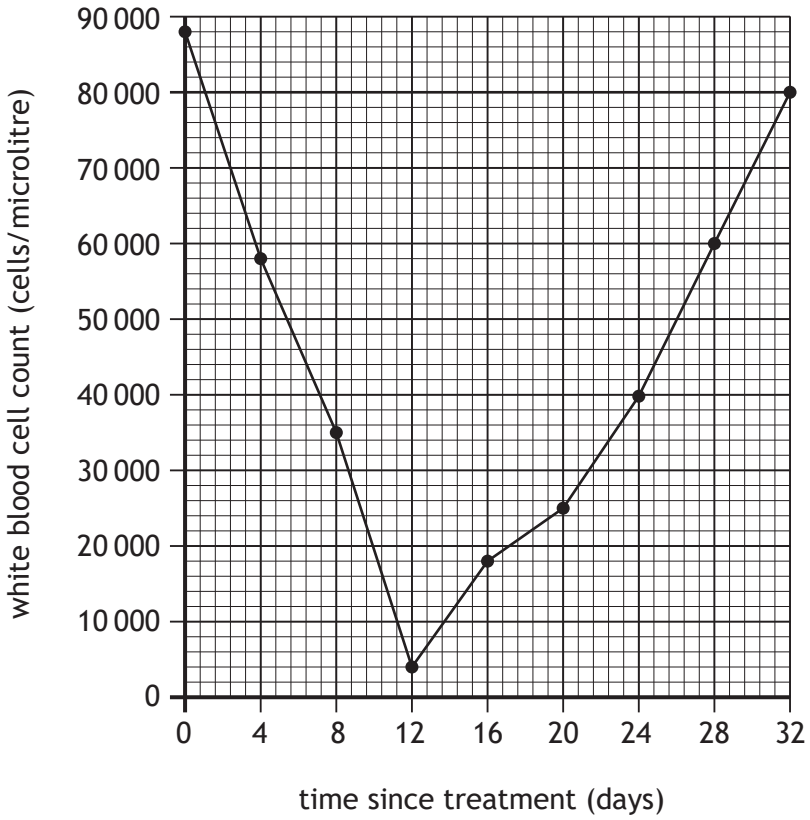
(i) Explain why these cancer cells divide excessively. **1**

1. (b) (continued)

MARKS DO NOT WRITE IN THIS MARGIN

A patient with cancer was treated with chemotherapy that affected their white blood cell count.

The graph shows the patient's white blood cell count over time following this treatment.



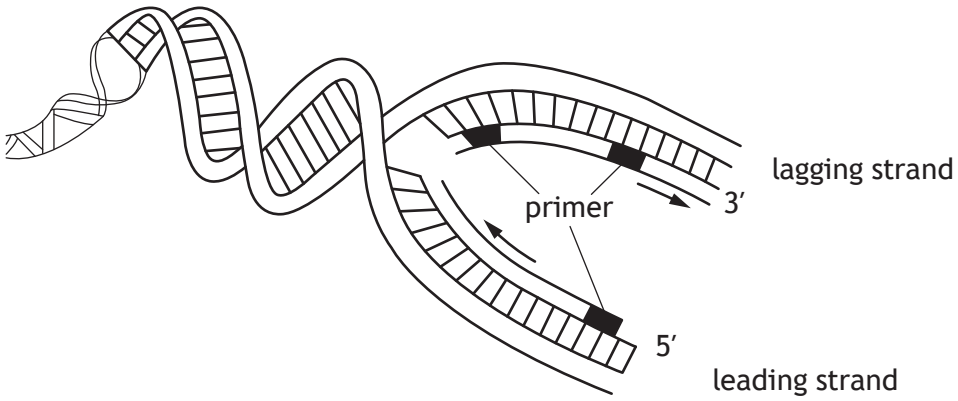
(ii) Use data from the graph to describe the changes that occur in the white blood cell count from days 0 to 32. **2**

(iii) Use data from the graph to predict the white blood cell count on day 34. **1**

_____ cells/microlitre

[Turn over

2. (a) The diagram shows the replication of DNA in a cell.



(i) Suggest what the arrows on the diagram indicate. **1**

(ii) Explain why the lagging strand is replicated in fragments. **1**

(iii) Name the enzyme that joins fragments together in the lagging strand. **1**

(b) DNA is involved in gene expression to allow cells to synthesise proteins.
The enzyme RNA polymerase is required in gene expression.
Describe the role of this enzyme in the synthesis of a protein. **2**

2. (continued)

(c) Transfer RNA (tRNA) is involved in gene expression.

Describe the structure of tRNA. **3**

[Turn over

3. Haemoglobin is a protein that transports oxygen in the blood.

Individuals with sickle cell anaemia have two copies of a single gene mutation that results in the production of the protein haemoglobin S. Although haemoglobin S is the same length as haemoglobin it folds into a different shape.

DNA sequences from the same region of the gene for haemoglobin and haemoglobin S are shown.

haemoglobin	**T G A G G A C T C**
haemoglobin S	**T G A G C A C T C**

(a) State the sequence of bases in the mRNA transcript that would be formed from the haemoglobin S sequence shown. **1**

(b) (i) Using information from the diagram, name the type of single gene mutation that causes sickle cell anaemia. **1**

(ii) Explain why the changed sequence of bases in the gene for haemoglobin S codes for a different shaped protein. **1**

(c) Suggest why individuals with sickle cell anaemia have increased blood lactate concentration and are more likely to experience muscle fatigue. **1**

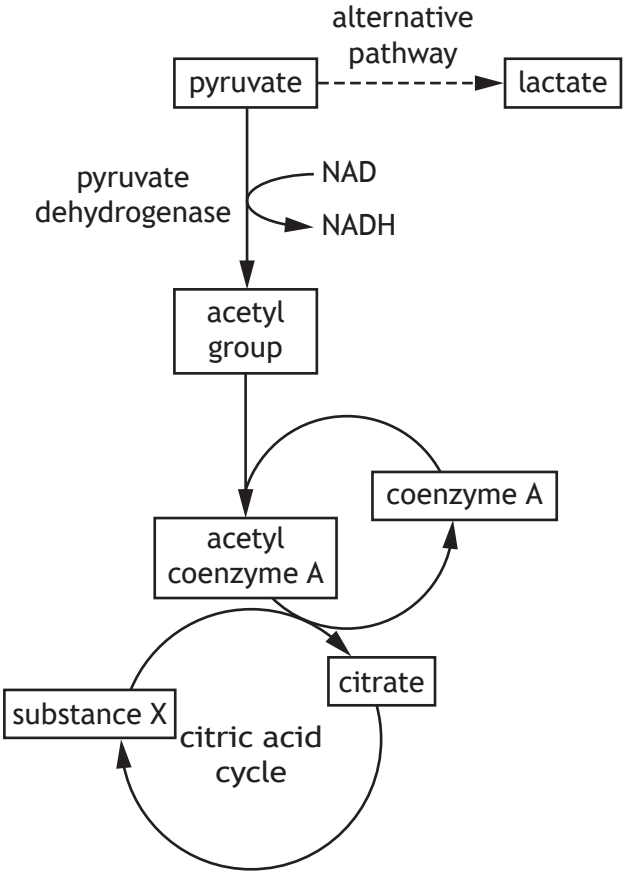
(d) In a country, 670 000 babies were born and tested for haemoglobin S.

Calculate the number of babies with haemoglobin S, if 1 in every 80 babies tests positive. **1**

Space for calculation

4. The diagram shows part of the metabolic pathway of cellular respiration.

The pathway includes the enzyme pyruvate dehydrogenase, which catalyses the conversion of pyruvate to an acetyl group.



(a) State the location of the production of pyruvate in a cell. **1**

_____

(b) Describe the role of pyruvate dehydrogenase in the production of NADH. **1**

_____

_____

(c) Name substance X and describe its role in the synthesis of citrate. **2**

Substance X _____

Role _____

[Turn over

4. (continued)

(d) Pyruvate dehydrogenase deficiency (PDHD) is caused by a mutation, which results in individuals with non-functional pyruvate dehydrogenase. This mutation affects cellular respiration.

Describe the effect of PDHD on ATP and lactate production. **2**

ATP production _____

Lactate production _____

(e) One type of PDHD is caused by an autosomal recessive allele.

A couple who are both unaffected by the condition have a child with PDHD.

(i) Pharmacogenetics could be used in the treatment of this child.

Give an advantage of using pharmacogenetics. **1**

_____

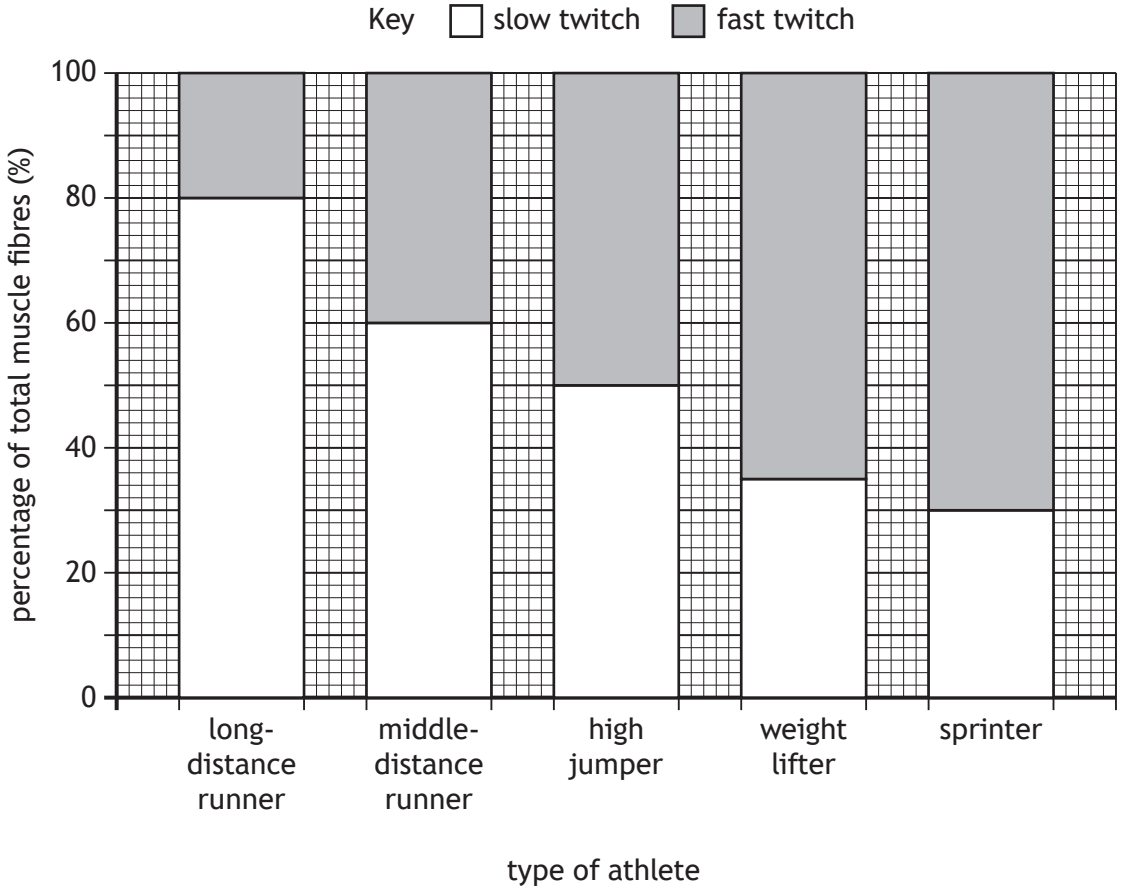
(ii) The couple are expecting a second child.

Calculate the percentage chance of this child having PDHD. **1**

Space for calculation

_____ %

5. The graph shows the percentage of slow twitch and fast twitch muscle fibres in different types of athletes.



(a) (i) Express, as a simple whole number ratio, the percentage of fast twitch muscle fibres in the long-distance runner, high jumper and weightlifter. **1**

Space for calculation

_____ : _____ : _____

long-distance runner | high jumper | weightlifter

5. (a) (continued)

(ii) Calculate how many times greater the percentage of fast twitch fibres in the muscles of the sprinter is compared to the muscles of the long-distance runner. **1**

Space for calculation

_____ times greater

(b) (i) State a feature of fast twitch muscle fibres that makes them useful for sprinting. **1**

_____

_____

(ii) Slow twitch muscle fibres have a higher concentration of myoglobin than fast twitch muscle fibres.

State the function of myoglobin and explain why slow twitch muscle fibres need to have a high concentration of myoglobin. **2**

Function _____

Explanation _____

_____

(iii) Describe one other structural difference between slow twitch and fast twitch muscle fibres. **1**

_____

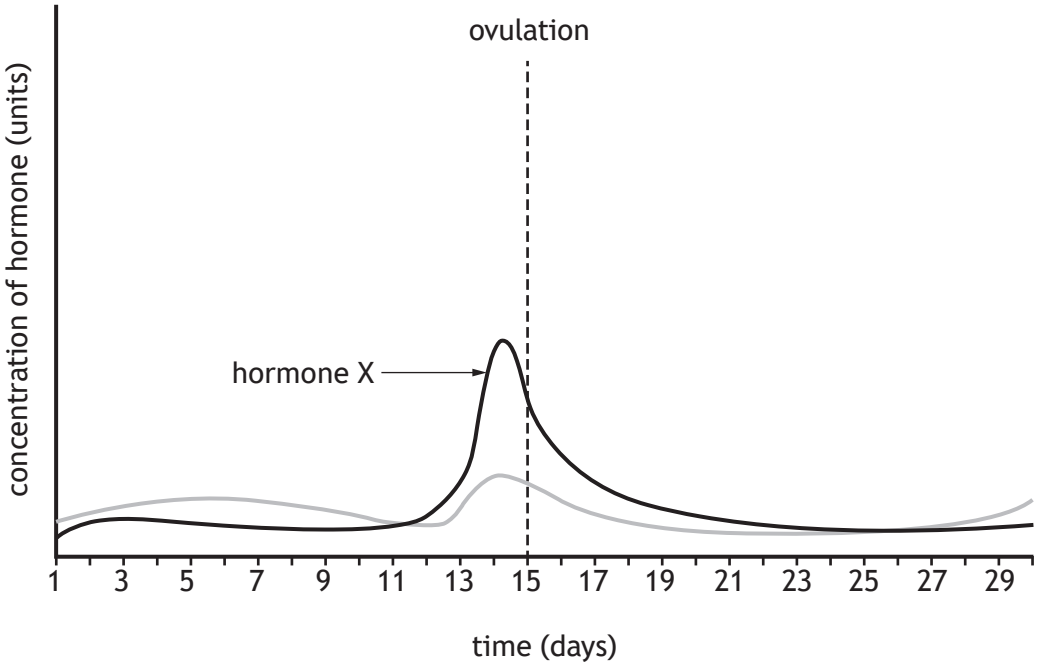
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[Turn over

6. The menstrual cycle is hormonally controlled.

(a) **Graph 1** shows the blood concentration of hormones produced by the pituitary gland in a female during one menstrual cycle.

Graph 1



(i) Name hormone X. **1**

_____

(ii) Body temperature and cervical mucus consistency change throughout the menstrual cycle.

Describe the difference in a female's body temperature and cervical mucus consistency on day 16 compared to day 3. **2**

Body temperature _____

_____

Cervical mucus consistency _____

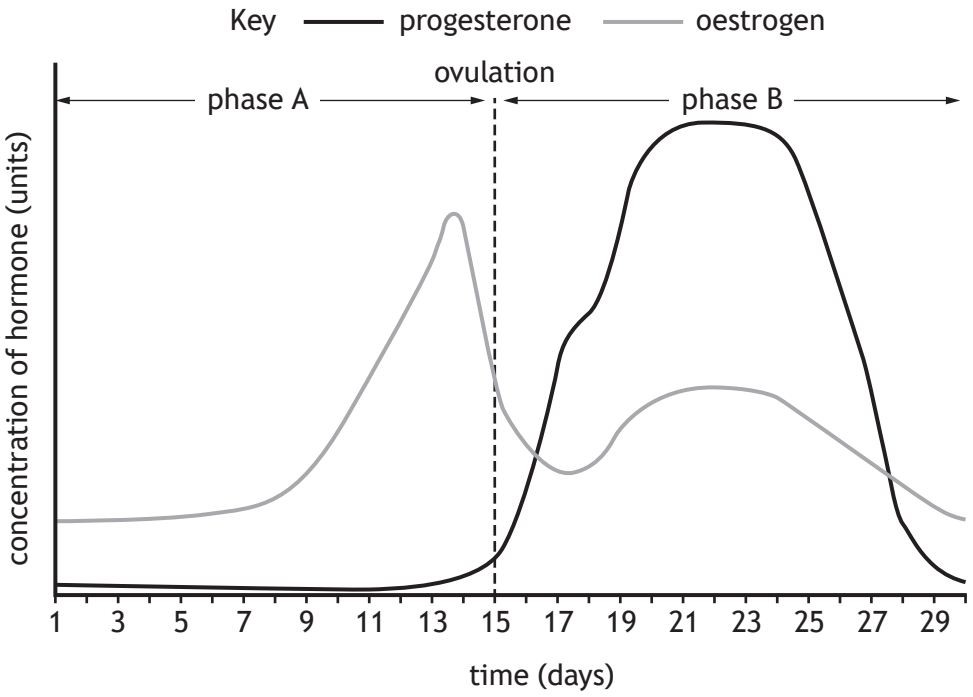
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MARKS

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(b) **Graph 2** shows the blood concentration of hormones produced by the ovaries in a female during one menstrual cycle.

Graph 2



(i) Name phase A and phase B. **2**

Phase A _____

Phase B _____

(ii) Describe the events that lead to an increase in progesterone levels during phase B. **2**

_____

_____

_____

(iii) In the next cycle, the female became pregnant.

Suggest one way in which the hormone concentration for this cycle would differ from Graph 2. **1**

_____

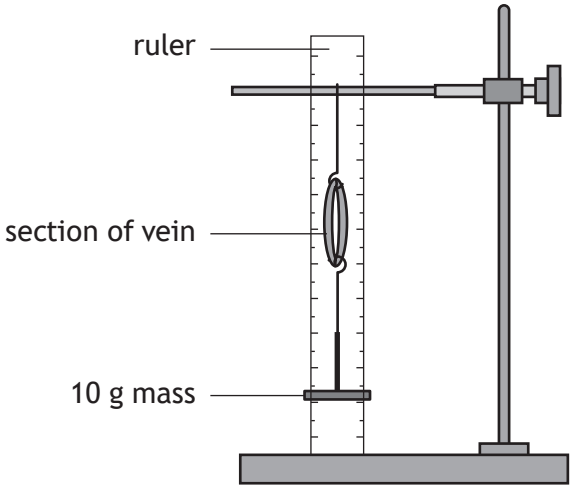
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[Turn over

MARKS DO NOT WRITE IN THIS MARGIN

7. An investigation was set up to compare the elasticity of arteries and veins.

A section of vein was suspended from a clamp stand as shown in the diagram.



The length of the vein was measured.

A mass of 10 g was attached to the vein and the length of the vein was remeasured using the ruler.

This was repeated with masses of 20, 30 and 40 g. The investigation was then repeated with a section of artery.

The results are shown in **Table 1**.

Table 1

Mass (g)	Vein		Artery	
	Length (mm)	Change in length (%)	Length (mm)	Change in length (%)
0	25	0	20	0
10	27	8	22	10
20	29	16	24	20
30	31	24	27	
40	31	24	30	50

(a) Calculate the percentage change in length for the artery with a 30 g mass attached. **1**

Space for calculation

_____ %

7. (continued)

MARKS | DO NOT WRITE IN THIS MARGIN

(b) (i) Explain why it was necessary to calculate the percentage change in the length of each blood vessel. **1**

(ii) Describe how the reliability of these results could be improved. **1**

(c) At the end of the investigation the masses were removed, and the lengths of the artery and vein were remeasured.

The values were recorded in **Table 2**.

Table 2

Blood vessel	Length (mm)	
	Start of investigation	End of investigation
Vein	25	30
Artery	20	20

(i) Suggest a difference between the structure of veins and arteries that leads to the difference in results shown in **Table 2**. **1**

(ii) Explain why it is important that the walls of arteries can stretch and recoil. **1**

(d) Describe a feature of artery walls that allows them to control blood flow. **1**

[Turn over

8. Studies have shown that the percentage of infertile males in the UK is increasing. One of the main causes of male infertility is a low sperm count.

Mice were used as a model organism in an investigation into the effect of regular alcohol consumption on sperm count over a 120-day period.

140 mice were divided into two groups.

Group 1 received water.

Group 2 received water and a daily dose of alcohol.

Semen samples were collected from the mice throughout the investigation and sperm counts recorded.

The results of the investigation are shown in the table.

Day semen sample taken	Average sperm count (million/cm^3)	
	Group 1	Group 2
0	45.2	45.0
30	44.7	42.2
60	45.3	40.2
90	44.5	38.2
120	45.0	36.2

(a) (i) State the purpose of including group 1 in this investigation. **1**

_____

(ii) Identify two variables, not already mentioned, that would need to be controlled to make the investigation valid. **2**

1. _____

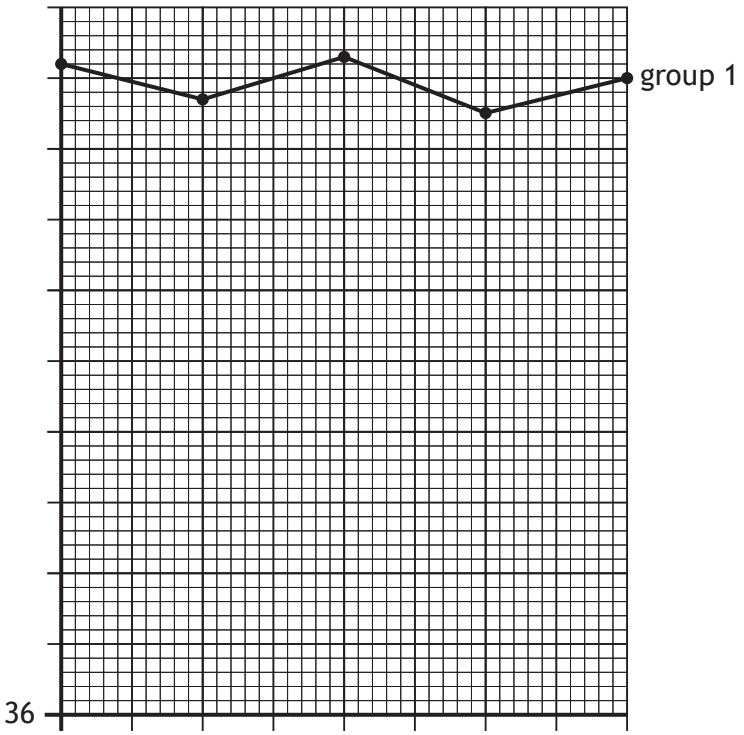
2. _____

8. (continued)

MARKS DO NOT WRITE IN THIS MARGIN

(b) (i) On the grid, complete the line graph to show the results of the investigation for **group 2**. The results of group 1 are shown. **2**

(Additional graph paper, if required, can be found on *page 28*.)



(ii) State the conclusion that can be made from these results. **1**

_____

_____

(iii) If the study continued for another 15 days, predict the average sperm count of the mice in group 2. **1**

_____ million/cm^3

(c) Artificial insemination can be used if a male has a low sperm count.

Describe a feature of artificial insemination that increases the chances of fertilisation. **1**

_____

_____

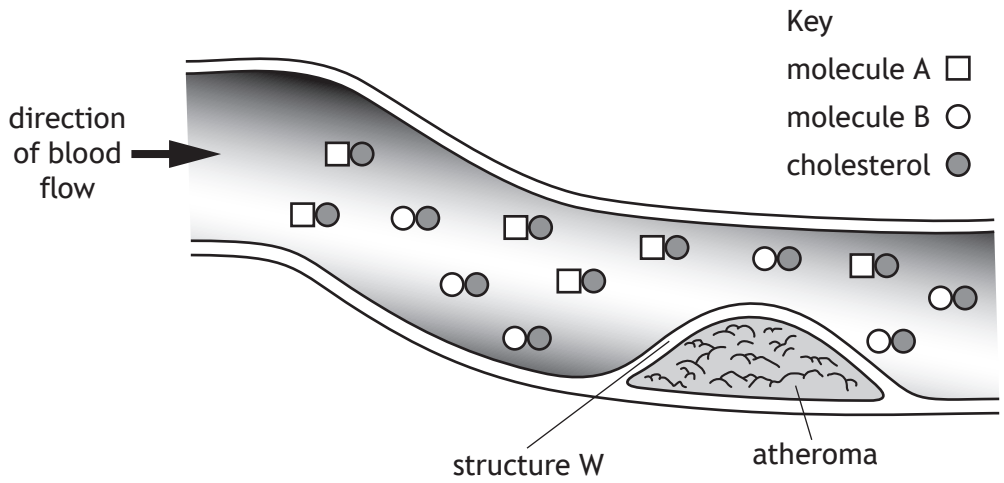
[Turn over

9. Cholesterol is essential for myelin sheath regeneration and maintenance.

(a) (i) State another function of cholesterol in the body. **1**

(ii) Describe the importance of the myelin sheath. **1**

(b) The diagram shows the transport of cholesterol through an artery by molecules A and B.



Molecule A transports cholesterol to body cells.

Molecule B prevents accumulation of cholesterol in the blood.

(i) Identify molecule B and describe how it prevents accumulation of cholesterol in the body. **2**

Molecule B _____

Description _____

9. (b) (continued)

(ii) Describe how cholesterol uptake is limited in body cells with sufficient cholesterol. **1**

(iii) Molecule A can also deposit cholesterol as part of the fatty material in an atheroma under structure W.

Name structure W and another substance that can be found in an atheroma. **2**

Structure W _____

Other substance _____

(c) If an atheroma ruptures, clotting factors lead to the conversion of the enzyme prothrombin to its active form thrombin.

Hirudin is a molecule that acts as a competitive inhibitor of thrombin.

(i) Suggest how the inhibitor hirudin prevents the formation of a clot. **2**

(ii) Suggest how the effect of hirudin on the clotting process could be reversed. **1**

[Turn over

10. Changes in an individual's heart rate and duration of stages in the cardiac cycle at rest and during exercise are shown in the table.

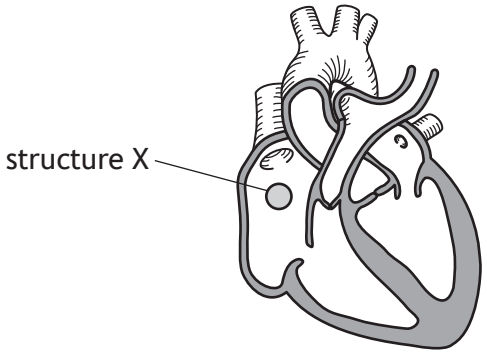
		Duration (s)		
Activity	**Heart rate (bpm)**	**Atrial systole**	**Ventricular systole**	**Diastole**
Rest	60	0.18	0.22	0.60
Exercise	120	0.10	0.14	

(a) Calculate the duration of diastole during exercise. **1**

Space for calculation

_____ s

(b) Atrial systole is brought about by structure X.



(i) Name structure X. **1**

_____

(ii) A stethoscope is used to listen to heart sounds during the cardiac cycle.

Describe the events during the cardiac cycle in the heart that are responsible for these sounds. **2**

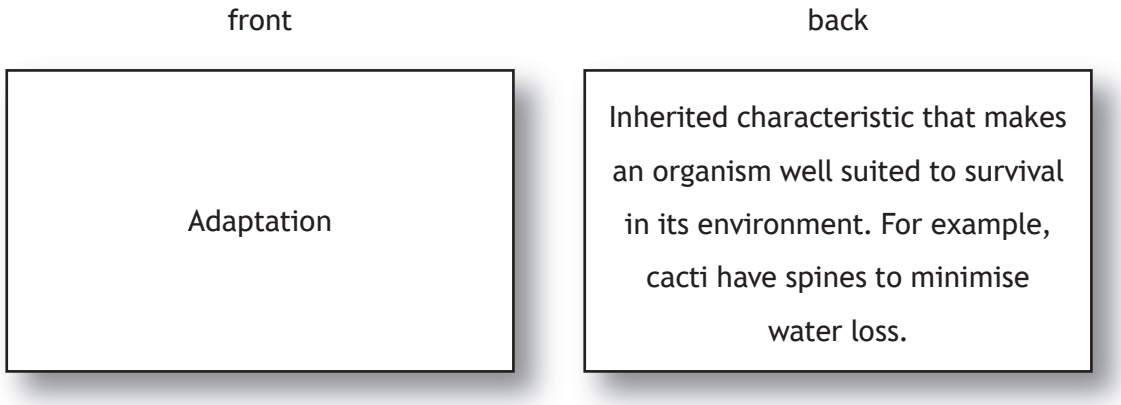
_____

_____

_____

11. Flashcards are often used in the rehearsal of information during revision.

A student created flashcards to revise for their N5 Biology exam. The flashcards had a biological term on the front and an explanation on the back. One of the flashcards is shown in the diagram.



(a) (i) The back of this flashcard shows another method that improves the transfer of information to the long-term memory.

Name this method. **1**

_____

(ii) Explain why this method leads to improved information retention compared to rehearsal of the information. **1**

_____

_____

(b) State what would happen to the information if it was not transferred from the short-term memory to long-term memory. **1**

_____

(c) The short-term memory has a limited capacity.

Name the method that increases the capacity of short-term memory and suggest how this method can increase the capacity. **2**

Name _____

Suggestion _____

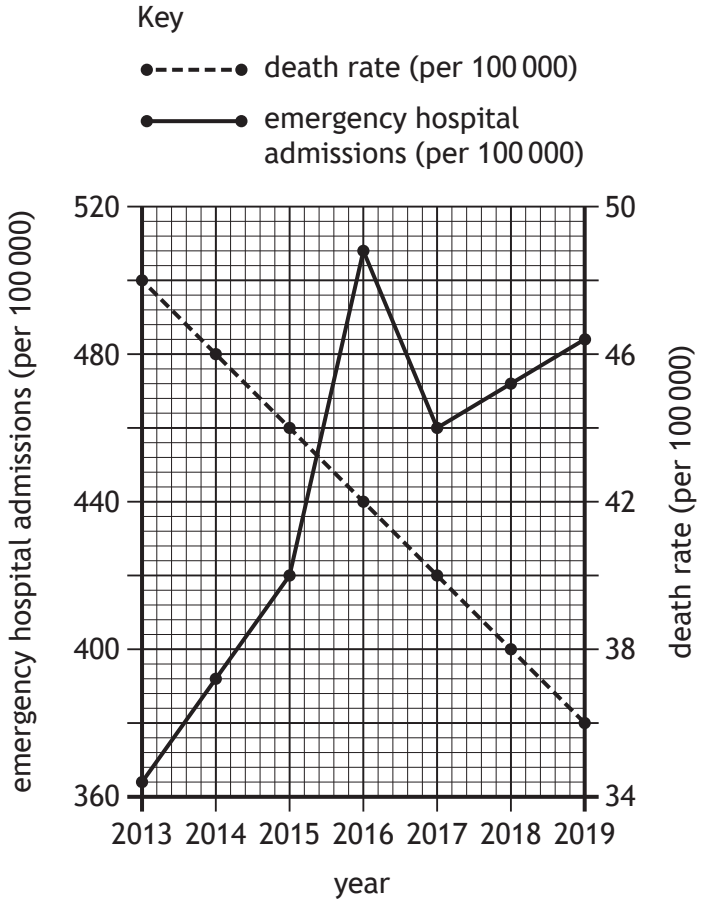
_____

[Turn over

12. A species of bacterium can be found in the upper respiratory tract of individuals where it is usually harmless and causes no symptoms.

If an individual contracts influenza, the normally harmless bacterium can move to the lower respiratory tract and cause pneumonia.

The graph shows the number of emergency hospital admissions and death rate for pneumonia in a country from 2013 to 2019.



(a) (i) Calculate the annual average increase in emergency hospital admissions for pneumonia over the six-year period from 2013 to 2019. **1**

Space for calculation

_____ per 100 000/year

(ii) State the death rate from pneumonia when the number of emergency hospital admissions was 420 per 100 000. **1**

_____ per 100 000

12 (a) (continued)

(iii) Calculate the difference between emergency hospital admissions and death rates for pneumonia in 2016. **1**

Space for calculation

_____ per 100 000

(iv) Using the information provided, suggest a reason for the sharp increase in the number of emergency hospital admissions for pneumonia from 2015 to 2016. **1**

_____

_____

(v) In 2017 the population of the country was 50 million and the total number of deaths in the country was 500 000.

Using data from the graph, calculate the percentage of the total number of deaths that were due to pneumonia. **1**

Space for calculation

_____ %

(b) Individuals in this country are offered the influenza vaccine.

(i) Name a source of antigens that can be used to produce a vaccine. **1**

_____

(ii) Explain why an adjuvant is added to a vaccine. **1**

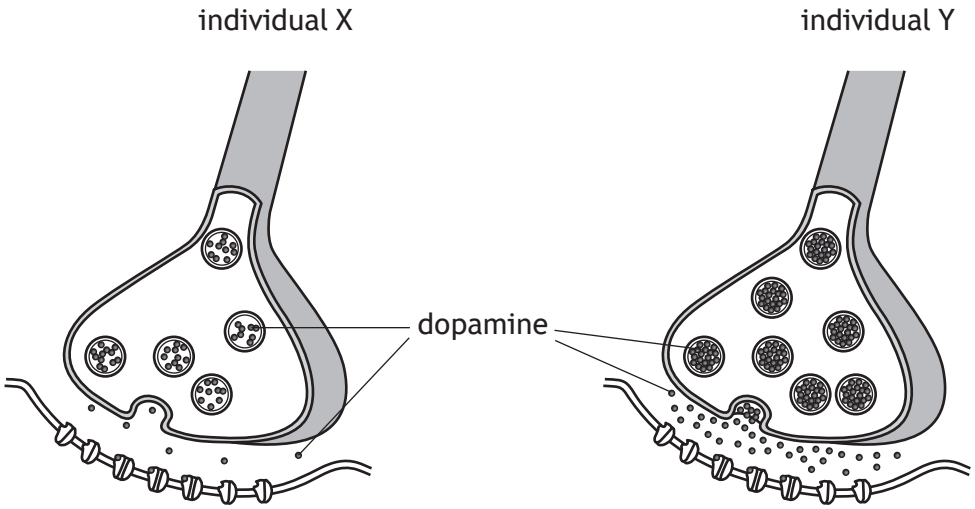
_____

_____

[Turn over

13. The diagram represents synapses from the brains of two individuals.

Individual X has Parkinson's disease and individual Y does not have Parkinson's disease.



(a) (i) Using information in the diagram, describe an effect of Parkinson's disease on the synapses of individual X. **1**

(ii) State an effect of the neurotransmitter dopamine. **1**

13. (continued)

(b) Parkinson's disease can be treated with a drug that mimics the action of dopamine at a synapse.

Name the type of drug that would mimic the action of dopamine. **1**

_____

(c) Describe how the release of neurotransmitters at a synapse triggers an impulse in the postsynaptic neuron. **2**

_____

_____

_____

_____

[Turn over

14. Attempt **either** A **or** B.

Write your answer in the space below and on *page 27*.

A Write notes on the role of B and T lymphocytes in the specific immune response to a pathogen. **8**

OR

B Write notes on the divisions of the nervous system to include the central nervous system and the peripheral nervous system. **8**

You may use labelled diagrams where appropriate.

National
Qualifications
2025

X840/76/12

Human Biology
Paper 1 — Multiple choice

TUESDAY, 27 MAY
9:00 AM – 9:40 AM

Total marks — 25

Attempt ALL questions.

You may use a calculator.

Instructions for the completion of Paper 1 are given on *page 02* of your answer booklet X840/76/02.

Record your answers on the answer grid on *page 03* of your answer booklet.

Space for rough work is provided at the end of this booklet.

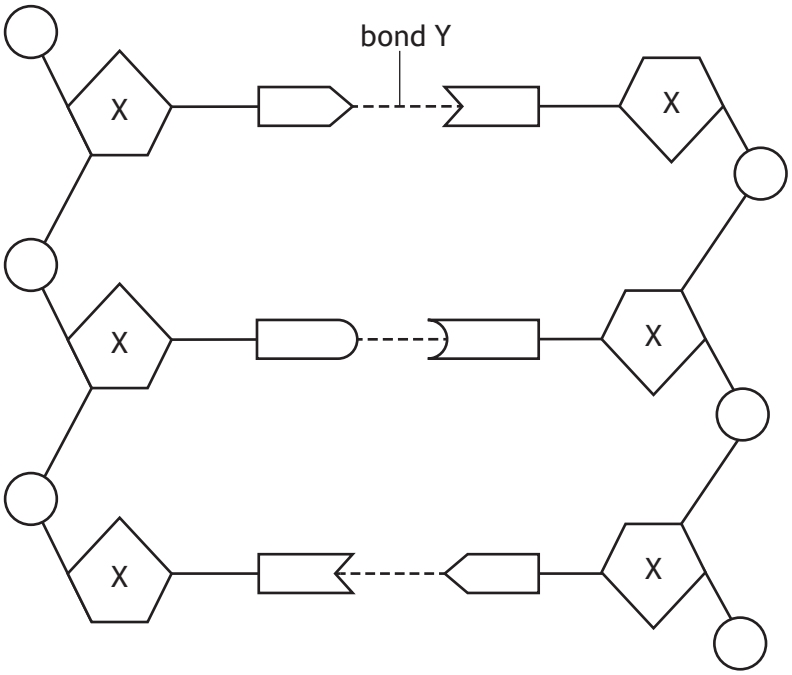
Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

Total marks — 25

Attempt ALL questions

1. The diagram shows part of a DNA molecule.



Which row in the table identifies sugar X and bond Y?

	Sugar X	**Bond Y**
A	deoxyribose	hydrogen
B	ribose	peptide
C	deoxyribose	peptide
D	ribose	hydrogen

2. The role of DNA polymerase is to

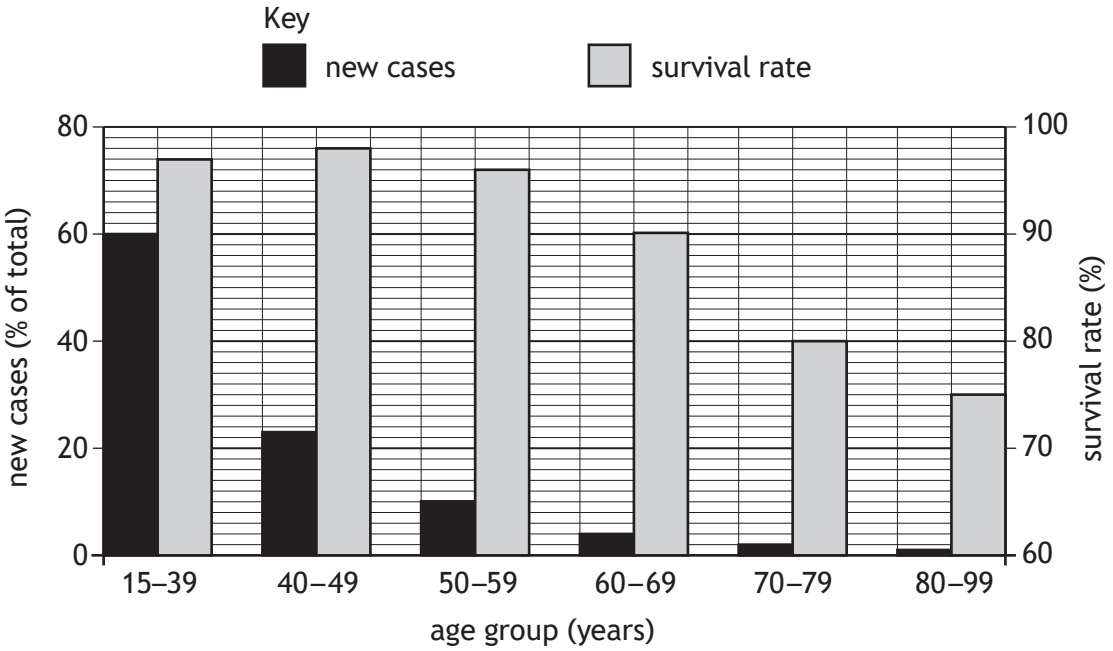
A remove introns from a primary mRNA transcript

B add DNA nucleotides to a new DNA strand being formed

C synthesise a primary mRNA transcript during transcription

D join DNA fragments together during replication of the lagging strand.

3. The graph shows the percentage of new cases of testicular cancer and the percentage survival rate across a range of age groups.



Which of the following is correct for this data?

A The ratio of survival rate to new cases for age group 50–59 is 48:5.

B The difference in survival rate between age groups 60–69 and 70–79 is 20%.

C The percentage of new cases at age group 40–49 is double that of age group 50–59.

D The age group with the highest percentage of new cases has the highest survival rate.

4. Which types of RNA can be found in the cytoplasm?

A mRNA only

B tRNA and rRNA only

C mRNA and rRNA only

D mRNA, tRNA and rRNA

5. The nucleic acids involved in gene expression are listed.

1. DNA
2. mRNA
3. tRNA
4. rRNA

Which nucleic acids are involved in the transcription stage of gene expression?

A 1 only

B 1 and 2 only

C 1, 2 and 3 only

D 1, 2, 3 and 4

6. PCR is a reaction that amplifies a section of DNA and involves several temperature changes per cycle.

The reaction stays at each temperature for 2 minutes.

Calculate the total number of minutes taken to produce 128 copies of one DNA section.

A 14

B 21

C 42

D 64

7. Pharmacogenetics is the use of

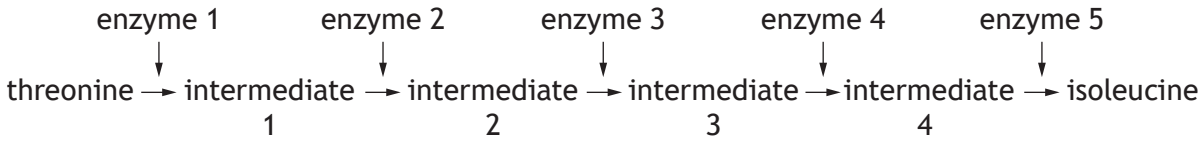
A gene sequences to assess the likelihood of passing on a disease

B computer programs to assess the likelihood of passing on a disease

C gene sequences to select appropriate drug and dosage to treat a disease

D computer programs to identify base sequences of known genetic diseases.

8. The diagram shows the metabolic pathway that converts the amino acid threonine into isoleucine.



An increased production of isoleucine would be caused by

A isoleucine acting as a competitive inhibitor of enzyme 2

B negative feedback inhibition of threonine by isoleucine

C an increased concentration of isoleucine

D an increased concentration of threonine.

9. During vigorous exercise not enough oxygen is delivered to cells to support the electron transport chain.

To allow glycolysis to continue, pyruvate is converted to

A acetyl, generating NAD

B lactate, generating NAD

C acetyl, generating NADH

D lactate, generating NADH.

10. A typical slow-twitch muscle fibre that took 100 milliseconds (ms) to complete one contraction was compared to a fast-twitch muscle fibre that took 75 ms.

Calculate how many more times a fast-twitch muscle fibre can contract in one minute compared to a slow-twitch muscle fibre.

(1000 ms = 1 second)

A 25

B 200

C 800

D 1500

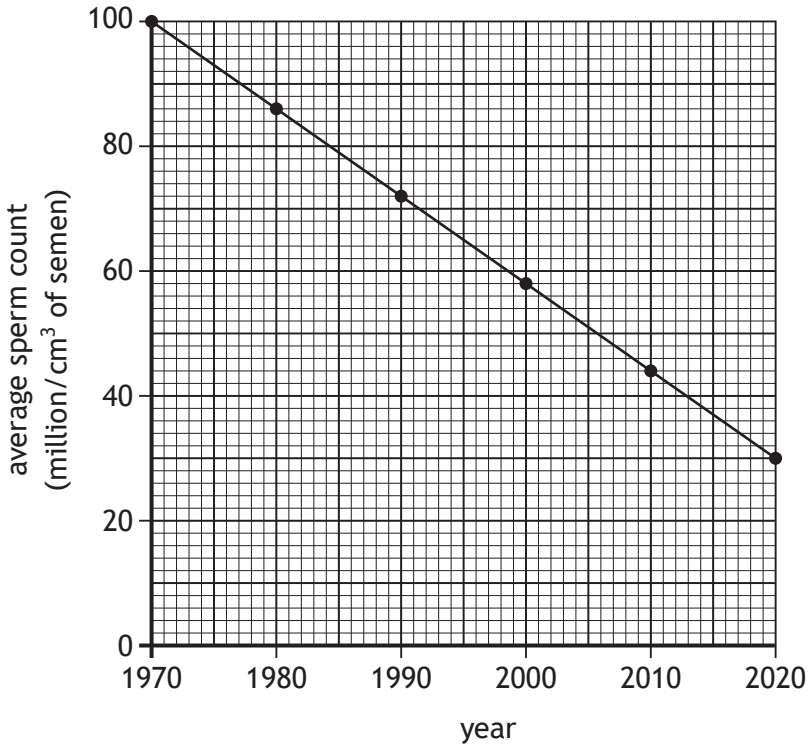
11. A sample of muscle fibres from the thigh of an Olympic sprinter was analysed and compared to a sample from an elite long-distance runner.

Which row in the table shows the expected results?

	Myoglobin concentration	**Glycogen concentration**	**Number of mitochondria**
A	lower	higher	lower
B	higher	lower	higher
C	lower	higher	higher
D	higher	lower	lower

[Turn over

12. The graph shows the changes in the average sperm count of males in a country.



Which of the following statements is correct?

A The average sperm count decreased by 14% between 1980 and 1990.

B The average sperm count decreased by 60 million/cm^3 between 1970 and 2010.

C The average sperm count will be 16 million/cm^3 in 2025 if the current trend continues.

D The average sperm count has decreased by 1.4 million/cm^3/year between 1970 and 2020.

13. A study was carried out to investigate the effects of different fertility drugs on sperm count.

12 000 males participated and they were randomly split into three groups of 4000.

Each group was given a different fertility drug.

It was discovered that many participants had also taken additional fertility supplements during the study.

From the information given, it was expected that the study would be

A valid and reliable

B valid but not reliable

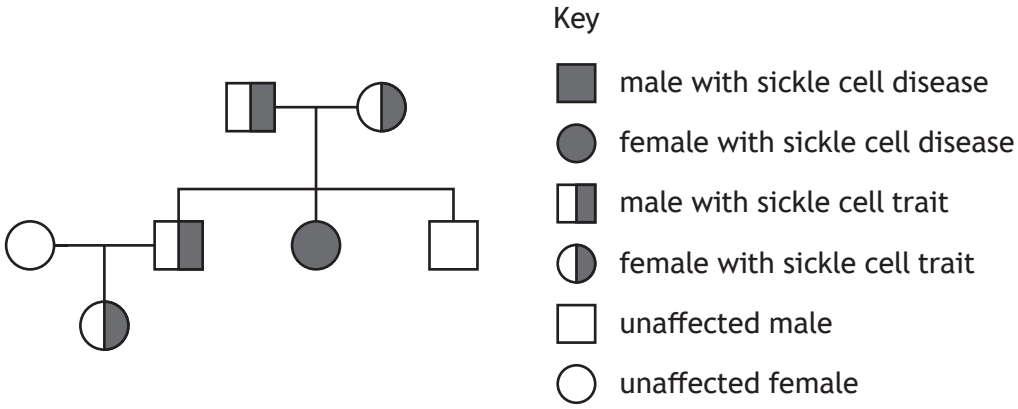
C not valid but reliable

D not valid and not reliable.

14. Which row in the table identifies a type of routine ultrasound scan and the stage in pregnancy when it is normally carried out?

	Scan	Stage in pregnancy (weeks)
A	anomaly	8–14
B	dating	14–18
C	anomaly	18–20
D	dating	18–20

15. The diagram shows the inheritance of sickle cell in three generations of a family.



Identify the type of inheritance for sickle cell.

A Autosomal recessive

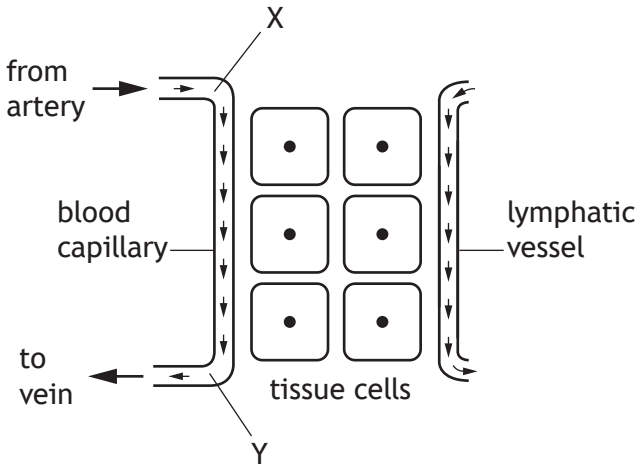
B Sex-linked recessive

C Autosomal dominant

D Incomplete dominance

[Turn over

16. The diagram shows a blood capillary, tissue cells and a lymphatic vessel.



Which of the following statements explains the difference in blood pressure between points X and Y?

A Blood pressure is higher at Y than X as plasma enters the blood between X and Y.

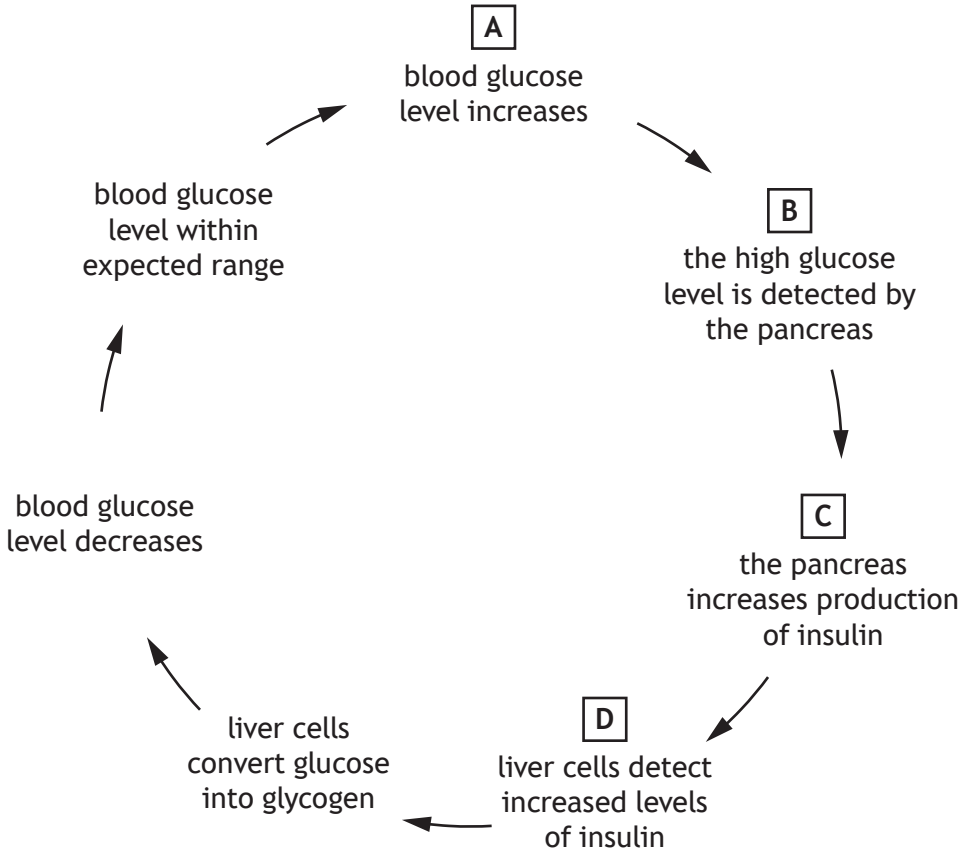
B Blood pressure is higher at X than Y as plasma leaves the blood between X and Y.

C Blood pressure is lower at X than Y as tissue fluid leaves the blood between X and Y.

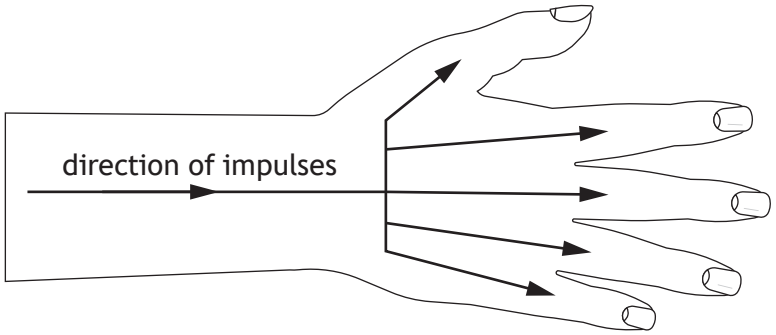
D Blood pressure is lower at Y than X as lymph enters the blood between X and Y.

17. The diagram shows some stages that occur during the control of blood glucose levels.

Identify the stage that may **not** occur in an individual with type 2 diabetes.



18. The diagram shows the direction of impulse transmission through a hand when the fingers are being used to type on a keyboard.



The fine motor control of the fingers occurs because of

A summation of weak stimuli

B diverging neural pathways

C converging neural pathways

D reverberating neural pathways.

19. The table shows the number of individuals of different age groups diagnosed with dementia in a country over 4 years.

	Number of individuals diagnosed with dementia		
Age group / Year	**51–65**	**66–80**	**Over 80**
2016	4421	5210	3256
2017	4821	4925	3111
2018	5012	5222	2965
2019	5123	5600	2709

Which of the following statements is correct?

A The total number of individuals diagnosed with dementia increased every year.

B More individuals were diagnosed with dementia in 2016 than in any other year.

C More individuals aged 51–80 were diagnosed with dementia in 2019 compared to 2016.

D More individuals were diagnosed with dementia between the ages of 51–65 than other age categories.

[Turn over

20. The table shows the concentration of a drug measured in the blood of an individual every hour for 4 hours after the drug was taken.

Time since taken (hours)	Concentration of drug in blood (mg/L)
0	36
1	34
2	30
3	24
4	16

Predict the concentration of the drug remaining in the blood at 5 hours.

A 6

B 8

C 10

D 16

21. The role of mast cells is to

A engulf pathogens

B release histamine

C produce antibodies

D cause vasoconstriction.

22. An individual's white blood cell count was measured during the course of an infection.

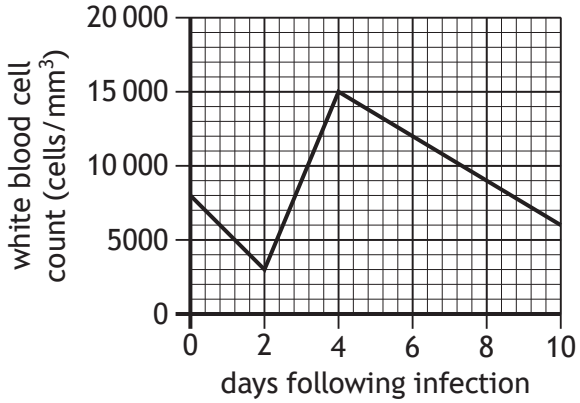
Initially the white blood cell count was within the normal range of 5000–10 000 cells/mm^3.

Two days later it had increased to 15 000 cells/mm^3.

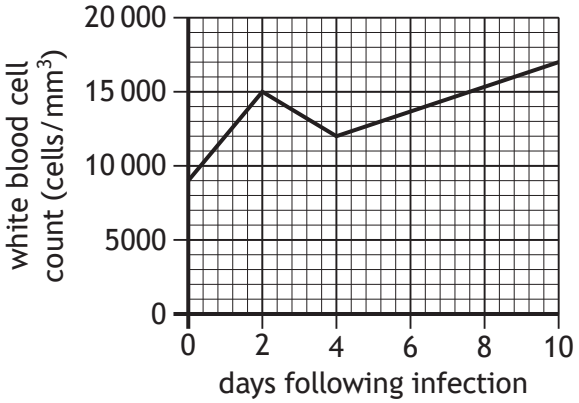
After four days it fell to 3000 cells/mm^3 before gradually increasing to within the normal range over the next six days.

Which graph shows these changes in the individual's white blood cell count?

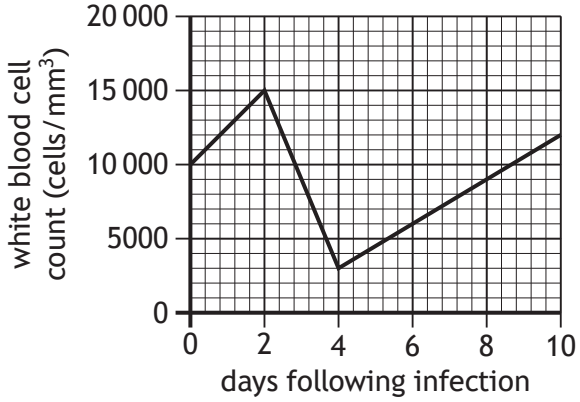
A



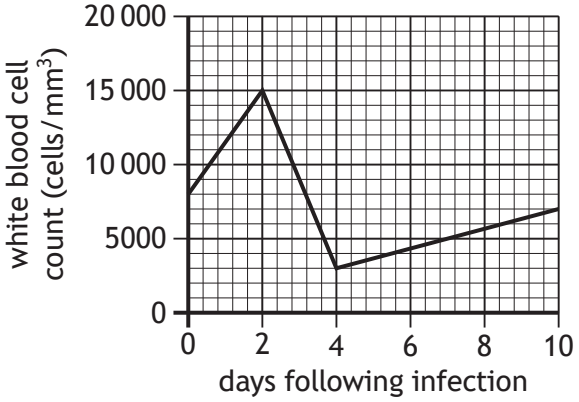
B



C

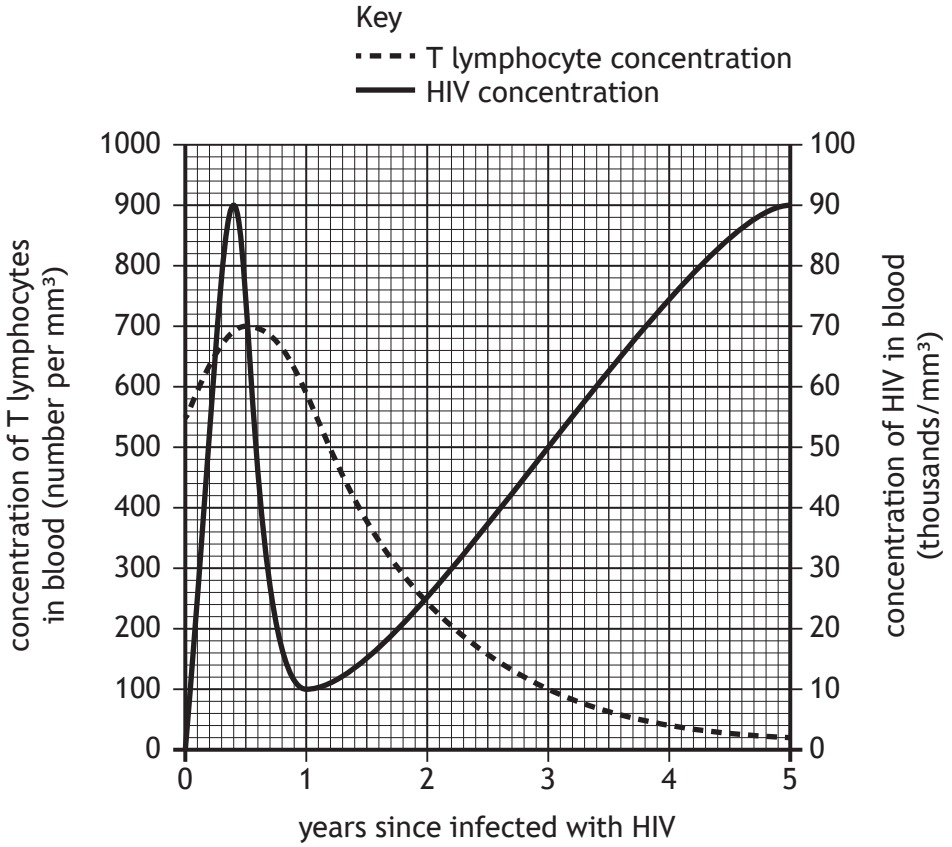


D



[Turn over

23. The graph shows concentrations of T lymphocytes and HIV in the blood of an individual following infection with HIV.



Which row in the table shows the concentration of T lymphocytes and HIV in the blood of the individual, three years since they were infected with HIV?

	Concentration of T lymphocytes in blood (number per mm^3)	**Concentration of HIV in blood (thousands/mm^3)**
A	100	50
B	500	10
C	100	500
D	10	50

24. A patient was found to have a white blood cell count of 7000 cells/cm^3 of blood.

The ratio of phagocytes to lymphocytes was 7:3.

Which row in the table shows the number of phagocytes and lymphocytes in a 10 cm^3 sample of this patient's blood?

	Phagocytes	**Lymphocytes**
A	4900	2100
B	2100	4900
C	49 000	21 000
D	21 000	49 000

25. Influenza is a public health problem every winter.

This is due to changes that occur in the

A antigens of the virus

B vaccine given to patients

C antibodies produced by the body

D memory cells present in the blood.

[END OF QUESTION PAPER]

FOR OFFICIAL USE

National Qualifications 2025

Mark

X840/76/01

Human Biology Paper 2

TUESDAY, 27 MAY
10:10 AM – 12:30 PM

X8407601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day | Month | Year

Scottish candidate number

Total marks — 95

Attempt ALL questions.

You may use a calculator.

Question 15 contains a choice.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. Score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

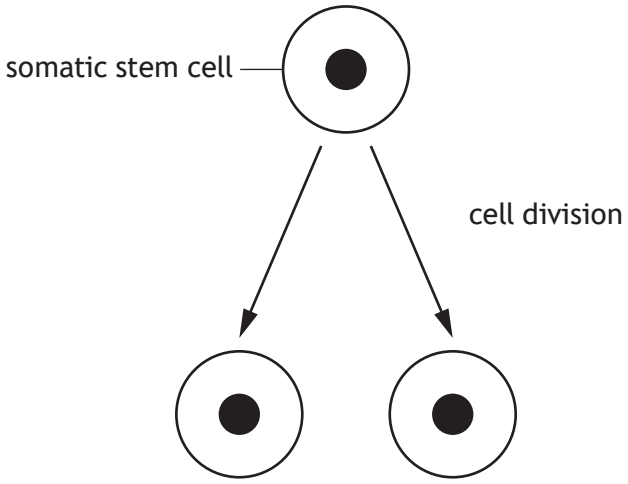
X840760101

Total marks — 95

Attempt ALL questions

Question 15 contains a choice

1. The diagram shows a somatic stem cell dividing.



(a) (i) Name the type of cell division shown in the diagram. **1**

_____

(ii) Germline stem cells undergo a different type of cell division to produce gametes.

State one location of the production of gametes. **1**

_____

(b) Somatic stem cells can be found in the cornea of the eye.

Describe how these stem cells could differentiate into specialised cells such as corneal cells. **1**

_____

_____

_____

1. (continued)

(c) Some conditions can cause severe damage to the cornea and somatic stem cells found in the eye. Stem cell transplants can be carried out to replace the damaged somatic stem cells with healthy stem cells.

The table shows the number of stem cell donors and the number of stem cell transplant recipients in one year.

Age group (years)	Number of stem cell donors	Number of stem cell transplant recipients
0–17	5	30
18–34	24	290
35–49	88	330
50–59	192	373
60–69	384	590
70–79	700	1021
80–99	394	656

(i) Use data from the table to describe the changes that occur in the number of stem cell donors as age increases from 0 to 99 years. **2**

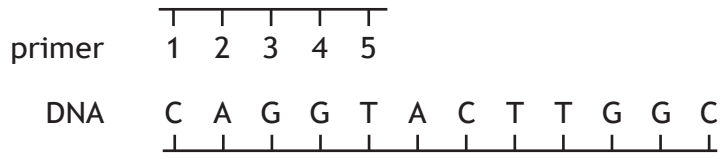
(ii) Using information in the table, suggest a benefit of using a patient's own stem cells for this treatment. **1**

(d) Corneal repair is a therapeutic use of stem cells.

Describe one use of stem cells in research. **1**

2. The polymerase chain reaction (PCR) is used to amplify DNA.

The diagram shows a primer attached to a single strand of DNA.



(a) Name base 3. **1**

_____

(b) PCR requires two primers.

Explain why the two primers have different base sequences. **2**

_____

_____

_____

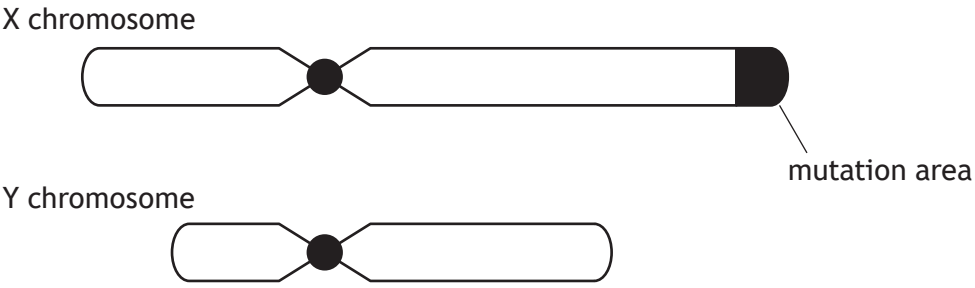
_____

(c) State a temperature used to allow primers to bind during PCR. **1**

_____ °C

3. A chromosome mutation on the X chromosome can result in the blood clotting disorder haemophilia A.

The diagram shows two chromosomes from an individual with haemophilia A.



(a) (i) Name the type of inheritance for haemophilia A. **1**

_____

(ii) In a population, 6300 males have haemophilia A. This represents 75% of the total number of individuals with this condition.

Calculate how many females have haemophilia A in this population. **1**

Space for calculation

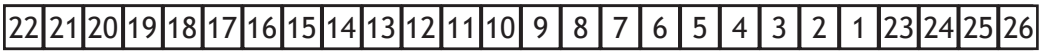
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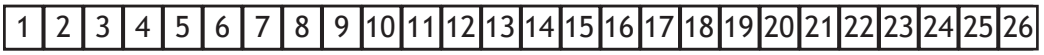
[Turn over

3. (continued)

(b) The diagram shows the order of exons in the mutation area of the X chromosome from an individual with haemophilia A and an individual without haemophilia A.

with haemophilia A





without haemophilia A

(i) Name the type of chromosome mutation that causes haemophilia A. **1**

(ii) State the function of an exon. **1**

(iii) During the expression of a gene for a blood clotting protein, mRNA is transcribed from exons 1 to 26.

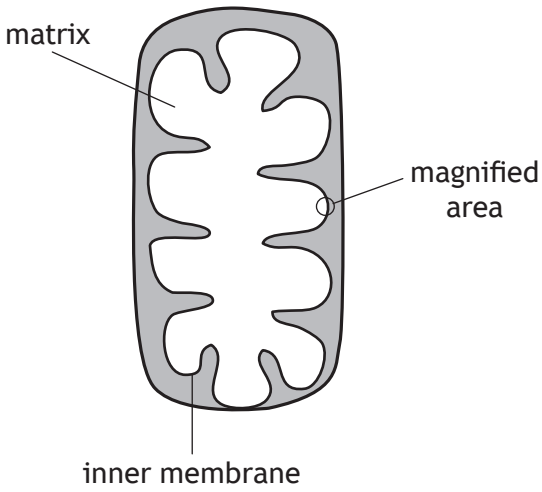
Explain why this protein is non-functional in individuals with haemophilia A. **2**

(c) Describe how alternative RNA splicing leads to different proteins being expressed from one gene. **1**

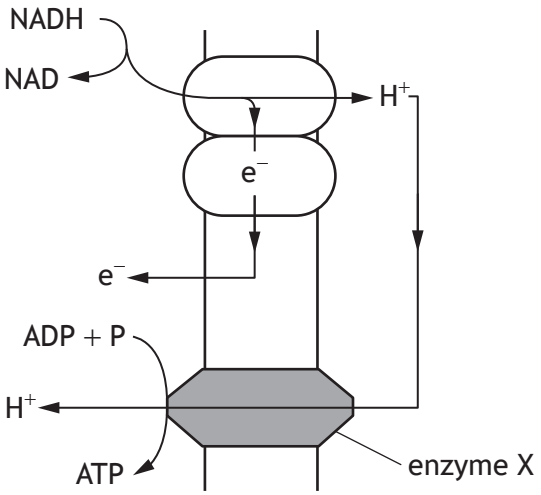
4. The diagram shows a mitochondrion and a magnified area.

Some of the reactions involved in an electron transport chain are shown in the magnified area.

mitochondrion



magnified area



(a) Use the mitochondrion diagram to describe how the inner membrane is adapted to maximise ATP production. **1**

_____

_____

(b) NADH is formed in the matrix of the mitochondrion.

State another location in the cell where NADH is formed. **1**

_____

(c) Describe one role of electrons (e^-) in the electron transport chain. **1**

_____

_____

(d) Name enzyme X. **1**

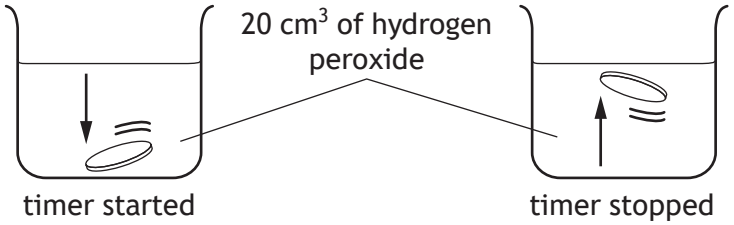
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[Turn over

5. An investigation was carried out into the effect of substrate concentration on the inhibition of catalase by copper nitrate. Catalase breaks down hydrogen peroxide into water and oxygen. **MARKS** DO NOT WRITE IN THIS MARGIN

10 cm^3 of catalase solution was mixed with 10 cm^3 copper nitrate solution and a paper disc was placed in this solution for 10 seconds.

The disc was then dropped into hydrogen peroxide. When the disc reached the bottom of the beaker, a timer was started and the time taken for the disc to float to the surface was recorded. Oxygen produced during the reaction caused the paper disc to float.



This was repeated using different concentrations of hydrogen peroxide.

A control experiment was carried out where copper nitrate solution was replaced with water.

The results are shown in the table.

Concentration of hydrogen peroxide (%)	Time for disc to float to surface (seconds)	
	Catalase + copper nitrate solution	Catalase solution
0.5	192	120
1.0	136	84
2.0	72	36
3.0	52	28
5.0	28	28

(a) Name the type of metabolic reaction catalysed by catalase. **1**

_____

(b) (i) State two variables, not already mentioned, that should be controlled to make the investigation valid. **2**

1. _____

2. _____

(ii) State the purpose of the control in **this** investigation. **1**

_____

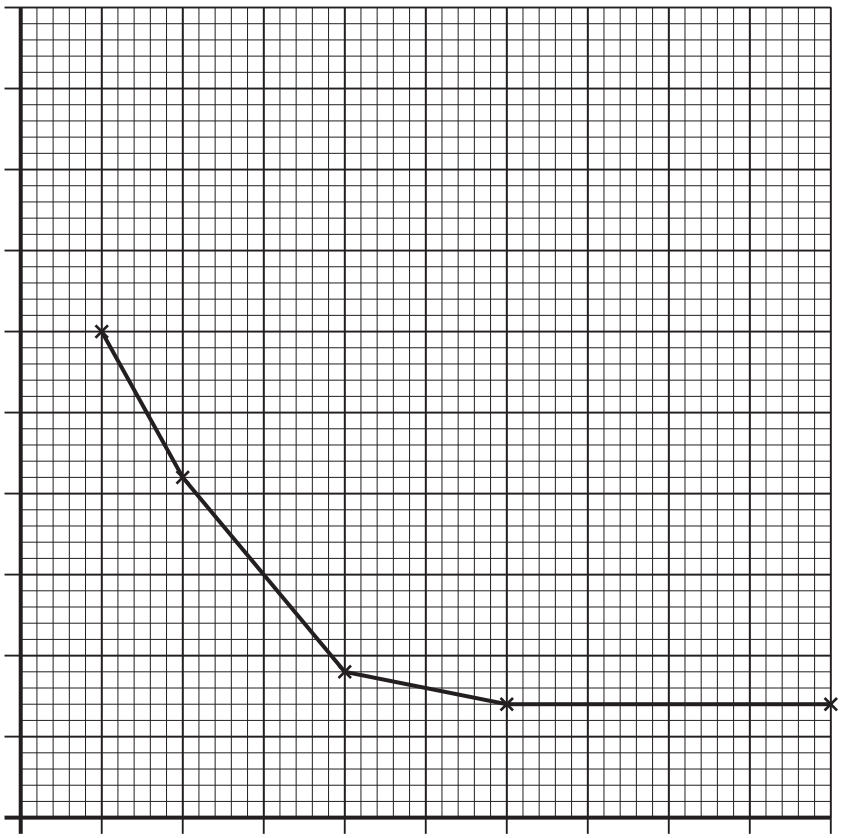
_____

5. (continued)

MARKS DO NOT WRITE IN THIS MARGIN

(c) On the grid, complete the line graph to show the results of the investigation using **catalase and copper nitrate**. **2**

(Additional graph paper, if required, can be found on *page 28*.)



(d) State the conclusion that can be made from these results. **1**

_____

_____

(e) Another disc was placed in the solution of catalase and copper nitrate then dropped into a beaker of 6.0% hydrogen peroxide.

Predict the average time for the disc to float to the surface. **1**

_____ seconds

[Turn over

6. A study was carried out to determine if the physical activity and body mass of females affects fertility.

The results are shown in the table.

Physical activity category	Body mass category	Chance of fertility problems (%)
Low	healthy	14
	overweight	17
	obese	27
Moderate	healthy	13
	overweight	16
	obese	18
High	healthy	11
	overweight	15
	obese	17

(a) State the physical activity and body mass category that resulted in the greatest chance of fertility problems in females. **1**

_____

(b) (i) BMI is used to determine body mass category.

Calculate the BMI of a female with a body mass of 77 kg and a height of 1.7 m. **1**

Space for calculation

_____

(ii) A female has a moderate level of physical activity and a BMI of 33.

State the chance that this female will have fertility problems. **1**

_____ %

6. (b) (continued)

(iii) The female and her partner are undergoing treatment for infertility because the partner has a low sperm count.

Name and describe a suitable treatment for this couple. **2**

Treatment _____

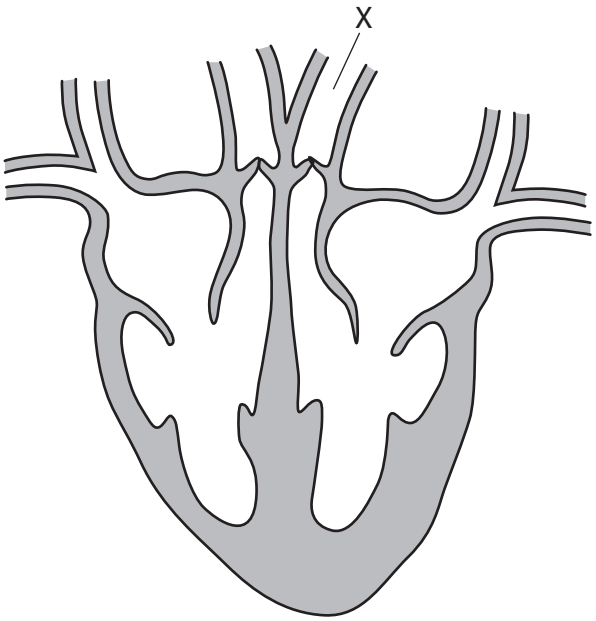
Description _____

_____

_____

[Turn over

7. The diagram shows a section through a human heart.



(a) (i) Name blood vessel X. **1**

_____

(ii) Describe the evidence shown in the diagram that suggests the heart is in atrial systole. **1**

_____

_____

(b) Describe how impulses from the sino-atrial node (SAN) cause ventricular systole. **2**

_____

_____

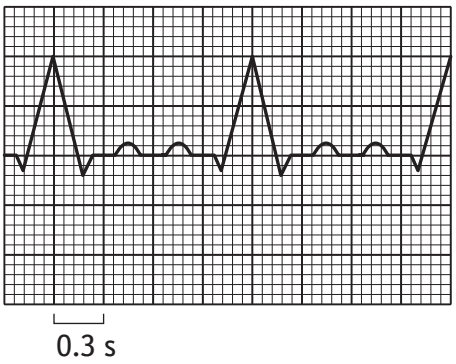
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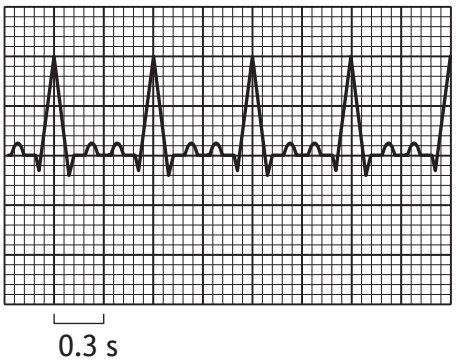
7. (continued)

(c) The diagrams show ECG traces from an individual before and after strenuous exercise.

before exercise



after exercise



Calculate the increase in heart rate during exercise. **1**

Space for calculation

_____ beats per minute

(d) Describe how the medulla lowers the heart rate following a period of exercise. **2**

_____

_____

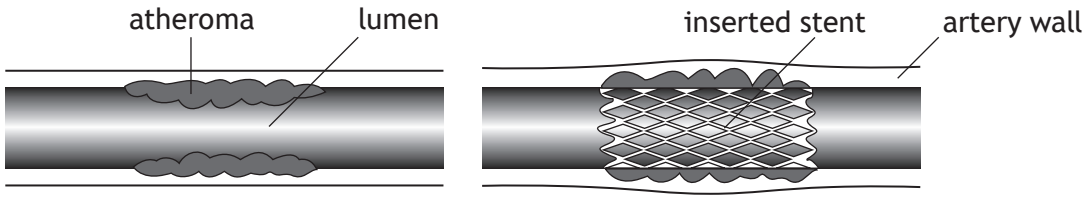
_____

_____

[Turn over

8. When an atheroma develops in an artery it can be treated by inserting a mesh tube called a stent.

The diagrams show an artery before and after a stent has been inserted.



before procedure after procedure

(a) (i) Describe the formation of an atheroma within an artery. **2**

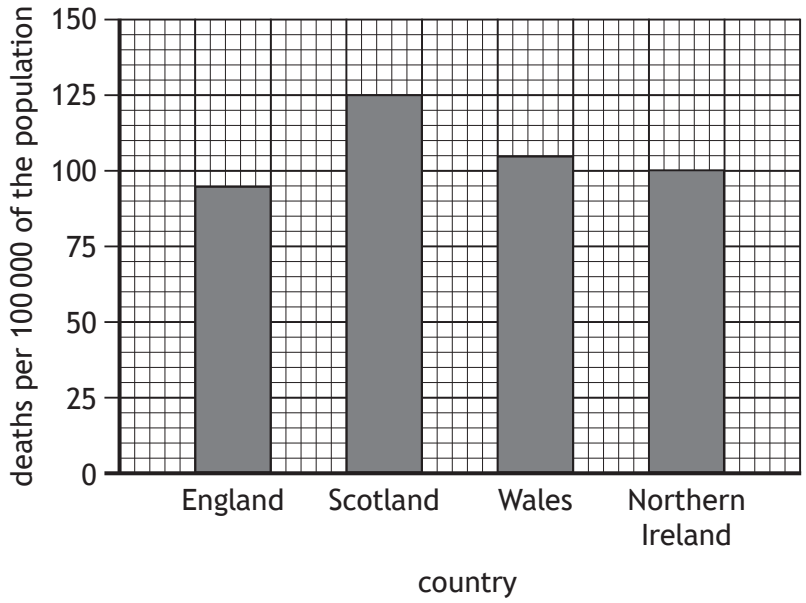
(ii) Use the diagrams to suggest how the insertion of a stent leads to increased blood flow in an artery. **1**

(b) A ruptured atheroma in an artery may lead to thrombosis.

Describe how thrombosis can cause a myocardial infarction (heart attack). **2**

8. (continued)

(c) The graph shows deaths from coronary heart disease in four countries.



(i) Using data from the graph, the risk of an individual dying from coronary heart disease in Northern Ireland was calculated to be 1 in 1000.

Use data from the graph to calculate the risk of an individual dying from coronary heart disease in Scotland. **1**

Space for calculation

1 in _____

(ii) Explain why deaths from coronary heart disease are expressed as per 100 000 of the population. **1**

_____

_____

[Turn over

9. High density lipoproteins (HDLs) and low density lipoproteins (LDLs) transport cholesterol around the body.

(a) (i) State a use of cholesterol in the body. **1**

_____

(ii) Name the organ to which HDL transports excess cholesterol for elimination. **1**

_____

(b) Familial hypercholesterolaemia (FH) is an inherited condition.

Individuals with FH have high levels of LDLs in their bloodstream.

(i) Statins are drugs, and can be prescribed to individuals with FH.

Statins act as competitive inhibitors of an enzyme involved in the synthesis of cholesterol.

Describe the mode of action of competitive inhibitors such as statins. **1**

_____

_____

(ii) FH is caused by an autosomal dominant allele.

Describe two ways in which the patterns of inheritance in a family history would indicate autosomal dominance. **2**

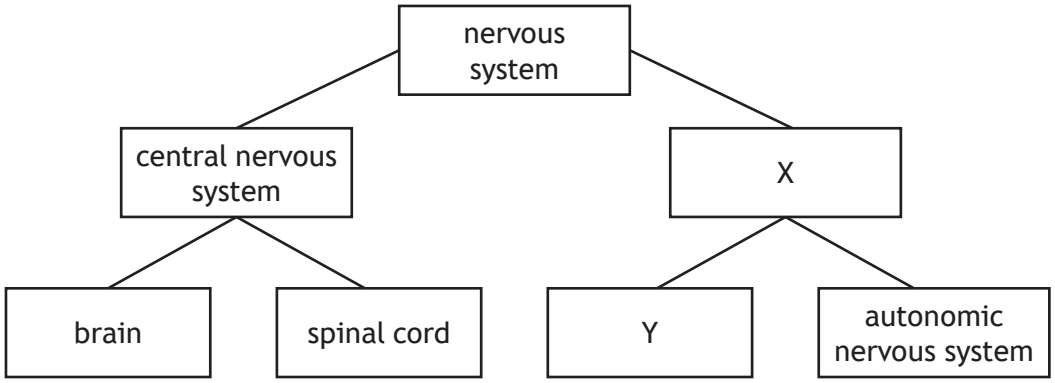
1. _____

_____

2. _____

_____

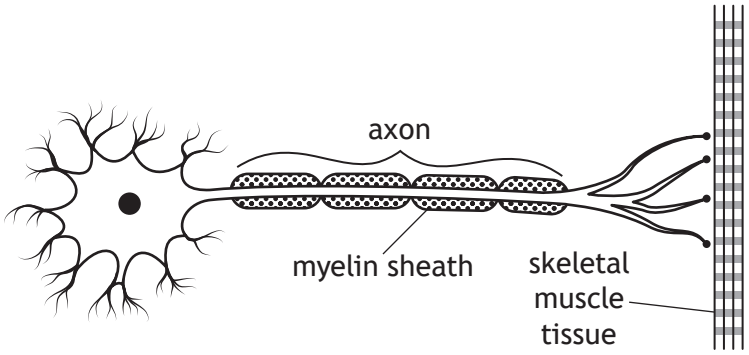
10. The diagram shows divisions of the nervous system.



(a) Identify X and Y. **2**

X _____ Y _____

(b) The diagram represents a neuron and its junction with skeletal muscle tissue.



Name the type of neuron shown in the diagram. **1**

_____

(c) Describe myelination and its importance in child development. **3**

_____

_____

_____

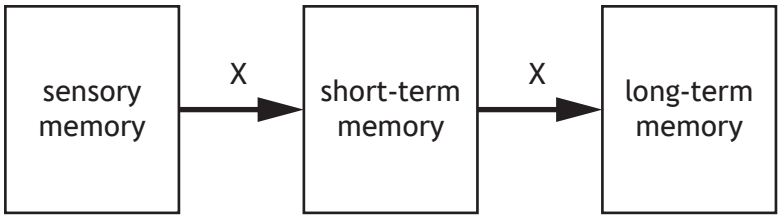
_____

_____

_____

11. The diagram represents a flow of information through memory.

MARKS | DO NOT WRITE IN THIS MARGIN



(a) (i) Name process X. **1**

_____

(ii) Name two methods that enable the transfer of information from short-term to long-term memory. **2**

1. _____

2. _____

(b) State the location in the brain where memories are stored. **1**

_____

(c) An investigation into the serial position effect on recall was carried out.

A list of words was read out to a group of students. Each student then wrote down the words that they could recall.

The table shows the position of the words in the list and the number of students who recalled each word.

Position of word in list	Number of students who recalled each word
1	18
2	17
3	16
4	14
5	10
6	8
7	7
8	8
9	16
10	17

11. (c) (continued)

(i) It was observed that more students recalled the words at the start and the end of the list compared to those in the middle.

Explain this observation. **2**

(ii) The investigation was evaluated as being unreliable.

State how the investigation could be made reliable. **1**

[Turn over

12. Dopamine is a neurotransmitter that activates the reward pathway in the brain.

(a) State one effect of activating the reward pathway. **1**

_____

(b) State one method of removal of dopamine from a synapse. **1**

_____

(c) (i) A dopamine agonist can be used in the treatment of Parkinson's disease.

State the effect of prolonged use of an agonist. **1**

_____

(ii) In a clinical trial of a potential new drug for the treatment of Parkinson's disease, individuals were divided into groups in a randomised way.

Name one other protocol that could be used during this clinical trial and explain its importance to the results. **2**

Protocol _____

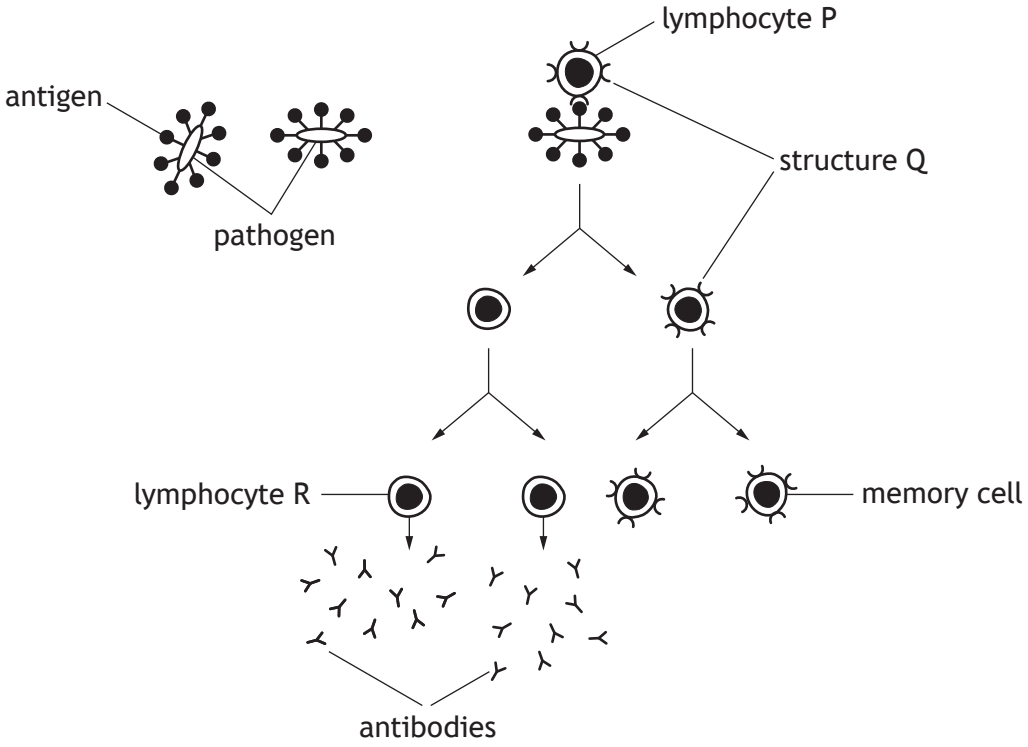
Importance _____

_____

(iii) State one other design factor that would allow the results of the clinical trial to have statistical significance. **1**

_____

13. The diagram represents lymphocyte responses to infection by a pathogen.



(a) (i) Structure Q is found on the surface of lymphocytes to allow them to bind with antigens.

Name structure Q. **1**

_____

(ii) Give evidence from the diagram that shows lymphocyte R is a B lymphocyte. **1**

_____

(b) Antibodies attach to and inactivate a pathogen, so the pathogen cannot spread through the body.

Describe the role of phagocytes following the inactivation of the pathogen. **1**

_____

_____

[Turn over

13. (continued)

(c) An individual contracted chicken pox during childhood. A few years later they were exposed to the same chicken pox antigen.

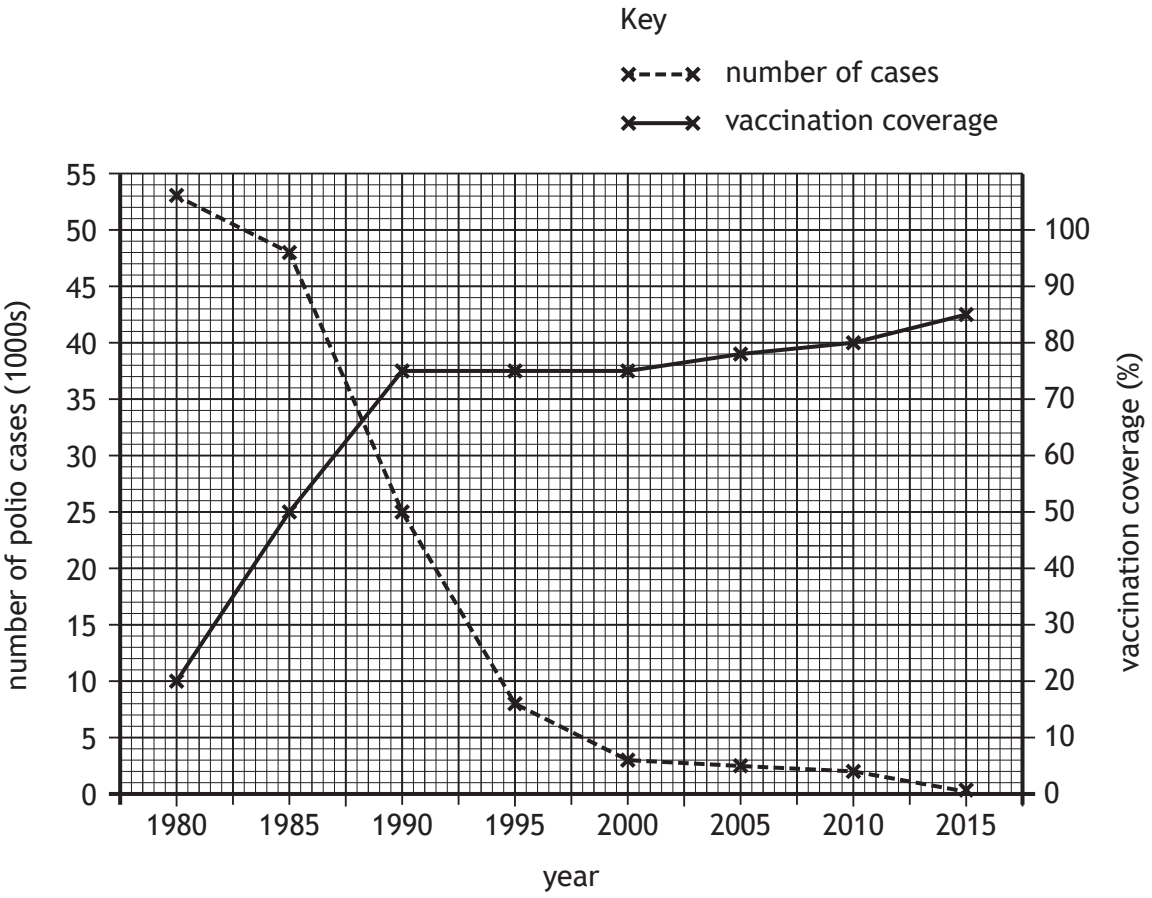
Describe the role of memory cells in protecting this individual from developing symptoms of chicken pox during this second exposure. **2**

(d) T lymphocytes are another type of lymphocyte that protect the body from infection.

Explain how T lymphocytes destroy infected cells. **1**

14. Polio is a viral disease that has been eradicated in most countries due to mass vaccination programmes.

The **graph** shows changes in the global number of polio cases and the percentage of the world population vaccinated between 1980 and 2015.



(a) (i) State the vaccination coverage achieved when the number of polio cases was 8000. **1**

_____ %

(ii) Calculate the percentage decrease in polio cases in the 30-year period from 1980 to 2010. **1**

Space for calculation

_____ %

MARKS

DO NOT WRITE IN THIS MARGIN

(iii) Use data from the graph to describe the changes in vaccination coverage between 1980 and 2000. **2**

(b) Polio is still present in some developing countries.

The **table** shows the number of polio cases in one of these countries between 2000 and 2010.

Year	2000	2002	2004	2006	2008	2010
Number of polio cases	90	50	20	15	10	15

(i) **Using data in the table and graph**, calculate the simple whole number ratio of the global number of cases of polio to the number of cases in this country in 2000. **1**

Space for calculation

_____ : _____

global country

(ii) In order to establish herd immunity for polio, there has to be at least 85% vaccination coverage.

If the population of this country is 2 million people, calculate the number of individuals that must be vaccinated to establish herd immunity for polio in this country. **1**

Space for calculation

(iii) Suggest why the herd immunity threshold may not be reached in this developing country despite a 10-year mass vaccination programme. **1**

15. Attempt **either** A **or** B.

Write your answer in the space below and on *page 27*.

A Write notes on the follicular and luteal phases of the menstrual cycle. **9**

OR

B Write notes on the biology of controlling fertility and the methods of contraception. **9**

You may use labelled diagrams where appropriate.

National
Qualifications
2024

X840/76/12

Human Biology
Paper 1 — Multiple choice

WEDNESDAY, 15 MAY
9:00 AM – 9:40 AM

Total marks — 25

Attempt ALL questions.

You may use a calculator.

Instructions for the completion of Paper 1 are given on *page 02* of your answer booklet X840/76/02.

Record your answers on the answer grid on *page 03* of your answer booklet.

Space for rough work is provided at the end of this booklet.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

Total marks — 25

Attempt ALL questions

1. Red blood cells are produced in the bone marrow by differentiation of

A multipotent tissue stem cells

B pluripotent tissue stem cells

C multipotent embryonic stem cells

D pluripotent embryonic stem cells.

2. A molecule of DNA has a thymine to guanine ratio of 3:2.

The molecule contains 1245 adenine bases.

The number of guanine bases in the molecule is

A 415

B 498

C 830

D 1660

3. Identify the catabolic reaction.

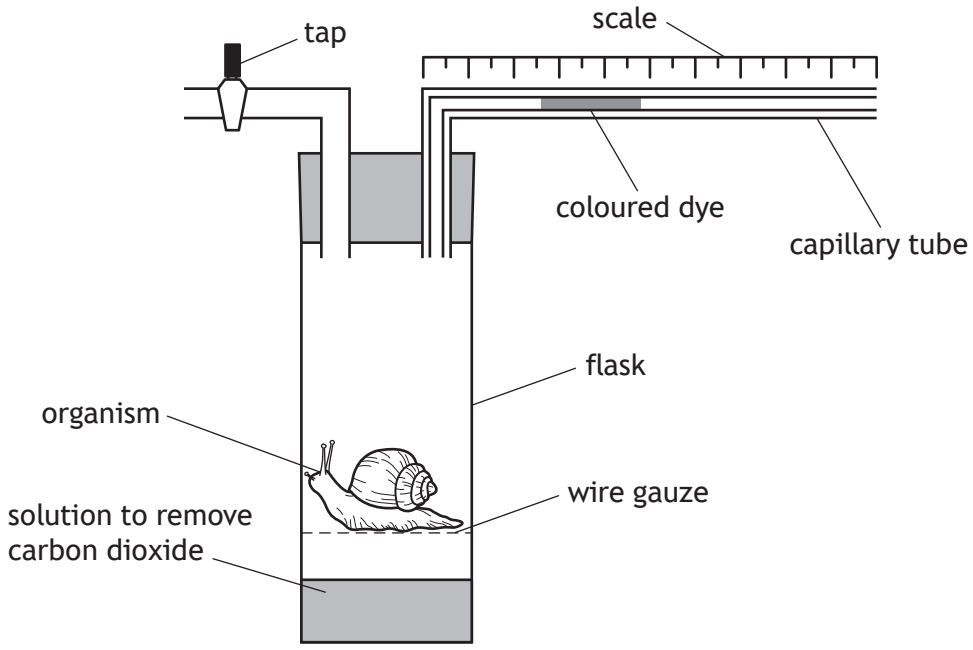
A small molecules —(requires energy)→ large molecule

B small molecules —(releases energy)→ large molecule

C large molecule —(requires energy)→ small molecules

D large molecule —(releases energy)→ small molecules

4. An investigation was carried out to compare the rate of respiration of an organism at different temperatures. The apparatus was set up as shown.



The investigation was repeated at four different temperatures. The organism was left in the apparatus for 30 minutes at each temperature.

A suitable control for this investigation would be to use the same set up with

A more organisms

B a wider range of temperatures

C glass beads in place of organism

D no solution to remove carbon dioxide.

5. What term describes the use of genome information in the choice of drugs that may be prescribed to a patient?

A Bioinformatics

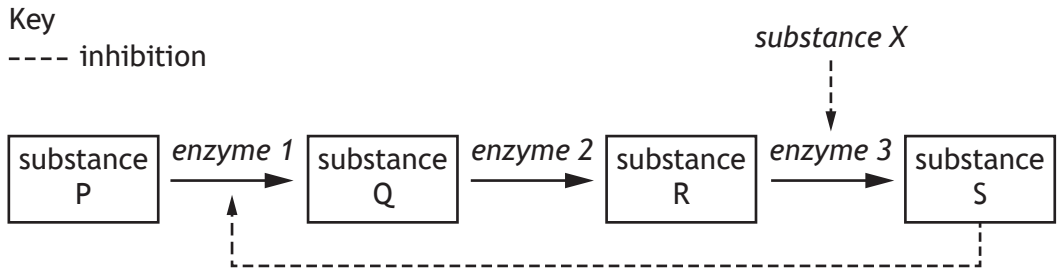
B Human genomics

C Pharmacogenetics

D Genomic sequencing

[Turn over

6. The diagram shows a metabolic pathway and two ways that it can be inhibited.



The effect of both inhibitors is reduced by increasing substrate concentration.

Which row in the table identifies the two types of inhibition?

	Inhibition due to substance S	**Inhibition due to substance X**
A	feedback	competitive
B	competitive	non-competitive
C	feedback	non-competitive
D	non-competitive	competitive

7. Which of the following mutations results in a premature stop codon causing a shorter protein to be produced?

A Missense

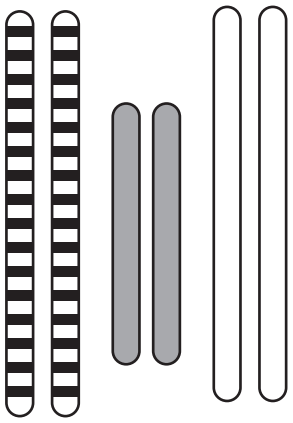
B Nonsense

C Splice-site

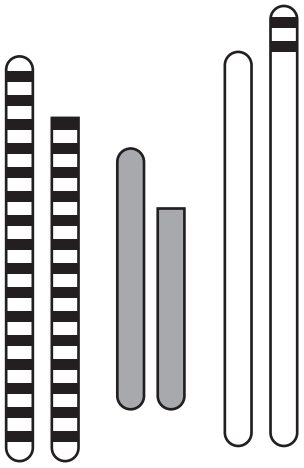
D Frameshift

8. The diagrams show three pairs of homologous chromosomes before and after chromosome mutations occur.

before mutation



after mutation



Which of the following types of chromosome mutation have occurred?

A Inversion and translocation

B Deletion and duplication

C Duplication and inversion

D Deletion and translocation

9. During exercise, an oxygen debt can build up.

When the oxygen debt is repaid, lactate is converted in the liver to

A pyruvate and glucose

B carbon dioxide and water

C pyruvate and water

D carbon dioxide and glucose.

[Turn over

10. PCR is a reaction that amplifies a section of DNA and involves three temperature changes per cycle.

Calculate how many temperature changes the reaction would undergo if there were 256 copies of the DNA section produced.

A 3

B 8

C 21

D 24

11. In the testes, sperm are produced by

A mitosis in the seminal vesicles

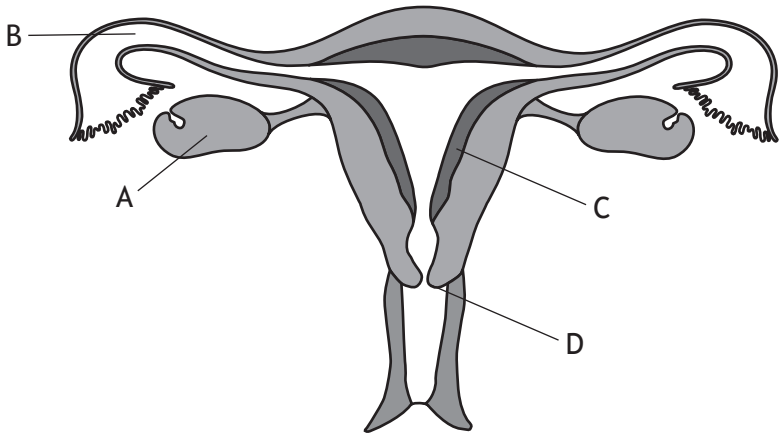
B meiosis in the seminal vesicles

C mitosis in the seminiferous tubules

D meiosis in the seminiferous tubules.

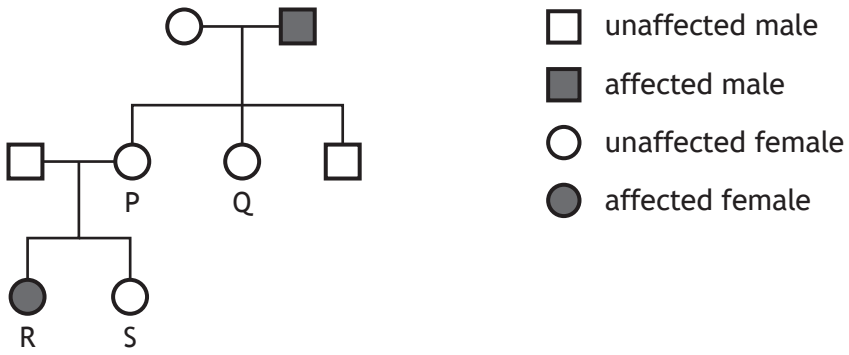
12. The diagram represents the female reproductive system.

Identify the structure affected by the progesterone-only (mini) pill as a means of contraception.



13. Thalassaemia is a recessive inherited blood condition in which individuals produce abnormal haemoglobin.

The diagram shows the children and grandchildren of a male with thalassaemia and a female who is homozygous.



Which of the labelled individuals confirms that thalassaemia is **not** a sex-linked condition?

A P

B Q

C R

D S

14. Artificial insemination requires

A PGD to identify single gene disorders

B several samples of semen to be collected

C surgical removal of eggs from the ovaries

D the head of a sperm to be drawn into a needle.

[Turn over

15. Phenylketonuria (PKU) is an inherited disorder caused by a mutation that affects a reaction in a metabolic pathway.

Which row in the table identifies the type of mutation and reaction affected?

	Type of mutation	Reaction affected
A	deletion	phenylalanine → tyrosine
B	deletion	tyrosine → phenylalanine
C	substitution	phenylalanine → tyrosine
D	substitution	tyrosine → phenylalanine

16. The table shows some measurements taken from an athlete before and after a 3-month training camp.

	Resting heart rate (bpm)	Stroke volume (l)
Before training	60	0.07
After training	51	0.09

Calculate the increase in cardiac output after training.

A 0.02

B 0.39

C 4.59

D 9.29

17. An atheroma may rupture, damaging the endothelium. This triggers the release of clotting factors, which activates a cascade of reactions.

Which of the following reactions comes first in the cascade?

A Fibrinogen → fibrin

B Fibrin → fibrinogen

C Prothrombin → thrombin

D Thrombin → prothrombin

18. Which of the following would be used to calculate the BMI of an individual?

A Body mass divided by height squared
B Body mass squared divided by height
C Height squared divided by body mass
D Height divided by body mass squared

19. The corpus callosum

A allows the transfer of information between cerebral hemispheres
B allows the localisation of brain functions in the cerebral cortex
C processes information from the opposite side of the cerebrum
D transfers information between short-term memory and long-term memory.

20. Which of the following is **not** a function of the association areas in the cerebral cortex?

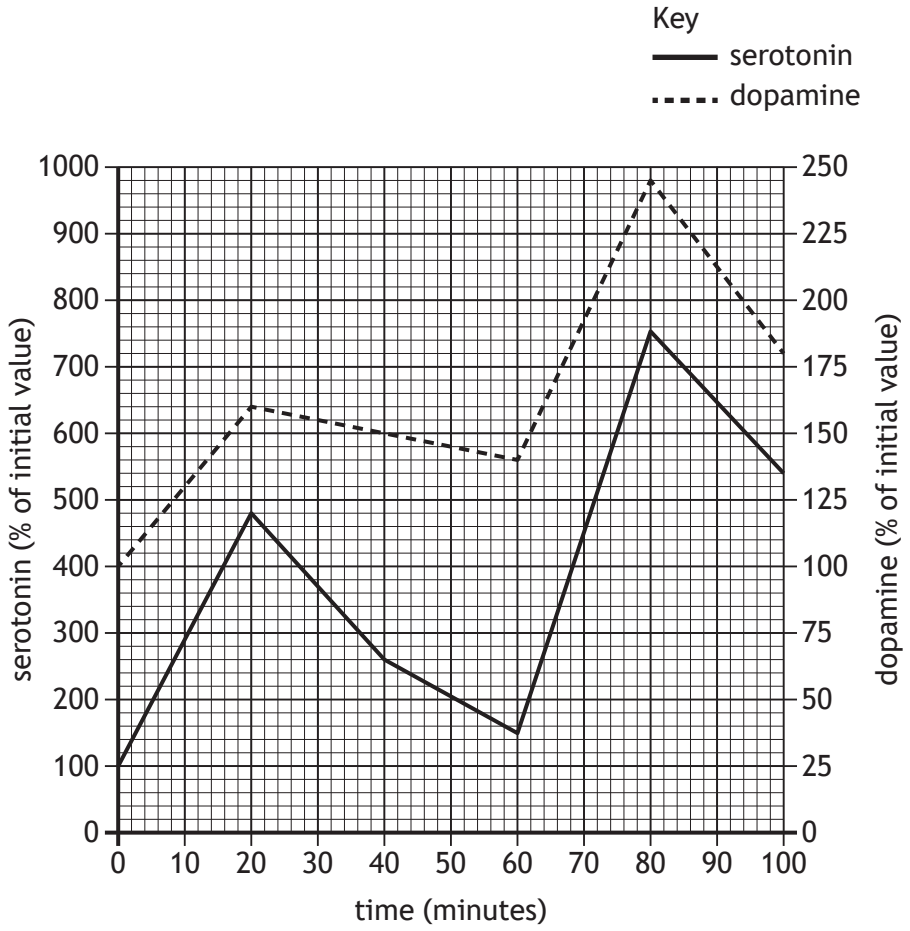
A Imagination
B Language processing
C Coordinating balance
D Determining personality and intelligence

21. A recreational drug can stimulate the reward pathway in the brain by

A blocking the dopamine reuptake protein
B blocking the endorphin reuptake protein
C acting as an antagonist at dopamine receptors
D acting as an antagonist at endorphin receptors.

[Turn over

22. The graph contains information about serotonin and dopamine levels in an individual, following injections of a drug at 0 and 60 minutes.



Predict what the individual's dopamine levels will be at 120 minutes.

A 80%

B 115%

C 320%

D 460%

23. The list shows cell types found in the immune system.

1. Mast cell
2. Phagocyte
3. Lymphocyte

Which of these cell types are involved in non-specific body defences against disease?

A 2 only

B 1 and 2 only

C 2 and 3 only

D 1, 2 and 3

24. An investigation was carried out into the use of different types of adjuvants on the effectiveness of a vaccine.

Four vaccines were prepared, each containing a different type of adjuvant and antigens against the pathogen.

The effectiveness of each vaccine was measured at 7 days and 14 days after immunisation.

Identify the independent variable in this investigation.

A Volume of the vaccine

B Type of adjuvant used

C Effectiveness of the vaccine

D Length of time left after immunisation

25. To establish herd immunity against measles 90% of a population must be successfully vaccinated.

The measles vaccine is not successful in some individuals who have a weakened immune system.

Which of the following populations is most at risk of a measles outbreak?

	Number of vaccinated individuals in the population	**Number of vaccinated individuals where vaccine was not successful**	**Population**
A	600 000	10 000	630 000
B	202 000	2000	220 000
C	410 000	5000	450 000
D	300 000	5000	340 000

[END OF QUESTION PAPER]

FOR OFFICIAL USE

National Qualifications 2024

Mark

X840/76/01

Human Biology Paper 2

WEDNESDAY, 15 MAY
10:10 AM – 12:30 PM

X8407601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day	Month	Year

Scottish candidate number

Total marks — 95

Attempt ALL questions.

You may use a calculator.

Question 15 contains a choice.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. Score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

X8407601

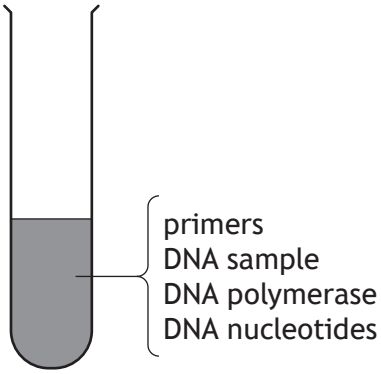
Total marks — 95

Attempt ALL questions

Question 15 contains a choice

1. To confirm an individual is infected with the herpes virus, a test to detect viral DNA can be carried out using the polymerase chain reaction (PCR) on a sample from the individual.

The diagram shows substances that are required to allow PCR to take place.



(a) Give the complementary DNA base sequence for the section of viral DNA shown. **1**



_____

(b) (i) State a temperature used to separate the DNA strands during PCR. **1**

_____ °C

(ii) State the role of primers and DNA polymerase in PCR. **2**

Primers _____

_____

DNA polymerase _____

_____

1. (continued)

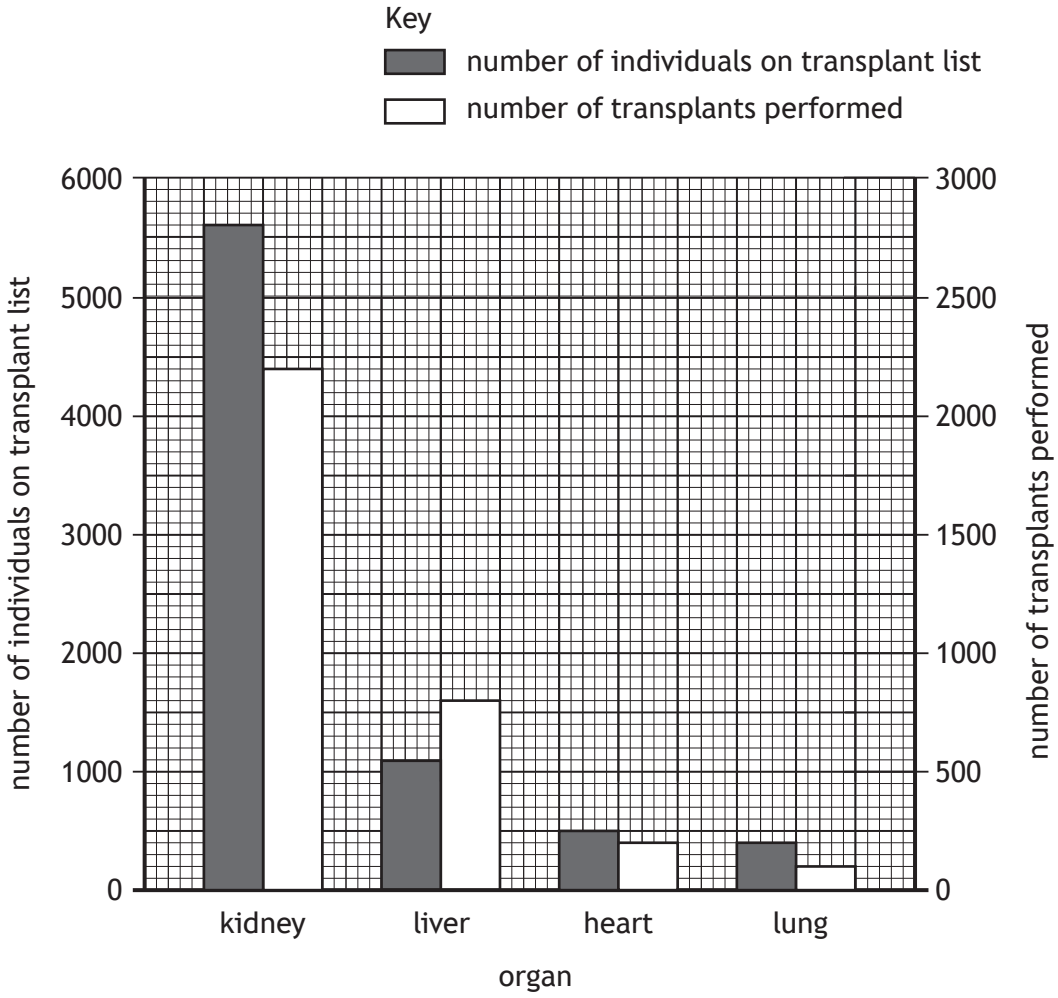
(c) State two uses of PCR, other than diagnosing viral infections. **2**

1. _____

2. _____

[Turn over

2. The graph contains information on some organ transplants performed during 2020 in the UK. It also shows the number of individuals waiting on the transplant list at the start of 2020.



(a) (i) State the number of lung transplants performed in 2020. **1**

_____

(ii) Use information from the graph to calculate the percentage of individuals who received a kidney transplant after being on the transplant list at the start of 2020. **1**

Space for calculation

_____%

2. (continued)

(b) The table shows the total number of transplants performed in three different countries in 2020.

Country	Number of transplants performed (per million of the population)
Scotland	72
England	47
Wales	40

(i) In 2020 the population of England was 57 million.

Calculate how many transplants were performed in England during 2020. **1**

Space for calculation

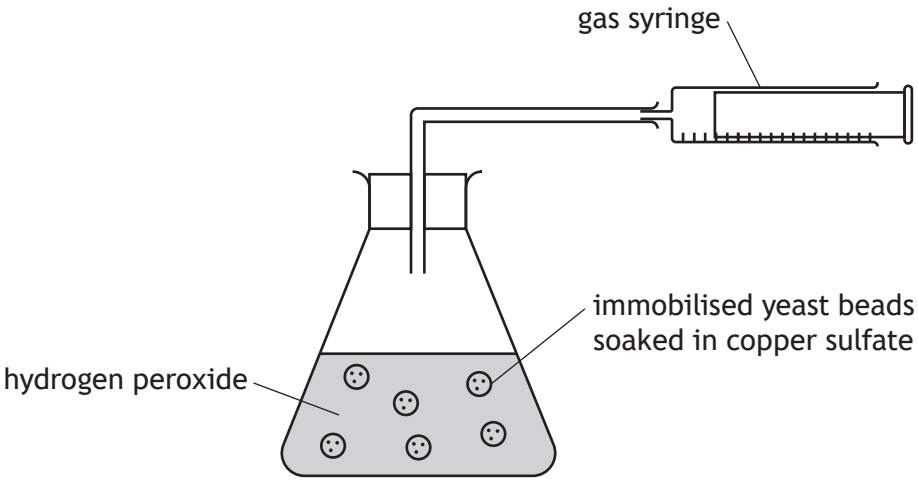
(ii) Explain why the data are presented as the number of transplants performed per million of the population. **1**

[Turn over

3. Catalase is an enzyme that breaks down hydrogen peroxide into oxygen and water. An investigation was carried out into the effect of inhibitor concentration on catalase activity.

Yeast cells containing catalase can be trapped in a gel substance to become 'immobilised' as gel beads. The catalase remains active within these beads.

Immobilised yeast beads were placed in different concentrations of the inhibitor copper sulfate for 24 hours. The beads were then added to a flask of hydrogen peroxide and the oxygen produced was collected in a gas syringe over a five-minute period.



The table shows the results of the investigation.

Concentration of copper sulfate (mmol/l)	Volume of oxygen collected (cm^3)		
	Experiment 1	Experiment 2	Average
0	80	84	82
5	65	71	68
10	45	45	45
25	26	24	25
50	15	17	16

(a) (i) State two variables, other than those shown above, that should be controlled so that a valid conclusion can be drawn. **2**

1. _____

2. _____

3. (a) (continued)

MARKS DO NOT WRITE IN THIS MARGIN

(ii) Suggest an advantage of using immobilised yeast beads rather than yeast in solution. **1**

_____

_____

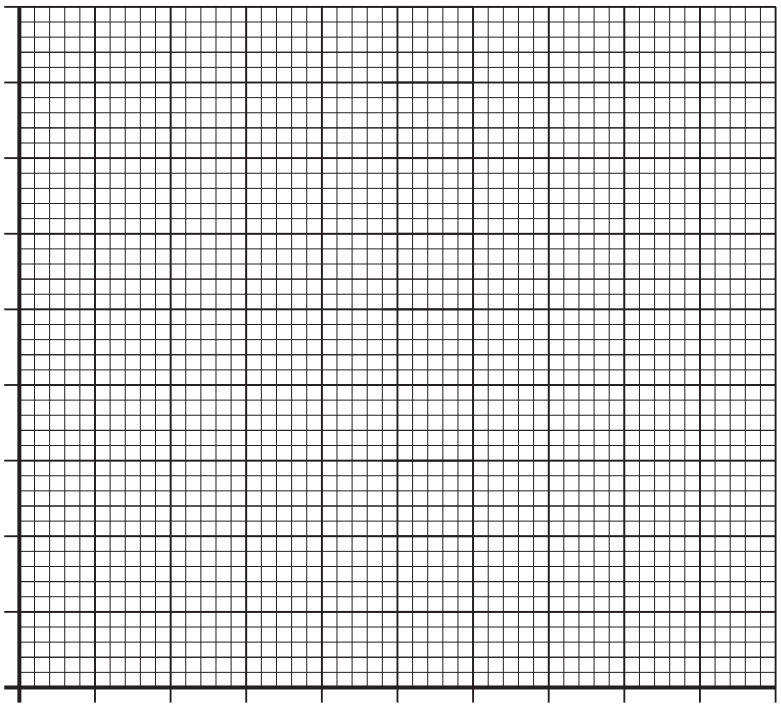
(iii) Suggest why the immobilised yeast beads were left in the copper sulfate solution for 24 hours before adding them to the hydrogen peroxide. **1**

_____

_____

(b) Using data from the table, draw a line graph to show the average volume of oxygen collected. **2**

(Additional graph paper, if required, can be found on *page 31*.)



(c) State the conclusion that can be drawn from this investigation. **1**

_____

_____

4. Cellular respiration takes place to generate ATP.

(a) The statements refer to events that occur during respiration in a muscle cell.

Letter	Statement
W	Carbon dioxide is released.
X	An acetyl group combines with coenzyme A.
Y	Glucose is broken down into pyruvate.
Z	Dehydrogenase enzymes remove hydrogen ions and electrons.

(i) Use letters from the table to identify the statements that apply to glycolysis. **1**

_____

(ii) Name the substance that is broken down to form the acetyl group that combines with coenzyme A. **1**

_____

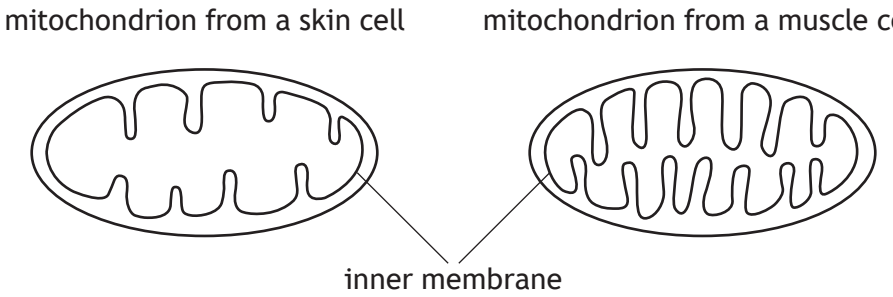
(iii) Oxaloacetate combines with an acetyl group to form another substance.

Name this substance. **1**

_____

4. (continued)

(b) The diagram shows the structure of mitochondria from a skin cell and a muscle cell.



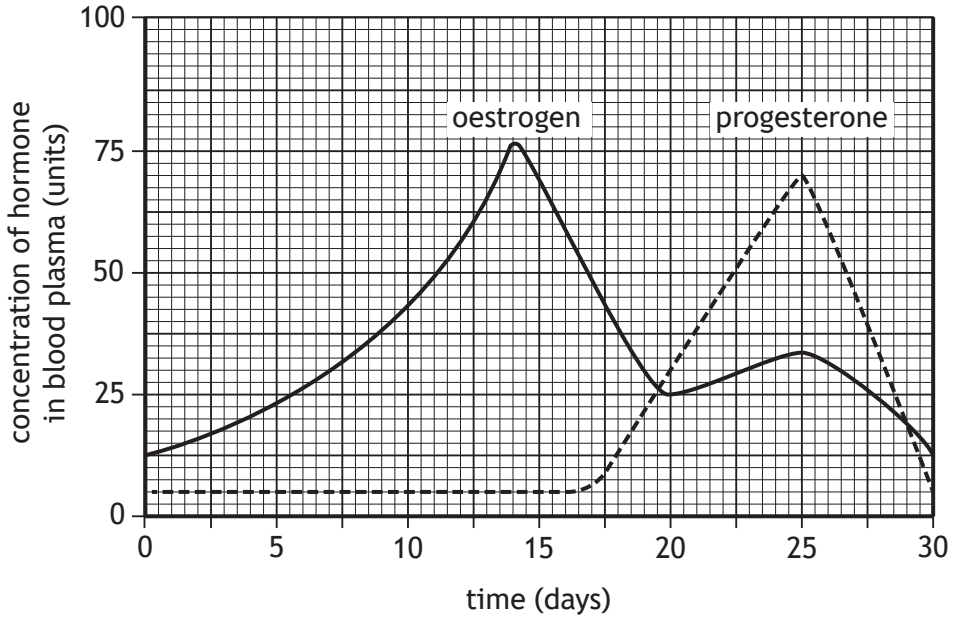
Use the diagram to suggest why mitochondria from muscle cells can generate more ATP than skin cells. **2**

(c) Slow-twitch muscle fibres are useful for endurance activities as they can sustain contractions for long periods of time.

Describe one structural feature of slow-twitch muscle fibres. **1**

[Turn over

5. The graph shows the concentrations of the ovarian hormones, oestrogen and progesterone, in a female's blood plasma during the menstrual cycle.



(a) During the first half of the cycle, the concentration of oestrogen in the blood plasma increases.

(i) Describe the role of follicle stimulating hormone (FSH) in causing this increase in oestrogen concentration. **2**

(ii) Describe how this increase in oestrogen concentration would affect the uterus. **1**

(iii) The high concentration of oestrogen at day 14 stimulates a surge in luteinising hormone (LH).

Describe the effect of this surge in LH on the ovary. **1**

5. (continued)

(b) During the second half of the cycle, the concentration of progesterone in the blood plasma increases.

(i) Calculate how many times greater the concentration of progesterone is on day 25 compared to its concentration on day 15. **1**

Space for calculation

_____ times greater

(ii) Name the structure within the ovary that produces progesterone. **1**

_____

(c) This female is receiving treatment for infertility.

(i) Describe evidence from the graph which indicates that she has **not** become pregnant during this menstrual cycle. **1**

_____

_____

(ii) It was discovered that her oviducts were blocked, reducing the chance of successful fertilisation.

Identify a suitable treatment and describe how the treatment would increase the chance of fertilisation. **2**

Treatment _____

Description _____

_____

[Turn over

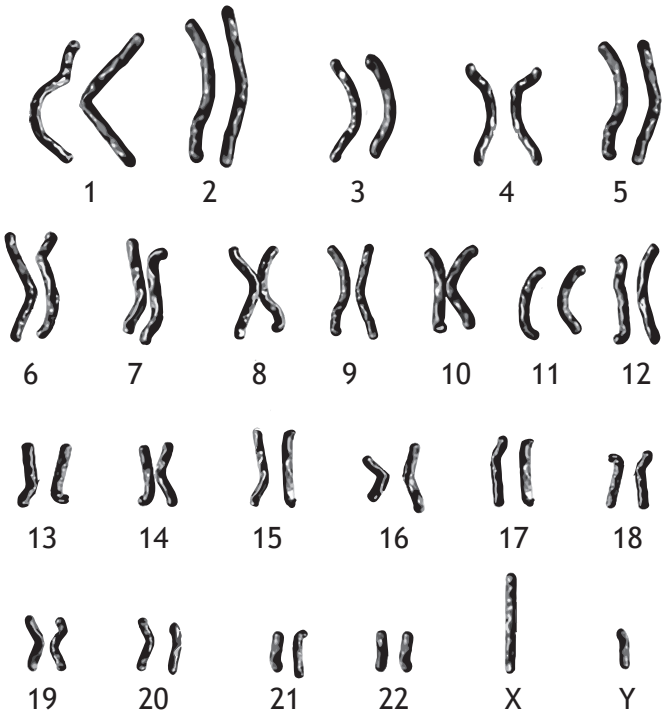
6. Routine blood tests are carried out throughout pregnancy to monitor the concentration of marker chemicals.

(a) Describe a problem with the results that could occur if a blood test is carried out at the wrong time during pregnancy. **1**

(b) A blood test showed unusually low levels of a protein in a pregnant female's blood. After medical advice, a diagnostic test called amniocentesis was carried out.

Suggest why amniocentesis was carried out instead of chorionic villus sampling (CVS). **1**

(c) Samples taken during amniocentesis were used to culture cells and the following image showing the fetal chromosomes was then produced.



(i) State the name given to an image of chromosomes arranged in this way. **1**

(ii) State the term used to describe chromosomes 1 to 22. **1**

7. The table shows an individual's blood flow rate to different parts of the body at rest and during exercise.

Part of body	Blood flow rate at rest (cm^3/min)	Blood flow rate during exercise (cm^3/min)
Brain	450	750
Heart muscle	250	1000
Skeletal muscles	1000	12 000
Intestines	1500	500

(a) (i) Calculate the percentage increase in the blood flow rate to the heart muscle from rest to exercise. **1**

Space for calculation

_____ %

(ii) Calculate the simple whole number ratio of the blood flow rate to the individual's intestines, skeletal muscles and brain during exercise. **1**

Space for calculation

_____ : _____ : _____

intestines skeletal muscles brain

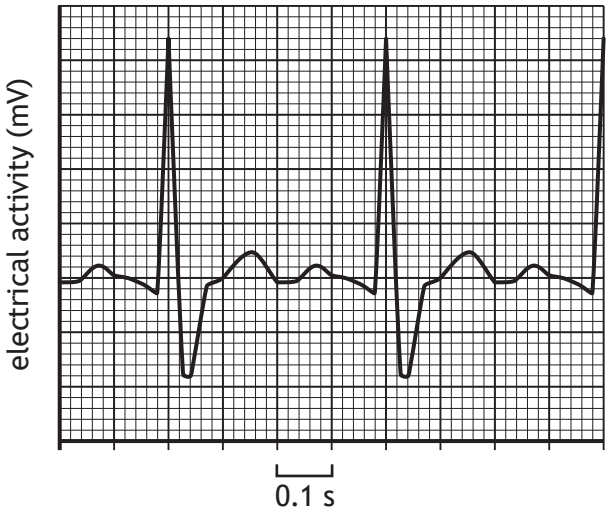
(iii) During exercise, blood flow to the intestines and skeletal muscles changes.

Complete the table to show how these changes would occur. **2**

Part of body	Change in blood flow rate	Process controlling blood flow
	decrease	
	increase	

7. (continued)

(b) (i) The graph shows part of an ECG trace taken when the individual was exercising.



Use information from the ECG trace to calculate their heart rate during exercise. **1**

Space for calculation

_____beats/minute

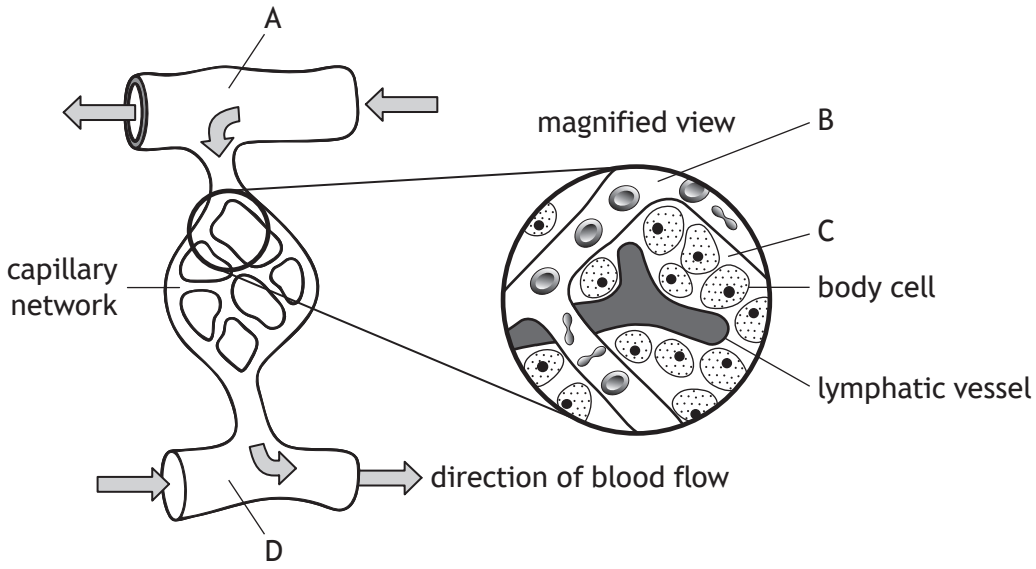
(ii) Describe the role of nerves in the autonomic nervous system to bring about an increase in heart rate. **1**

_____

_____

[Turn over

8. The diagram represents a capillary network and its associated blood vessels.



(a) (i) Name liquid B. **1**

_____

(ii) Name the process that forms liquid C. **1**

_____

(b) Explain why blood vessel A requires a thicker muscular wall than blood vessel D. **1**

_____

_____

(c) (i) Describe a feature of blood capillaries that allows the rapid diffusion of molecules into and out of the bloodstream to occur. **1**

_____

_____

(ii) Name a type of molecule that is normally unable to leave the bloodstream in a capillary network. **1**

_____

8. (continued)

(d) (i) Describe the function of the lymphatic vessel shown in the diagram. **1**

(ii) Suggest a reason why lymphatic vessels contain valves. **1**

[Turn over

9. A medical investigation was carried out into the effect of energy drinks on blood pressure. Energy drinks contain glucose and caffeine.

20 participants were divided into two groups.

Group 1 participants consumed 250 cm^3 of an energy drink.

Group 2 participants consumed 250 cm^3 of water.

Each participant's blood pressure was measured before consuming the drink and again three hours later.

(a) Describe how a sphygmomanometer is used to measure systolic blood pressure. **2**

(b) **Table 1** shows the average blood pressures of both groups.

Table 1

	Average blood pressure (mmHg)	
Group	**Before consuming drink**	**Three hours after consuming drink**
1	123/75	123/84
2	122/74	122/74

(i) Identify the dependent variable in this investigation. **1**

(ii) State one conclusion that can be drawn from these results. **1**

9. (b) (continued)

(iii) The mean arterial blood pressure (MAP) can be calculated using the formula:

$$\text{MAP} = \text{diastolic pressure} + \left(\frac{\text{pulse pressure}}{3}\right)$$

Pulse pressure is the difference between systolic and diastolic pressure.

Three calculated MAP values are shown in **Table 2**.

Table 2

	MAP (mmHg)	
Group	**Before consuming drink**	**Three hours after consuming drink**
1	91	
2	90	90

Use the information in **Table 1** to complete **Table 2** to show the MAP for group 1 after consuming the energy drink. **1**

Space for calculation

(c) Name the condition in which an individual has sustained high blood pressure. **1**

_____

[Turn over

10. A study was carried out to determine if there is a link between low levels of vitamin D and type 1 diabetes.

Vitamin D blood concentrations were measured in a group of individuals with type 1 diabetes and in a control group. There were equal numbers in both groups.

The results are shown in the table.

Group	Individuals with low concentrations of vitamin D (%)
Individuals with type 1 diabetes	91
Control	59

(a) (i) State one variable that would have to be taken into account when allocating individuals to the groups in this study. **1**

(ii) A student concluded from these results that low concentrations of vitamin D increases the risk of developing type 1 diabetes.

Suggest why this conclusion may be incorrect. **1**

(iii) The total number of individuals involved in the study was 400.

Calculate the number of individuals in the control group who had low concentrations of vitamin D. **1**

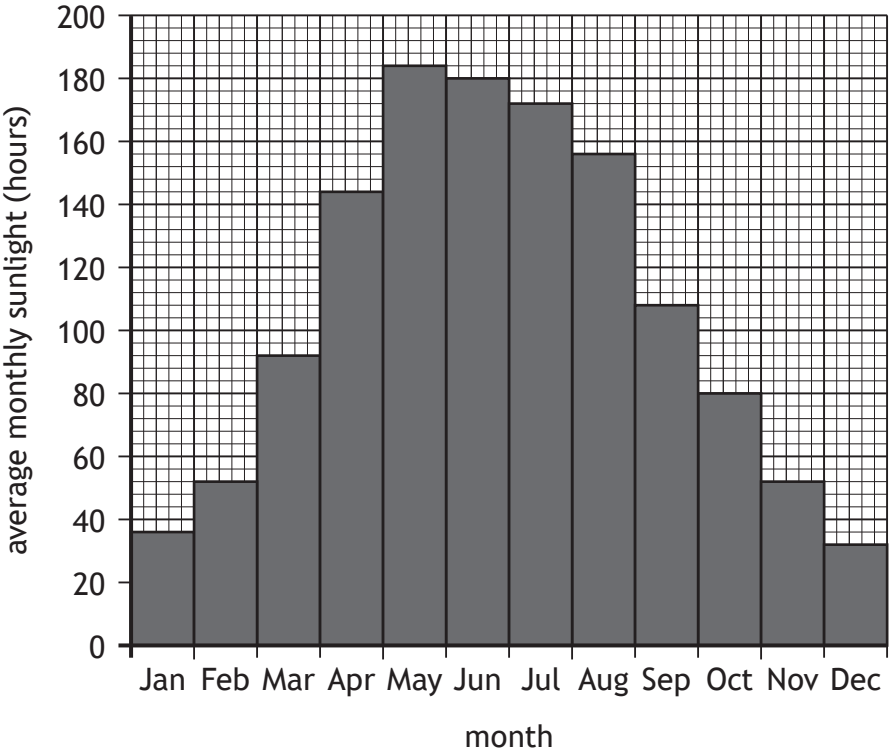
Space for calculation

(iv) Describe how the reliability of the study could be increased. **1**

10. (continued)

(b) When skin is exposed to sunlight the production of vitamin D increases.

The graph shows the average monthly hours of sunlight in a city in Scotland throughout the year.



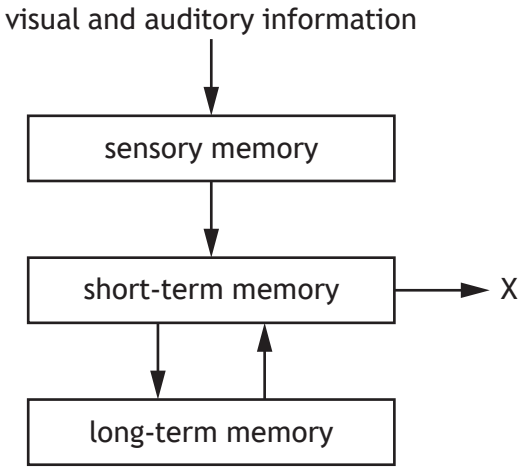
(i) Use data from the graph to describe the changes that occur in the average monthly hours of sunlight throughout the year. **2**

(ii) Most individuals produce enough vitamin D in their skin when the levels of sunlight are higher than 100 hours per month.

During which months would an individual living in this city need to obtain more of their vitamin D from their diet? **1**

[Turn over

11. The diagram represents the flow of information from the environment through memory.



(a) Name process X. **1**

_____

(b) Name the model that is used to explain the ability of the short-term memory to perform simple cognitive tasks. **1**

_____

(c) A mobile phone number typically consists of 11 numbers.

Explain why it is difficult to store a mobile phone number in the short-term memory. **1**

_____

_____

11. (continued)

(d) In an investigation into the recall of information from long term memory, a large group of students was divided into two sub-groups.

The students were then given a list of 20 words to memorise in one minute.

Group 1 was given a list of the words arranged into four different categories.

Group 2 was given a list that contained the same words but in a random order.

Five minutes later the students had to write down all the words they could recall from the list.

(i) Suggest a method that could be used to randomly allocate the students to each sub-group. **1**

_____

_____

(ii) Explain why students in group 1 recalled more words than students in group 2. **1**

_____

_____

_____

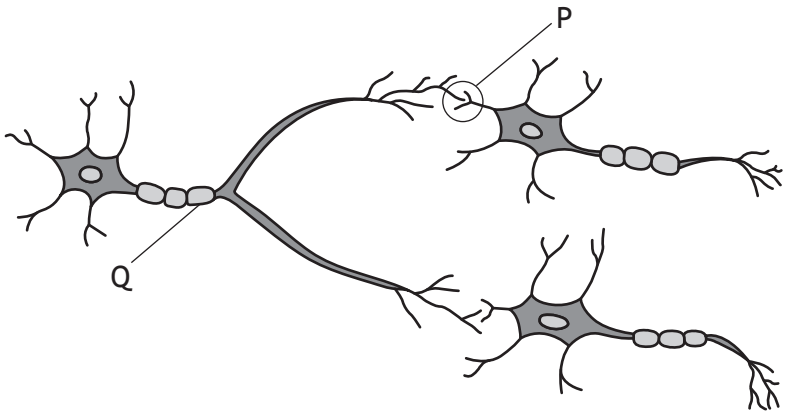
(e) Recalling the events that occurred when information was encoded into long term memory can help the later retrieval of the information.

Name the term that is used to describe these events. **1**

_____

[Turn over

12. The diagram shows the arrangement of some motor neurons.



(a) Name structure P. **1**

_____

(b) (i) Name the type of neural pathway shown in the diagram. **1**

_____

(ii) Explain how this arrangement of neurons helps with fine motor control when writing. **1**

_____

_____

(c) (i) Name structure Q. **1**

_____

(ii) Structure Q develops from birth to adolescence.

Describe why this development allows children to become more coordinated as they get older. **1**

_____

_____

12. (continued)

(d) (i) Myasthenia gravis is a disease where the neurotransmitter receptors on skeletal muscles are destroyed.

Suggest how this results in problems with movement. **1**

(ii) Myasthenia gravis is an autoimmune disease.

Describe the immune response that results in an autoimmune disease. **2**

(iii) The populations of three countries are shown in the table.

Country	Population (million)
Scotland	5.4
England	57.0
Wales	3.1

The incidence of Myasthenia gravis throughout the UK is 20 per 100 000.

Calculate how many more people suffer from the disease in England compared to Scotland. **1**

Space for calculation

[Turn over

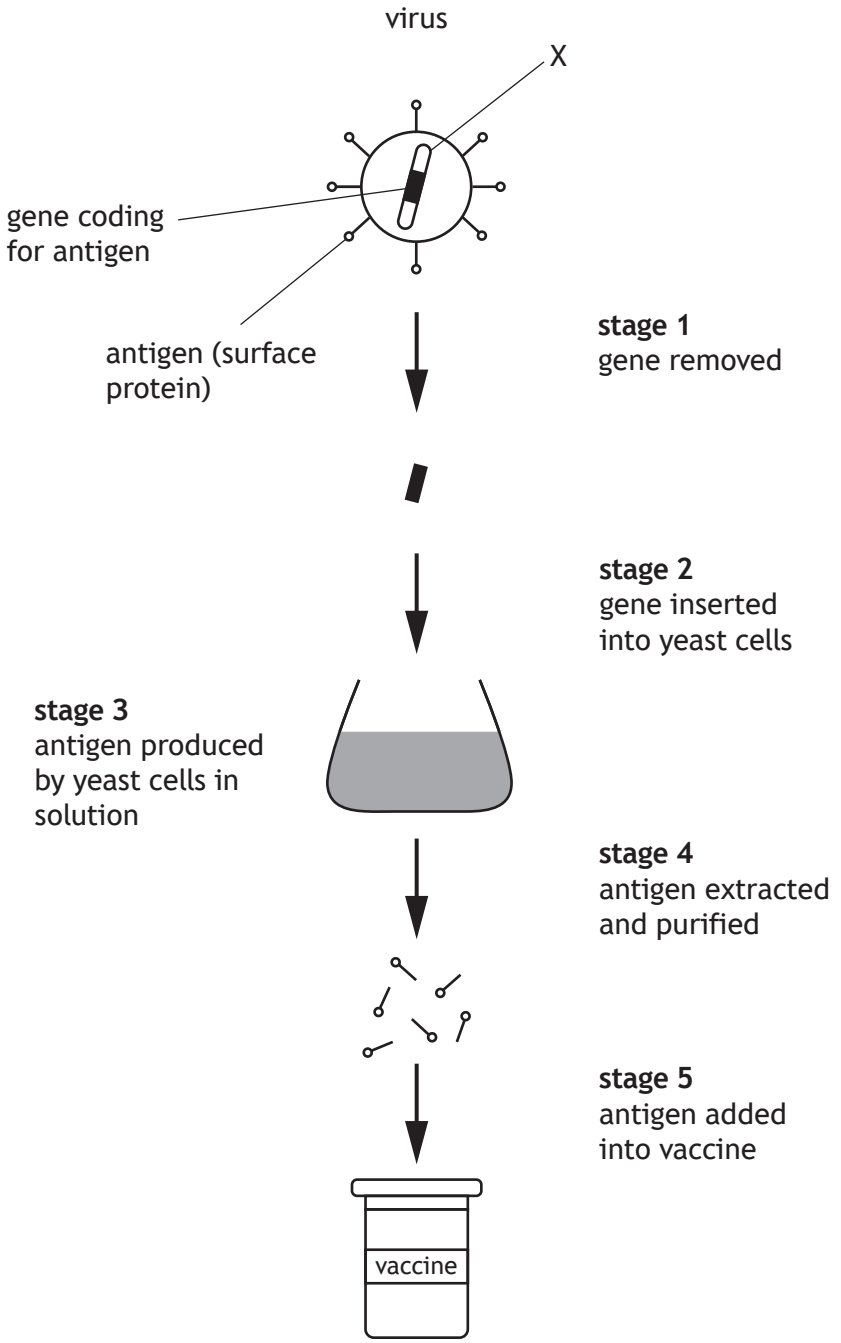
13. B lymphocytes form part of the specific immune system.

(a) Describe the mechanism of action of B lymphocytes against pathogens. **3**

(b) The specific immune system can respond to substances that are harmless to the body.

State the term used for this response. **1**

14. The diagram shows stages in the development of a vaccine against a virus.



(a) Suggest what structure X represents. **1**

_____

[Turn over

14. (continued)

(b) The development of this vaccine involves using part of the virus as an antigen.

State another source of antigens that can be used to produce vaccines. **1**

(c) (i) Explain why new vaccines must be subjected to clinical trials before being licensed for use. **1**

(ii) Describe how a double-blind procedure prevents a biased interpretation of the results from a clinical trial. **1**

(d) A new influenza vaccine is developed every year.

Explain why this is required to protect the body from the influenza virus. **2**

15. Attempt **either** A **or** B.

Write your answer in the space below and on *page 30*.

A Write notes on somatic and germline cells, including cell divison in both these cell types. **8**

OR

B Write notes on the production of the primary and mature mRNA transcripts. **8**

You may use labelled diagrams where appropriate.

National
Qualifications
2023

X840/76/12

Human Biology
Paper 1 — Multiple choice

THURSDAY, 27 APRIL
9:00 AM – 9:40 AM

Total marks — 25

Attempt ALL questions.

You may use a calculator.

Instructions for the completion of Paper 1 are given on *page 02* of your answer booklet X840/76/02.

Record your answers on the answer grid on *page 03* of your answer booklet.

Space for rough work is provided at the end of this booklet.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

Total marks — 25 marks

Attempt ALL questions

1. Which row in the table matches the type of cell division that occurs in germline cells with a description of the gametes produced?

	Type of division	Gametes produced
A	mitosis	diploid
B	mitosis	haploid
C	meiosis	diploid
D	meiosis	haploid

2. Which of the following statements about DNA replication is correct?

Fragments are formed on the

A leading strand and are joined by DNA polymerase

B lagging strand and are joined by ligase

C lagging strand and are joined by DNA polymerase

D leading strand and are joined by ligase.

3. PCR was used to amplify a region of DNA. After 5 cycles 32 copies were present.

Calculate the number of **additional copies** present after 4 further cycles.

A 224

B 256

C 480

D 512

4. The list shows the nucleic acid molecules involved in gene expression within a cell.

1. rRNA
2. mRNA
3. DNA
4. tRNA

Which of these molecules are required for translation?

A 1 only

B 2 and 4 only

C 1, 2 and 4 only

D 1, 2, 3 and 4

5. A student set up a test tube containing 5 cm^3 of milk and 1 cm^3 of the enzyme trypsin.

The milk became clear and the student concluded that the white milk protein had been broken down by trypsin.

To show that trypsin caused the milk to become clear, a control tube should contain

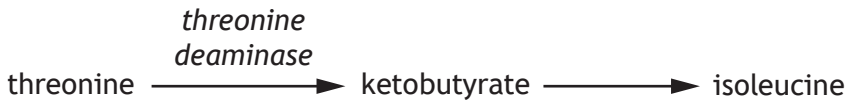
A 5 cm^3 of milk and 1 cm^3 of distilled water

B 5 cm^3 of distilled water and 1 cm^3 of trypsin

C 5 cm^3 of boiled and cooled milk and 1 cm^3 of trypsin

D 5 cm^3 of boiled and cooled milk and 1 cm^3 of distilled water.

6. Part of a metabolic pathway used to produce the amino acid isoleucine is shown.



Isoleucine is a feedback inhibitor of the enzyme threonine deaminase.

The statements refer to substances in the metabolic pathway.

1. Isoleucine binds to threonine deaminase.
2. Threonine deaminase lowers the activation energy required to convert threonine into ketobutyrate.
3. Ketobutyrate is the substrate of isoleucine.

Which of these statements are correct?

A 1 only

B 1 and 2 only

C 2 and 3 only

D 1, 2 and 3

[Turn over

7. The table shows the effect of a competitive inhibitor on an enzyme's activity.

Concentration of inhibitor (mM)	Enzyme activity (%)
0.00	100
0.10	60
0.20	40
0.30	20
0.40	0
0.50	0

Predict the concentration of inhibitor when half of the active sites of the enzyme are occupied by inhibitor.

A 0.15 mM

B 0.25 mM

C 0.40 mM

D 0.50 mM

8. An investigation was carried out into the effect of substrate concentration on the rate of a reaction in the presence of a competitive inhibitor.

Which of the following would ensure the results were valid?

A Repeating the investigation using a non-competitive inhibitor.

B Keeping the inhibitor concentration constant.

C Repeating the investigation with each concentration of substrate.

D Keeping the substrate concentration constant.

9. During aerobic respiration in a cell, one molecule of glucose yielded 38 molecules of ATP.

During respiration without oxygen in the same cell, one molecule of glucose yielded 2 molecules of ATP.

Calculate the percentage decrease in ATP yield per glucose molecule when this cell carries out respiration without oxygen compared to when it respires aerobically.

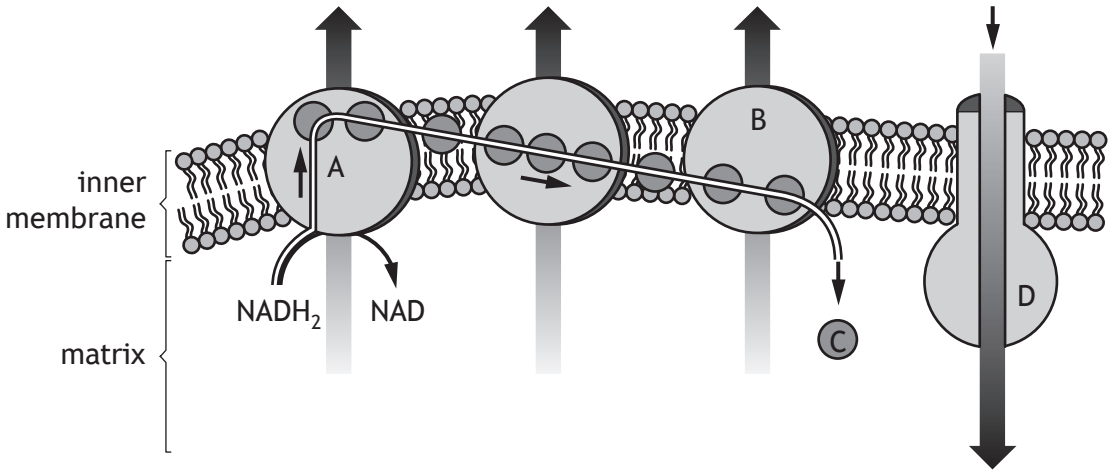
A 5%

B 6%

C 95%

D 1800%

10. The diagram represents some of the processes that occur in a mitochondrion.



Which letter represents ATP synthase?

11. Which of the following applies to the skeletal muscle fibres that are most useful during a cross-country ski race?

	Number of mitochondria	**Blood supply**	**Generation of ATP**
A	low	low	glycolysis only
B	high	large	glycolysis only
C	low	low	aerobic respiration
D	high	large	aerobic respiration

[Turn over

12. Which hormone, found in contraceptive pills, causes thickening of the cervical mucus?

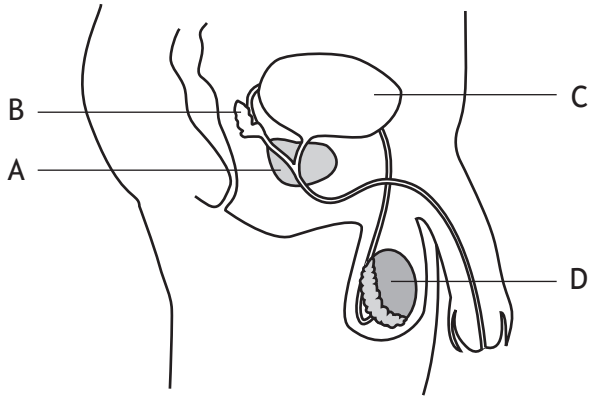
A Oestrogen

B Progesterone

C Luteinising Hormone (LH)

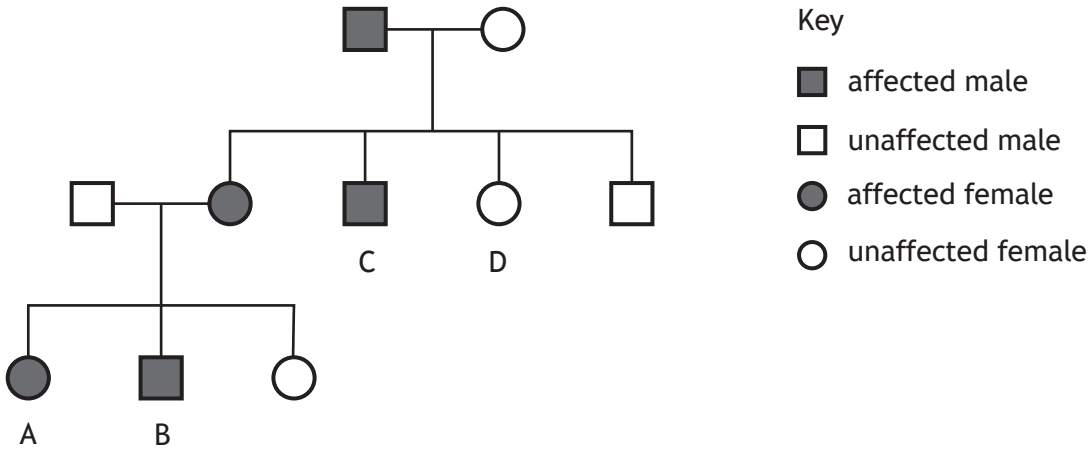
D Follicle Stimulating Hormone (FSH)

13. Which letter in the diagram indicates the site of testosterone production?



14. Cystic fibrosis is caused by a recessive allele.

The diagram shows the inheritance of cystic fibrosis in three generations of a family.



Which individual confirms that cystic fibrosis is **not** a sex-linked condition?

15. The following steps are involved in the process of in vitro fertilisation (IVF):

1. Zygotes are incubated.
2. Woman is given drugs to stimulate ovulation.
3. Eggs are mixed with sperm in a culture dish.
4. Eggs are surgically removed.
5. Embryos are implanted.
6. Fertilisation occurs.

Which of the following sequences shows the order in which these events take place?

A 4, 3, 6, 1, 2, 5
B 2, 4, 3, 6, 1, 5
C 4, 3, 1, 6, 5, 2
D 2, 4, 3, 6, 5, 1

16. Measurement of a child's cardiac cycle showed that systole lasted for 0.2 seconds while diastole lasted for 0.4 seconds.

What was the heart rate of this child?

A 60 beats per minute
B 75 beats per minute
C 100 beats per minute
D 150 beats per minute

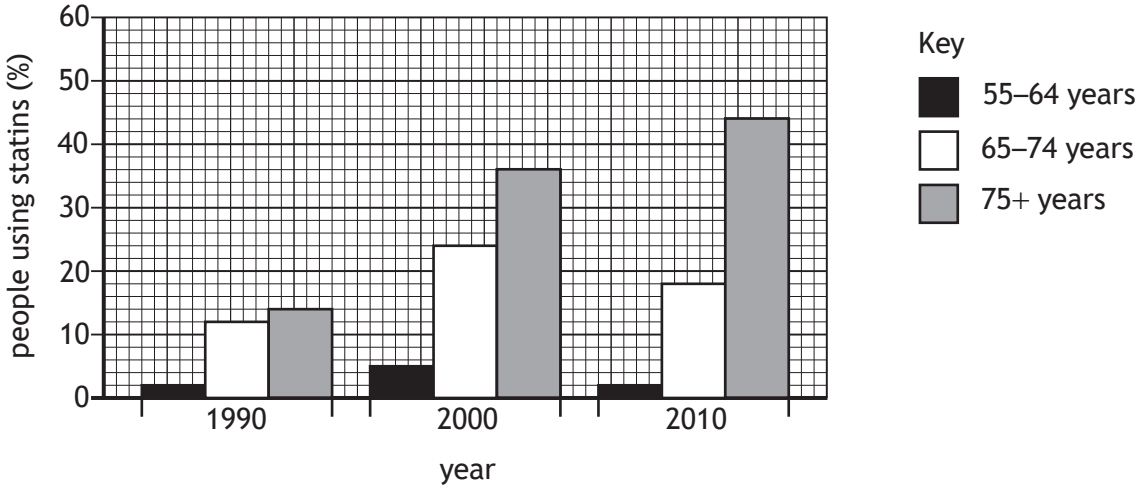
17. Peripheral vascular disease most commonly occurs in the arteries to the

A leg
B lung
C brain
D heart.

[Turn over

18. In a study, the percentage of people aged 55 years and older who use statins was recorded in 1990, 2000 and 2010.

The results are shown in the bar graph.



Which of the following statements is correct?

A Between 2000 and 2010, statin use increased in all age groups.

B Between 1990 and 2010, statin use in the 75+ age group tripled.

C Between 1990 and 2000, statin use in the 65–74 age group doubled.

D Between 1990 and 2010, statin use in the 55–64 age group decreased.

19. Which row in the table matches the branch of the autonomic nervous system (ANS) with its effect?

	Branch	**Effect**
A	sympathetic	increases intestinal secretions
B	sympathetic	decreases breathing rate
C	parasympathetic	decreases intestinal secretions
D	parasympathetic	increases peristalsis

20. Which of the following is **not** a method of encoding information into long term memory?

A Chunking

B Rehearsal

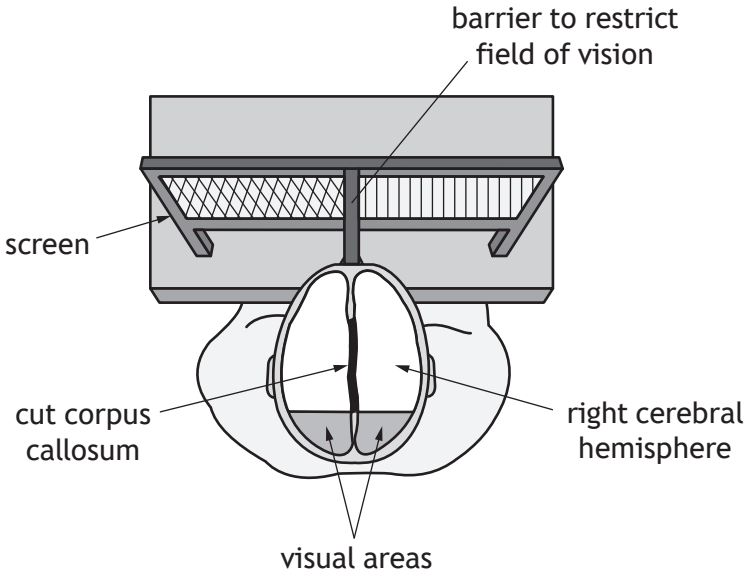
C Elaboration

D Organisation

21. Split-brain patients cannot transfer information between the right and left hemispheres of the cerebral cortex because the corpus callosum has been cut during surgery.

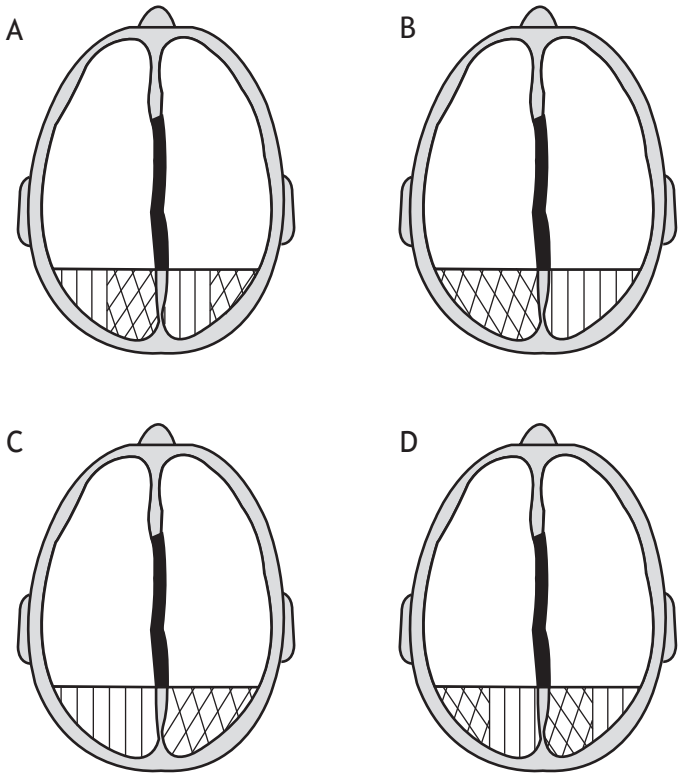
In a study a split-brain patient was asked to press their forehead against a barrier so that their left eye could only see to the left of the barrier and their right eye to the right of the barrier.

The diagram shows the setup of the study.



The patient was asked to look straight ahead and then two patterns were flashed briefly on the screen as shown.

Which of the following diagrams represents where the patterns are interpreted in the visual areas of this patient's cortex?



22. An individual displayed a hypersensitive response to a foreign substance.

Which row in the table identifies the cause of this response and the type of lymphocyte involved?

	Cause of response	**Type of lymphocyte involved**
A	autoimmune disease	B
B	allergy	B
C	autoimmune disease	T
D	allergy	T

23. During an epidemic, the R number refers to the number of cases of a disease that are directly caused by contact with one infected individual.

During an epidemic, an individual becomes infected when the R number is 2. This means that during the first level of transmission, two further people will be infected who will each infect two more people during the second level of transmission.

Predict the **total** number of people infected after four levels of transmission.

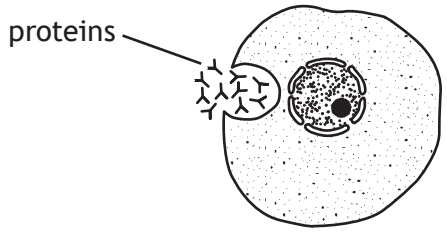
A 8

B 15

C 16

D 31

24. The diagram shows a lymphocyte releasing proteins it has produced.



Which row in the table identifies the type of lymphocyte and the proteins produced?

	Lymphocyte	**Protein**
A	B	antibody
B	T	antigen
C	T	antibody
D	B	antigen

25. The list contains statements about factors considered during the vaccination of a population.

1. The type of disease
2. The population density
3. The effectiveness of the vaccine
4. The percentage of non-immune individuals

Which of the statements refer to factors that can affect the herd immunity threshold?

A 1 only

B 3 only

C 1, 2 and 3 only

D 1, 2, 3 and 4

[END OF QUESTION PAPER]

FOR OFFICIAL USE

National Qualifications 2023

Mark

X840/76/01

Human Biology Paper 2

THURSDAY, 27 APRIL

10:10 AM – 12:30 PM

X8407601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day | Month | Year

Scottish candidate number

Total marks — 95

Attempt ALL questions.

You may use a calculator.

Question 15 contains a choice.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. Score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

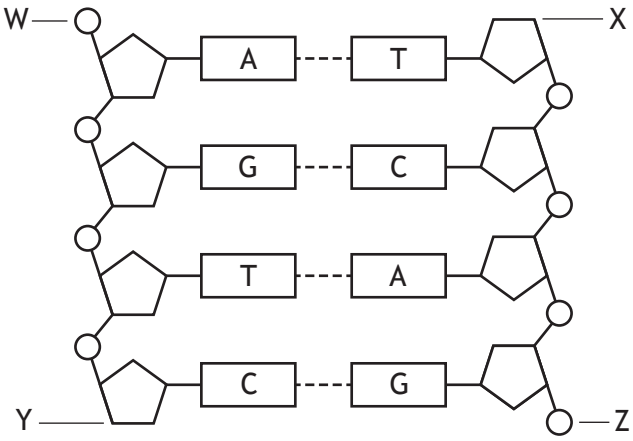
X8407601

Total marks — 95

Attempt ALL questions

Question 15 contains a choice

1. The diagram represents the structure of a section of DNA.



(a) (i) Name the repeating units that make up DNA. **1**

_____

(ii) Name the parts of these units, joined by strong bonds, that make up the backbone of DNA. **1**

_____ and _____

(iii) Identify one letter from the diagram that labels the 3′ end of a DNA strand. **1**

_____

1. (continued)

(b) (i) The first stage in the production of a protein involves the synthesis of mRNA from a section of DNA.

Name this stage. **1**

_____

(ii) Describe how alternative RNA splicing can produce different proteins from one gene. **1**

_____

_____

(iii) Describe the function of tRNA in protein synthesis. **2**

_____

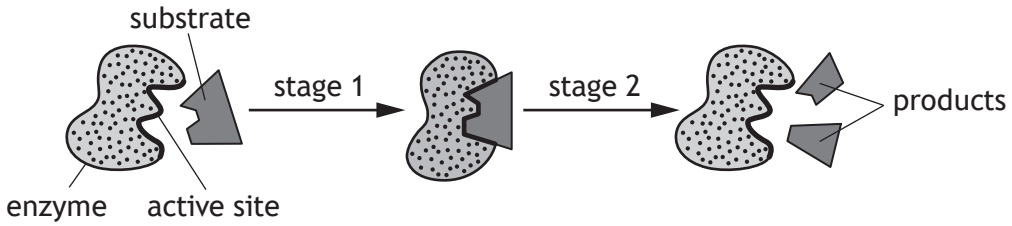
_____

_____

_____

[Turn over

2. (a) The diagram represents stages of an enzyme-catalysed reaction.



(i) Use the diagram to explain why this reaction can be described as catabolic. **1**

(ii) The diagram shows induced fit occurring between the enzyme and its substrate molecule.

Describe what happens during induced fit. **1**

(iii) Once the reaction is complete the products are released from the active site and the enzyme can be reused.

Explain why the products leave the active site. **1**

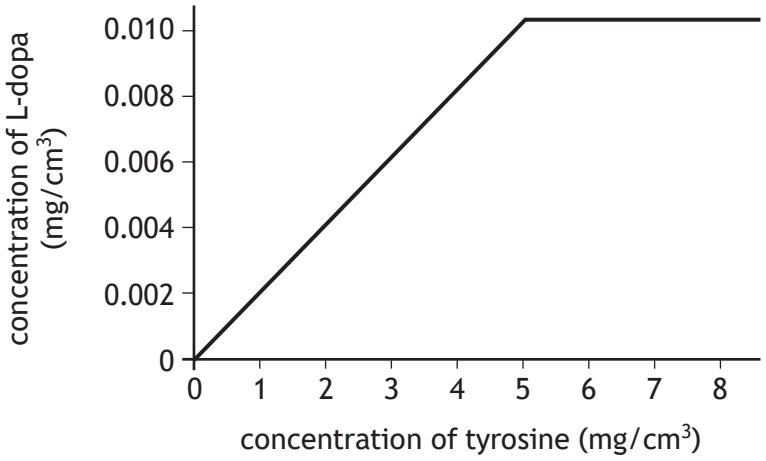
2. (continued)

(b) Parkinson's disease is caused by low levels of dopamine in the brain resulting in poor muscle co-ordination. Parkinson's disease can be treated using the drug L-dopa, which is converted to dopamine in the brain.

L-dopa is produced commercially in a reaction vessel using the enzyme tyrosinase as shown.

$$\text{tyrosine} \xrightarrow{\textit{tyrosinase}} \text{L-dopa}$$

The graph shows the results of using different concentrations of the substrate tyrosine on the concentration of L-dopa produced in the reaction vessel.



Using information from the graph, suggest why a tyrosine concentration of 5 mg/cm^3 is used in the commercial production of L-dopa. **1**

(c) Many enzymes are used in commercial processes. When using enzymes in these processes the end-product is regularly removed.

Suggest why end-products are removed during production processes. **1**

[Turn over

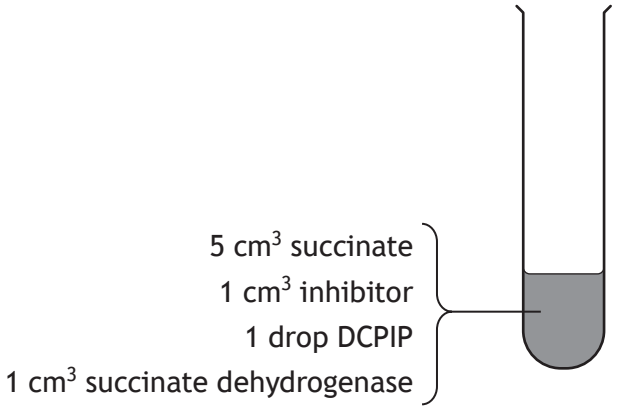
3. Succinate is converted to fumarate by a dehydrogenase enzyme as shown.

$$\text{succinate} \xrightarrow{\textit{succinate dehydrogenase}} \text{fumarate} + \text{hydrogen}$$

The rate of this reaction can be measured using the indicator DCPIP, which changes colour from blue to colourless when it combines with hydrogen.

An investigation was carried out into the effect of an inhibitor on the rate of this reaction at different concentrations of succinate.

Five test tubes were set up as shown in the diagram, each test tube contained a different concentration of succinate.



The time to decolourise the DCPIP indicator in each tube was measured.

The investigation was repeated without the inhibitor.

The results of the investigation are shown in the table.

	Time to decolourise DCPIP (s)	
Concentration of succinate (M)	**With inhibitor**	**Without inhibitor**
0.2	94	72
0.4	48	30
0.6	24	16
0.8	14	8
1.0	8	8

(a) (i) State one variable, other than those mentioned, that would need to be controlled for a valid conclusion to be drawn. **1**

_____

3. (a) (continued)

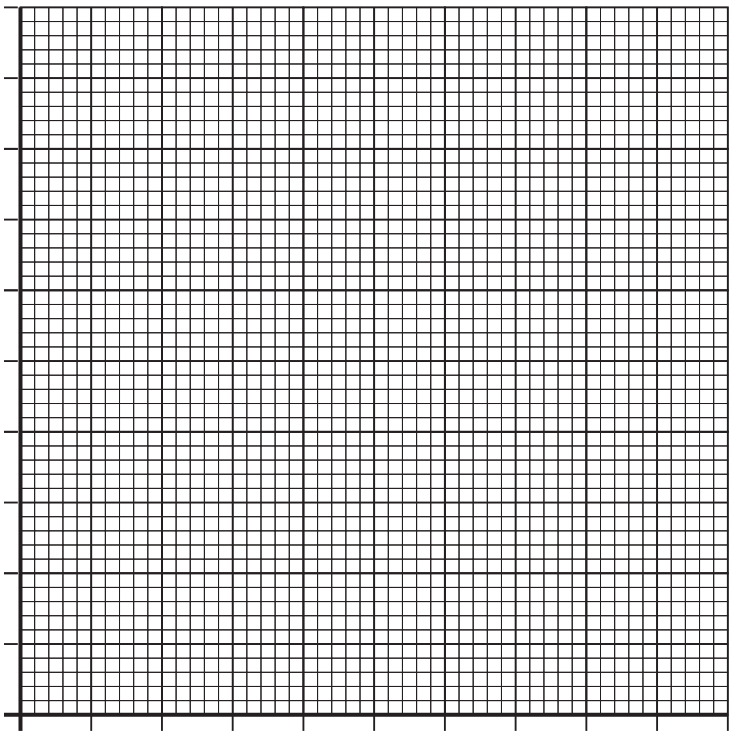
MARKS DO NOT WRITE IN THIS MARGIN

(ii) Describe one possible source of error when using DCPIP in this investigation. **1**

(iii) Suggest how the reliability of the results in this investigation could be improved. **1**

(b) Draw a line graph to show the results of the investigation **with inhibitor**. **2**

(Additional graph paper, if required, can be found on *page 30*.)



(c) Describe how the results show that the inhibitor used is a competitive inhibitor of succinate dehydrogenase. **1**

3. (continued)

(d) Succinate is a metabolite in the citric acid cycle.

(i) State the exact location of the citric acid cycle in a cell. **1**

_____

(ii) Describe the role of dehydrogenase enzymes in the citric acid cycle. **1**

_____

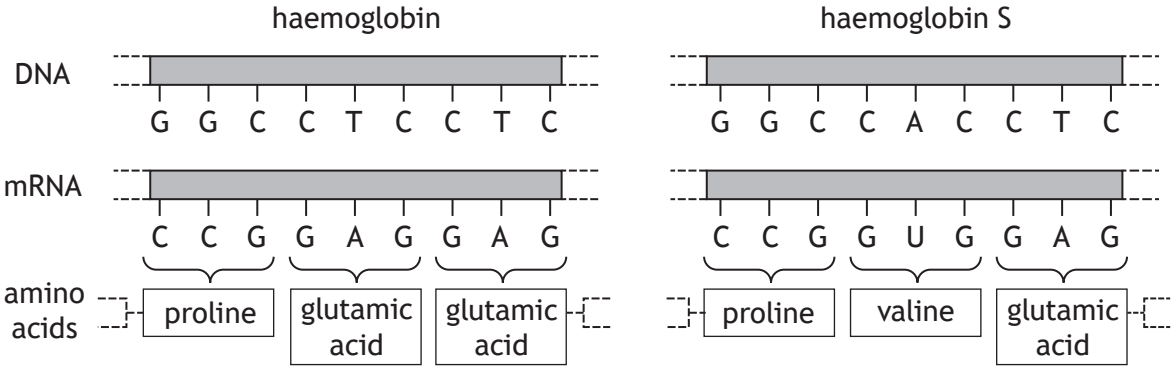
_____

(iii) Name the substance that combines with an acetyl group to form citrate during the citric acid cycle. **1**

_____

4. (a) Individuals with sickle cell disease have a form of haemoglobin called haemoglobin S, which is less efficient at carrying oxygen.

The diagram shows how a mutation in the gene coding for haemoglobin causes the production of haemoglobin S.



(i) Sickle cell disease is caused by a substitution mutation.

Describe evidence from the diagram that supports this statement. **1**

(ii) Suggest how changing one amino acid could cause the haemoglobin S protein to function less effectively than haemoglobin. **1**

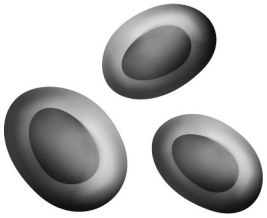
(b) Compare the effects of frame-shift and missense mutations on the resulting sequence of amino acids in a protein. **2**

[Turn over

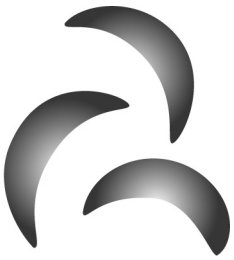
4. (continued)

(c) An individual with sickle cell disease has the homozygous genotype SS.

All their haemoglobin is type S. Their red blood cells can develop a distorted sickle shape and are stickier than normal red blood cells.



normal red blood cells



sickle shaped red blood cells

An individual with sickle cell trait has the heterozygous genotype HS.

Their red blood cells contain both forms of haemoglobin.

(i) Explain why the inheritance of sickle cell trait is an example of incomplete dominance. **1**

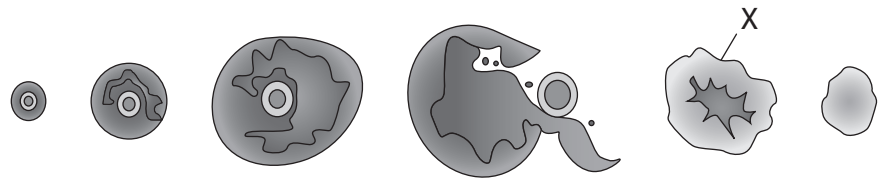
(ii) Suggest why individuals with sickle cell disease have an increased risk of having a stroke. **1**

(iii) When the population of the UK was 67 620 000, one in every 4600 people had sickle cell disease.

Calculate how many people had sickle cell disease. **1**

Space for calculation

5. The diagram represents six developmental stages that may be seen within an ovary during a menstrual cycle.



(a) (i) State which of the numbered stages represents ovulation. **1**

_____

(ii) Name structure X. **1**

_____

(b) At the end of the menstrual cycle progesterone concentration decreases.

(i) Describe the process that leads to this decrease. **2**

_____

_____

_____

(ii) Describe one effect of the decrease in progesterone concentration. **1**

_____

_____

(c) One treatment for infertility in women is to administer drugs to stimulate ovulation.

Describe the action of these drugs. **1**

_____

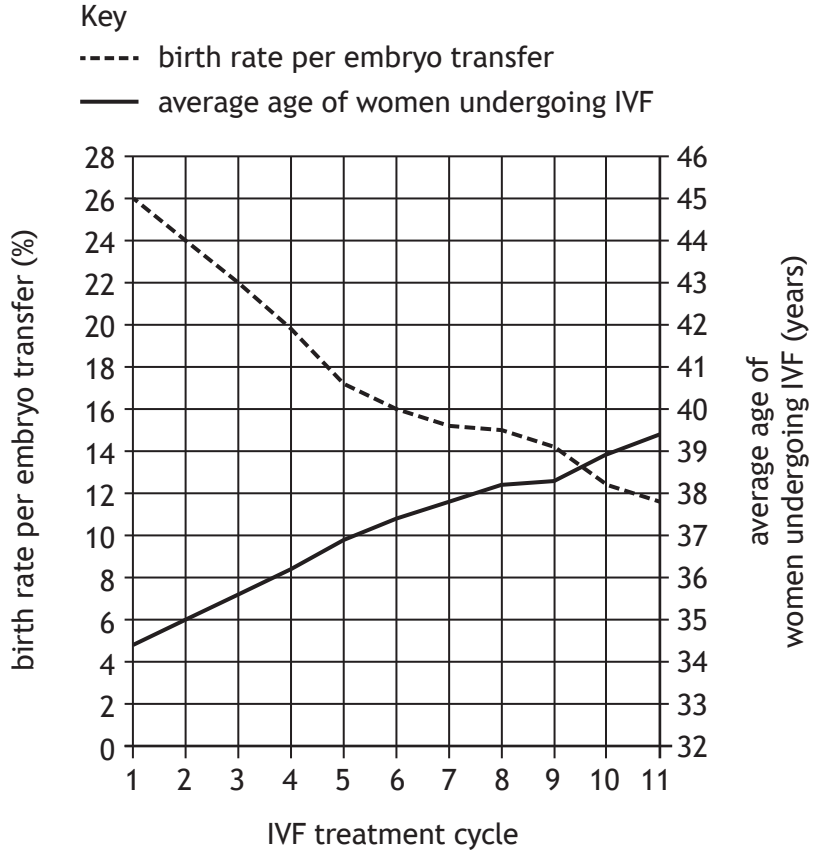
_____

[Turn over

6. In vitro fertilisation (IVF) is a fertility treatment, which may involve several treatment cycles.

Graph 1 contains information about the success rate of different treatment cycles of IVF in the UK in a year. It also shows the average age of women undergoing each treatment cycle.

Graph 1



(a) (i) Describe the two key trends shown in the graph. **2**

1. _____

_____

2. _____

_____

(ii) State the average age of women undergoing IVF when the birth rate per embryo transfer is 24%. **1**

_____ years

MARKS

DO NOT WRITE IN THIS MARGIN

(b) The table shows the number of women undergoing IVF and the number of embryos transferred during each treatment cycle.

IVF treatment cycle	Number of women undergoing IVF	Number of embryos transferred
1	23 400	24 000
3	9815	12 143
5	2800	3694
7	800	1119
9	300	341
11	220	300

(i) Calculate the percentage decrease in the number of women undergoing 11 treatment cycles of IVF compared to the number undergoing 1 treatment cycle of IVF. **1**

Space for calculation

_____ %

(ii) **Using data from the table and Graph 1**, calculate the number of babies born after 1 treatment cycle of IVF. **1**

Space for calculation

_____

(iii) Explain why the birth rate per embryo transfer for each IVF treatment cycle is expressed as a percentage in **Graph 1**. **1**

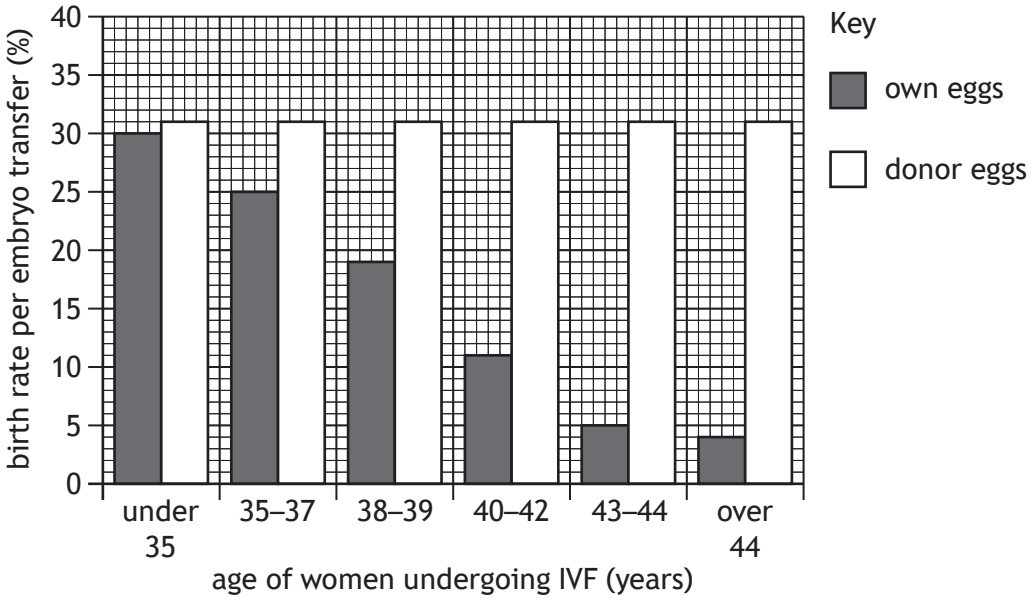
_____

_____

6. (continued)

(c) **Graph 2** shows the relationship between the age of women undergoing IVF and the birth rate per embryo transfer for women using their own eggs and donor eggs.

Graph 2



(i) State the difference in the birth rate per embryo transfer for women over 44 who use donor eggs instead of their own. **1**

_____ %

(ii) Describe how age affects the birth rate per embryo transfer when women's own eggs are used compared to using donor eggs. **1**

_____

_____

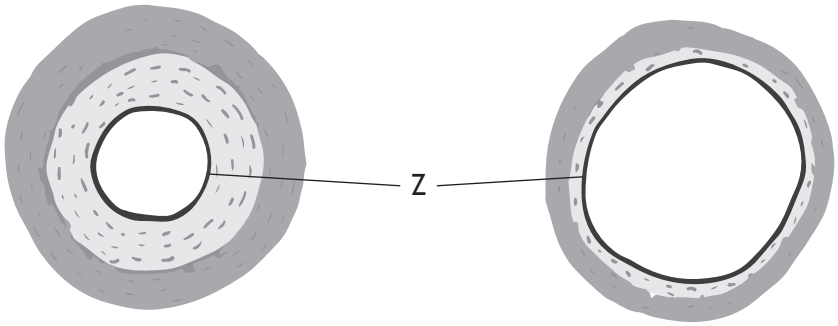
(iii) Suggest why the birth rate per embryo transfer is higher for donor eggs. **1**

_____

_____

7. The three main types of blood vessels are arteries, capillaries and veins.

(a) The diagram shows cross sections through two of these blood vessels.



blood vessel X blood vessel Y

(i) Blood vessel X is an artery and Y is a vein.

Describe one feature shown in the diagram that confirms this. **1**

_____

_____

(ii) Name layer Z. **1**

_____

(b) Describe the exchange of materials between blood plasma in capillaries and body cells. **3**

_____

_____

_____

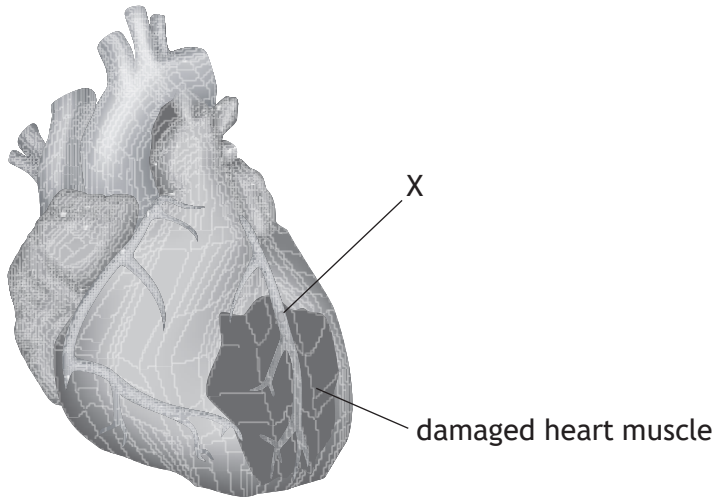
_____

_____

_____

[Turn over

8. The diagram shows muscle damage in the heart of an individual who has had a heart attack.



(a) (i) Name blood vessel X. **1**

_____

(ii) Explain how the formation of a thrombus in blood vessel X results in heart muscle damage. **1**

_____

_____

(b) Suggest why statins are prescribed to individuals to reduce the risk of a heart attack. **2**

_____

_____

_____

_____

8. (continued)

(c) Obesity is a major risk factor in the development of cardiovascular disease. BMI can be used to measure obesity.

(i) State the minimum BMI value that indicates obesity. **1**

_____

(ii) Calculate the mass of an individual who is 1.8 m tall with a BMI of 35.8. **1**

Space for calculation

_____ kg

[Turn over

9. Research has shown that some individuals who are obese can prevent the onset of type 2 diabetes by reducing their body mass.

(a) Name the test used to diagnose diabetes. **1**

_____

(b) An individual who weighed 120 kg was advised that they were at risk of developing type 2 diabetes. It was recommended that they reduce their body mass by 5%.

Calculate the individual's target body mass. **1**

Space for calculation

_____ kg

(c) Describe how type 2 diabetes affects liver cells. **2**

_____

_____

_____

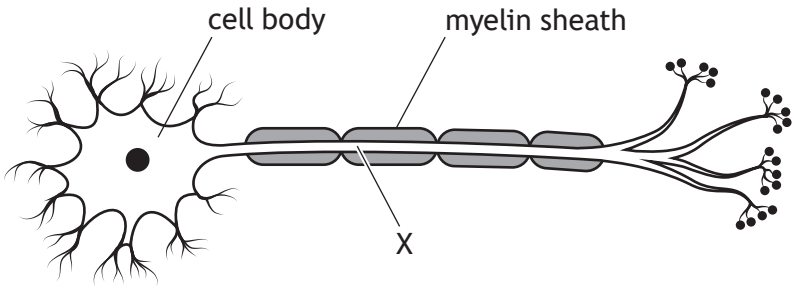
_____

(d) Describe how untreated diabetes may damage the retina. **1**

_____

_____

10. The diagram shows a motor neuron from an adult.



(a) (i) Name structure X. **1**

_____

(ii) State why it is important that structure X is insulated by the myelin sheath. **1**

_____

_____

(iii) Name the type of cell that produces the myelin sheath. **1**

_____

(b) Explain how the structure of motor neurons causes a one year old child to be less coordinated than an adult. **1**

_____

_____

_____

(c) Motor neurons are often part of diverging neural pathways.

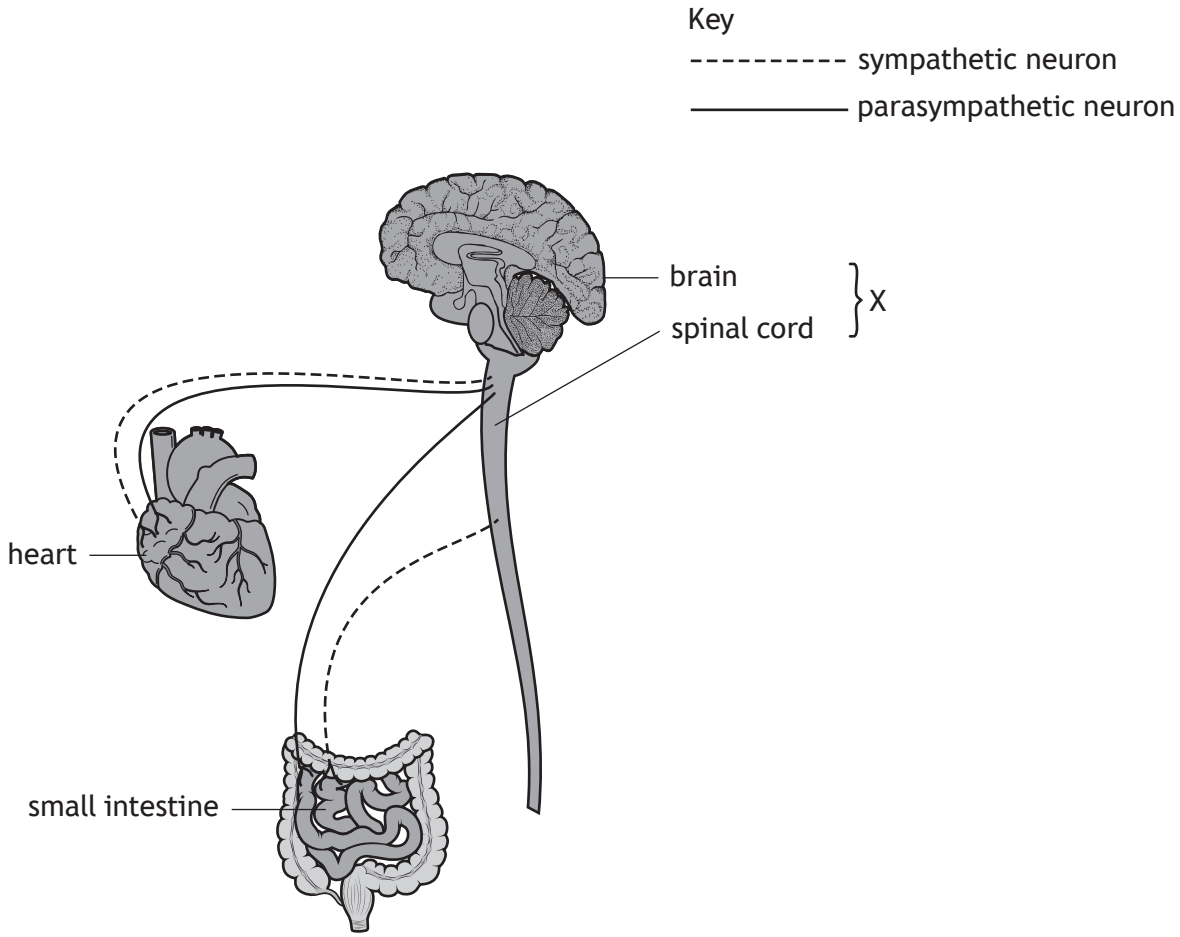
State one advantage of neurons being arranged in a diverging pathway. **1**

_____

_____

[Turn over

11. The diagram shows how neurons of the autonomic nervous system connect to some body organs.



(a) Name the system identified by X. **1**

_____

(b) (i) State the term that describes the opposing effect of the sympathetic and parasympathetic nervous systems on body organs. **1**

_____

11. (b) (continued)

(ii) During exercise, stimulation by sympathetic neurons increases heart rate and causes vasoconstriction of arteries in the small intestine.

Explain the importance of increased heart rate and vasoconstriction of arteries in the small intestine during exercise. **2**

Increased heart rate _____

_____

_____

Vasoconstriction of arteries in small intestine _____

_____

_____

[Turn over

12. An investigation was carried out into the effect of background rock music on the ability to recall words in a list.

MARKS DO NOT WRITE IN THIS MARGIN

Students of the same age were arranged into two groups of 20. At the same time of day, each group listened to a list of 10 words being read aloud.

While the words were being read out, one group was played rock music. After the words were read out the music was stopped, and each student was asked to write down the words they had heard.

The other group was a control group.

The results are shown in the tables.

Rock music group	
Position of word in list	**Number of students recalling word**
1	19
2	18
3	15
4	9
5	5
6	3
7	3
8	12
9	17
10	19

Control group	
Position of word in list	**Number of students recalling word**
1	20
2	19
3	17
4	12
5	7
6	6
7	6
8	14
9	19
10	20

(a) State two variables, other than those described above, that would need to be controlled during this investigation. **2**

1. _____

2. _____

(b) Calculate the difference between the two groups in the average number of words recalled per student. **1**

Space for calculation

_____

12. (continued)

(c) Describe the control that was set up for this investigation. **1**

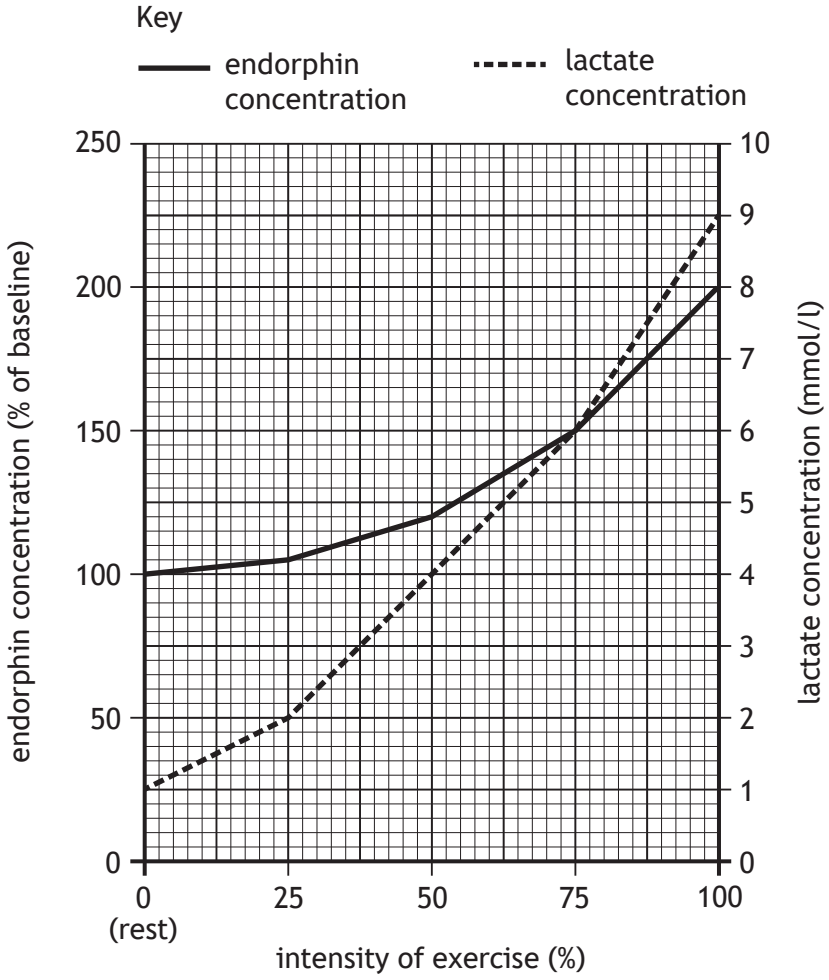
(d) State the conclusion that can be drawn from the results of the investigation. **1**

(e) Explain why there is an increase in the number of students recalling the words towards the end of the list. **1**

[Turn over

13. After exercise, some individuals experience a pleasant feeling known as 'runner's high' due to the production of endorphins.

(a) The graph shows the effect of the intensity of exercise on endorphin and lactate concentrations in the blood of an individual.



(i) State the endorphin concentration when the lactate concentration was 2 mmol/l. **1**

_____ % of baseline

(ii) Calculate the percentage increase in lactate concentration as the intensity of exercise increased from 50% to 100%. **1**

Space for calculation

_____%

13. (a) (continued)

(iii) Explain why lactate concentration increases as the intensity of exercise increases. **2**

(b) State one activity, other than exercise, that may increase endorphin levels. **1**

[Turn over

13. (continued)

(c) The table shows the results of an investigation to compare endorphin concentrations in two individuals of different fitness levels after exercise.

	Endorphin concentration (% of baseline)	
Duration of exercise (minutes)	**Fit individual**	**Unfit individual**
0	100	100
30	110	112
60	140	138
90	160	160

(i) Express, as a simple whole number ratio, the endorphin concentration after 30 minutes of exercise compared to after 90 minutes in an unfit individual. **1**

Space for calculation

_____ : _____

30 minutes 90 minutes

(ii) The investigation shows that endorphin levels increase as the duration of exercise increases.

State one other conclusion that can be drawn from this investigation. **1**

_____

_____

14. Tetanus is a disease caused by a type of bacteria. Tetanus bacteria live in soil and enter the body from the soil through wounds in the skin.

(a) State the term that describes disease-causing organisms such as tetanus bacteria. **1**

_____

(b) When bacteria enter the body, phagocytes often destroy them.

Describe how phagocytes destroy bacteria. **2**

_____

_____

_____

_____

(c) Phagocytes release chemicals called cytokines at the wound site.

Describe one function of cytokines. **1**

_____

_____

(d) Tetanus bacteria produce a toxin. One of the actions of this toxin is to bind to pre-synaptic membranes in synapses between neurons and muscle fibres.

Suggest why binding of this toxin to pre-synaptic membranes would decrease the transmission of impulses to muscle fibres. **1**

_____

_____

(e) Explain why herd immunity would **not** protect unvaccinated individuals against tetanus. **1**

_____

_____

[Turn over

15. Attempt **either** A **or** B.

Write your answer in the space below and on *page 29*.

A Describe the mode of action of recreational drugs and the effects on the body of their repeated use. **8**

OR

B Describe the contents of vaccines and the design of clinical trials used to establish their effectiveness. **8**

You may use labelled diagrams where appropriate.

National
Qualifications
2022

X840/76/12

Human Biology
Paper 1 — Multiple choice

THURSDAY, 19 MAY
9:00 AM – 9:40 AM

Total marks — 25

Attempt ALL questions.

You may use a calculator.

Instructions for the completion of Paper 1 are given on *page 02* of your answer booklet X840/76/02.

Record your answers on the answer grid on *page 03* of your answer booklet.

Space for rough work is provided at the end of this booklet.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

X8407612

Total marks — 25

Attempt ALL questions

1. The following list shows some procedures in which stem cells can be used:

1. Corneal repair
2. Drug testing
3. Skin regeneration.

Which procedures involve the therapeutic use of stem cells?

A 1 only

B 2 only

C 1 and 3 only

D 1, 2 and 3

2. The table shows the average cost of treating individuals with different types of cancer in the UK depending on their stage of diagnosis.

Type of cancer	Average cost of treatment (£)	
	Early stage diagnosis	Late stage diagnosis
Colon	3000	13 000
Ovarian	5000	15 000
Rectal	4000	12 000
Lung	8000	13 000

Which of the following statements is correct?

A Colon cancer is always the least expensive cancer to treat.

B Lung cancer is always the most expensive cancer to treat.

C Late stage diagnosis of ovarian cancer results in a 300% increase in the cost of treatment compared to early stage diagnosis.

D Early stage diagnosis of rectal cancer results in a 67% decrease in the cost of treatment compared to late stage diagnosis.

3. A section of a molecule of DNA has 12 000 bases with a 1:3 ratio of adenine to cytosine.

The number of guanine bases in this section is:

A 1500

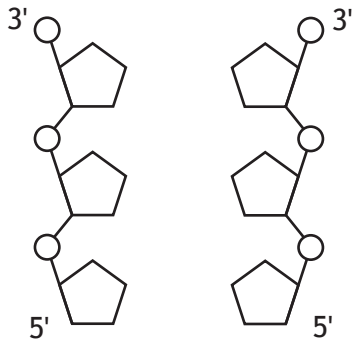
B 3000

C 4500

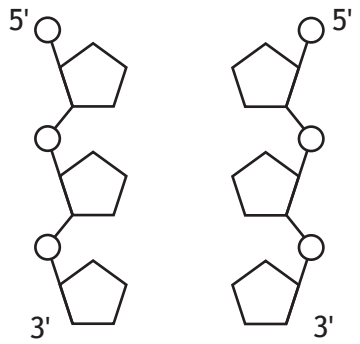
D 9000.

4. Which diagram shows the correct arrangement and labelling of the strands found in a section of a molecule of DNA?

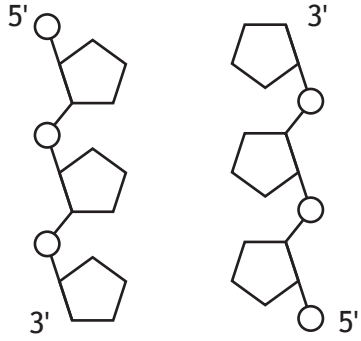
A



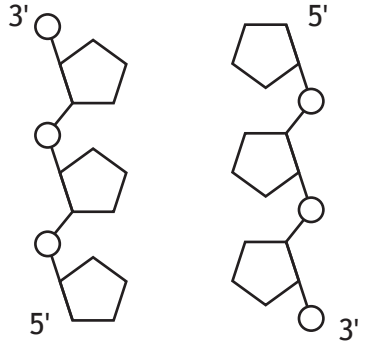
B



C

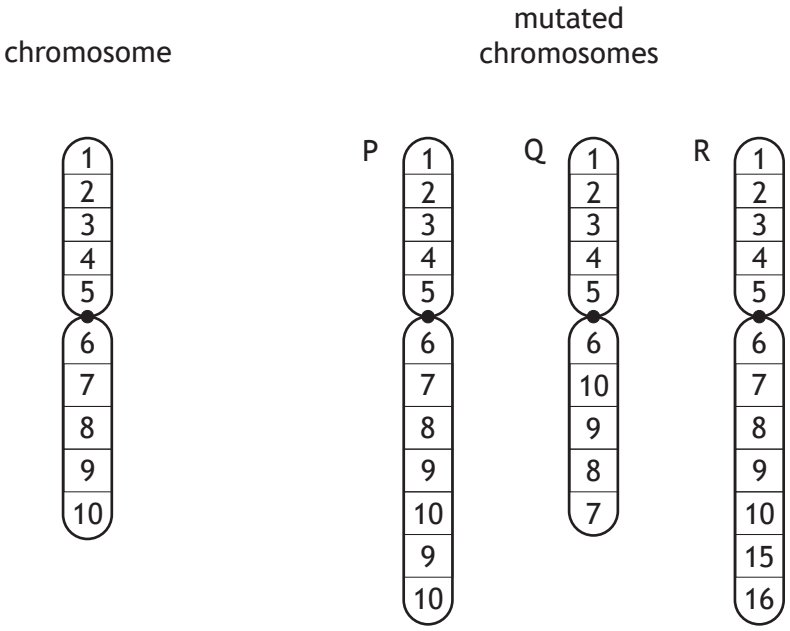


D



[Turn over

5. The diagram shows a chromosome and mutated versions of the same chromosome.
Each numbered segment on the chromosomes represents a gene.



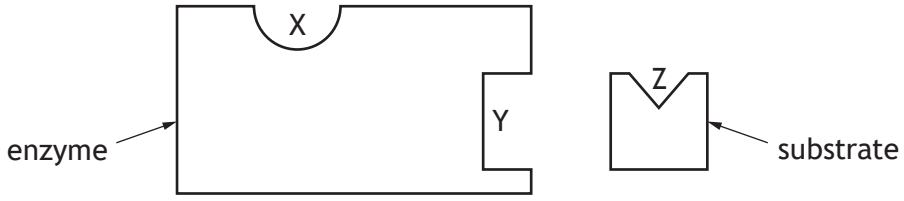
Which row in the table shows the type of mutations that have occurred?

	Chromosome P	**Chromosome Q**	**Chromosome R**
A	duplication	inversion	translocation
B	duplication	translocation	insertion
C	translocation	duplication	inversion
D	insertion	inversion	translocation

6. Which statement describes induced fit between an enzyme and its substrate?

A The active site changes shape after the substrate binds.
B The substrate changes shape after the enzyme binds.
C The active site changes shape before the substrate binds.
D The substrate changes shape before the enzyme binds.

7. The diagram shows an enzyme and its substrate.



Which row in the table identifies the sites where a competitive and a non-competitive inhibitor could bind?

	Competitive inhibitor	**Non-competitive inhibitor**
A	X	Y
B	X	Z
C	Y	Z
D	Y	X

8. Saliva contains the enzyme amylase, which breaks down starch into maltose.

The presence of starch can be tested for by adding iodine solution, which turns blue/black if starch is present.

In an investigation, four test tubes were set up in a water bath at 37 °C. Each test tube contained 10 cm^3 of starch solution and 2 cm^3 of amylase. 2 cm^3 of buffer solutions of different pH values were added to each test tube.

A sample of the contents of each test tube was removed every 30 seconds for 10 minutes and tested with iodine.

Identify the independent variable in this investigation.

A pH of solution in each test tube

B Volume of starch solution in each test tube

C Temperature of the test tubes in the water bath

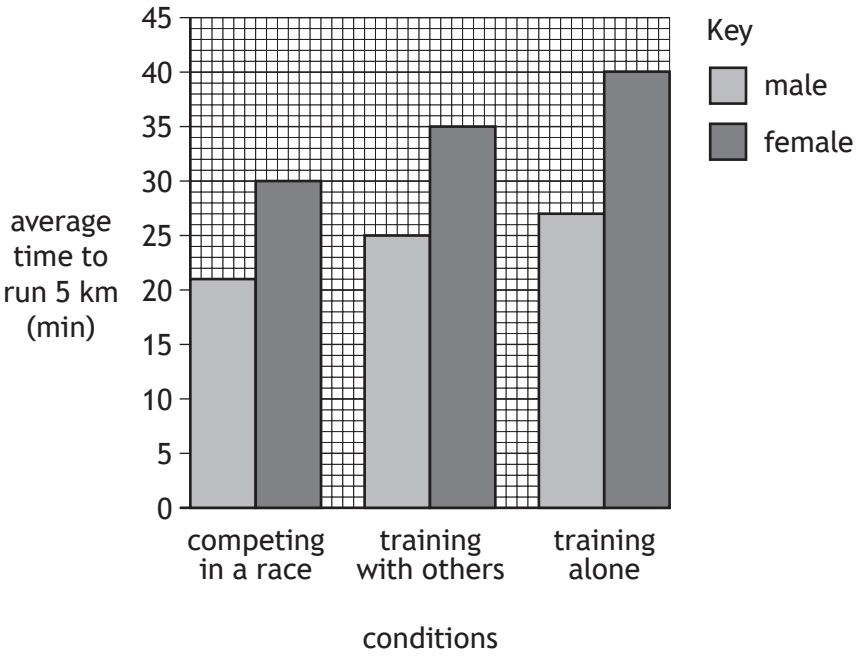
D Time taken for iodine solution to no longer turn blue/black

[Turn over

9. Which row in the table matches a substance with the stage of respiration in which it is involved?

	Substance	Stage
A	pyruvate	citric acid cycle
B	oxaloacetate	citric acid cycle
C	oxaloacetate	electron transport chain
D	pyruvate	electron transport chain

10. The graph shows the results of a survey carried out on members of a running club who ran 5 km under three different conditions.



What is the percentage improvement shown by females competing in a race compared to training alone?

A 10%

B 12.5%

C 25%

D 33.3%

11. The onset of puberty in males is triggered by a secretion from the:

A pituitary gland

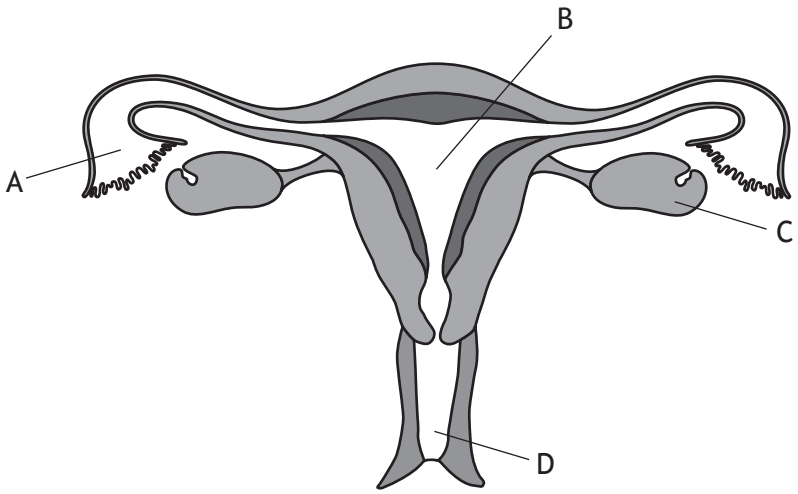
B hypothalamus

C interstitial cells

D seminal vesicles.

12. During IVF a fertilised egg is incubated until at least eight cells are formed.

Which letter indicates the location in the reproductive system into which this ball of cells would be transferred?



13. Red-green colour blindness is caused by an allele that is sex-linked and recessive.

A woman's father has the allele for colour blindness, but her mother does not.

The woman has a son with a man who is not colour blind.

What is the percentage chance that their son will be colour blind?

A 0%

B 25%

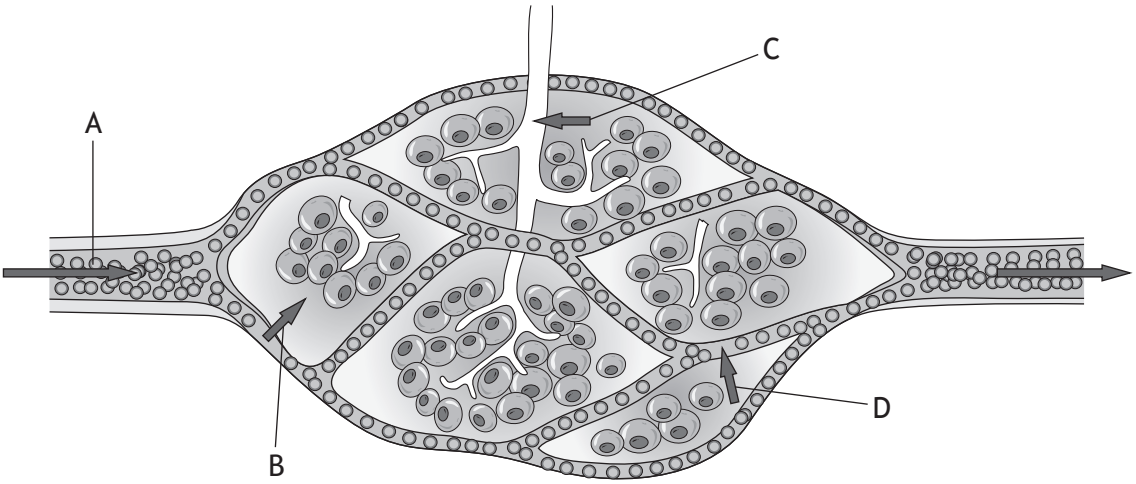
C 50%

D 75%

[Turn over

14. The diagram shows a capillary network within a tissue.

Which arrow represents pressure filtration of plasma?



15. Which row in the table shows the typical blood pressure in a blood vessel of a young adult during the cardiac cycle?

	Blood pressure (mmHg)	Blood vessel	Cardiac cycle stage
A	80	vein	diastole
B	80	artery	systole
C	120	vein	diastole
D	120	artery	systole

16. Which row in the table describes features typical of type 2 diabetes?

	Onset	Effect
A	Occurs in childhood	Cells unable to produce insulin
B	Occurs in childhood	Cells less sensitive to insulin
C	Develops later in life	Cells unable to produce insulin
D	Develops later in life	Cells less sensitive to insulin

17. The table shows the number of new cases of diabetes diagnosed in the UK in 2013 and 2018.

	Number of new cases of diabetes (thousands)	
Year / Location	**2013**	**2018**
England	2700	3200
Scotland	250	300
Wales	170	200
Northern Ireland	80	100

Which statement is correct for the number of new cases between 2013 and 2018?

A Scotland had a 50% increase in new cases.

B Wales had the lowest increase in new cases.

C England had a yearly average increase of 100 new cases.

D Northern Ireland had a 25% increase in new cases.

18. The increase in an athlete's heart rate and breathing rate during a race involves:

A sympathetic neurons of the autonomic nervous system

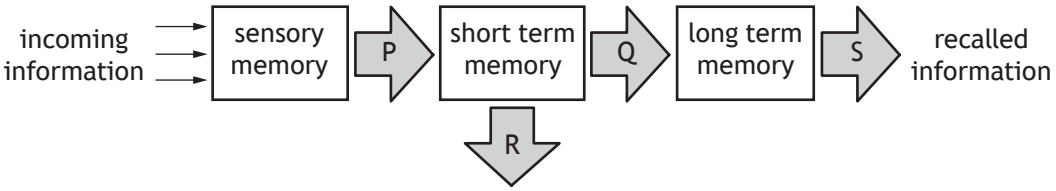
B parasympathetic neurons of the somatic nervous system

C sympathetic neurons of the somatic nervous system

D parasympathetic neurons of the autonomic nervous system.

[Turn over

19. The diagram shows the processing of information within memory.



Which row in the table identifies the memory processes shown in the diagram?

	Memory process		
	Retrieval	**Encoding**	**Displacement**
A	S	Q	R
B	P	R	Q
C	S	P	Q
D	Q	P	R

20. The following are methods used to aid memory:

1. Chunking
2. Elaboration
3. Organisation.

Which of these methods can be used to improve the transfer of information from short term to long term memory?

A 1 only

B 1 and 3 only

C 2 and 3 only

D 1, 2 and 3

21. The following steps occur during the inflammatory response:

1. Blood flow increases.
2. Histamine is released by mast cells.
3. Phagocytes accumulate at the site of infection.
4. Vasodilation occurs and capillary permeability increases.

In which sequence do these steps occur?

A 2, 1, 4, 3
B 4, 2, 1, 3
C 4, 1, 3, 2
D 2, 4, 1, 3

22. Autoimmune diseases are a result of:

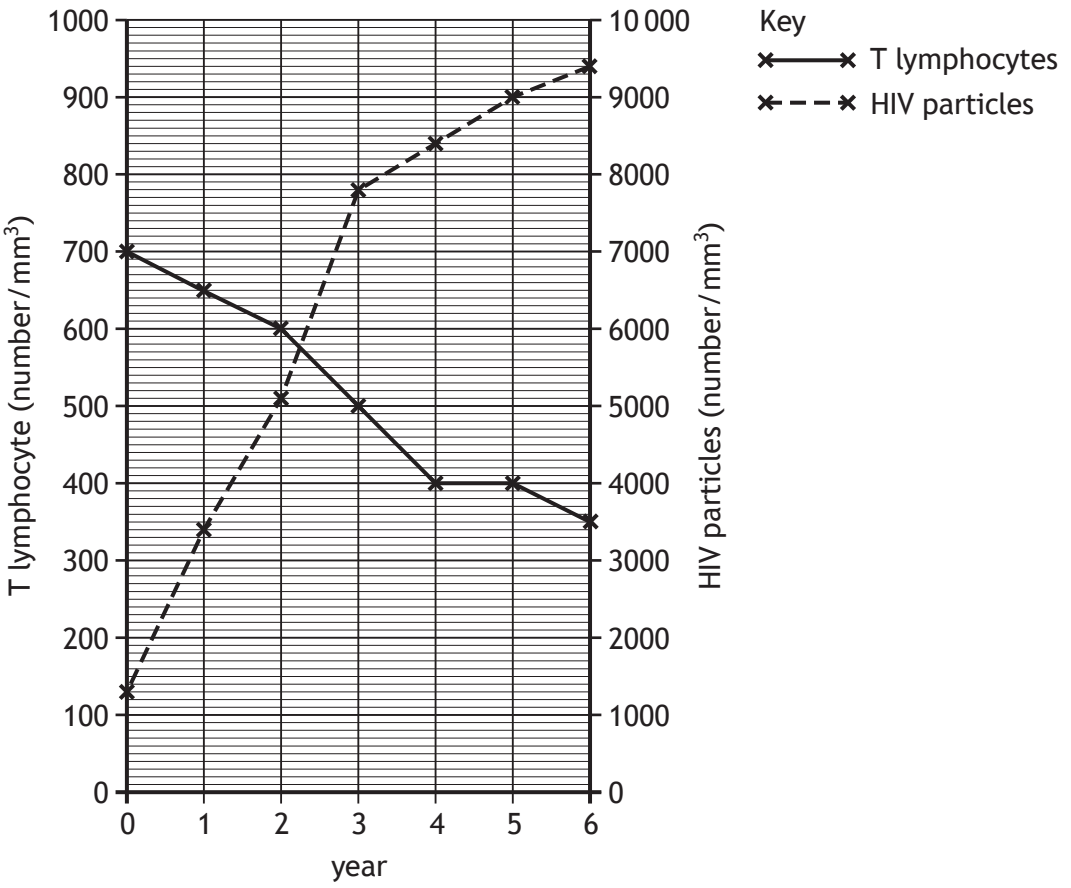
A T lymphocytes responding to pathogens
B T lymphocytes responding to self-antigens
C B lymphocytes responding to pathogens
D B lymphocytes responding to self-antigens.

23. Some individuals have allergies, which mean they cannot receive certain vaccines. These individuals may benefit from vaccination programmes through:

A herd immunity
B antigenic variation
C non-specific immunity
D personalised medicine.

[Turn over

24. The graph shows the relationship between the number of T lymphocytes and the number of HIV particles present in the blood of an infected individual over a 6-year period.



Which of the following statements is **not** correct?

A The ratio of T lymphocytes to HIV particles at year 4 is 1:21.

B The HIV particle number increased fastest between years 2 and 3.

C The T lymphocyte number decreases continuously over the 6-year period.

D The HIV particle number is always higher than the T lymphocyte number.

25. Which experimental design feature reduces the magnitude of experimental error in a clinical trial?

A A placebo control

B A suitable group size

C Double-blind protocols

D Using randomised groups

[END OF QUESTION PAPER]

FOR OFFICIAL USE

National Qualifications 2022

Mark

X840/76/01

Human Biology Paper 2

THURSDAY, 19 MAY

10:10 AM – 12:30 PM

X8407601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day | Month | Year

Scottish candidate number

Total marks — 95

Attempt ALL questions.

You may use a calculator.

Question 15 contains a choice.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. Score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

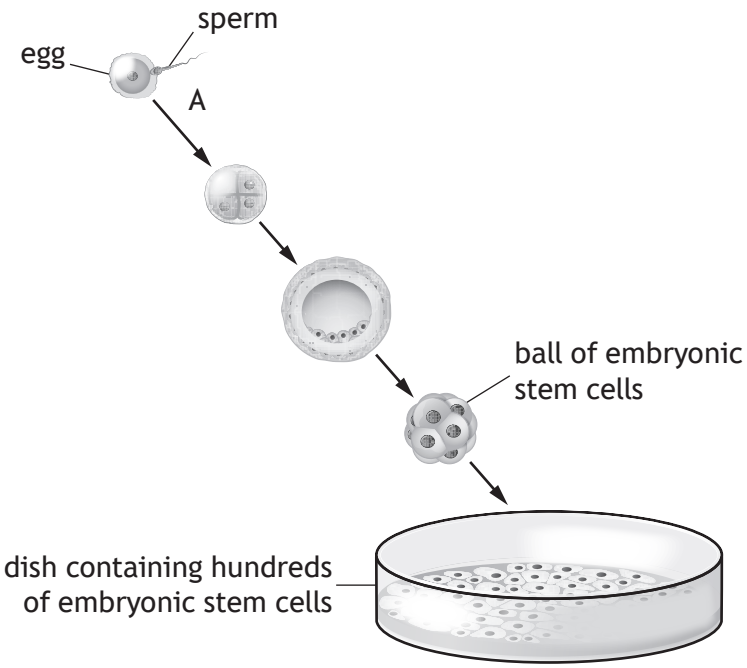
X8407601 01

Total marks — 95

Attempt ALL questions

Question 15 contains a choice

1. The diagram shows some of the stages involved in the formation of embryonic stem cells for use in a laboratory.



(a) (i) Name the type of cells that divide to form sperm and eggs. **1**

_____

(ii) Name the type of cell division occurring at A. **1**

_____

(iii) Explain why embryonic stem cells are described as being pluripotent. **1**

_____

_____

1. (continued)

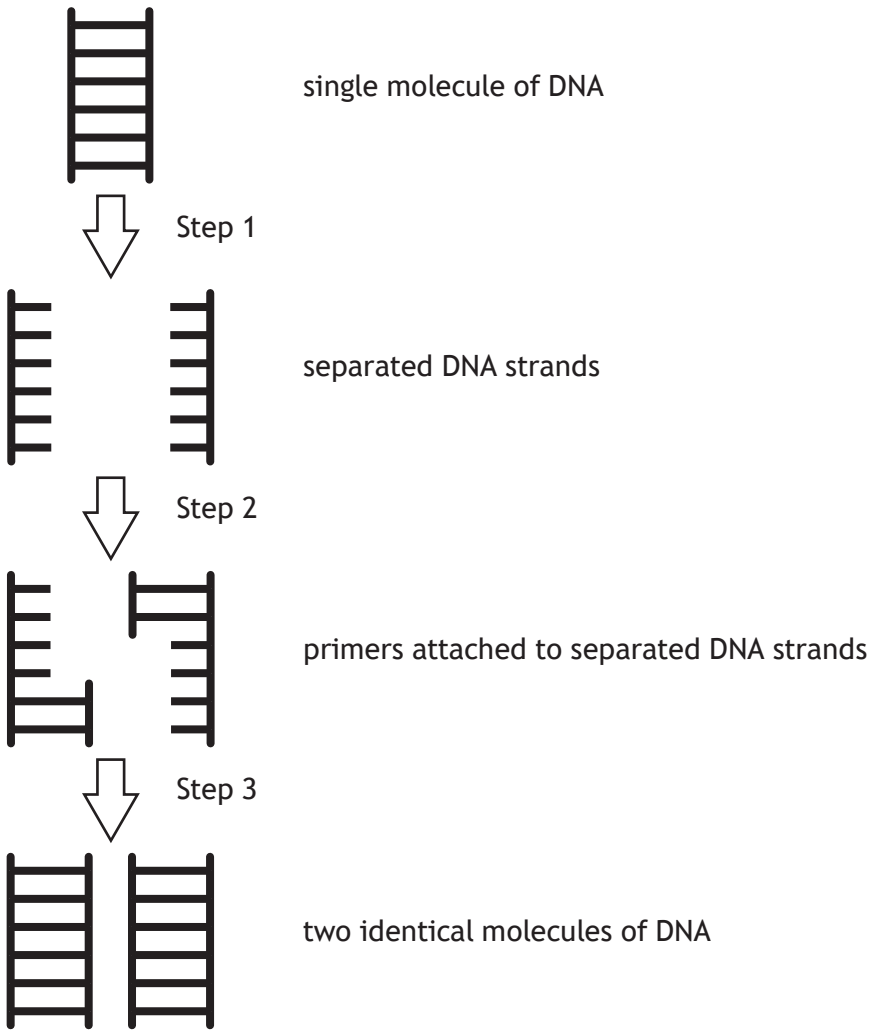
(b) Explain how differentiation of tissue stem cells leads to the production of specialised cells such as red blood cells. **1**

(c) Research has developed a type of stem cell that can be cultured in a laboratory directly from a patient's own somatic cells.

Suggest a benefit to the patient of using these stem cells in therapeutic treatments. **1**

[Turn over

2. The diagram represents three steps during one cycle of the Polymerase Chain Reaction (PCR).



(a) Give values to describe the change in the temperature that occurs in step 2. **1**

2. (continued)

(b) (i) Name the enzyme used in step 3. **1**

_____

(ii) Suggest an advantage of using a heat tolerant form of this enzyme during PCR. **1**

_____

_____

(c) Calculate the number of DNA molecules produced from a single molecule of DNA after 10 cycles of PCR. **1**

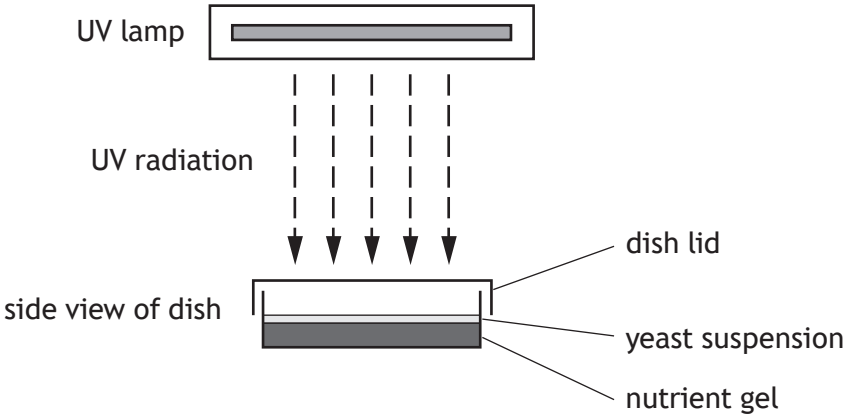
Space for calculation

_____DNA molecules

[Turn over

3. An investigation was carried out into the effect of UV radiation exposure time on the survival of yeast cell colonies.

20 cm^3 of a concentrated yeast cell suspension was diluted with 80 cm^3 of water. 6 dishes containing a nutrient gel had 2 cm^3 of the diluted yeast cell suspension added to them. Each dish was then exposed to UV radiation for different periods of time. UV radiation induces mutation in these yeast cells.



The dishes were then transferred to an incubator for 48 hours before the number of yeast cell colonies in each dish was counted.

(a) State two variables, other than those described above, that would need to be controlled when setting up this investigation. **2**

1. _____

_____

2. _____

_____

(b) Suggest why the concentrated yeast cell suspension was diluted with water before it was added to the dishes. **1**

_____

_____

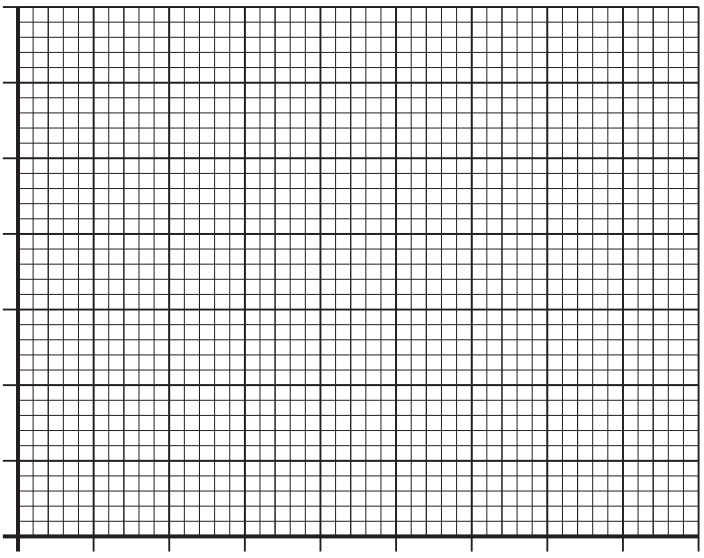
3. (continued)

(c) The results of the investigation are shown in the table.

Length of UV exposure time (min)	Number of yeast cell colonies
1	70
3	66
5	58
7	46
9	30

(i) Draw a line graph to show the data in the table. **2**

(Additional graph paper, if required, can be found on *page 31*.)



(ii) State the conclusion that can be drawn from these results. **1**

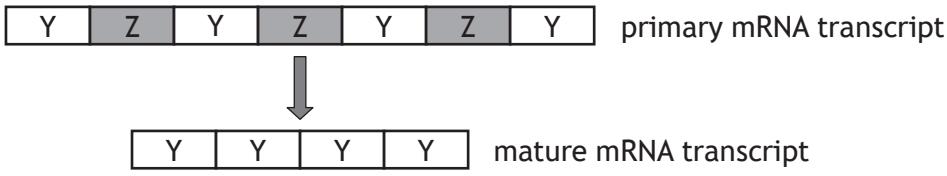
_____

_____

(iii) Predict the number of yeast cell colonies that would be present if the exposure time was 11 minutes. **1**

_____

4. The diagram shows a strand of mRNA undergoing splicing.



(a) Name the regions labelled Y. **1**

_____

(b) The primary transcript contained 3150 bases. The total number of bases removed by the splicing process was 600. The mature transcript includes one start and one stop codon, which do not code for amino acids in the final protein.

Calculate how many amino acids are present in the protein coded for by the mature transcript. **1**

Space for calculation

_____amino acids

(c) (i) Name the process that results in different proteins being expressed from a single gene. **1**

_____

(ii) Describe how this process leads to the formation of different proteins. **1**

_____

_____

_____

4. (continued)

(d) A splice-site mutation resulted in the following mature mRNA transcript.

Y	Z	Y	Y	Y

(i) Describe the effect that this mutation has had on the mature mRNA transcript. **1**

(ii) Suggest how this mutation may affect the structure of the protein formed. **1**

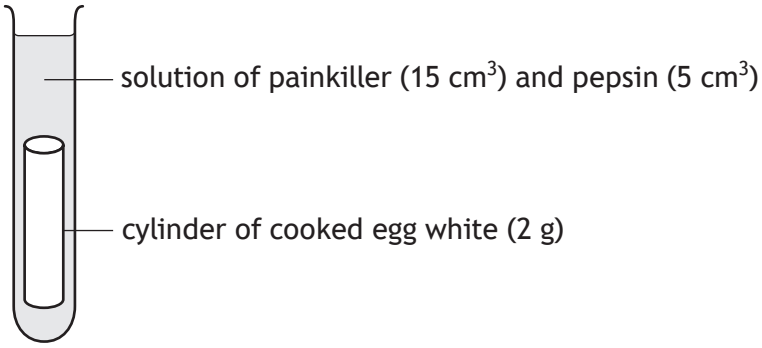
[Turn over

5. Some painkillers are recommended to be taken after a meal. However, painkillers can inhibit the action of digestive enzymes.

An investigation was carried out into the effect of different painkillers on the inhibition of the digestive enzyme, pepsin.

Cooked egg white is composed of protein, which can be broken down by pepsin.

Test tubes containing different painkiller solutions were set up as shown. A control test tube was also set up.



The test tubes were left for 24 hours at 37 °C and then the mass of egg white broken down was calculated.

The table shows the results of the investigation.

Painkiller	Mass of egg white broken down (g)
Paracetamol	1.4
Aspirin	1.1
Ibuprofen	1.3

(a) Two tablets of each painkiller were used to make up the solutions.

Suggest why this may not have allowed a valid comparison of the effects of the different painkillers. **1**

_____

_____

5. (continued)

(b) Describe the contents of the solution in the control tube. **1**

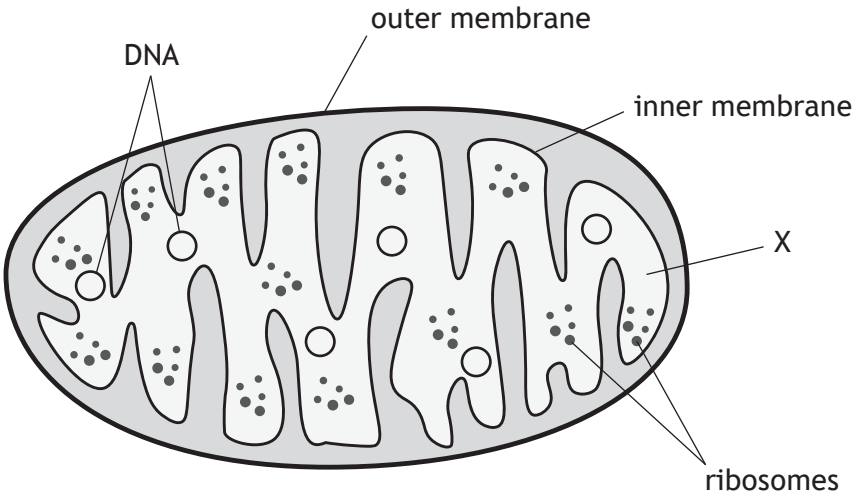
(c) Describe how the results were calculated in this investigation. **1**

(d) State which painkiller had the greatest inhibitory effect on pepsin activity. **1**

(e) Describe how the reliability of the results from this investigation could be improved. **1**

[Turn over

6. The diagram represents a mitochondrion from a cell.



(a) (i) Name the region of the mitochondrion labelled X. **1**

_____

(ii) Name the stage of respiration that occurs at the inner membrane. **1**

_____

(iii) Explain why this mitochondrion can synthesise proteins. **1**

_____

_____

_____

6. (continued)

(b) Mitochondrial disease is a condition caused by mutations in the genes needed for mitochondria to function effectively.

(i) Suggest why muscle is one of the main tissues affected by mitochondrial disease. **1**

(ii) Name the type of muscle fibre most likely to be affected by mitochondrial disease. **1**

(iii) Explain why some individuals with mitochondrial disease are unable to carry out endurance activities such as long distance running. **1**

(c) In another form of mitochondrial disease, affected individuals produce a non-functional form of an enzyme, which results in large quantities of lactate in their cells.

Suggest the function of this enzyme in **unaffected** individuals. **1**

[Turn over

7. A number of hormonal changes occur in a woman's body during the menstrual cycle.

(a) (i) State one function of each of the following hormones in the menstrual cycle. **2**

FSH _____

_____

Oestrogen _____

_____

(ii) Name the structure within an ovary that produces progesterone. **1**

_____

(b) A woman took a daily oral contraceptive pill.

(i) Explain how taking this pill would affect the FSH concentrations in her blood during her menstrual cycle. **2**

_____

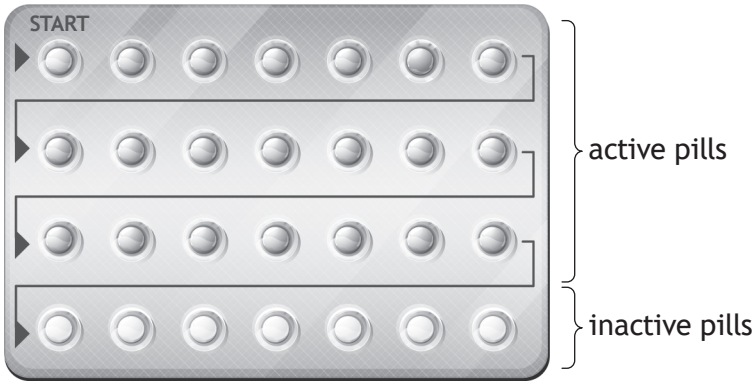
_____

_____

_____

7. (b) (continued)

(ii) One type of oral contraceptive is made up of 21 active pills and 7 inactive pills.



The inactive pills are normally identical to the active pills but contain no active ingredients.

Explain why menstruation usually occurs during the days that the woman takes the inactive pills. **1**

(c) The 'morning after pill' is an emergency hormonal contraceptive pill.

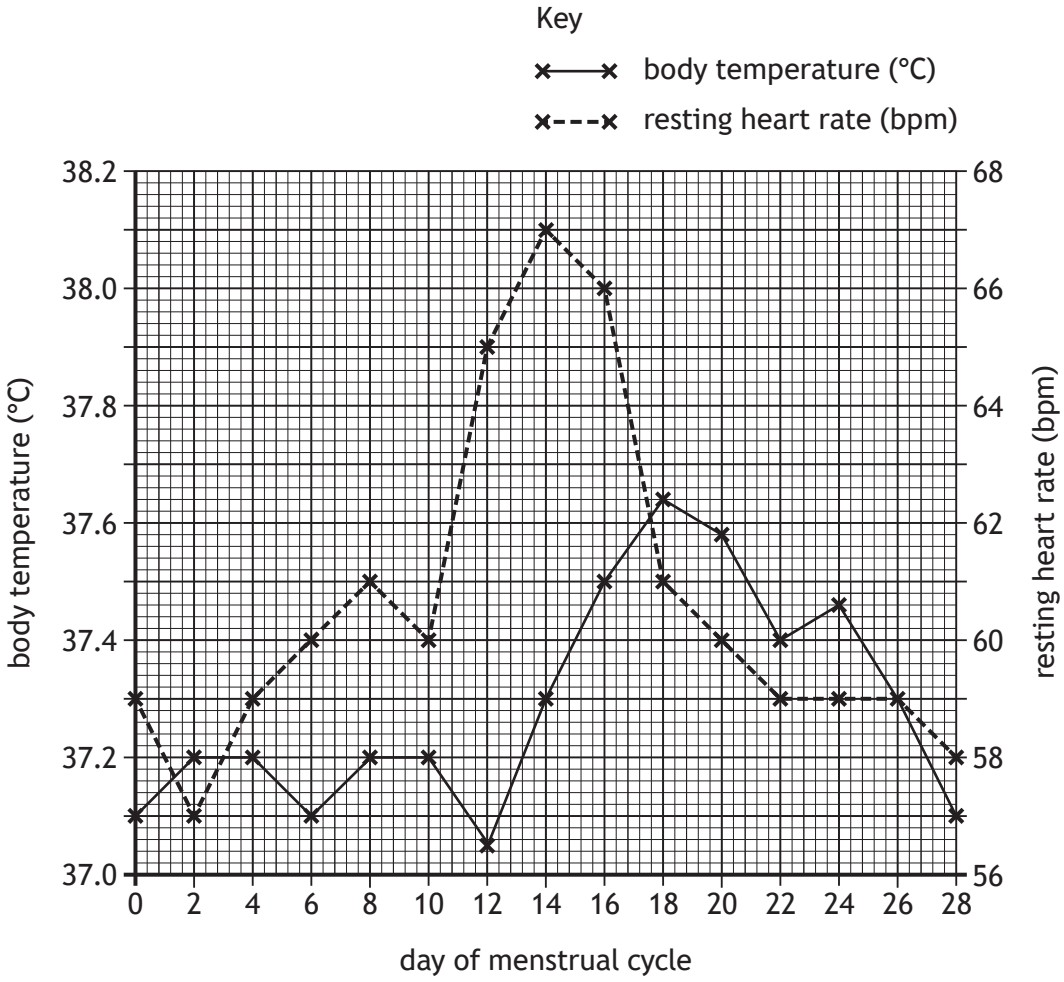
Explain how this type of pill prevents pregnancy. **1**

[Turn over

8. Body temperature and heart rate change during the menstrual cycle.

Graph 1 shows the body temperature and heart rate of a woman during a 28 day menstrual cycle. Readings were taken at the same time every day.

Graph 1



(a) State the body temperature when the resting heart rate was 57 bpm. **1**

_____°C

(b) Calculate the decrease in resting heart rate that occurs between days 14 and 28 of the cycle. **1**

Space for calculation

_____bpm

8. (continued)

(c) (i) Calculate the increase in body temperature that occurs between days 12 and 18 of the cycle. **1**

Space for calculation

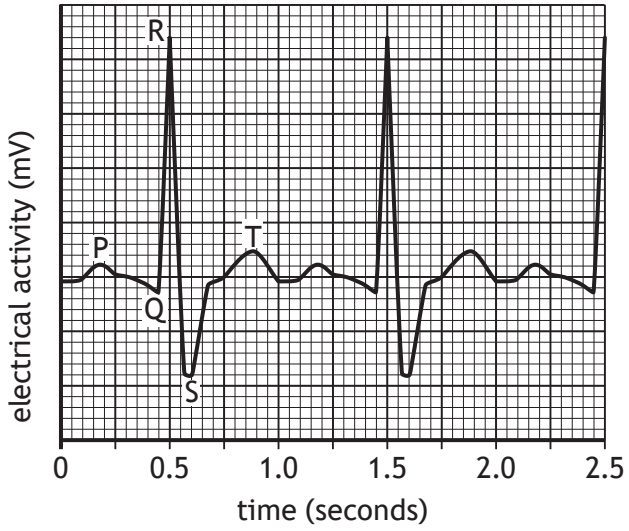
_____°C

(ii) State the event that occurs prior to this increase in body temperature. **1**

_____

(d) **Graph 2** shows an electrocardiogram (ECG) trace of this woman's heartbeat.

Graph 2



(i) Describe what happens to the heart as a result of the electrical activity between Q and S. **1**

_____

(ii) **Use the information in Graph 1 and Graph 2** to determine on how many days this electrocardiogram (ECG) could have been taken. **1**

Space for calculation

_____days

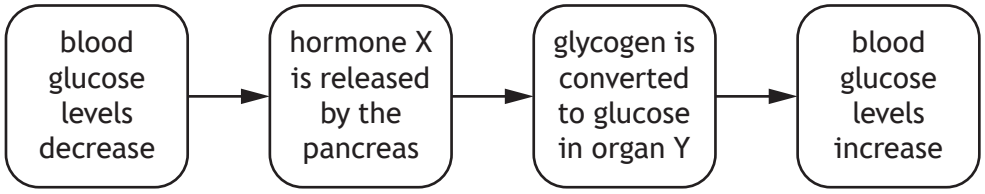
8. (continued)

(e) Women are only fertile for a few days during the menstrual cycle.

Apart from body temperature and heart rate, state one other indicator of a woman's fertile period. **1**

_____

9. The diagram shows part of the hormonal regulation of blood glucose levels.



(a) Name hormone X and organ Y. **2**

Hormone X _____

Organ Y _____

(b) Describe how the glucose tolerance test is carried out and how the results can indicate if an individual has diabetes. **3**

_____

_____

_____

_____

_____

_____

_____

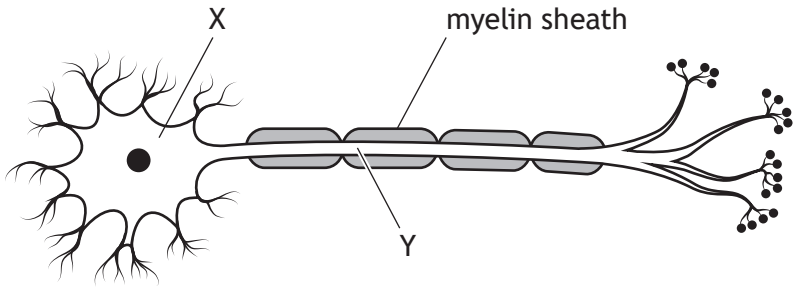
_____

_____

_____

[Turn over

10. The diagram shows a neuron.



(a) Name the parts labelled X and Y. **2**

X _____

Y _____

(b) Neurons connect with other neurons at a synaptic cleft.

(i) State one way that neurotransmitters are removed from a synaptic cleft. **1**

_____

_____

(ii) Explain why neurotransmitters must be removed from a synaptic cleft. **1**

_____

_____

(c) (i) State the function of the myelin sheath. **1**

_____

_____

(ii) Name the type of cells that produce myelin. **1**

_____

10. (continued)

(d) Multiple Sclerosis (MS) is a degenerative disease that leads to the destruction of the myelin sheath.

The table shows the number of cases of MS in males and females in the UK in 2016.

Age (years)	Number of cases (per 100 000)	
	Males	Females
0–14	0	10
15–24	15	20
25–34	70	210
35–44	200	480
45–54	210	590
55–64	270	405
65–74	150	350
75+	80	110

Describe two differences in the trends for the number of cases of MS in males and females. **2**

1. _____

_____

_____

2. _____

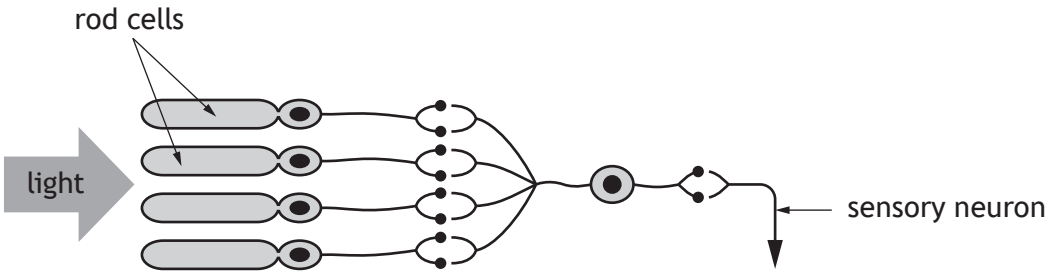
_____

_____

[Turn over

11. The retina in the eye contains specialised receptor cells called rods, that can detect light. These are connected to sensory neurons, which carry impulses out of the eye.

The diagram represents part of a neural pathway in the retina.



(a) Describe the function of sensory neurons. **1**

(b) (i) Use the diagram to explain why this pathway can be described as a converging neural pathway. **1**

(ii) Suggest how this converging arrangement of rod cells increases sensitivity to allow vision in dim light. **2**

(c) A genetic disorder, which does not show sex-linked inheritance, can lead to the gradual loss of rod cells in the retina.

State the location of the allele that causes this genetic disorder. **1**

12. The photograph shows an individual skiing in the mountains.



(a) State how prolonged activities like skiing can affect an individual's endorphin production. **1**

(b) The individual falls and breaks a bone in their leg.

Explain the benefit of endorphin release immediately after the injury. **1**

(c) The injured individual is given an injection of the drug morphine, which acts as an agonist of endorphins.

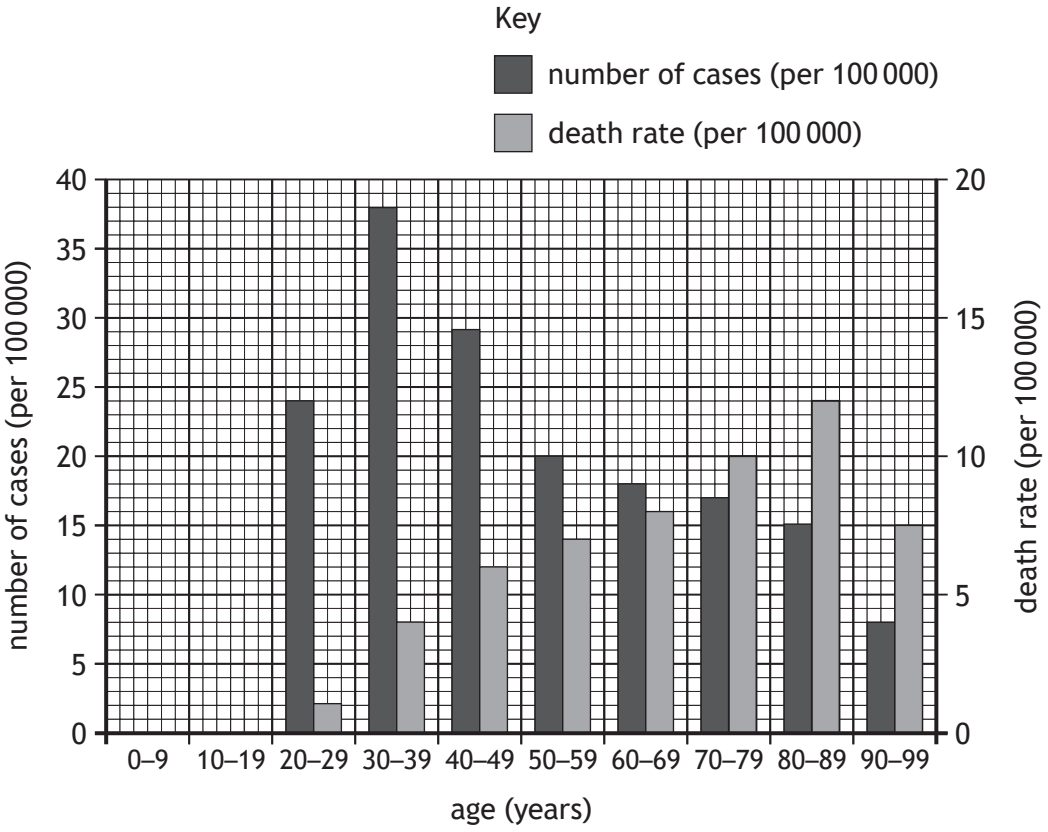
(i) Describe how morphine acts at a synapse to relieve pain. **1**

(ii) Heroin is a recreational drug that is converted to morphine in the body.

Describe how the repeated use of heroin can result in an individual developing a tolerance to it. **1**

[Turn over

13. The graph shows the number of cases of cervical cancer in females of different ages in the UK in one year. It also shows the death rate from cervical cancer at each age.



(a) Use data from the graph to describe changes that occur in the number of cases of cervical cancer from the ages of 20–29 to 90–99. **2**

(b) (i) Calculate the decrease in death rate from 80–89 to 90–99 years. **1**

Space for calculation

_____ per 100 000

13. (b) (continued)

(ii) Suggest a reason for the decrease in the death rate from cervical cancer between ages 80–89 and 90–99. **1**

(c) Express, as a simple whole number ratio, the number of cases compared to death rate at ages 30–39. **1**

Space for calculation

number of cases : death rate

(d) Some females with cervical cancer develop secondary tumours in their body.

Explain how these secondary tumours occur. **1**

(e) The human papilloma virus (HPV) is commonly associated with cases of cervical cancer.

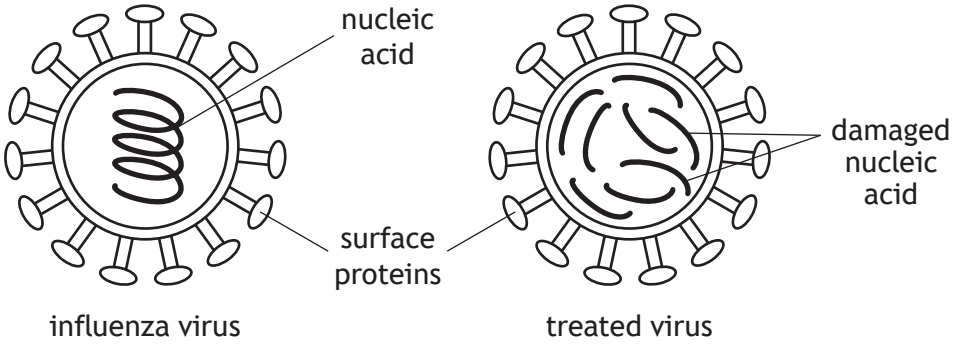
The UK has a vaccination programme against HPV.

Use information from the graph to suggest why females are given the vaccine when they are teenagers. **1**

[Turn over

14. Many people are vaccinated against the influenza virus each year.

(a) The diagram represents the influenza virus and a treated form of the same virus, which is contained in a vaccine.



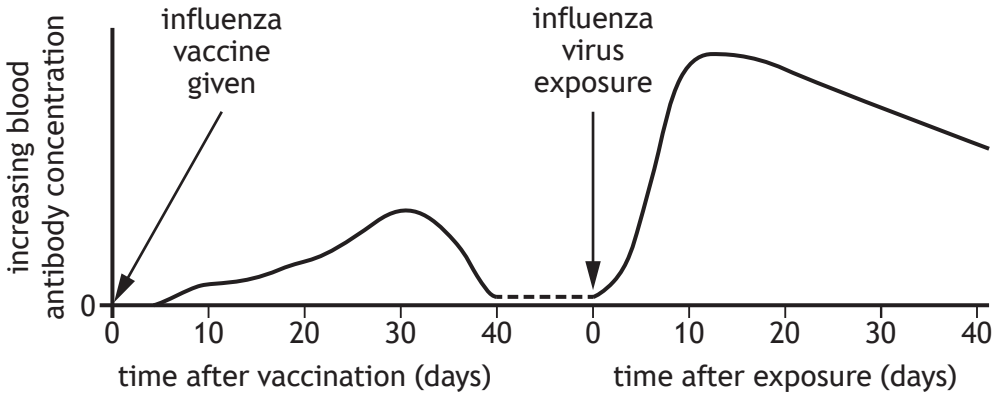
(i) Suggest why the nucleic acid molecule must be damaged in a virus used in a vaccine. **1**

(ii) Explain why the surface proteins of the virus used in the vaccine must remain undamaged. **2**

(iii) Describe why an adjuvant is often added to a vaccine. **1**

14. (continued)

(b) The graph shows the blood antibody concentration of an individual after an influenza vaccination and after exposure to the influenza virus a few weeks later.



(i) The individual produces more antibodies after exposure to the influenza virus than after vaccination.

Use the graph to describe two other ways in which the individual's blood antibody concentration differs after exposure to the influenza virus compared to after vaccination. **2**

1. _____

_____

2. _____

_____

(ii) Describe how memory cells lead to an increase in antibody production after the individual has been exposed to the influenza virus. **1**

_____

_____

[Turn over

14. (continued)

(c) Antibodies inactivate viruses, which are then removed by phagocytosis.

Describe the process of phagocytosis. **2**

MARKS | DO NOT WRITE IN THIS MARGIN

15. Attempt **either** A **or** B. Write your answer in the space below and on *page 30*.

A Discuss the formation of a thrombus and the damaging effects it can cause in the body. **9**

OR

B Discuss the production, transport, and role of cholesterol in the body. **9**

You may use labelled diagrams where appropriate.

National Qualifications

X840/76/12

Human Biology
Paper 1 — Multiple choice

Duration — 40 minutes

Total marks — 25

Attempt ALL questions.

You may use a calculator.

Instructions for the completion of Paper 1 are given on *page 02* of your answer booklet X840/76/02.

Record your answers on the answer grid on *page 03* of your answer booklet.

Space for rough work is provided at the end of this booklet.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

X8407612

Total marks — 25

Attempt ALL questions

1. Which row in the table describes one type of cell division?

	Type of division	Parental cell type	Cells produced
A	mitosis	germline	somatic
B	meiosis	somatic	somatic
C	meiosis	germline	gametes
D	mitosis	somatic	gametes

2. Single gene mutations involve the alteration of a DNA nucleotide sequence as a result of

A deletion, substitution or insertion

B duplication, insertion or substitution

C duplication, substitution or inversion

D deletion, inversion or translocation.

3. Personalised medicine researchers identified three mutated sequences of mRNA.

Mutated sequence	Effects of mutation
AUUG	non-harmful
ACUU	harmful
ACUG	non-harmful

A drug was designed to bind to the mutated mRNA preventing its harmful effects.

Which of the following DNA sequences would be found in individuals who could be prescribed this drug as part of their treatment?

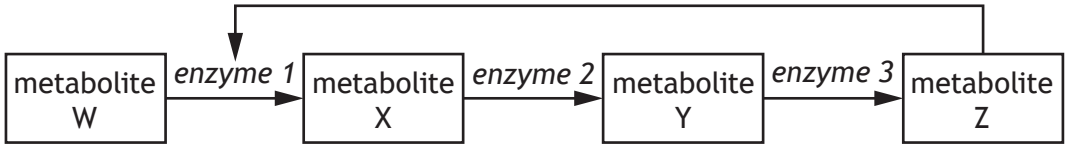
A CGACUUC

B TAACGAT

C GCTGAAT

D TCTTGAC

4. This metabolic pathway is regulated by feedback inhibition.



Which of the following would occur if a mutation caused enzyme 3 to be non-functional?

A Metabolite Z would inhibit enzyme 1

B Metabolite Z would increase in concentration

C Metabolite Y would not be converted to metabolite Z

D Metabolite W would not be converted to metabolite X

[Turn over

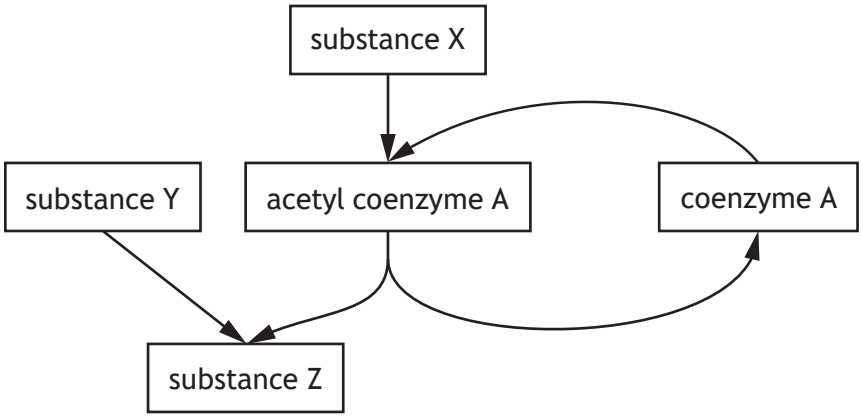
5. Three stages of respiration are listed.

1. Glycolysis
2. Citric acid cycle
3. Electron transport chain

Oxygen is used in

A stage 2 only
B stage 3 only
C stages 2 and 3 only
D stages 1, 2 and 3.

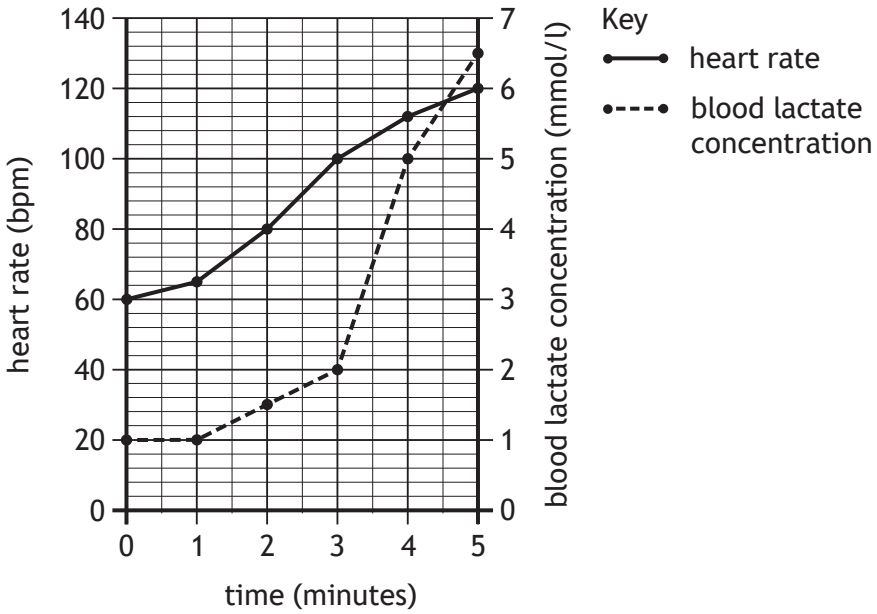
6. The diagram represents part of the respiratory pathway.



Which row in the table identifies substances X, Y and Z?

	Substance X	**Substance Y**	**Substance Z**
A	acetyl group	oxaloacetate	citrate
B	acetyl group	citrate	oxaloacetate
C	pyruvate	oxaloacetate	citrate
D	pyruvate	citrate	oxaloacetate

7. The graph shows an athlete's heart rate and blood lactate concentration while cycling for 5 minutes.



Which of the following statements is correct?

A When the blood lactate concentration was 5 mmol/l the heart rate was 100 bpm

B The greatest increase in blood lactate concentration occurs between minutes 2 and 3

C Both heart rate and blood lactate concentration increased during every minute of exercise

D Over the 5 minutes, the average increase in heart rate was 12 bpm

8. A sample of leg muscle from an Olympic 100 metre sprinter was analysed.

Which row in the table is most likely to indicate the results of this analysis?

	Most common muscle fibre type	**Mitochondria per cell**
A	fast-twitch	2000
B	slow-twitch	2000
C	fast-twitch	500
D	slow-twitch	500

[Turn over

10. Which row in the table matches a method of contraception with its effect?

	Method of contraception	Effect
A	combined oral contraceptive pill	prevents release of FSH
B	progesterone-only pill	prevents implantation
C	combined oral contraceptive pill	prevents implantation
D	progesterone-only pill	prevents release of FSH

11. The table shows information a woman used to monitor her fertility.

Date	Temperature (°C)	Consistency of cervical mucus
14 November	37·3	thick
18 November	37·3	thin
22 November	37·8	thin
26 November	37·8	thick

On which date is ovulation most likely to have occurred?

A 14 November
B 17 November
C 21 November
D 26 November

[Turn over

12. Duchenne muscular dystrophy is a sex-linked recessive condition that results in weakening of skeletal muscles over time.

An unaffected man and a carrier woman have a child.

The percentage chance that the child will have Duchenne muscular dystrophy is

A 0%

B 25%

C 50%

D 100%

13. A study was carried out into the effect of caffeine on blood pressure.

The blood pressure of four individuals was measured before and after consumption of a drink containing caffeine.

The results are shown in the table.

Individual	Initial blood pressure (mmHg)	Final blood pressure (mmHg)
1	120/75	146/97
2	115/79	132/99
3	127/86	159/100
4	118/80	139/96

The average increase in systolic blood pressure was

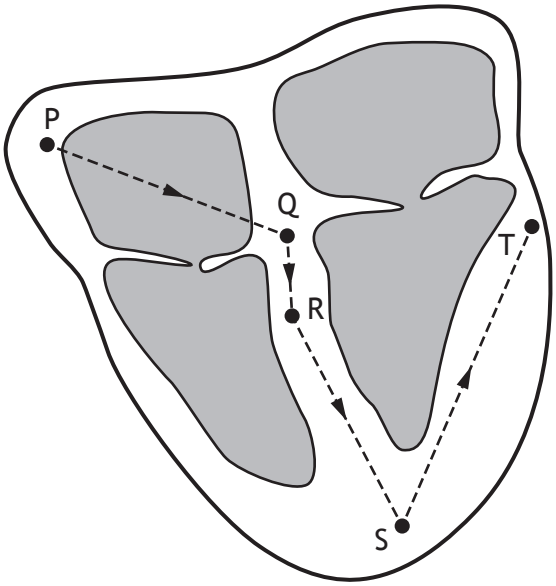
A 18

B 24

C 72

D 96.

14. The diagram shows a section through a heart and the pathway an impulse takes through the heart muscle.



Key

● location in heart muscle

---- pathway of impulse

The table shows the distance and the time taken for the impulse to travel through each section of the pathway.

Section of pathway	Distance travelled (mm)	Time taken (s)
P to Q	40	0·07
Q to R	20	0·09
R to S	85	0·04
S to T	115	0·05

Through which section of the pathway does an impulse travel fastest?

A P to Q

B Q to R

C R to S

D S to T

[Turn over

15. A group of individuals took part in an investigation into the effect of a drug on their heart rate.

Which of the following procedures should have been carried out to ensure the results were valid?

A There should have been a larger number of individuals in the group

B The results should have been regularly recorded using a heart rate monitor

C The investigation should have been repeated with another group of individuals

D The activity levels of the individuals during the investigation should have been the same

16. Atheromas can rupture, damaging the endothelium and triggering the following events.

1. A thrombus forms
2. Clotting factors are released
3. Fibrinogen is converted into fibrin
4. Prothrombin is converted into thrombin

Which of the following sequences shows the order in which these events occur?

A 1, 3, 4, 2

B 2, 4, 3, 1

C 2, 3, 4, 1

D 1, 2, 4, 3

17. The table shows the result of a glucose tolerance test carried out on two individuals. One of the individuals was found to have diabetes.

Time after drinking glucose solution (minutes)	Blood glucose concentration (mmol/l)	
	Individual 1	Individual 2
0	5·5	7·5
20	7·5	12·0
40	8·0	15·0
60	7·5	15·0
80	7·0	13·5
100	6·5	12·0
120	6·0	10·5

The blood glucose concentration of the individual with diabetes would be expected to return to its initial value after a further

A 20 minutes
B 40 minutes
C 140 minutes
D 160 minutes.

18. A neural pathway in which neurons later in the pathway link with earlier neurons is a

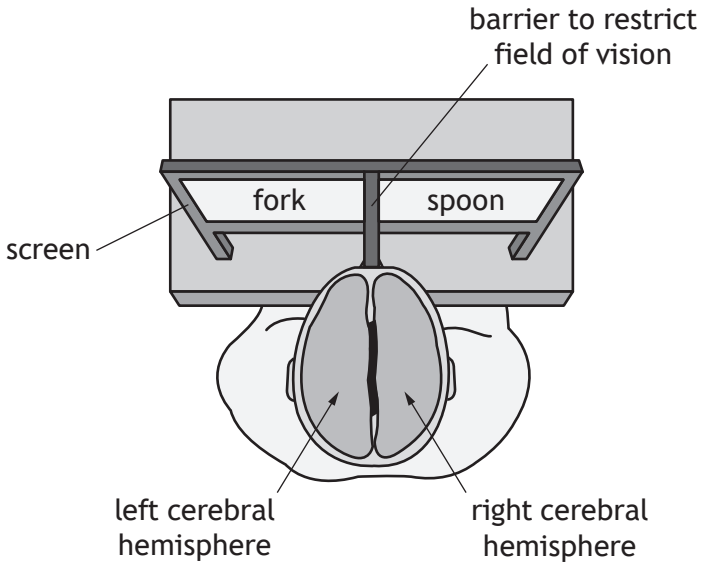
A diverging pathway
B converging pathway
C summation pathway
D reverberating pathway.

[Turn over

19. The left cerebral hemisphere controls speech production and processes information from the right eye while the right cerebral hemisphere processes information from the left eye.

An individual whose corpus callosum had been cut for medical reasons took part in a study. They had to press their forehead against a barrier so that their left eye could only see to the left of the barrier and their right eye to the right of the barrier.

The diagram shows the setup of the study.



The individual was asked to look straight ahead and then the words 'fork' and 'spoon' appeared briefly on the screen as shown. The individual was then asked to say what they had just seen.

The individual would be most likely to say

A fork only

B spoon only

C nothing

D fork and spoon.

20. A group of 18 people took part in an experiment on memory.

The table shows the number of people who could recall words from a list that had been read out to them.

Place of word in list	Number of people who could recall word
1st	18
2nd	18
3rd	18
4th	13
5th	12
6th	10
7th	12
8th	16
9th	17
10th	18

The change in the number of people who recalled the words in the middle of the list was due to

A chunking

B rehearsal

C displacement

D elaborative encoding.

21. Drug addiction affects neurotransmitter receptors by

A increasing their sensitivity as a result of exposure to antagonists

B decreasing their sensitivity as a result of exposure to antagonists

C increasing their sensitivity as a result of exposure to agonists

D decreasing their sensitivity as a result of exposure to agonists.

[Turn over

22. Parkinson's disease results in reduced levels of the neurotransmitter dopamine in the brain.

Which of the following drugs would be a suitable treatment for this disease?

A Dopamine agonists

B Dopamine inhibitors

C Dopamine antagonists

D Dopamine reuptake activators

23. Cytokines act at the site of infection by

A signalling the production of phagocytes

B stimulating phagocytes to release antibodies

C carrying out phagocytosis to engulf bacteria

D causing the accumulation of phagocytes.

24. The herd immunity threshold for measles in a particular population is 90%.

The table shows the number of individuals vaccinated within the population in a four year period.

Year	Population	Number of individuals vaccinated
2012	300 000	270 000
2013	310 000	280 000
2014	325 000	285 000
2015	335 000	310 000

A measles outbreak would most likely have occurred in

A 2012

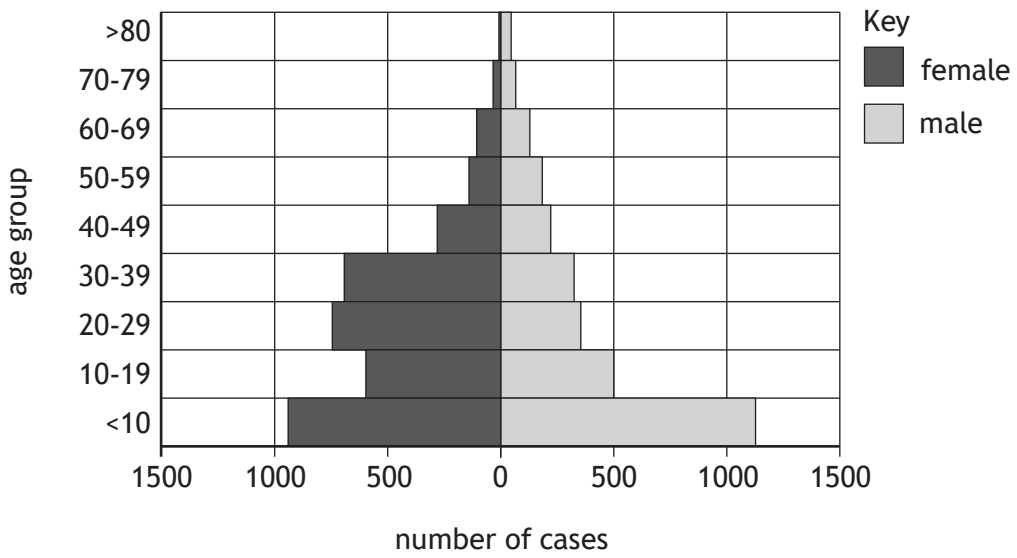
B 2014

C 2012 and 2014

D 2012, 2013 and 2015.

25. Cryptosporidium is a parasite that causes individuals to suffer from severe diarrhoea.

The graph shows details of the number of cases of cryptosporidium infection in a country in 2016.



What conclusion can be drawn from the graph?

A In all age groups there are fewer cases in females than males

B In all age groups there are fewer cases in males than females

C As the age group of females increases the number of cases always decreases

D As the age group of males increases the number of cases always decreases

[END OF QUESTION PAPER]

FOR OFFICIAL USE

National Qualifications

Mark

X840/76/01

Human Biology Paper 2

Duration — 2 hours 20 minutes

X8407601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day Month Year

Scottish candidate number

Total marks — 95

Attempt ALL questions.

You may use a calculator

Question 14 contains a choice.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. Score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

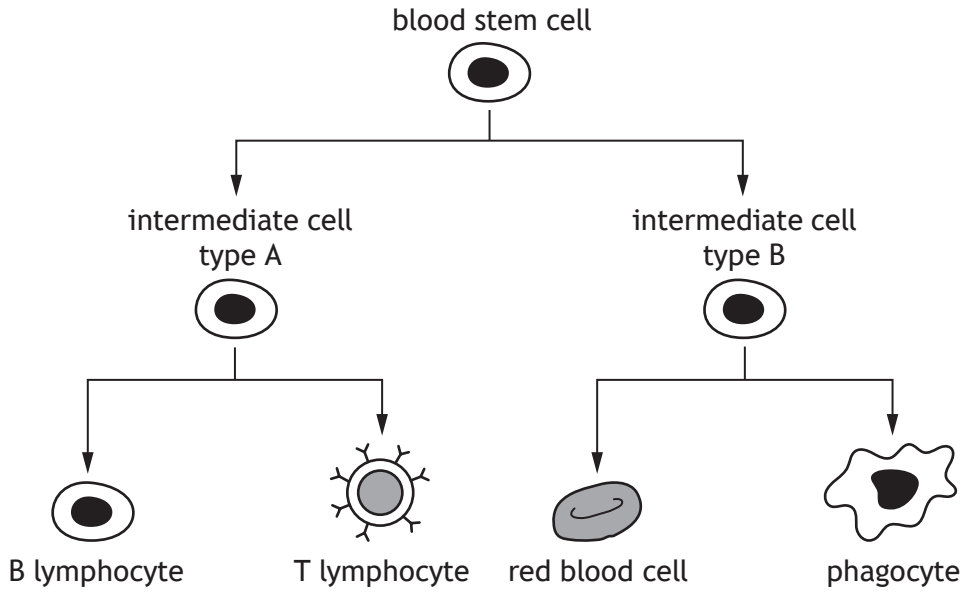
X8407601

Total marks — 95

Attempt ALL questions

Question 14 contains a choice

1. The diagram shows a range of cell types produced by blood stem cells.



(a) State the location of blood stem cells in the body. **1**

_____

(b) Use information in the diagram to explain why blood stem cells are described as tissue stem cells. **1**

_____

_____

(c) Describe one therapeutic use and one research use of stem cells. **2**

Therapeutic use _____

_____

Research use _____

_____

2. (a) The tables show the number of deaths from some common types of cancer in males and females in Scotland in 2016.

The tables also show the percentage change in the number of deaths since 2007 due to these types of cancer.

MARKS DO NOT WRITE IN THIS MARGIN

Table 1 Males

Type of cancer	Number of deaths in 2016	Change in number of deaths since 2007 (%)
Lung	2036	–24·3
Prostate	986	–6·2
Bowel	884	–15·1
Head and neck	377	+17·1
Liver	375	+47·4
Bladder	311	–7·7
Stomach	273	–32·8
Other	3085	—
Total	**8327**	**–12·4**

Table 2 Females

Type of cancer	Number of deaths in 2016	Change in number of deaths since 2007 (%)
Lung	2033	–7·1
Breast	946	–17·2
Bowel	803	–5·8
Ovary	400	
Liver	256	+92·8
Uterus		+59·1
Bladder	166	–8·5
Other	2138	—
Total	**7774**	**–7·5**

(i) State which type of cancer in males shows the largest percentage decrease in the number of deaths since 2007. **1**

_____

2. (a) (continued)

MARKS DO NOT WRITE IN THIS MARGIN

(ii) Calculate the number of female deaths from cancer of the uterus in 2016. **1**

Space for calculation

_____

(iii) There were 500 deaths from cancer of the ovary in 2007.

Calculate the percentage decrease in the number of deaths due to this type of cancer since 2007. **1**

Space for calculation

_____ %

(iv) **Using information from Table 1**, explain why it may **not** be correct to state that there were no male deaths from breast cancer in 2016. **1**

_____

_____

(v) Suggest a reason for the difference in the percentage change in the number of deaths due to lung cancer between males and females since 2007. **1**

_____

_____

(b) Since 2007, all people in Scotland between the ages of 50 and 74 have been offered regular screening for bowel cancer.

Suggest how this screening programme may have contributed to the percentage decrease in the number of deaths from bowel cancer between 2007 to 2016. **1**

_____

_____

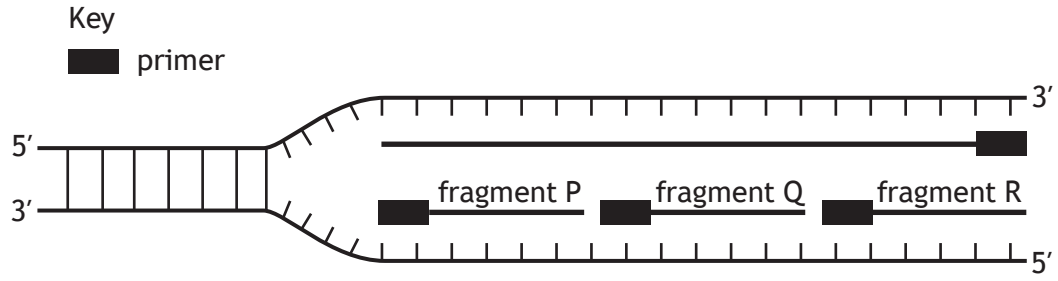
(c) Cancer cells may divide excessively to form a tumour.

Describe how secondary tumours can then form from this tumour. **1**

_____

_____

3. (a) The diagram represents a section of DNA from a chromosome undergoing replication.



(i) Describe the role of primers in allowing DNA replication to start. **2**

(ii) Fragments P, Q and R are joined together to replicate the lagging strand.

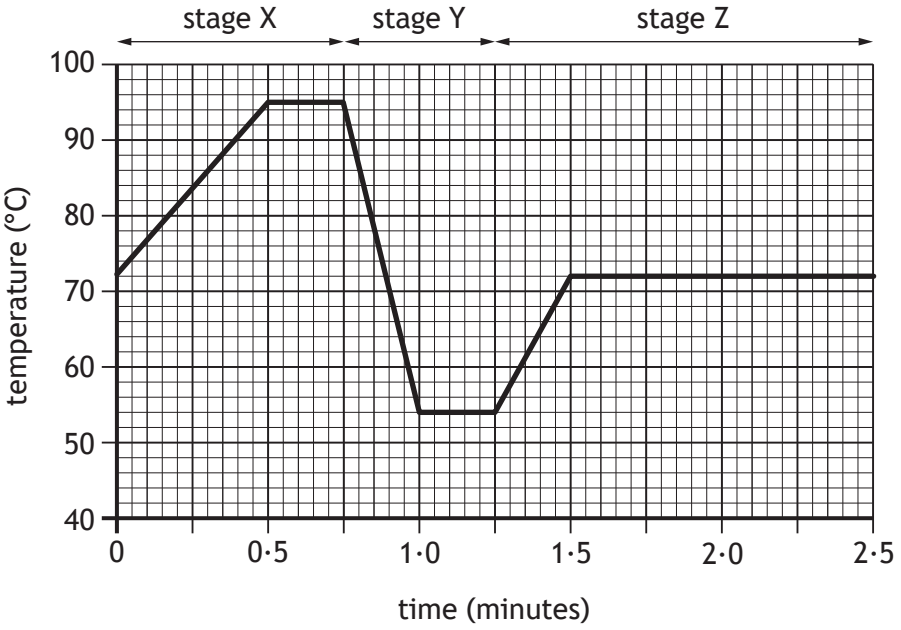
Explain why the lagging strand is replicated in fragments. **1**

(iii) Identify the first fragment produced from the lagging strand. **1**

(iv) Name the enzyme that joins the fragments together. **1**

3. (continued)

(b) The graph shows changes in the temperature of a reaction tube during one cycle of the polymerase chain reaction (PCR).



(i) Before the reaction began there were 1000 copies of a DNA fragment in the reaction tube.

Calculate the time it would take until there were at least one million copies of this DNA fragment present. **1**

Space for calculation

_____ minutes

(ii) Explain why the reaction tube is heated in stage X. **1**

_____

_____

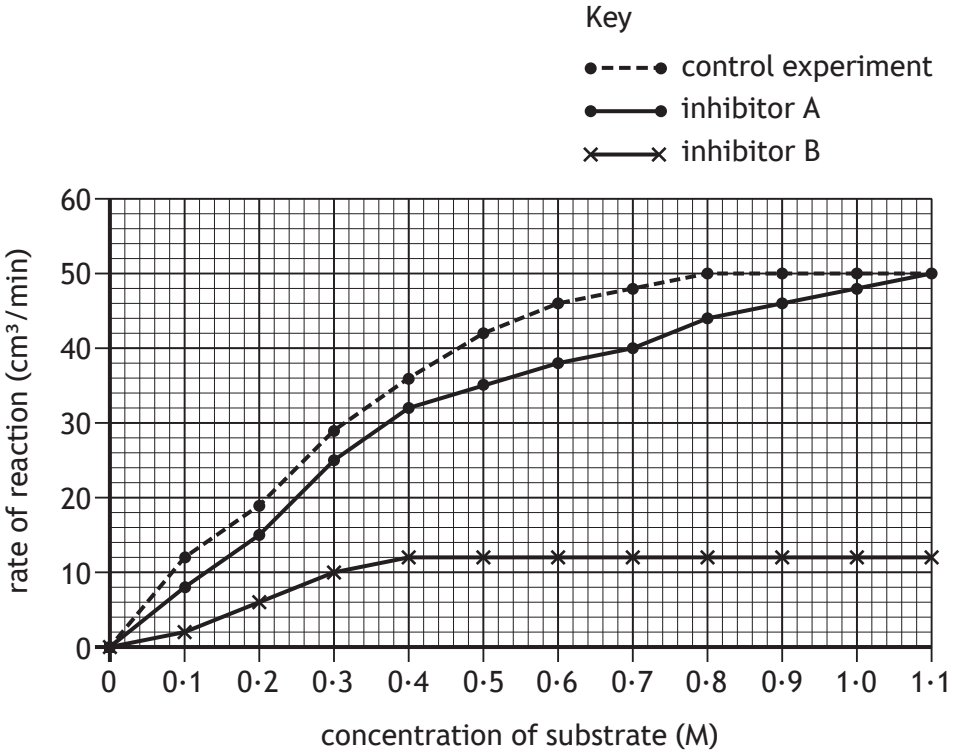
(iii) Explain why the reaction tube is cooled in stage Y. **1**

_____

_____

4. An investigation was carried out to show the effect of two different inhibitors on the rate of a reaction, catalysed by an enzyme.

The graph shows the results of this investigation.



(a) Name the substances present in the control experiment. **1**

_____

(b) Use the information in the graph to select which inhibitor is non-competitive and give a reason for your choice. **1**

Inhibitor _____

Reason _____

_____

(c) Calculate the decrease in the rate of reaction caused by inhibitor B at 0·7 M substrate concentration. **1**

Space for calculation

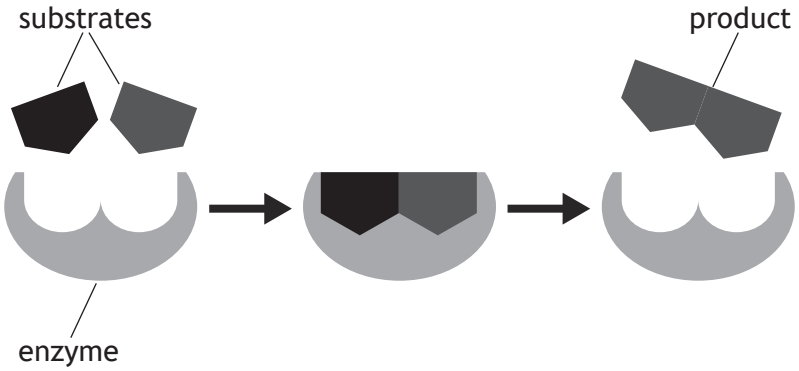
_____ cm^3/min

4. (continued)

(d) Predict the rate of reaction using inhibitor A at a substrate concentration of 1·2 M. **1**

_____ cm^3/min

(e) The diagram represents a reaction in a metabolic pathway.



(i) Name the type of reaction shown in the diagram and give a reason for your answer. **2**

Type of reaction _____

Reason _____

_____

(ii) Describe the role of induced fit in this enzyme-catalysed reaction. **2**

_____

_____

_____

_____

[Turn over

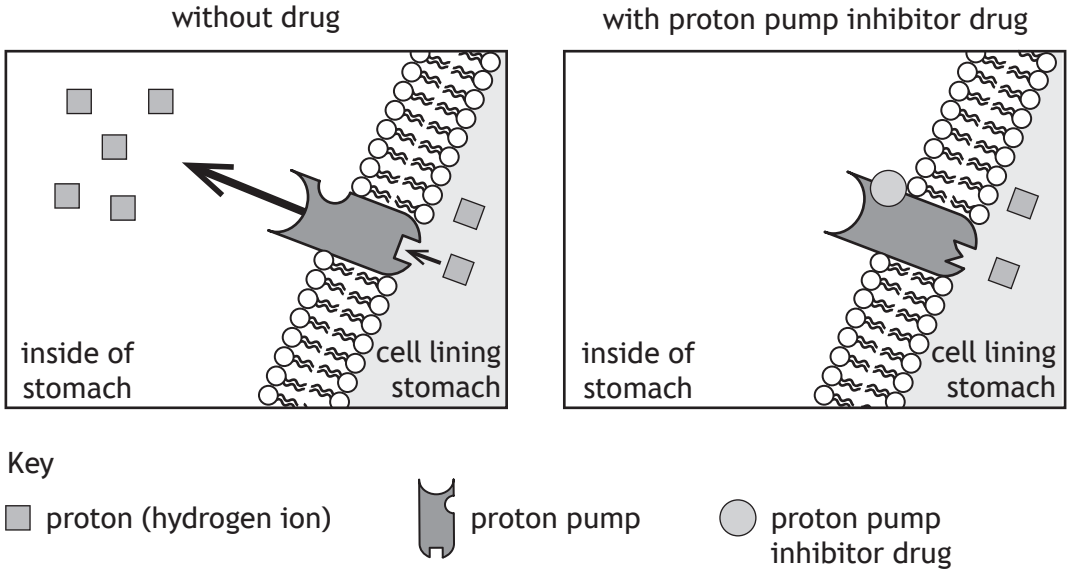
5. Proton pumps are proteins found in the membranes of cells lining the stomach.

Protons are hydrogen ions that increase acidity.

An investigation was carried out to determine the effect of a drug that acts as a proton pump inhibitor.

The inhibitor drug is usually given to people who produce too much stomach acid.

The diagrams show how the drug works.



(a) Use the diagrams to describe how this proton pump inhibitor drug works. **2**

(b) A large number of individuals who produced too much acid in their stomachs were divided into two groups.

Individuals in Group A consumed a tablet containing the proton pump inhibitor drug at the start of the investigation.

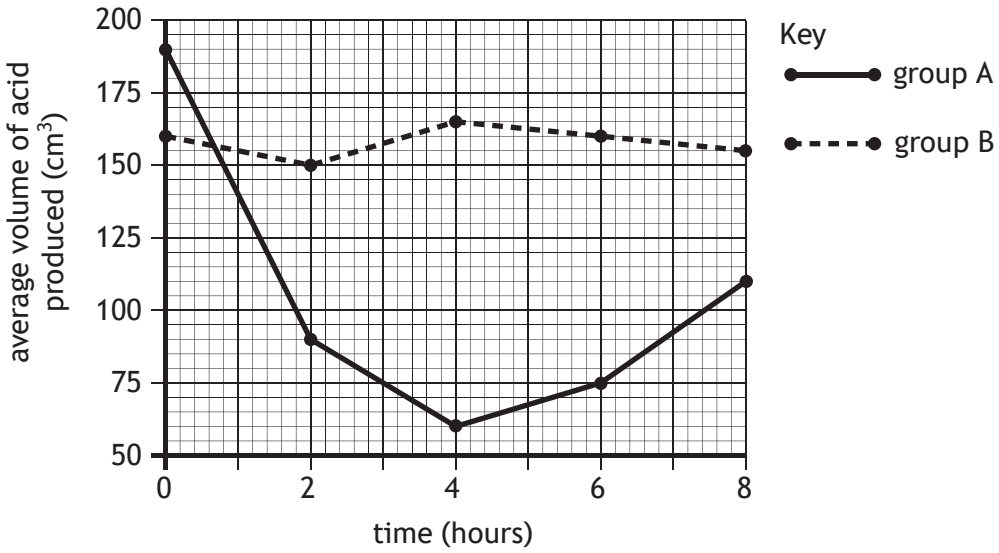
Individuals in Group B formed the control group.

(i) State how the individuals would be assigned to each group in order to reduce bias. **1**

MARKS

DO NOT WRITE IN THIS MARGIN

(ii) The average acid production of each group was measured every two hours over an eight hour period and the results are shown in the graph.



Use data from the graph to describe the changes that occurred in the acid production of group A during the investigation. **2**

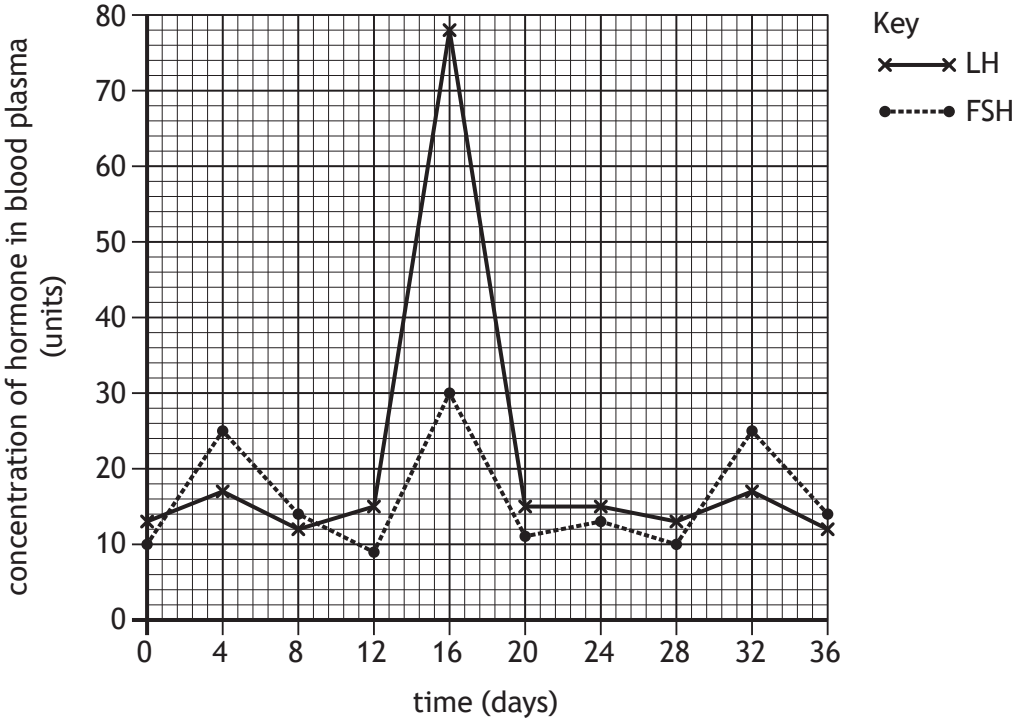
(iii) A student analysed these results and concluded that the drug was most effective at four hours.

Explain why this conclusion may **not** be correct. **1**

(c) Apart from their role in making the stomach more acidic, hydrogen ions are required for ATP synthesis in the mitochondria.

Describe the role of hydrogen ions in ATP synthesis. **2**

6. The graph shows changes in a female's FSH and LH concentrations over 36 days.



(a) Name the part of the brain that produces the releaser hormone, which triggers FSH and LH secretion at puberty. **1**

_____

(b) Describe **one** role of FSH in females. **1**

_____

_____

(c) Calculate the increase that occurs in the concentration of LH between day 12 and day 16. **1**

Space for calculation

_____ units

6. (continued)

(d) A surge in LH triggers ovulation.

Name the phase of the menstrual cycle that takes place after ovulation. **1**

_____

(e) Describe how a decrease in LH concentration leads to menstruation. **2**

_____

_____

_____

_____

(f) Explain how the information in the graph indicates that this female has cyclical fertility. **1**

_____

_____

_____

(g) Explain one way that fertility drugs stimulate ovulation. **1**

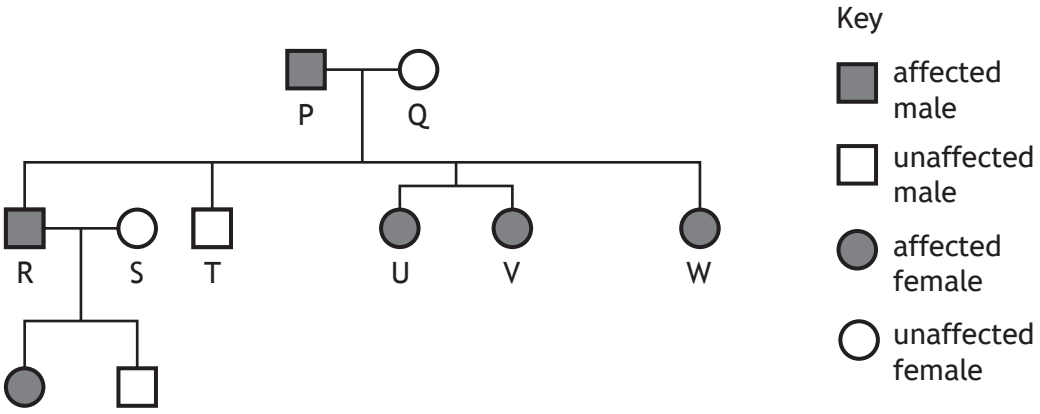
_____

_____

[Turn over

7. Familial hypercholesterolaemia (FH) is a genetic disorder caused by a **dominant** allele that raises blood cholesterol concentrations.

(a) The diagram shows the family history of this condition over three generations.



(i) FH is caused by a mutation on chromosome 19.

State the name for a chromosome that is **not** a sex chromosome. **1**

_____

(ii) Referring to individuals labelled in the diagram, describe how the inheritance pattern shown indicates that this condition cannot be caused by a sex-linked allele. **1**

_____

_____

_____

(iii) If daughter W had a child with a man who is heterozygous, calculate the percentage chance of the child inheriting FH. **1**

Space for calculation

_____ %

(iv) State the relationship between sisters U and V. **1**

_____

7. (continued)

(b) FH results in non-functional LDL receptors.

Explain why non-functional LDL receptors can lead to an individual having high blood pressure. **2**

(c) (i) Name a type of medication used to reduce blood cholesterol concentrations. **1**

(ii) Explain the benefit of regular physical activity to individuals suffering from high blood cholesterol. **2**

[Turn over

8. An investigation was set up to study the effect of the intensity of exercise on cardiac output.

An individual was asked to run on a treadmill and their heart rate and stroke volume were measured.

At the start of the investigation the gradient of the treadmill was set at 0.

To increase the intensity of exercise the gradient was increased in evenly stepped stages from 0 through to 10.



Heart rate and stroke volume were measured and used to calculate the cardiac output.

(a) (i) State **one** variable that should be kept constant during this investigation. **1**

(ii) Describe how the reliability of the results from this investigation could be increased. **1**

8. (continued)

MARKS DO NOT WRITE IN THIS MARGIN

(b) The results of this investigation are shown in the table.

Treadmill gradient	Heart rate (beats/min)	Stroke volume (cm^3)	Cardiac output (litres/min)
0	100	86	8·6
2	109	90	9·8
4	124	100	12·4
6	151	110	16·6
8	174	100	17·4
10	185		17·6

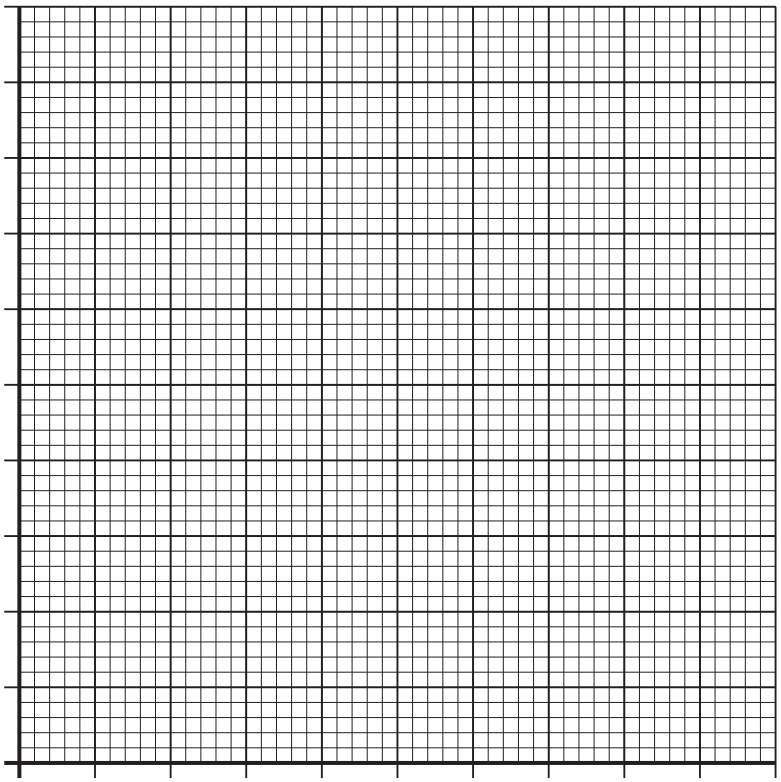
(i) Calculate the stroke volume when the treadmill gradient was set at 10. **1**

Space for calculation

_____ cm^3

(ii) Draw a line graph to show the effect of treadmill gradient on cardiac output. **2**

(Additional graph paper, if required, can be found on *page 28*.)



8. (b) (continued)

(iii) State a conclusion that can be drawn from the results of this investigation. **1**

_____

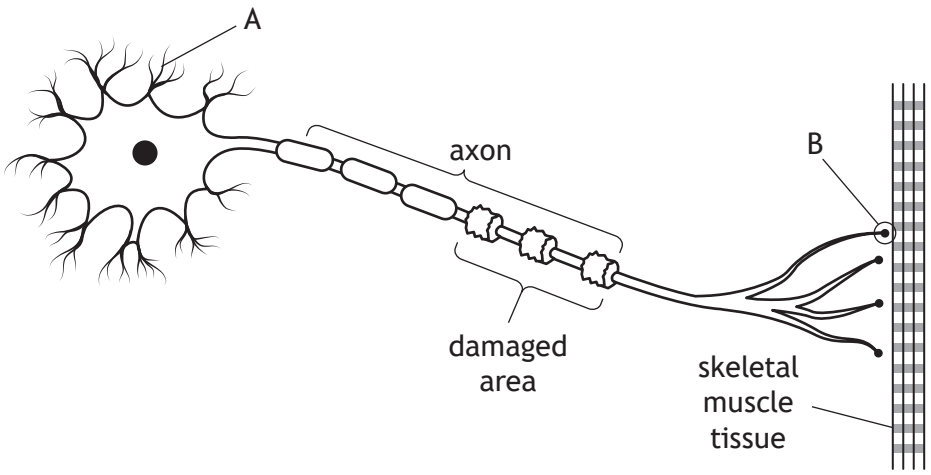
_____

(c) Suggest why stroke volume was observed to decrease after treadmill gradient 6. **1**

_____

_____

9. The diagram represents a neuron from an individual who has an autoimmune disease.



(a) Name A and B. **2**

A _____

B _____

(b) Describe the role of the immune system in causing this autoimmune disease. **2**

_____

_____

_____

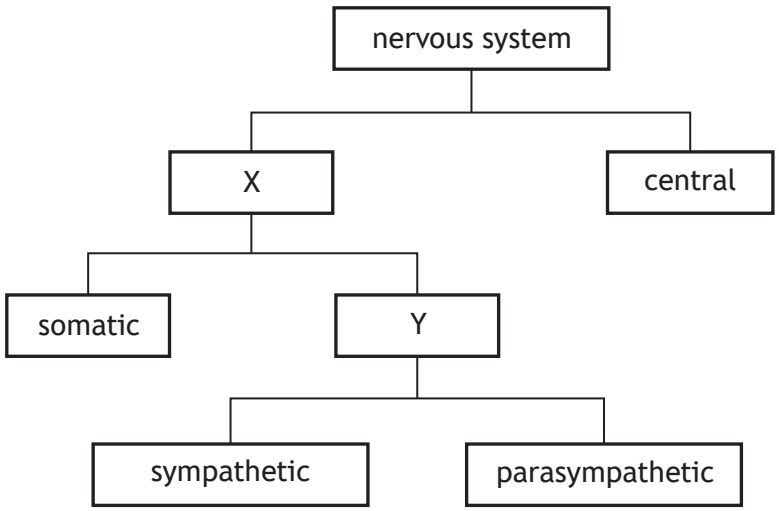
(c) Explain why this individual has a loss of muscle coordination. **1**

_____

_____

[Turn over

10. The diagram shows divisions of the nervous system.



(a) Name the divisions of the nervous system labelled X and Y. **2**

X _____

Y _____

(b) The somatic nervous system contains sensory neurons.

Describe the function of sensory neurons. **1**

_____

_____

(c) Describe an effect of the parasympathetic nervous system on breathing and the digestive system. **2**

Breathing _____

_____

Digestive system _____

_____

MARKS | DO NOT WRITE IN THIS MARGIN

11. (a) A biology teacher was teaching a lesson about the heart.

(i) Name the level of memory involved when students first saw and heard information on the heart. **1**

_____

(ii) As part of the lesson, students coloured in and labelled a diagram of the heart.

Suggest how a diagram of the heart in an exam might provide a contextual cue to this activity. **1**

_____

_____

(b) Describe **three** methods that help transfer information from short-term to long-term memory during learning. **3**

_____

_____

_____

_____

_____

_____

_____

_____

[Turn over

12. The number of excess winter deaths is calculated by subtracting the average number of deaths in autumn and spring from the number of deaths in winter.

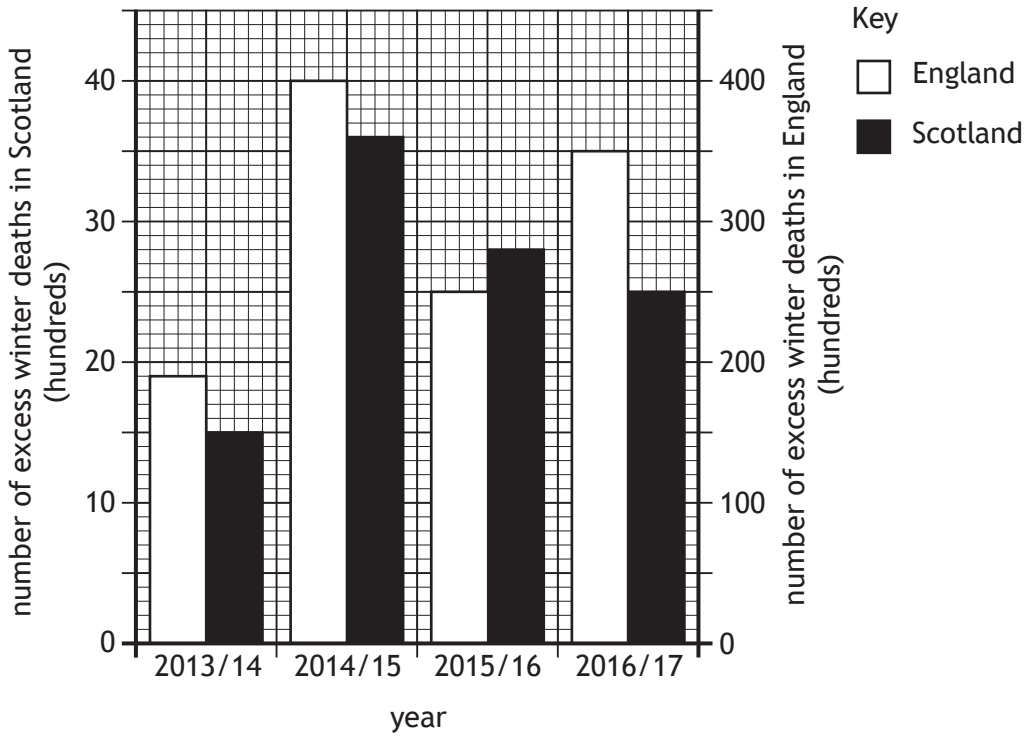
(a) The table shows the number of deaths in Scotland in 2017/18.

Season	Number of deaths
autumn 2017	18 694
winter 2017/18	23 137
spring 2018	17 986

Calculate the number of excess winter deaths in 2017/18. **1**

Space for calculation

(b) The graph shows the number of excess winter deaths in Scotland and England between 2013/14 and 2016/17.



12. (b) (continued)

MARKS DO NOT WRITE IN THIS MARGIN

(i) Describe one similarity and one difference in the trends in the number of excess winter deaths between the two countries. **2**

Similarity _____

_____

Difference _____

_____

(ii) Explain how the data should be expressed to allow a valid comparison of excess winter deaths in Scotland compared to England in each year. **1**

_____

_____

(iii) Express, as a simple whole number ratio, the number of excess winter deaths in Scotland compared to those in England in the winter of 2014/15. **1**

Space for calculation

_____ : _____
Scotland England

(c) The table shows the percentage of excess winter deaths caused by respiratory and circulatory diseases in England in the winter of 2015/16.

Diseases	Excess winter deaths (%)
circulatory	18
respiratory	39

Use information in the table and graph to calculate the number of excess winter deaths in England caused by respiratory diseases in 2015/16. **1**

Space for calculation

_____

12. (continued)

(d) The influenza virus can cause respiratory diseases.

Describe how the influenza virus can change from one winter to the next. **1**

_____

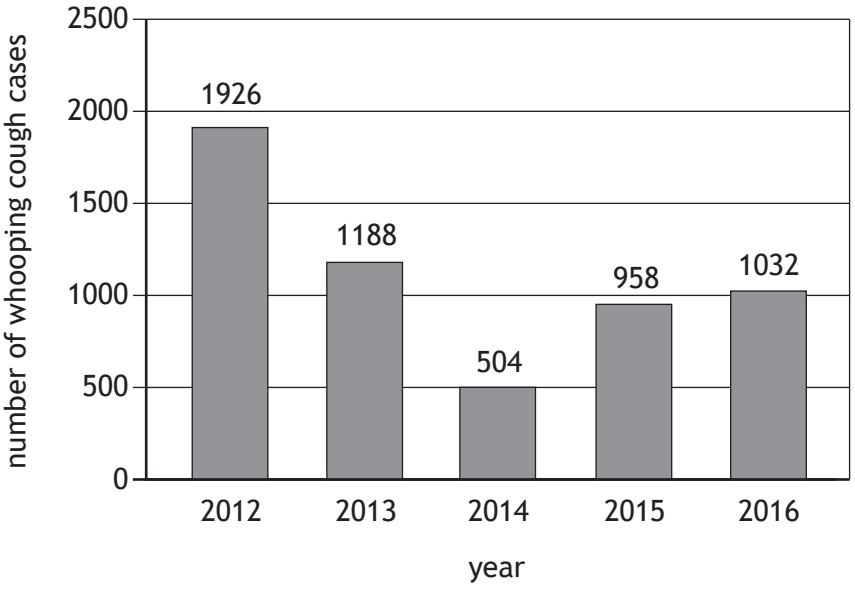
_____

(e) Name the type of cells that form a physical barrier in the inner lining of the respiratory system. **1**

_____

13. Whooping cough is an infectious disease caused by a bacterium.

The graph shows the number of whooping cough cases in Scotland between 2012 and 2016.



(a) A vaccine for whooping cough was first introduced for pregnant women in Scotland in 2012.

(i) Suggest a reason for the increase in the number of cases of whooping cough between 2014 and 2016. **1**

_____

_____

(ii) Calculate the percentage increase in cases from 2014 to 2016. **1**

Space for calculation

_____%

(b) Explain why vaccines do not cause disease in vaccinated individuals. **1**

_____

_____

(c) Name a substance that is added to a vaccine to make it more effective. **1**

_____

14. Attempt **either** A **or** B. Write your answer in the space below and on *page 27*.

A Discuss the use of antenatal screening and diagnostic testing to monitor the health of the developing fetus during pregnancy. **9**

OR

B Discuss the structure and function of arteries, veins and capillaries in the circulatory system. **9**

You may use labelled diagrams where appropriate.

National
Qualifications
2019

X840/76/12

Human Biology
Paper 1 — Multiple choice

TUESDAY, 30 APRIL
9:00 AM – 9:40 AM

Total marks — 25

Attempt ALL questions.

You may use a calculator.

Instructions for the completion of Paper 1 are given on *page 02* of your answer booklet X840/76/02.

Record your answers on the answer grid on *page 03* of your answer booklet.

Space for rough work is provided at the end of this booklet.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

Total marks — 25

Attempt ALL questions

1. A section of DNA containing 8000 bases has 1200 cytosine bases.
The number of adenine bases in this section of DNA is

A 800
B 1200
C 1400
D 2800.

2. Identify the RNA codon and anticodon for the DNA base sequence GAT.

	RNA codon	**RNA anticodon**
A	CTA	GAT
B	GAT	CTA
C	CUA	GAU
D	GAU	CUA

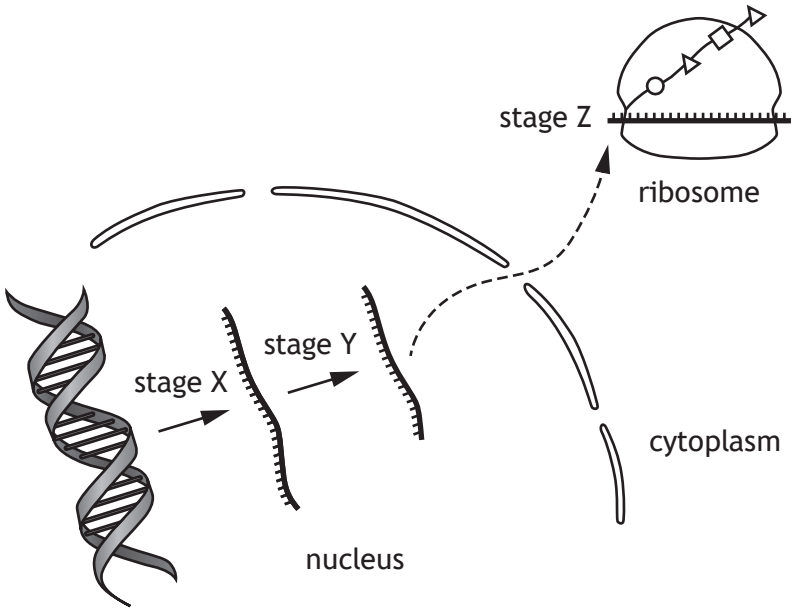
3. The following sequence shows the order of bases in a segment of mRNA.

A U G A U C G C U A U G A A G A C C G C A G C U

How many different amino acid molecules are coded for by this segment of mRNA?

A 6
B 7
C 8
D 24

4. The diagram shows three of the stages involved in gene expression.



Which row in the table identifies these stages?

	Stage		
	X	**Y**	**Z**
A	translation	RNA splicing	transcription
B	RNA splicing	transcription	translation
C	transcription	translation	RNA splicing
D	transcription	RNA splicing	translation

5. The diagram shows the base sequence for a section of a gene and a mutated version of this section.

section of gene C G T A T A T C C T A T G C C C C T G A C

mutated version C G T A T A T C C T A T C T C C C T G A C

This is an example of

A a deletion mutation

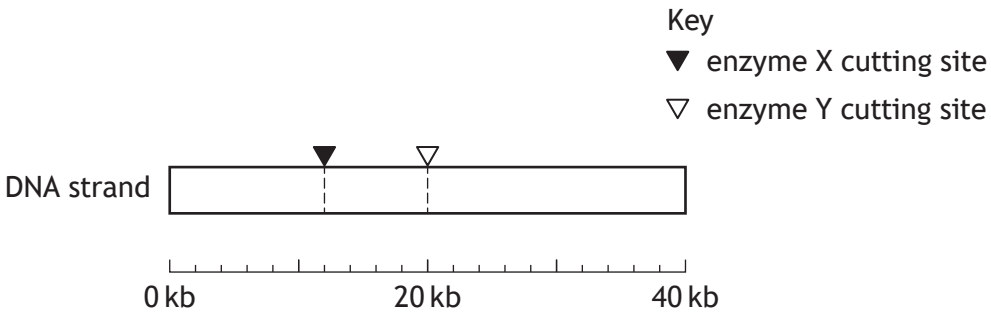
B an insertion mutation

C a substitution mutation

D a translocation mutation.

6. Enzymes can be used to cut strands of DNA into fragments.

The diagram shows the cutting sites of two different enzymes on a DNA strand that is 40 kilobases (kb) long.



What length of fragments (kb) would be produced if the DNA strand was cut with enzyme X only?

A 12, 20

B 12, 28

C 20, 20

D 12, 8, 20

7. Three applications of human genomics are listed.

1. Comparing genomes to settle paternity disputes
2. Screening a genome for a particular sequence to diagnose disease
3. Using genome information in the choice of effective drug treatments

Which of these applications are examples of pharmacogenetics?

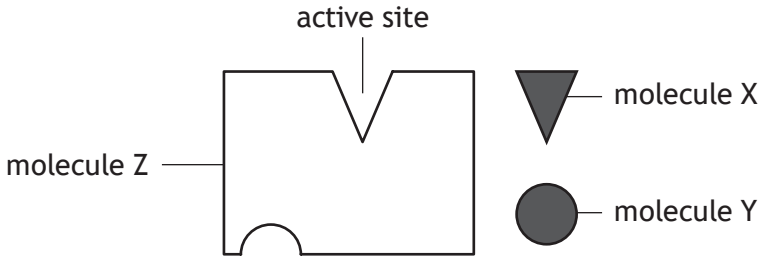
A 1 only

B 3 only

C 2 and 3 only

D 1, 2 and 3

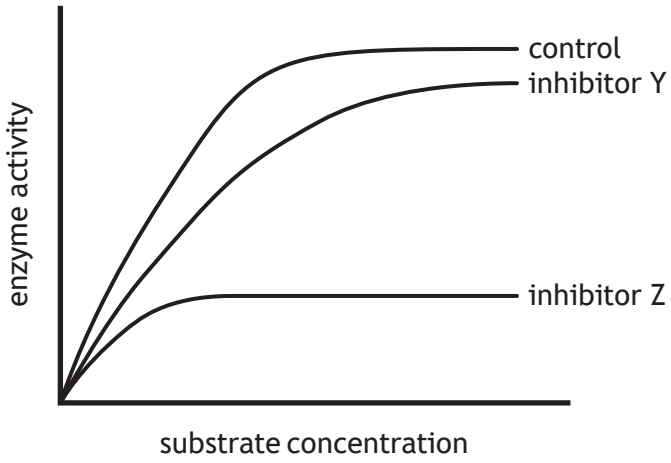
8. The diagram shows molecules involved in an enzyme controlled reaction.



Which row in the table identifies these molecules?

	Molecule X	**Molecule Y**	**Molecule Z**
A	substrate	non-competitive inhibitor	enzyme
B	non-competitive inhibitor	enzyme	substrate
C	substrate	competitive inhibitor	enzyme
D	competitive inhibitor	non-competitive inhibitor	substrate

9. The graph shows the results of an investigation into the effects of two different inhibitors on enzyme activity.



Which of the following conclusions can be drawn from this graph?

A Inhibitor Z is a competitive inhibitor

B There is a steady increase in enzyme activity with no inhibitor

C Inhibitor Y has less effect on enzyme activity than inhibitor Z

D Increasing substrate concentration increases the effect of both inhibitors

[Turn over

10. At the end of the electron transport chain hydrogen ions and electrons combine with

A NAD

B ATP

C water

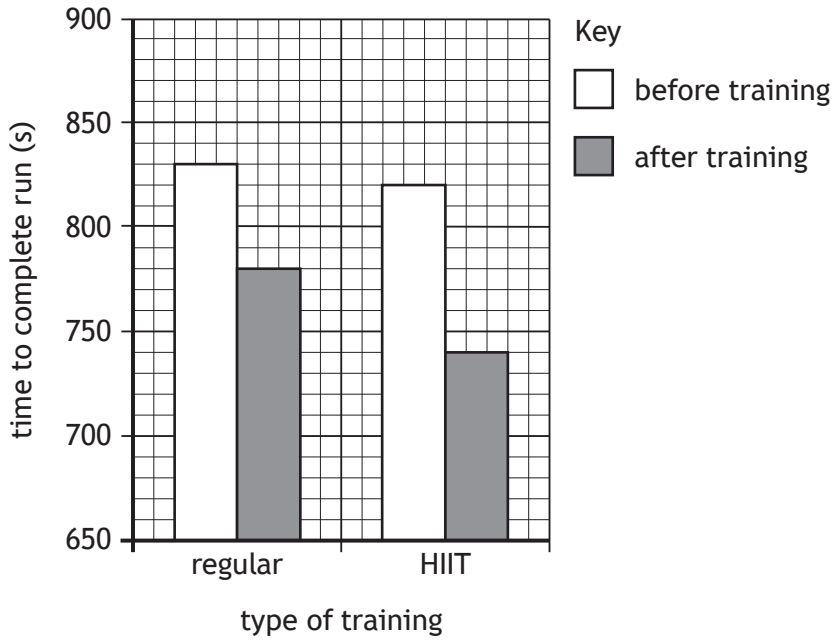
D oxygen.

11. Which row in the table describes the generation of ATP in slow-twitch muscle fibres?

	Generation of ATP	**Relative number of mitochondria**
A	aerobic respiration	many
B	aerobic respiration	few
C	glycolysis only	many
D	glycolysis only	few

12. Two athletes took part in different training programmes. One undertook regular training sessions while the other undertook high intensity interval training (HIIT) sessions.

The graph shows the time taken to complete a run before and after each training programme.



The improvement in performance as a result of HIIT training compared to regular training is

A 30 s

B 40 s

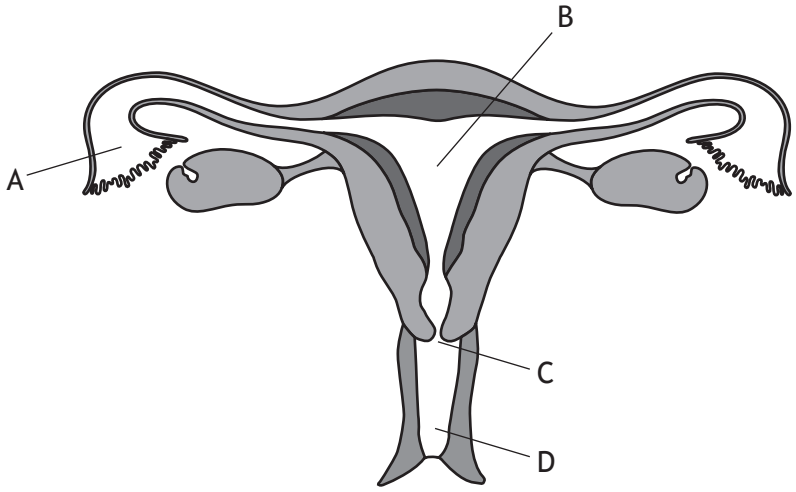
C 50 s

D 80 s.

[Turn over

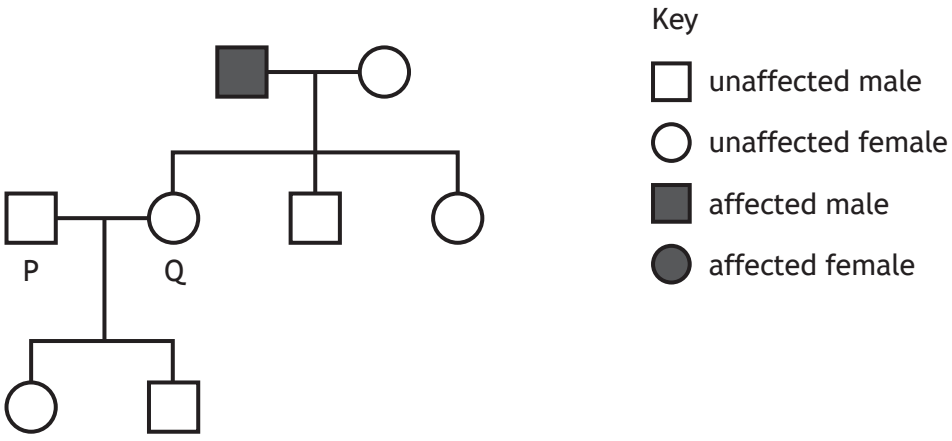
13. The diagram represents the female reproductive system.

Which letter indicates the location where an intra-uterine device would be positioned to prevent pregnancy?



14. The diagram shows the inheritance of haemophilia, a blood clotting disorder, in three generations of a family.

Haemophilia is caused by a sex-linked recessive allele.



Individuals P and Q are expecting a third child.

What is the percentage chance that this child will have haemophilia?

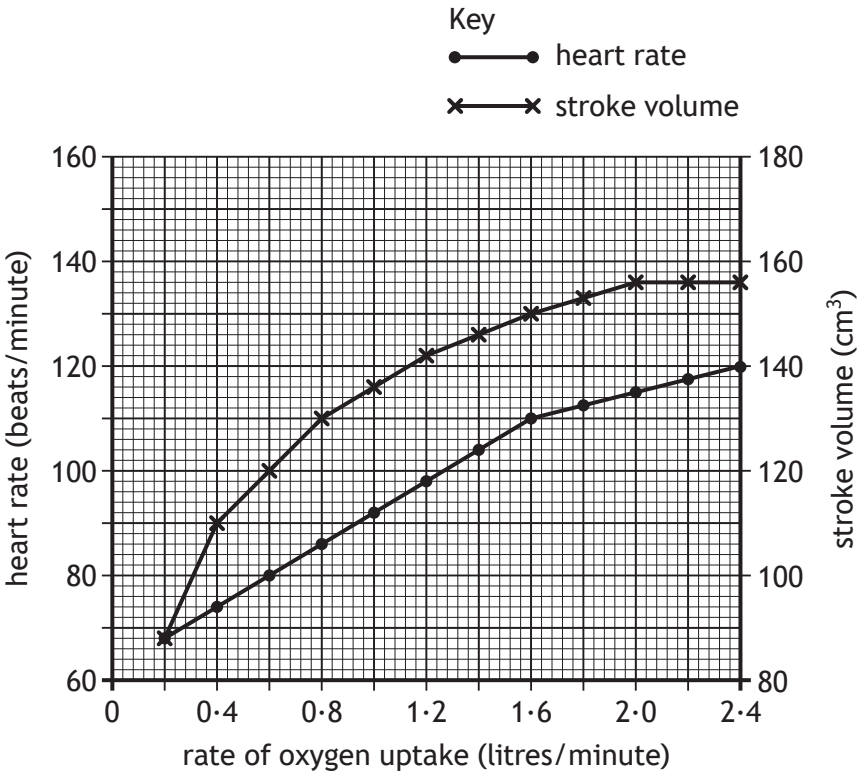
A 0 %

B 25 %

C 50 %

D 75 %

15. The graph shows how increasing oxygen uptake affects heart rate and stroke volume.



What is the cardiac output when the rate of oxygen uptake is 0·6 litres/minute?

A 1·50 cm^3/min

B 8000 cm^3/min

C 9600 cm^3/min

D 10 000 cm^3/min

16. Increased blood flow within an artery is the result of smooth muscle

A relaxation causing vasodilation

B contraction causing vasodilation

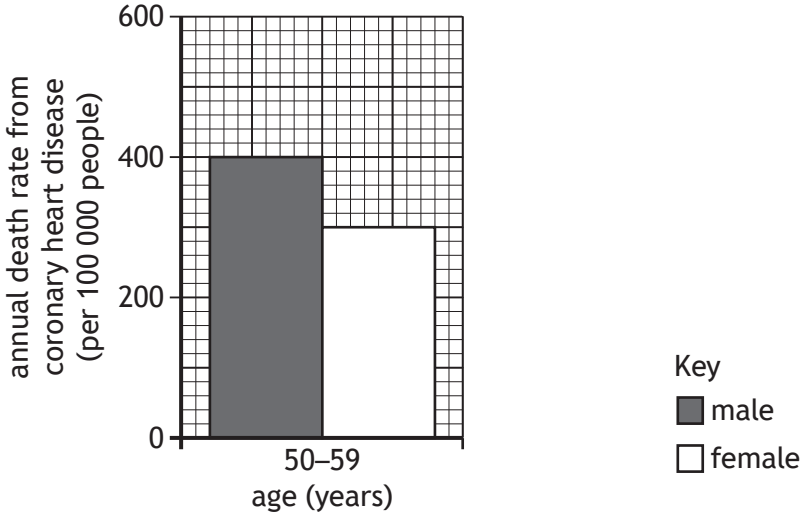
C relaxation causing vasoconstriction

D contraction causing vasoconstriction.

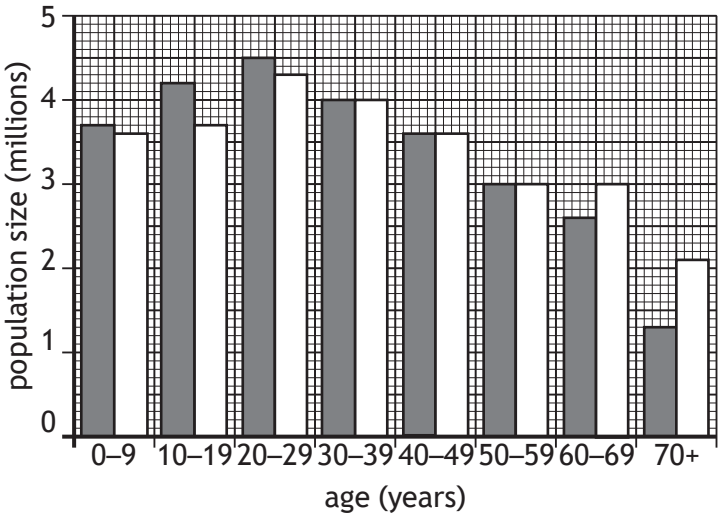
[Turn over

17. The graphs contain information about the population of the United Kingdom.

Graph 1



Graph 2



How many males aged between 50–59 years die from coronary heart disease annually in the United Kingdom?

A 400

B 9000

C 12 000

D 21 000

18. Which row in the table identifies components of the central and peripheral nervous systems?

	Central nervous system		Peripheral nervous system	
A	spinal cord	brain	somatic	medulla
B	brain	spinal cord	sympathetic	somatic
C	parasympathetic	brain	spinal cord	sympathetic
D	sympathetic	parasympathetic	brain	spinal cord

19. Transfer of information between cerebral hemispheres occurs through the

A medulla
B glial cells
C myelin sheath
D corpus callosum.

20. The following statements refer to events that occur at a synapse.

1. An impulse is generated
2. Acetylcholine diffuses across the synaptic cleft
3. Acetylcholine is broken down by an enzyme
4. Acetylcholine is released from storage vesicles

In which sequence do these events occur?

A 4 → 2 → 3 → 1
B 1 → 2 → 4 → 3
C 4 → 2 → 1 → 3
D 2 → 4 → 3 → 1

21. A function of glial cells is the production of

A axons
B myelin
C dopamine
D noradrenaline.

[Turn over

22. A drug trial was carried out to study the effect of a drug on mice.

A group of mice, with an average body mass of 20 g, were given 1 mg per kg of their body mass of the drug. The drug was given twice a day for 15 days.

What was the total mass of drug given to each mouse over the 15 days?

A 0·02 mg

B 0·04 mg

C 0·30 mg

D 0·60 mg

23. Multiple sclerosis (MS) is an autoimmune disease in which lymphocytes destroy the myelin sheath.

Which row in the table shows the type of lymphocyte involved and the effect on the speed of nerve impulse transmission?

	Type of lymphocyte	**Speed of nerve impulse transmission**
A	B	increased
B	B	decreased
C	T	increased
D	T	decreased

24. The table shows the number of cases of a disease and the number of deaths resulting from this disease in a country over four years.

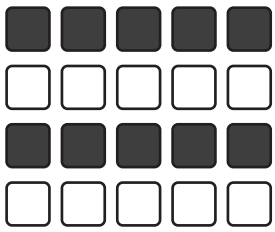
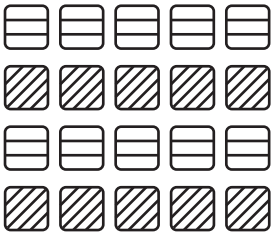
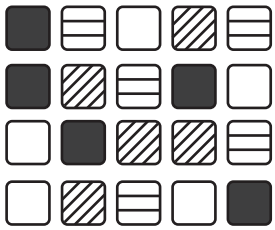
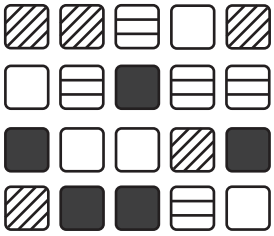
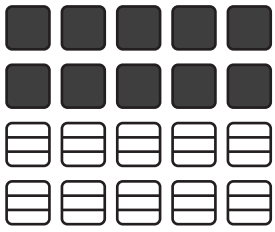
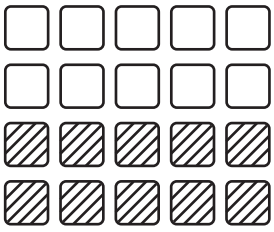
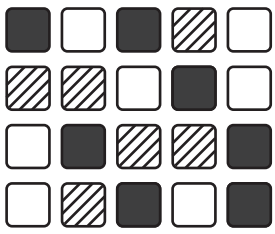
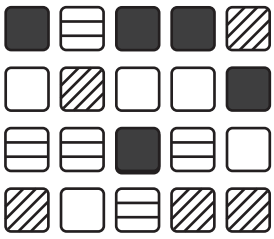
In which year did the greatest percentage of deaths occur?

	Year	**Number of cases**	**Number of deaths**
A	2014	1070	45
B	2015	420	12
C	2016	1960	60
D	2017	2290	90

25. Forty individuals were selected to take part in a clinical trial for a new drug to treat a lung condition.

They were split into two groups. One group was treated with the drug while the other group was treated with a placebo.

Which row in the table would have allowed a valid comparison of the treatments?

	Group treated with drug	**Group treated with placebo**
A		
B		
C		
D		

Key

male smoker

male non-smoker

female smoker

female non-smoker

[END OF QUESTION PAPER]

FOR OFFICIAL USE

National Qualifications 2019

Mark

X840/76/01

Human Biology Paper 2

TUESDAY, 30 APRIL

10:10 AM – 12:30 PM

X840761

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day | Month | Year

Scottish candidate number

Total marks — 95

Attempt ALL questions.

You may use a calculator.

Question 14 contains a choice.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. Score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

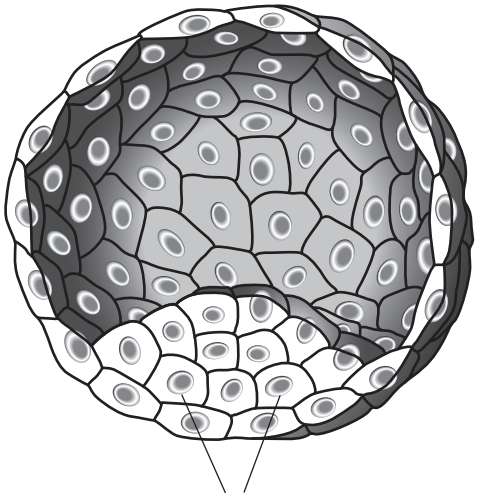
X840760101

Total marks — 95

Attempt ALL questions

Question 14 contains a choice

1. The diagram shows an early stage in the development of a human embryo.



(a) (i) The embryonic stem cells shown can differentiate into all body cell types.

State the term that describes this feature of these cells. **1**

_____

(ii) Explain why embryonic stem cells can differentiate into all cell types. **1**

_____

_____

(b) State the number of chromosomes in the nuclei of cells produced when germline stem cells divide by meiosis. **1**

_____

1. (continued)

(c) Stem cells can be cultured in the laboratory for research purposes.

State **one** way in which stem cells are used in research. **1**

_____

_____

(d) Scientists have recently found a way of converting skin cells into embryonic stem cells.

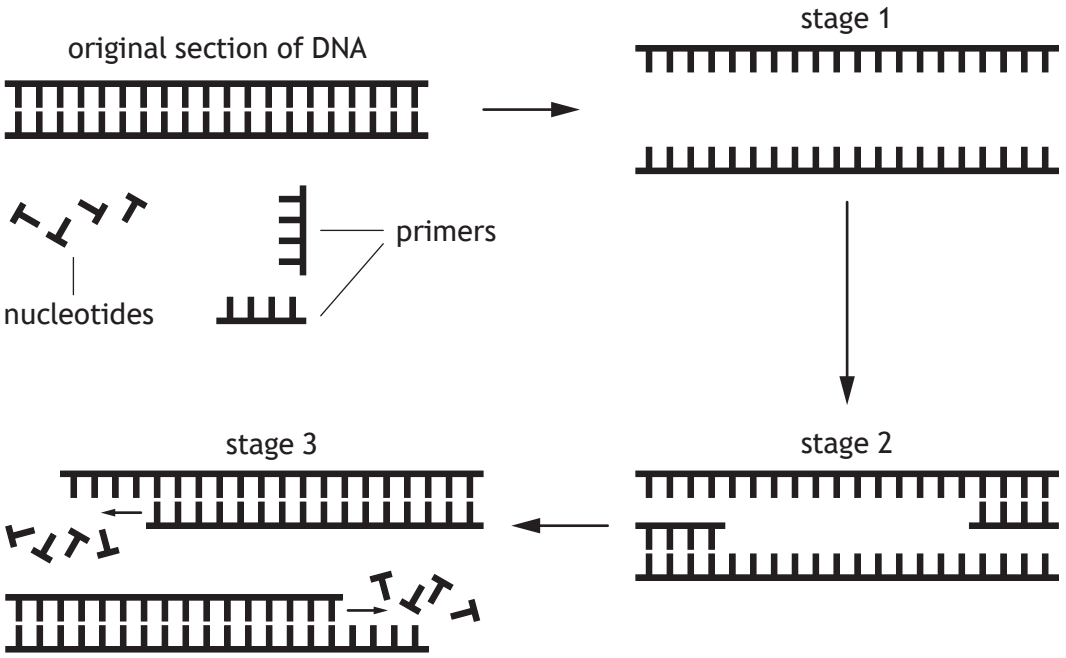
Suggest why this is an advantage from an ethical viewpoint. **1**

_____

_____

[Turn over

2. The polymerase chain reaction (PCR) is a process used to help solve crimes.
The diagram shows the stages in one PCR cycle.



(a) Each stage of PCR is temperature dependent.
Complete the table for stages 1 and 3. **2**

Stage	Temperature (°C)	Reason
1	95	
3		To allow replication of DNA

2. (continued)

(b) Each PCR cycle produces two copies of a section of DNA.

This PCR cycle takes 3 seconds.

Calculate how long it would take for at least 2000 copies of the original section to be produced. **1**

Space for calculation

_____ s

(c) Describe the role of primers in stage 2 and stage 3. **2**

Stage 2 _____

_____

Stage 3 _____

_____

(d) PCR was first used to help solve a crime in 1986.

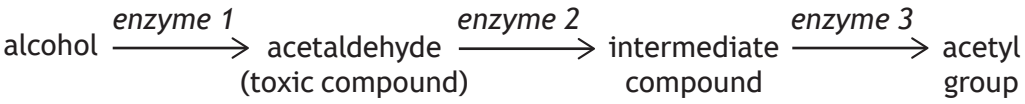
Suggest why PCR can now be used to help solve a crime committed in 1980, where only a small blood spot was found as evidence. **1**

_____

_____

[Turn over

3. The diagram shows some of the stages in the breakdown of alcohol by the liver.



(a) Use information from the diagram to explain how alcohol can be used as a substrate in respiration. **2**

(b) Some individuals cannot produce enough fully functioning enzyme 2.

(i) In these individuals an altered form of enzyme 2 is produced due to a missense mutation.

Suggest why this altered form of enzyme 2 works less well than the unaltered form. **1**

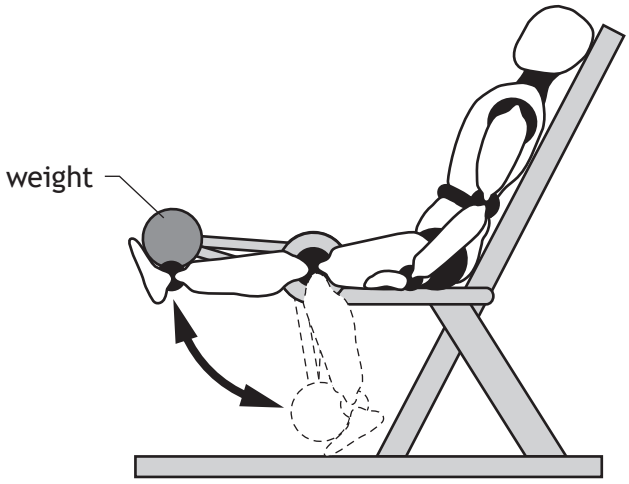
(ii) Explain why these individuals are less tolerant of alcohol. **1**

(c) A drug that acts as a competitive inhibitor of enzyme 2 can be prescribed to treat alcoholism.

Explain how this drug will affect the enzyme. **1**

4. A protein supplement is used by some people who take part in sport. They believe its use increases muscle mass and therefore improves performance in sporting activities such as weightlifting.

An investigation was carried out to test if protein supplements improved the ability of the upper leg muscles to raise weights.



A class of 20 students was divided into two groups, A and B. The groups were balanced for age, gender, fitness level and body mass.

Each student carried out 10 weeks of regular training on the apparatus.

Students in group A took a daily dose of a protein supplement.

Students in group B took a daily dose of a placebo.

All students had their results recorded every 2 weeks for 10 weeks, by measuring the maximum weight that could be raised using the apparatus shown in the diagram.

(a) State **two** additional variables, other than those described above, that would need to be controlled during this investigation. **2**

1 _____

2 _____

[Turn over

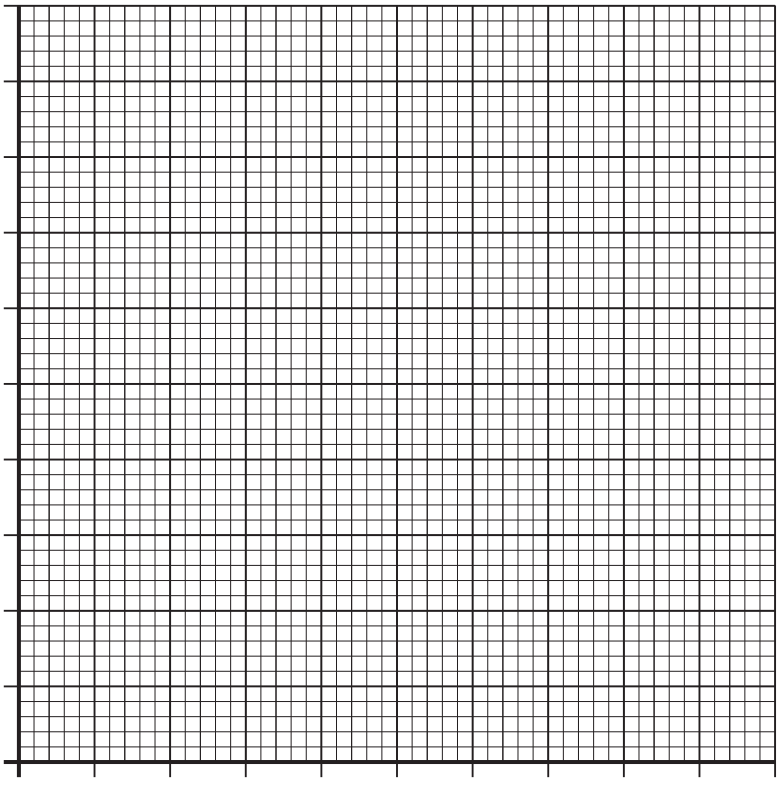
4. (continued)

(b) The results from the investigation are shown in **Table 1**.

Table 1

Time(weeks) / Group	Average maximum weight raised (kg)					
	0	**2**	**4**	**6**	**8**	**10**
A (protein supplement)	52	57	64	72	86	95
B (placebo)	50	55	60	68	74	80

(i) Draw a line graph to show **all** the data in **Table 1**.
(Additional graph paper, if required, can be found on *page 30*) **3**



(ii) State the conclusion that can be drawn from these results. **1**

_____

_____

4. (b) (continued)

(iii) Suggest a reason for the increase in performance of Group B. **1**

(c) The average body mass and percentage body fat of the students was measured at the start of the investigation and after 10 weeks.

The measurements for each group are shown in **Table 2**.

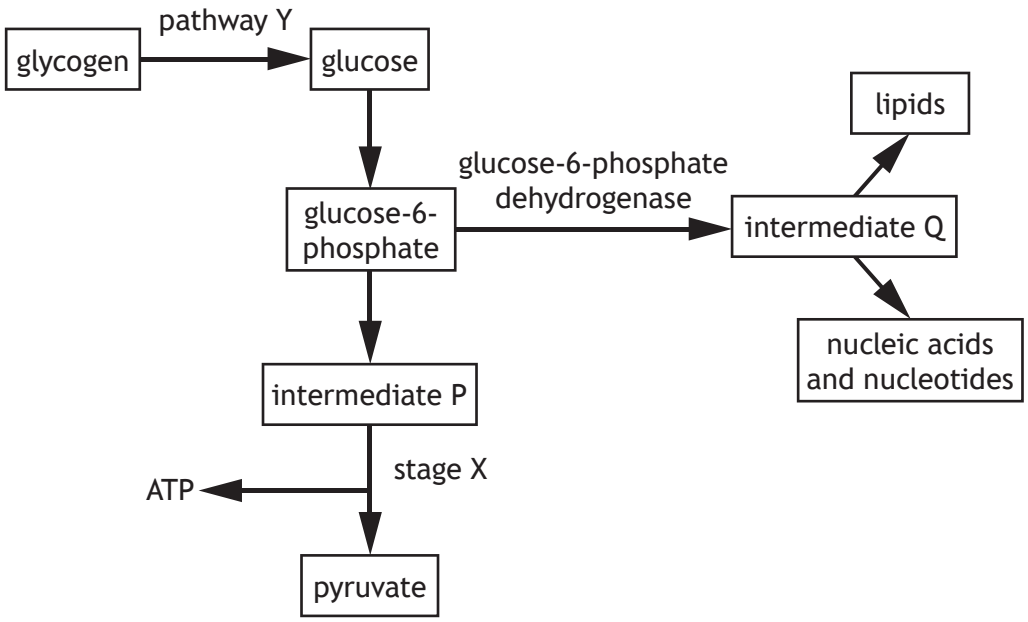
Table 2

Group	Measurement	Start (0 weeks)	After 10 weeks
A (protein supplement)	body mass (kg)	60·7	62·5
	body fat (%)	28·3	23·8
B (placebo)	body mass (kg)	60·5	61·3
	body fat (%)	27·9	25·8

Use the data to explain why taking protein supplements resulted in a greater increase in muscle mass. **2**

[Turn over

5. The diagram is a summary of part of a respiratory pathway in a cell.



(a) Give evidence from the diagram that confirms stage X is the energy pay-off phase. **1**

_____

(b) State the function of dehydrogenase enzymes. **1**

_____

_____

(c) Name a hormone that is involved in pathway Y. **1**

_____

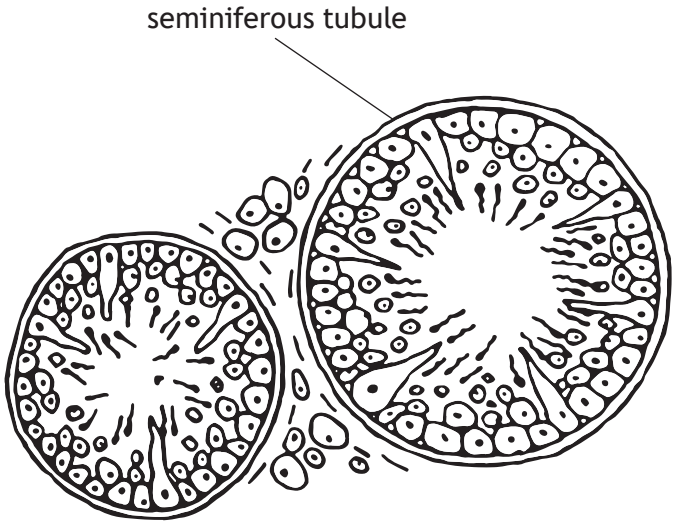
5. (continued)

(d) Some individuals are unable to produce the enzyme glucose-6-phosphate dehydrogenase.

Use information from the diagram to explain why this would reduce cell division in these individuals. **2**

[Turn over

6. (a) The diagram shows part of a testis.



(i) **Use the letter T** to label a cell in the diagram that produces testosterone. **1**

(ii) Describe two functions of testosterone. **2**

1 _____

_____

2 _____

_____

(iii) Describe how negative feedback control raises the concentration of testosterone in the blood if it has fallen to a low level. **2**

_____

_____

_____

_____

6. (continued)

(b) Artificial insemination (AI) and intra-cytoplasmic sperm injection (ICSI) are fertility treatments that may be used if a man has a low sperm count.

Describe how each of these treatments increases the chance of fertilisation. **2**

AI _____

_____

_____

ICSI _____

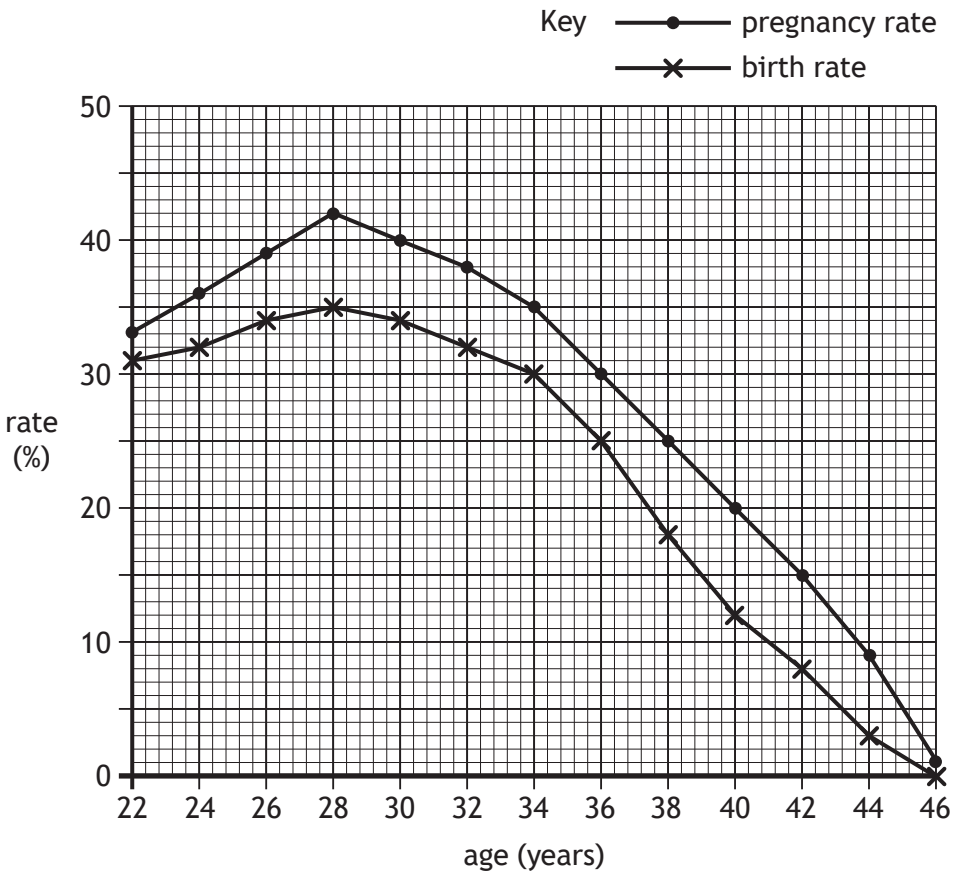
_____

_____

[Turn over

7. Data from *in vitro* fertilisation (IVF) clinics is used to indicate how a woman's age can affect the success rate of IVF.

(a) The graph shows the effect of age on pregnancy and birth rate after using IVF. The data refers to women using their own eggs.



(i) State **one** similarity and **one** difference in the trends for pregnancy and birth rate. **2**

Similarity _____

_____

_____

Difference _____

_____

_____

7. (a) (continued)

(ii) Calculate how many 34 year old women in a sample of 1000 would be predicted to get pregnant using IVF. **1**

Space for calculation

_____

(b) Some women are given the option to use eggs from a donor when undergoing IVF. This can increase their chances of becoming pregnant and giving birth.

(i) The table shows the birth rates for women of different ages after undergoing IVF using donor eggs.

Age (years)	22	26	30	34	38	42	46
Birth rate (%)	46	64	60	58	54	48	40

Using data in the table and graph, calculate the difference in birth rate for women aged 38 when using donor eggs rather than their own eggs. **1**

Space for calculation

_____ %

(ii) Suggest why older women undergoing IVF are more likely to produce a child when using donor eggs rather than their own eggs. **1**

_____

_____

(c) Some women undergoing IVF consent to pre-implantation genetic diagnosis (PGD) of their embryos.

Explain why PGD is offered to some women. **1**

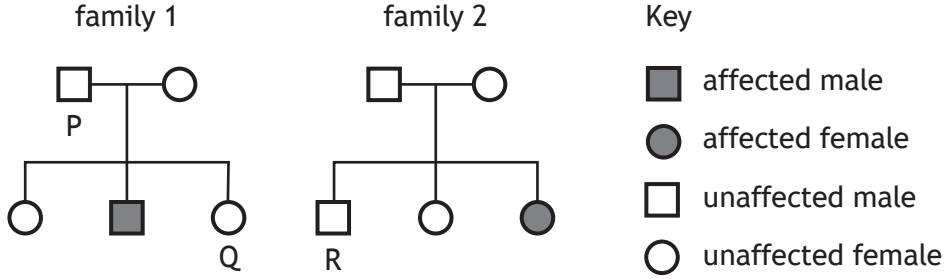
_____

_____

8. Adenosine deaminase (ADA) deficiency is an autosomal recessive disorder that affects the immune system. It is caused by a mutation in the gene that codes for this enzyme.

(a) Describe the difference between an autosomal disorder and a sex-linked disorder. **1**

(b) The chart shows the inheritance of ADA deficiency in two families.



(i) State the genotype of individual P using 'a' to represent the allele for ADA deficiency. **1**

(ii) Individuals Q and R plan to have a child together.

Use information from the chart to explain why it is difficult to predict the chances of their child having the disorder. **1**

8. (continued)

(c) Individuals with ADA deficiency have severely reduced numbers of lymphocytes. They can be treated by a bone marrow transplant.

Suggest how this treatment can help to restore a functional immune system. **1**

(d) Amniocentesis and chorionic villus sampling (CVS) are examples of antenatal tests that can be carried out.

Describe an advantage and a disadvantage of using amniocentesis rather than chorionic villus sampling (CVS). **2**

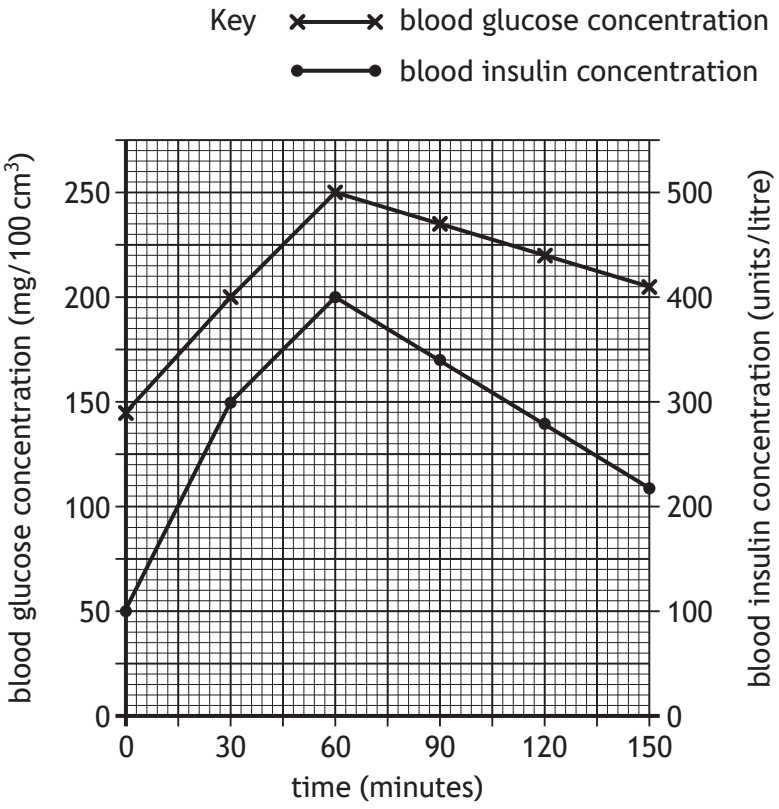
Advantage

Disadvantage

[Turn over

9. A man had a glucose tolerance test to indicate if he had type 2 diabetes.

The graph shows changes in the concentrations of glucose and insulin in his blood during 150 minutes, after drinking the glucose solution.



(a) State the man's blood insulin concentration when his blood glucose concentration was 200 mg/100 cm^3. **1**

_____ units/litre

(b) The man's blood glucose concentration will eventually return to its original value.

Predict how much longer this will take after 150 minutes. **1**

Space for calculation

_____ minutes

9. (continued)

(c) This man's body contains 5 litres of blood.

Calculate the total mass of glucose in his bloodstream at 60 minutes. **1**

Space for calculation

_____ mg

(d) (i) The glucose tolerance test indicated that this man had type 2 diabetes.

Explain why production of insulin did not lower his blood glucose concentration in the first hour of the test. **2**

_____

_____

_____

_____

(ii) Suggest **one** reason why this man's blood glucose concentration started to decrease after 60 minutes. **1**

_____

_____

(e) Describe evidence from the graph that indicates this man does **not** have type 1 diabetes. **1**

_____

_____

(f) Apart from reducing the sugar intake in his diet, suggest another way in which this man could control his blood glucose levels. **1**

_____

_____

[Turn over

10. (a) The table shows the death rate from cardiovascular disease (CVD) in groups of men and women in England and Scotland between 2001 and 2013.

	Death rate from CVD (per 100 000)			
	England		Scotland	
Year	Men	Women	Men	Women
2001	603	402	684	476
2005	490	337	577	406
2009	392	271	450	316
2013	324	221	396	275

(i) Identify the group that shows the greatest percentage decrease in death rate from CVD between 2001 and 2013. **1**

Space for calculation

Group _____

(ii) Death rates have decreased in all groups between 2001 and 2013.

State another conclusion that can be drawn from the data in the table. **1**

_____

_____

(iii) Express, as a simple whole number ratio, the death rate from CVD in English men compared to Scottish men in 2013. **1**

Space for calculation

_____ : _____

English men Scottish men

10. (a) (continued)

(iv) Explain how the data in the table allows a valid comparison of deaths from CVD between England and Scotland to be made, despite their populations being different sizes. **1**

(b) Three thousand men took part in a clinical trial to investigate the effect of a cholesterol reducing drug. Half the men were given the cholesterol reducing drug while half were given a placebo.

(i) Describe how a double-blind design could be achieved when setting up the clinical trial. **1**

(ii) State what aspect of the design of the study increased the reliability of the results. **1**

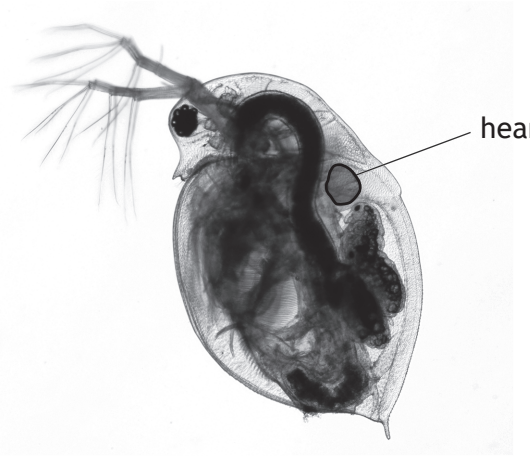
(iii) Researchers concluded that taking the cholesterol reducing drug decreased risk of death from CVD, as it lowered the concentration of LDLs in the blood.

Explain why having a lower LDL concentration in the blood decreases the risk of death from CVD. **2**

[Turn over

11. (a) The water flea, *Daphnia pulex*, is a small invertebrate animal that lives in ponds.

Water fleas can be used as model organisms to investigate the effect of chemicals on heart rate.



A student carried out an investigation to find out how caffeine concentration affects the heart rate of water fleas.

A water flea was placed in a small container of pond water and left for 5 minutes. The container was then placed under a microscope and the water flea videoed for a period of time. The video was analysed and the heart rate of the water flea calculated. This procedure was then repeated using different concentrations of caffeine solutions.

The results of the investigation are shown in the table.

Caffeine concentration (g/l)	Heart rate of water flea (bpm)
0	135
0·2	185
0·4	220
0·6	245
0·8	270
1·0	270

(i) Suggest why the student left the water flea in the solution for 5 minutes before videoing its heart rate. **1**

_____

_____

11. (a) (continued)

(ii) Suggest why the student videoed the water flea rather than simply counting its heart beat at the time. **1**

(iii) Use data from the table to describe the changes that occur in the heart rate as the caffeine concentration increases. **2**

(iv) State how the reliability of the results from this investigation could be improved. **1**

(b) In humans, describe how the autonomic nervous system increases the heart rate. **3**

12. Endorphins and dopamine are neurotransmitters that affect mood and behaviour.

(a) (i) State **one** activity that increases endorphin production. **1**

_____

_____

(ii) State **one** function of endorphins. **1**

_____

_____

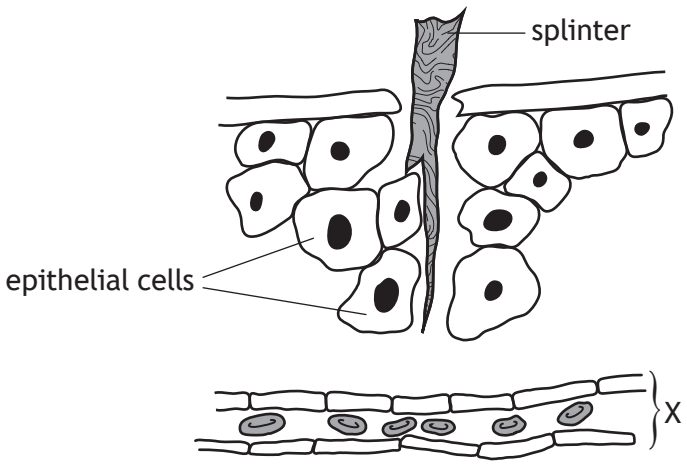
(b) Parkinson's disease is associated with a loss of dopamine-producing neurons in the brain. It cannot be treated by taking dopamine.

Describe a possible mode of action of a drug that could be used to treat this disease. **1**

_____

_____

13. The diagram represents a section of human skin punctured by a splinter of wood.



(a) (i) Describe **one** way in which these epithelial cells defend the body against pathogens. **1**

_____

_____

(ii) Name structure X. **1**

_____

(b) The inflammatory response involves the release of the chemical histamine.

(i) Name the cells that release histamine. **1**

_____

(ii) Explain why the skin around a puncture wound often becomes red and swollen. **1**

_____

_____

(c) Lymphocytes are part of the specific defence system of the body.

(i) Explain how a lymphocyte is able to recognise a particular pathogen. **1**

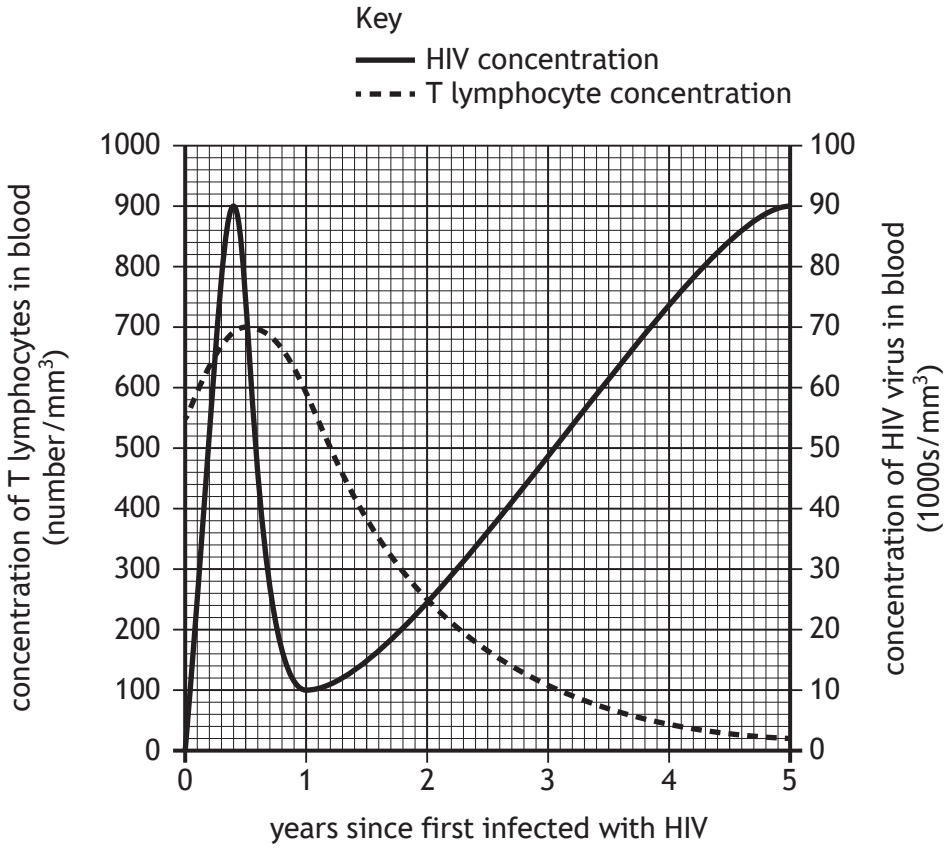
_____

_____

13. (c) (continued)

(ii) Describe how T lymphocytes destroy infected body cells. **1**

(d) The graph shows the concentrations of the HIV virus and T lymphocytes in the blood of an individual in the years following HIV infection.



(i) Suggest a reason for the decrease in HIV concentration during the first year after infection. **1**

(ii) Explain why the T lymphocyte concentration is decreasing between 1 and 5 years after infection. **1**

14. Attempt **either** A **or** B. Write your answer in the space below and on *page 29*.

A Discuss the encoding, storage and retrieval of information in memory. **9**

OR

B Describe vaccination and discuss its role in establishing herd immunity in a population. **9**

You may use labelled diagrams where appropriate.

National
Qualifications
2018

X740/76/02

Human Biology
Section 1 — Questions

TUESDAY, 15 MAY
1:00 PM – 3:30 PM

Instructions for the completion of Section 1 are given on *page 02* of your question and answer booklet X740/76/01.

Record your answers on the answer grid on *page 03* of your question and answer booklet.

Before leaving the examination room you must give your question and answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

©

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SECTION 1 — 20 marks

Attempt ALL questions

1. Each type of human cell has a different structure and function because

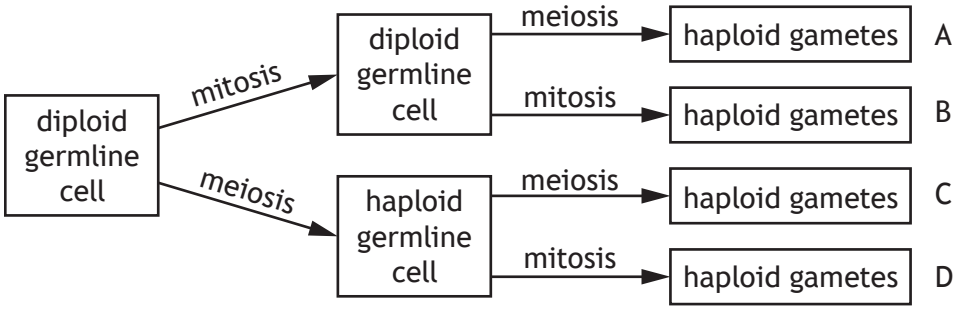
A they contain different genes

B different genes are expressed in each

C some genes are lost during differentiation

D some genes are gained during differentiation.

2. Which pathway describes the production of haploid gametes from diploid germline cells?



3. The table shows the number of dividing and non-dividing cells in samples of three types of tissue.

Type of tissue	*Number of cells dividing*	*Number of cells not dividing*
nerve	8	17
blood	4	16
muscle	1	19

The percentage of connective tissue cells which are dividing is

A 5%

B 20%

C 25%

D 32%

4. A fragment of DNA contained 144 nucleotide base pairs.

What is the total number of deoxyribose sugars in this fragment?

A 48

B 72

C 144

D 288

5. The table shows the positions of bases in the mRNA codons for specific amino acids.

	Second position				
First position	U	C	A	G	*Third position*
U	phenylalanine	serine	tyrosine	cysteine	U
					C
	leucine		stop	stop	A
			stop	tryptophan	G
C	leucine	proline	histidine	arginine	U
					C
			glutamine		A
					G
A	isoleucine	threonine	asparagine	serine	U
					C
	start/ methionine		lysine	arginine	A
					G
G	valine	alanine	aspartic acid	glycine	U
					C
			glutamic acid		A
					G

Which of the following mutations in a section of mRNA would result in the production of a shortened protein?

	Original mRNA codons	*Mutated mRNA codons*
A	..... AUG GCC CAU	..... AUG GCA CAU .....
B	..... CAG UAC GUG	..... CAG UAG GUG .....
C	..... AAU UGG CCA	..... AAU UGU CCA .....
D	..... GUC AAC UCG	..... GUC AAG UCG .....

6. A mature mRNA transcript is produced from a primary mRNA transcript by

A adding exons
B adding introns
C removing exons
D removing introns.

7. DNA probes are short fragments of DNA that

A allow RNA polymerase to begin transcription
B allow DNA polymerase to begin DNA replication
C are used to detect specific sequences in samples of DNA
D bind to specific target sequences in the PCR reaction to amplify DNA.

8. The list shows some of the substances produced during the respiration of glucose in the presence of oxygen.

1 acetyl group
2 pyruvate
3 citrate
4 ATP

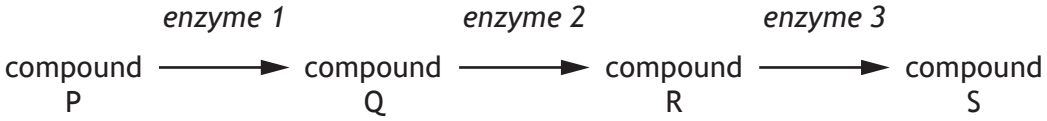
Which of the following sequences shows the order in which these substances are produced?

A 4, 2, 1, 3
B 4, 2, 3, 1
C 2, 1, 4, 3
D 2, 3, 1, 4

9. In cellular respiration, the products of the electron transport chain are

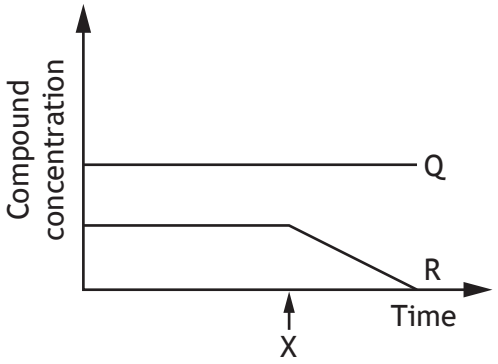
A water and ATP
B oxygen and ATP
C NADH and $FADH_2$
D carbon dioxide and water.

10. The following diagram shows an enzyme-controlled metabolic pathway.

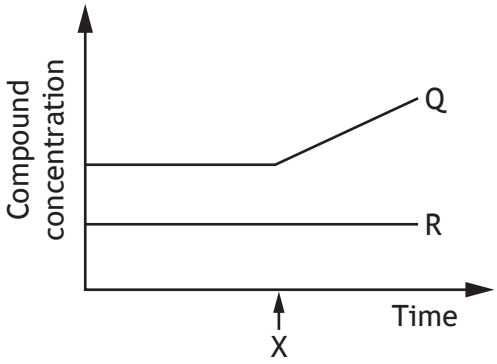


If enzyme 2 is inhibited at time X, which graph predicts the resulting concentrations of compounds Q and R?

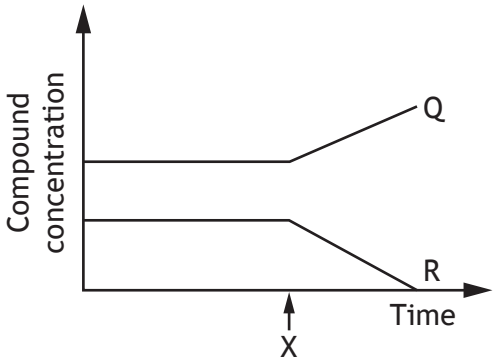
A



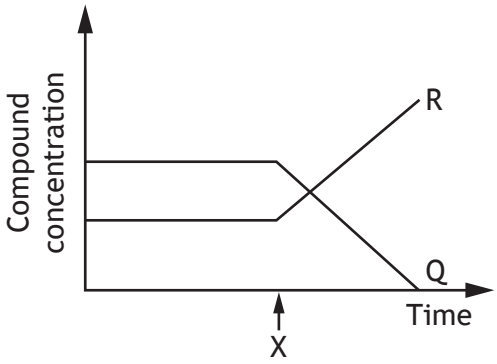
B



C

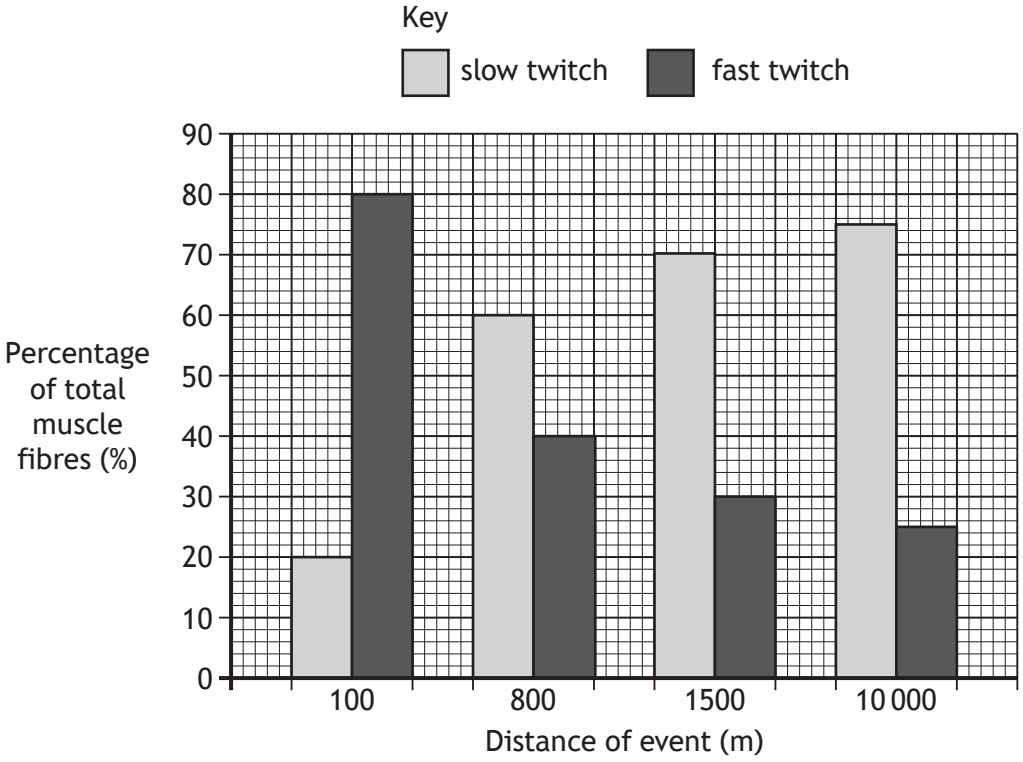


D



[Turn over

11. The graph shows the percentage of slow and fast twitch muscle fibres present in athletes who trained for events of different distances.



Which of the following conclusions can be drawn from this graph?

A Athletes who trained for the 100 m event have 5 times more fast twitch muscle fibres than slow twitch muscle fibres.

B Athletes who trained for the 10 000 m event have 4 times more slow twitch muscle fibres than fast twitch muscle fibres.

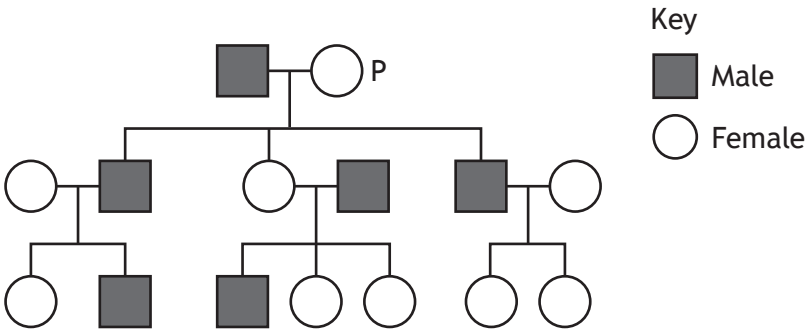
C Athletes who trained for the 800 m event have twice as many slow twitch muscle fibres as athletes in the 1500 m event.

D Athletes who trained for the 100 m event have twice as many fast twitch muscle fibres as athletes in the 800 m event.

12. The mitochondria of human cells contain DNA.

Women can pass mitochondrial DNA to their offspring but men cannot.

The diagram shows a family tree.



Identify the number of individuals in the family tree that have inherited mitochondrial DNA which originated from P.

A 3

B 4

C 5

D 6

13. Thalassaemia is an inherited condition that affects the ability of haemoglobin to carry oxygen. The condition is **not** sex-linked.

The table shows genotypes and phenotypes associated with thalassaemia.

Genotype	*Phenotype*
AA	unaffected
AT	thalassaemia trait
TT	severe thalassaemia

An unaffected man and a woman with thalassaemia trait have a child.

The chance that the child will also have thalassaemia trait is

A 0%

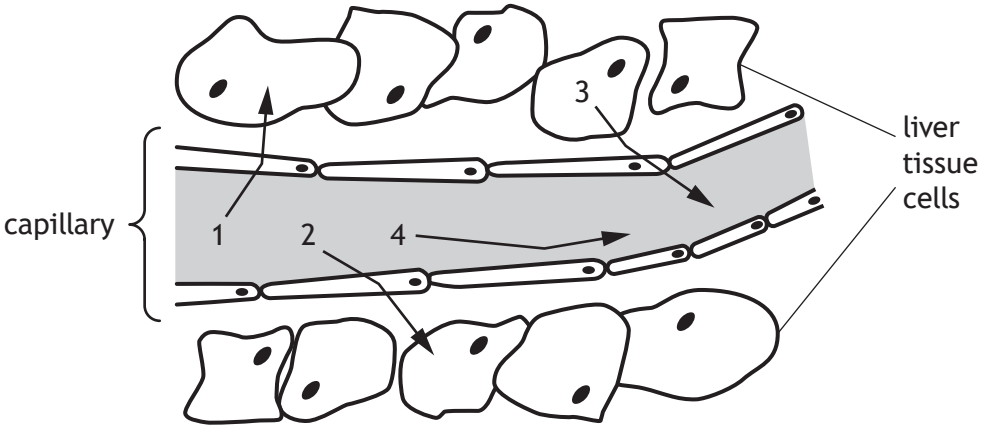
B 25%

C 50%

D 100%

[Turn over

14. The diagram shows the movement of substances between a capillary and the surrounding liver tissue cells.



Which row in the table identifies the substances in the diagram?

	Substance			
	1	2	3	4
A	glucose	carbon dioxide	oxygen	protein
B	oxygen	glucose	carbon dioxide	protein
C	protein	glucose	oxygen	carbon dioxide
D	protein	oxygen	carbon dioxide	glucose

15. During the formation of a thrombus, fibrin

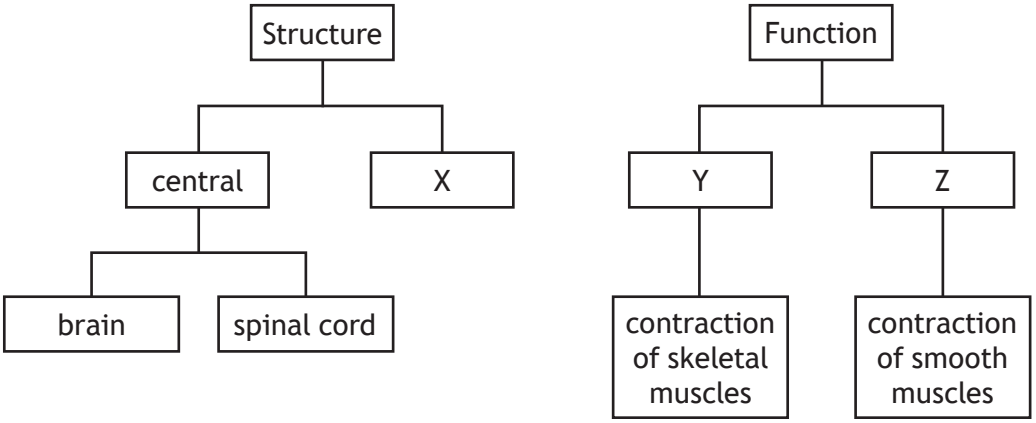
A converts prothrombin to thrombin

B causes the formation of fibrinogen

C forms a meshwork to clot the blood

D causes the release of clotting factors.

16. The diagrams show two ways to classify the nervous system.



Which row in the table identifies X, Y and Z?

	Nervous System		
	X	Y	Z
A	peripheral	somatic	autonomic
B	somatic	autonomic	peripheral
C	autonomic	peripheral	somatic
D	peripheral	autonomic	somatic

17. A child was stung by a wasp. This led to them being afraid of all flying insects.
This is an example of

A discrimination
B generalisation
C internalisation
D reinforcement.

[Turn over

18. An investigation was carried out into the effect of colour on the recognition of shapes. The procedure included the following features.

- Two groups of 20 students were selected.
- Each group was shown the same sequence of shapes but in a different colour.
- One group was shown blue shapes and the other group was shown red shapes.
- The time taken by each student to identify each shape was noted.

Which of the following would improve the reliability of the results?

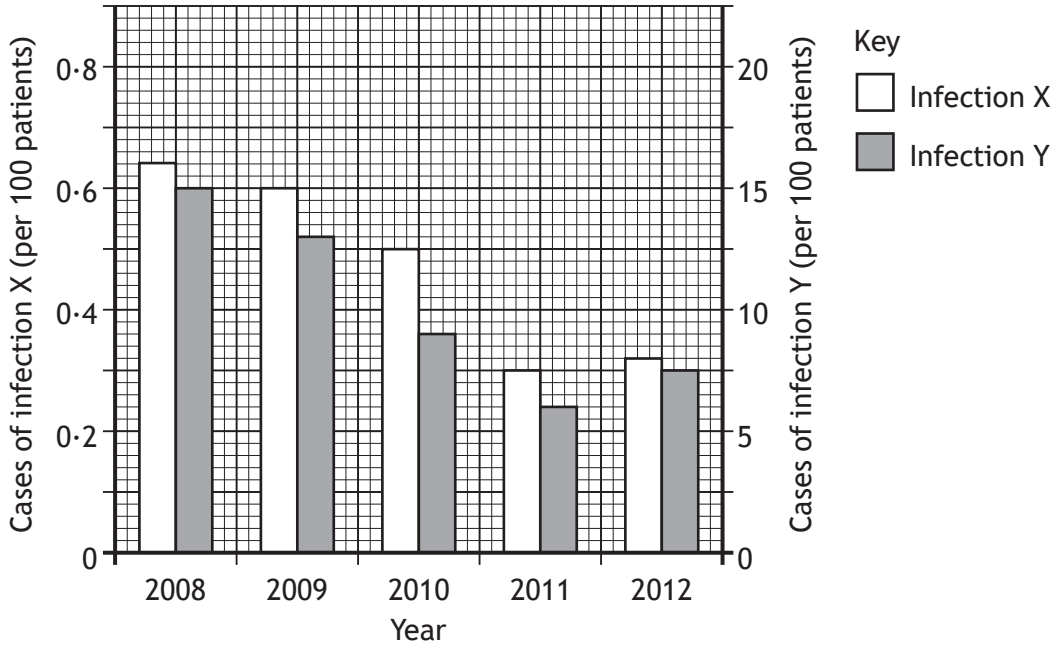
A Showing the same set of shapes to each group.
B Ensuring all the selected students were the same age.
C Having the same number of males and females in each group.
D Repeating the whole procedure with two more groups of students.

19. The virus that causes influenza can evade the specific immune response by

A attacking phagocytes
B attacking lymphocytes
C surviving within phagocytes
D showing antigenic variation.

20. A hospital introduced a programme of handwashing in 2008.

The graph shows the impact of this on the number of cases of two infections.



Which of the following statements is **not** correct?

A The cases of both infections fell by 50% over the 5 year period.

B The number of cases of infection Y was always greater than the number of cases of infection X.

C The highest number of cases of infection X was 0·62 per 100 patients while the highest number of cases of infection Y was 15 per 100 patients.

D The lowest number of cases of infection X was 0·3 per 100 patients while the lowest number of cases of infection Y was 6 per 100 patients.

[END OF SECTION 1. NOW ATTEMPT THE QUESTIONS IN SECTION 2 OF YOUR QUESTION AND ANSWER BOOKLET.]

FOR OFFICIAL USE

National Qualifications 2018

Mark

X740/76/01

Human Biology
Section 1 — Answer Grid and Section 2

TUESDAY, 15 MAY
1:00 PM – 3:30 PM

X7407601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day	Month	Year

Scottish candidate number

Total marks — 100

SECTION 1 — 20 marks

Attempt ALL questions.

Instructions for the completion of Section 1 are given on *page 02*.

SECTION 2 — 80 marks

Attempt ALL questions.

Question 13 contains a choice.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. You should score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

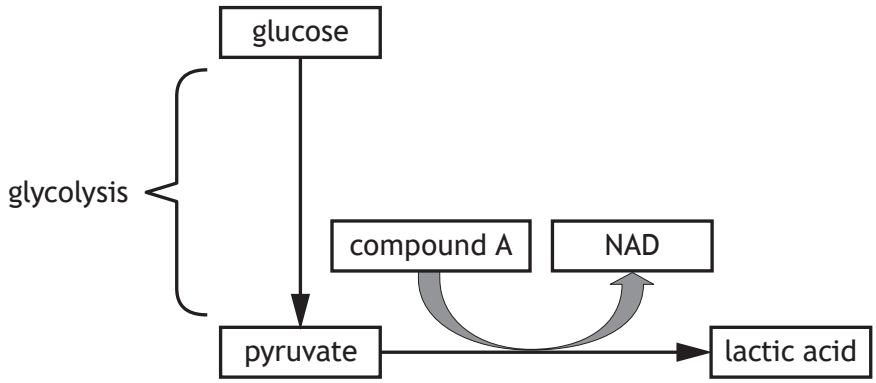
X7407601 01

SECTION 2 — 80 marks

Attempt ALL questions

Question 13 contains a choice

1. The diagram represents glycolysis and the metabolic pathway which synthesises lactic acid.



(a) (i) State where glycolysis occurs within a cell. **1**

_____

(ii) Describe what happens during the energy investment phase of glycolysis. **1**

_____

_____

(b) During lactic acid synthesis NAD is regenerated.

(i) Name compound A. **1**

_____

(ii) Explain the importance of the regeneration of NAD for glycolysis. **1**

_____

_____

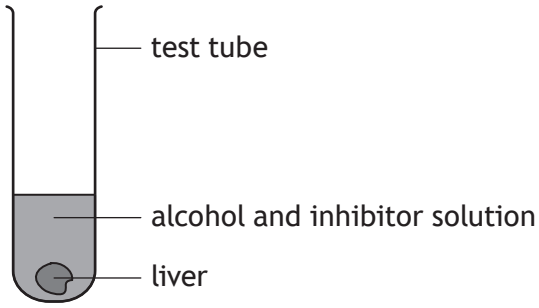
(iii) State the reason why muscle cells produce lactic acid during vigorous exercise. **1**

_____

_____

2. An investigation was carried out into the effect of inhibitor concentration on the activity of an enzyme which breaks down alcohol in liver cells.

Six test tubes were set up, each containing a piece of liver, alcohol and a different concentration of inhibitor, as shown in the diagram.



The test tubes were left for 30 minutes at 37 °C.

The final concentration of alcohol in each tube was then measured.

(a) State **two** variables, not already mentioned, which should be kept constant to make this investigation valid. **2**

1 _____

2 _____

(b) The inhibitor used in this investigation was non-competitive.

Describe how a non-competitive inhibitor works. **1**

_____

_____

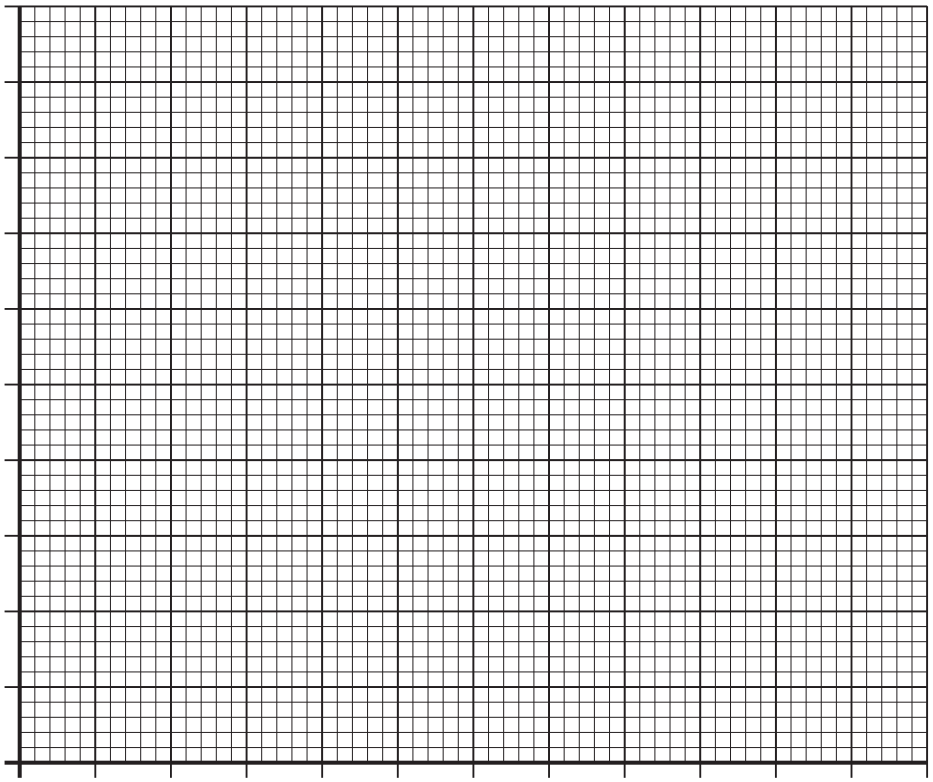
(c) The results of the investigation are shown in the table.

Inhibitor concentration (mM)	*Final alcohol concentration* (% of initial concentration)
0·5	20
1·5	28
2·5	60
3·5	96
4·5	100
5·5	100

2. (c) (continued)

(i) Construct a line graph to show the data in the table. **2**

(Additional graph paper, if required, can be found on *page 27*)



(ii) **Use the data** to describe the relationship between the concentration of inhibitor and enzyme activity. **2**

(d) A second experiment, using increasing concentrations of alcohol, was carried out to show that the inhibitor was non-competitive.

Six test tubes were set up, each containing a piece of liver, an inhibitor concentration of 4·5 mM and different concentrations of alcohol.

Suggest how the results would confirm that the inhibitor was non-competitive. **1**

3. Olfactory genes code for receptors in the nose that detect smells.

The base sequences from the same region of a rat olfactory gene and a human olfactory gene are shown.

Rat	... A T A C G A T T G C A T G C C G A T...
Human	... A T A C G A T T G C A T C C G A T...

The rat olfactory gene codes for a functional protein but the protein coded for by the human olfactory gene is non-functional.

(a) (i) Name the type of single gene mutation that has occurred to change the human base sequence. **1**

_____

(ii) Suggest why the changed sequence of bases in the human gene codes for a non-functional protein. **1**

_____

_____

_____

(b) State the term which describes the comparison of human genome sequence data with the genomes of other species. **1**

_____

3. (continued)

(c) The table shows the number of functional olfactory genes identified in rats, humans and chickens.

Animal	*Number of functional olfactory genes*
rat	1200
human	400
chicken	80

(i) Express, as a simple whole number ratio, the number of functional olfactory genes found in the animals. **1**

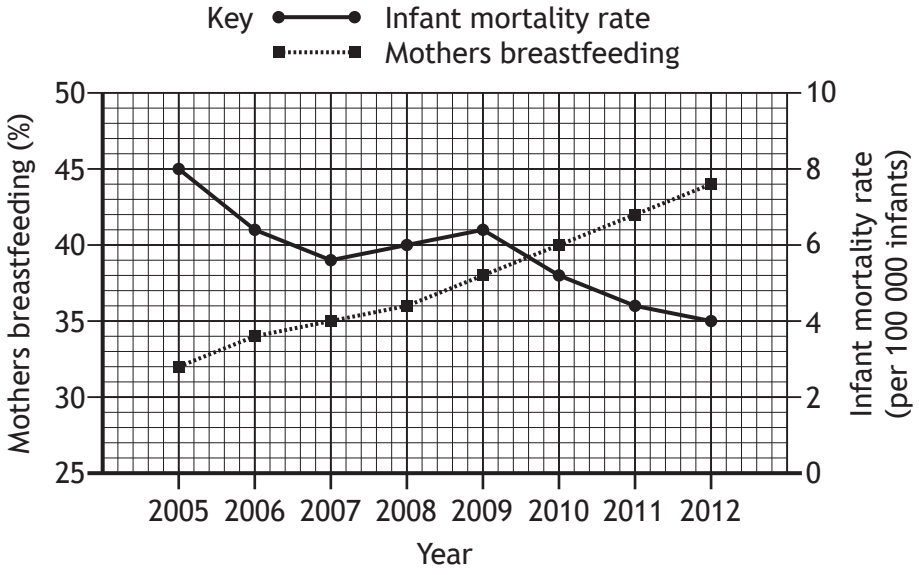
Space for calculation

_____ : _____ : _____
rat human chicken

(ii) Suggest what the number of functional olfactory genes indicates about the sense of smell of these animals. **1**

(d) Describe **two** chromosome structure mutations and the overall consequence of these to the individual. **3**

4. The graph contains information about breastfeeding and infant mortality in a country between 2005 and 2012.



(a) (i) State the infant mortality rate when 35% of mothers were breastfeeding. **1**

_____

(ii) Calculate the percentage decrease in the infant mortality rate between 2009 and 2012. **1**

Space for calculation

_____%

(iii) Predict the percentage of mothers who would be breastfeeding in 2013. **1**

_____%

(iv) Describe evidence from the graph which indicates that the increase in breastfeeding mothers cannot be the only reason for the decrease in infant mortality. **1**

_____

_____

_____

4. (continued)

(b) The table contains information from a Scottish survey comparing the incidence of diarrhoea in breastfed and bottle fed babies.

	Age of baby (months)			
	0–3		4–6	
Feeding method	Breast	Bottle	Breast	Bottle
Incidence of diarrhoea (%)	3·6	21·6	10·2	20·4

(i) Calculate how many times greater the incidence of diarrhoea is when 0–3 month old babies are bottle fed rather than breastfed. **1**

Space for calculation

_____ times greater

(ii) Diarrhoea is a symptom of an intestinal infection.

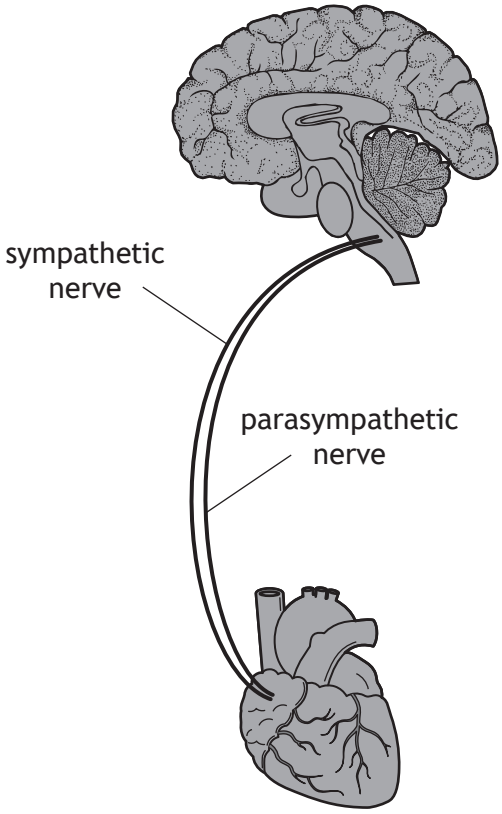
Suggest why there is a greater incidence of diarrhoea in bottle fed babies. **1**

_____

_____

[Turn over

5. The diagram represents part of the autonomic nervous system which links the brain to the heart.



(a) Name the parts of the brain and heart which are linked by the nerves shown in the diagram. **2**

Brain _____

Heart _____

(b) Explain how the sympathetic and parasympathetic nerves control heart rate. **2**

_____

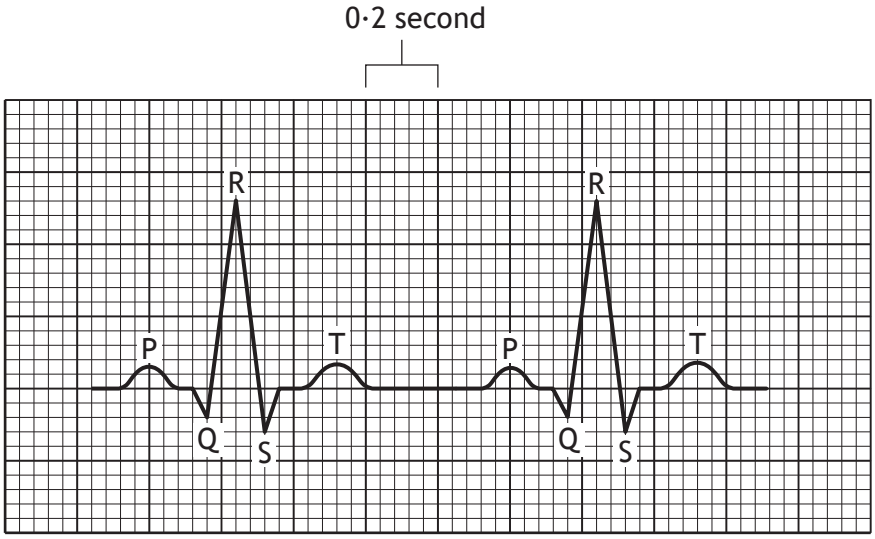
_____

_____

_____

5. (continued)

(c) The diagram shows an electrocardiogram (ECG) of an individual's heart.



(i) Use the diagram to calculate the individual's heart rate. **1**

Space for calculation

_____ beats/min

(ii) Describe what happens in the heart between points Q and S. **1**

_____

_____

[Turn over

6. The table shows the number of males in different age groups with raised cholesterol levels in Scotland between 2004 and 2013.

	Number of males with raised cholesterol levels (per 1000)									
Age group (years)	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013
Under 15	1	1	0	1	0	0	0	2	0	1
15–44	33	33	31	29	21	21	22	16	18	14
45–64	236	232	217	209	132	139	134	137	125	128
Over 64	274	316	295	274	167	166	164	167	171	167

(a) Calculate which age group had the greatest percentage decrease in the number of males with raised cholesterol levels between 2004 and 2013. **1**

Space for calculation

_____ years

(b) (i) Name a type of drug that is used to control cholesterol levels. **1**

_____

(ii) Use the data in the table to identify the year in which this type of drug became widely available. **1**

_____

(c) State **one** role of cholesterol in the body. **1**

_____

_____

7. An office worker and an Olympic swimmer were found to have the same BMI. They each weighed 105 kg and were 1·85 m in height.

(a) (i) Calculate their BMI. **1**

Space for calculation

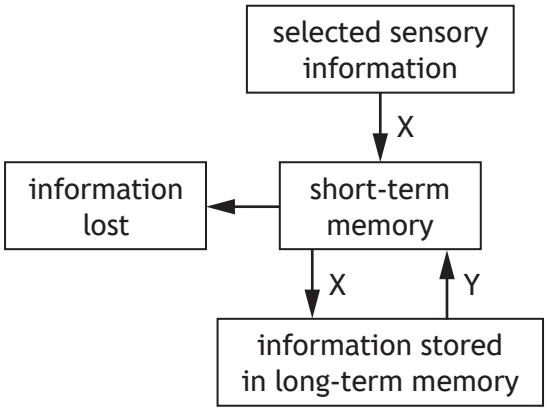
(ii) Suggest why, after calculating their BMI, a health professional advised only the office worker to lose weight. **1**

(b) The office worker developed Type 2 diabetes.

Explain why this condition causes the blood glucose concentration to remain high. **2**

[Turn over

8. (a) The diagram represents the flow of information through memory.



(i) Name process X. **1**

_____

(ii) Explain why information can be lost by displacement from short-term memory. **1**

_____

_____

(iii) Rehearsal increases the chance of information being transferred from short-term to long-term memory.

Describe **one** other way that information can be transferred from short-term to long-term memory. **1**

_____

_____

_____

(iv) Y represents the retrieval of information from long-term memory.

Describe how contextual cues aid the retrieval of information. **1**

_____

_____

_____

(b) State where semantic memories are stored in the brain. **1**

_____

9. (a) The photograph shows cars parked in a street.



The judgement of distance depends on visual cues.

Explain how the following visual cues allow a person to judge how far away each car is from them in the street.

(i) Size _____ **1**

_____

_____

(ii) Superimposition _____ **1**

_____

_____

(b) State the term used for the ability of the brain to judge the distance of an object based on the different images received by each eye. **1**

_____

[Turn over

MARKS

DO NOT WRITE IN THIS MARGIN

(c) Students carried out the following experiment on the judgement of distance.

1 One student raised their right arm and closed both eyes.

2 A coin was randomly placed on the bench in front of them.

3 The student opened both eyes and immediately lowered their arm trying to touch the coin with their index finger.

4 The distance between the coin and the spot where the index finger landed was measured.

5 Steps 1–4 were repeated, firstly **only** opening the left eye and secondly **only** opening the right eye.

6 Then steps 1–5 were repeated another nine times.

The results are shown in the table.

Attempt	*Distance between coin and spot where finger landed* (mm)		
	both eyes open	*left eye open*	*right eye open*
1	2	4	12
2	2	6	15
3	1	6	16
4	1	8	14
5	1	9	13
6	0	5	14
7	0	4	17
8	1	10	18
9	0	7	
10	1	8	15
Average	0·9	6·7	15·0

(i) Calculate the missing distance for attempt 9. **1**

Space for calculation

_____ mm

9. (c) (continued)

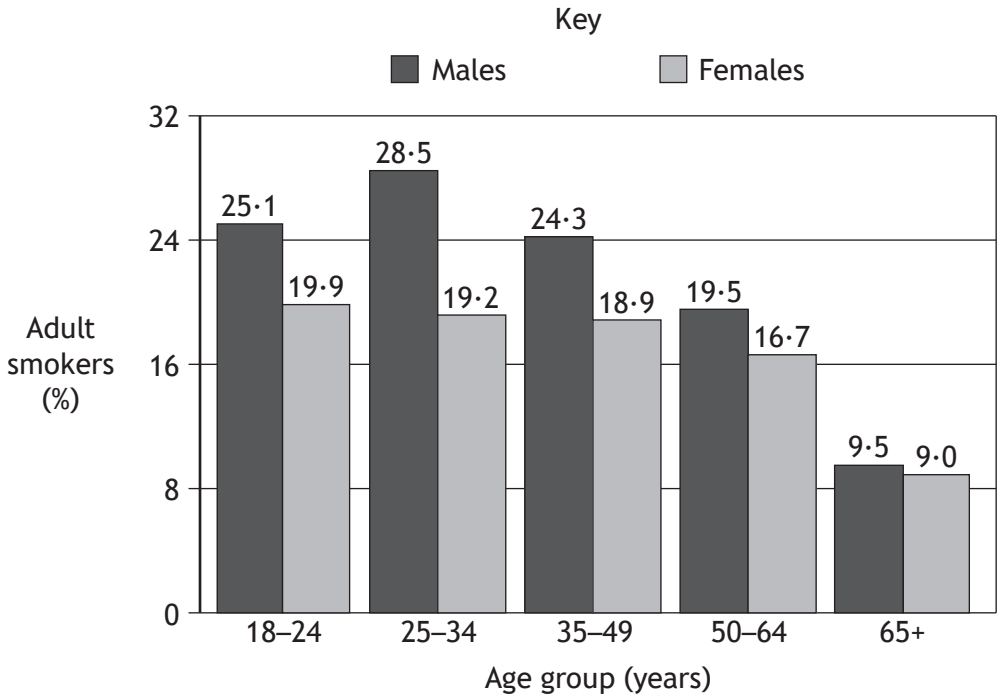
(ii) Suggest a reason why the coin was randomly placed for each attempt. **1**

(iii) The results of the experiment show that two eyes are more accurate than one for judging distance.

State another conclusion which can be drawn from the results. **1**

[Turn over

10. The graph shows the percentage of adult smokers in different age groups in the UK in 2011.



(a) (i) Describe **two** trends shown in the graph. **2**

1 _____

_____

_____

2 _____

_____

_____

(ii) Apart from individuals giving up smoking, suggest a reason for the difference in the percentage of 25–34 year olds and 65+ year olds smoking. **1**

_____

_____

10. (continued)

(b) Smokers can become addicted to the nicotine in tobacco.

Nicotine acts as an agonist of acetylcholine causing an increase in the levels of dopamine.

(i) Describe how nicotine acts as an agonist at a synapse. **1**

(ii) Describe how dopamine reinforces smoking behaviour. **1**

(iii) Describe how repeated exposure to nicotine can lead to nicotine tolerance by desensitisation. **1**

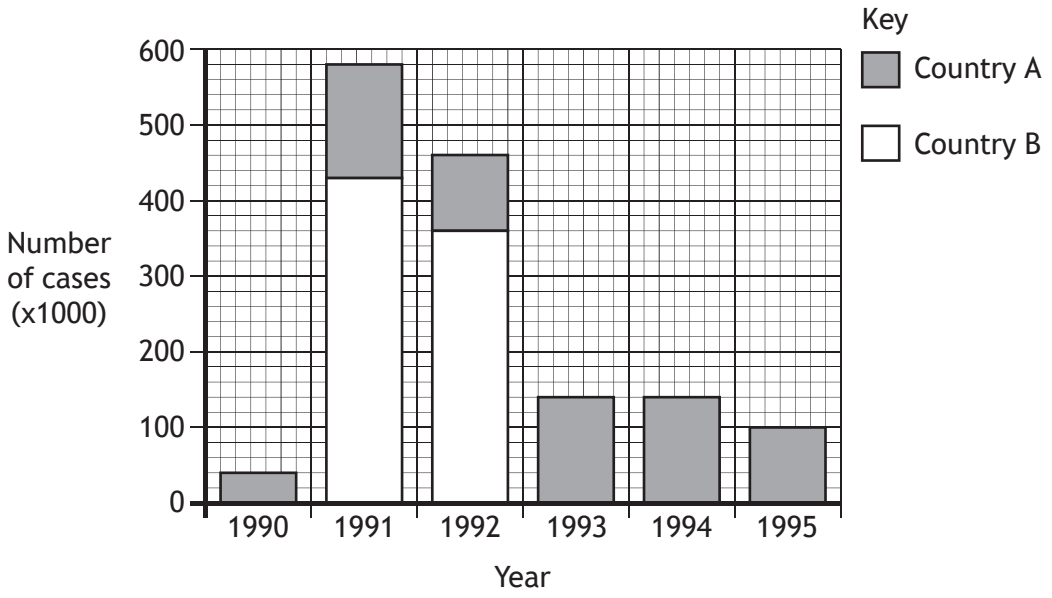
(c) Explain why anti-smoking campaigns often feature a celebrity. **1**

(d) In 2011 there were 36 980 cases of lung cancer in the UK linked to smoking. This was 86% of all lung cancer cases in the UK in 2011.

Calculate the total number of lung cancer cases in the UK in 2011. **1**

Space for calculation

11. The graph compares the number of cases of a water-borne disease in two countries between 1990 and 1995.



(a) (i) Describe how the graph demonstrates that the pattern of the disease was epidemic in country B. **1**

(ii) State the term which describes the pattern of disease in country A. **1**

(iii) Suggest how the pattern of this disease in country A would differ if the disease was sporadic. **1**

11. (continued)

(b) In 1996, a programme of immunisation against this disease was started in country B. Herd immunity was established once 85% of the population was immunised.

(i) Explain why this level of immunisation protected the whole population. **1**

_____

_____

_____

(ii) State **one** reason why it is difficult to immunise 100% of a population against a disease. **1**

_____

_____

[Turn over

12. The non-specific immune system provides resistance to infection by physical, chemical and cellular means.

(a) (i) Name the type of cell which forms a physical barrier in the skin. **1**

_____

(ii) In addition to forming a physical barrier, state another way in which these cells resist infection. **1**

_____

_____

(b) Mast cells initiate the inflammatory response.

Name the chemical which they release and explain how it increases the supply of fluid into the infected tissue. **2**

Chemical _____

Explanation _____

_____

(c) (i) The cellular aspect of the non-specific response is provided by two types of white blood cell.

Complete the table by describing how these cells destroy pathogens. **2**

Type of cell	*How cell destroys pathogens*
Phagocyte	
NK cell	

(ii) State how phagocytes and NK cells stimulate the specific immune response. **1**

_____

_____

13. Answer **either** A **or** B in the space below.

Labelled diagrams may be used where appropriate.

A Give an account of hormonal control of puberty and sperm production in males. **8**

OR

B Give an account of treatments for male and female infertility. **8**

National
Qualifications
2017

X740/76/02

Human Biology
Section 1 — Questions

TUESDAY, 23 MAY
1:00 PM – 3:30 PM

Instructions for the completion of Section 1 are given on *Page 02* of your question and answer booklet X740/76/01.

Record your answers on the answer grid on *Page 03* of your question and answer booklet.

Before leaving the examination room you must give your question and answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

©

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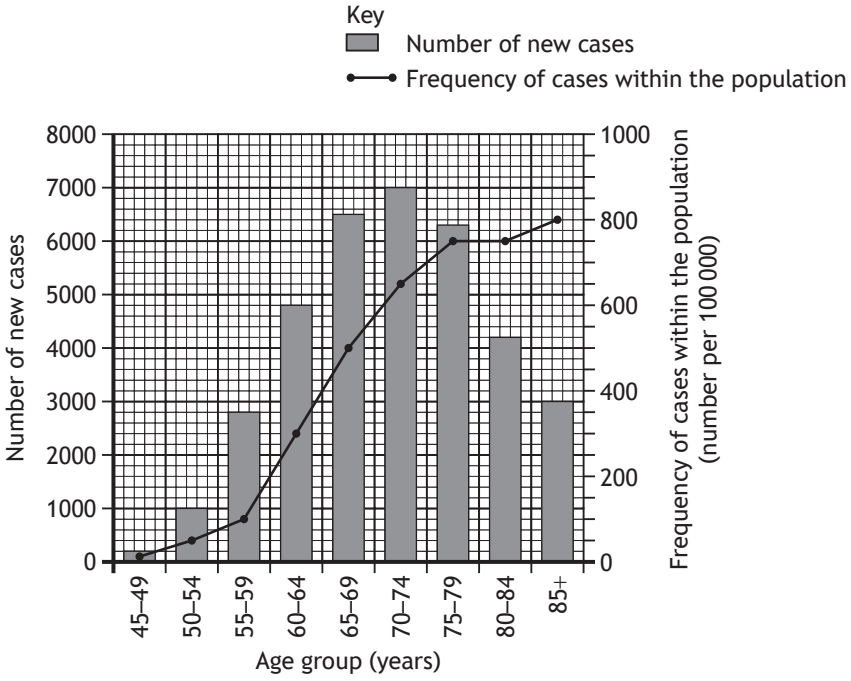
SECTION 1 — 20 marks

Attempt ALL questions

1. Which row in the table shows the type of stem cell that has the potential to form the greatest variety of specialised cells?

	Type of stem cell	*State of differentiation*
A	embryonic	differentiated
B	tissue	differentiated
C	embryonic	undifferentiated
D	tissue	undifferentiated

2. The graph contains information about prostate cancer in the UK in 2006.



Which of the following conclusions can be drawn from the graph?

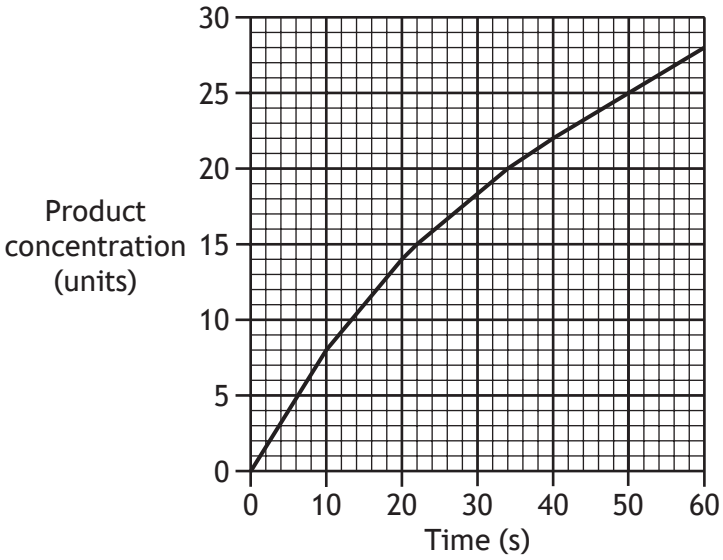
A The highest frequency of cases within the population was in the 70–74 year old age group.

B As the age group increases, the frequency of cases within the population always increases.

C When there were 4800 new cases, the frequency of cases within the population was 600 per 100 000.

D The greatest increase in the number of new cases, between consecutive age groups, occurred between 55–59 and 60–64.

3. The graph shows how the concentration of product changes during an enzyme-controlled reaction.



How long does it take the product concentration to reach 50% of its final concentration?

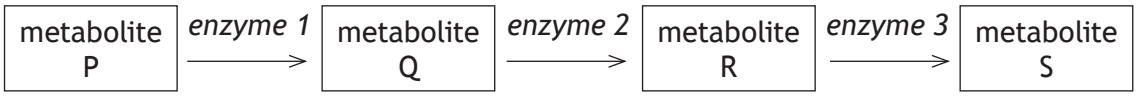
A 20 s

B 22 s

C 25 s

D 28 s

4. A metabolic pathway is shown.



In end-product inhibition

A enzyme 3 binds to enzyme 1

B enzyme 3 binds to metabolite P

C metabolite S binds to enzyme 1

D metabolite S binds to metabolite P.

[Turn over

5. Mature red blood cells have no nucleus and no mitochondria.

Which of the following processes can be carried out by a mature red blood cell?

A Glycolysis

B Cell division

C Protein synthesis

D Electron transport chain

6. During respiration most ATP is produced when

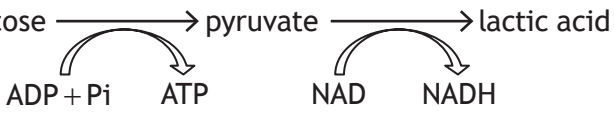
A electrons are passed through the membrane protein ATP synthase

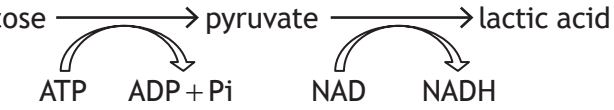
B hydrogen ions are passed through the membrane protein ATP synthase

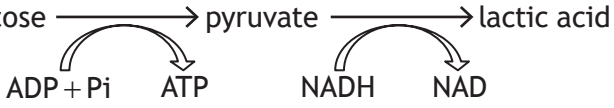
C electrons are pumped through the outer membrane of the mitochondrion

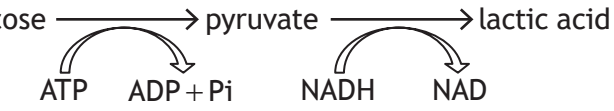
D hydrogen ions are moved along carriers in the inner membrane of the mitochondrion.

7. Which of the following equations summarises the conversion of glucose to lactic acid?

A 

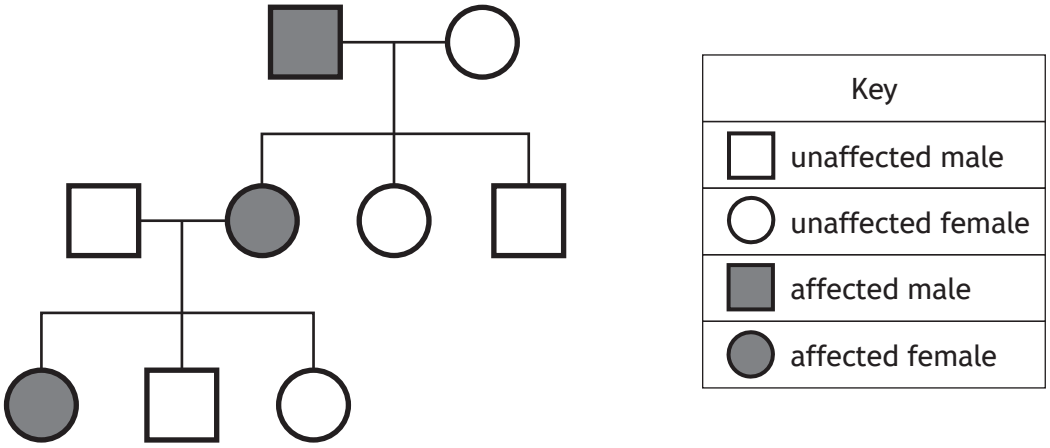
B 

C 

D 

8. The diagram shows the inheritance of familial hypercholesterolaemia (FH) in three generations of a family.

FH is caused by an autosomal dominant allele.



How many individuals in this family are homozygous dominant for this condition?

A 0

B 1

C 2

D 3

9. Red-green colour vision deficiency is a sex-linked recessive condition.

Females heterozygous for the condition are described as being 'carriers'.

A colour vision deficient woman and an unaffected man have children.

Which of the following show the expected phenotypic ratio of the children?

A 1 carrier daughter : 1 colour vision deficient son

B 1 unaffected daughter : 1 colour vision deficient son

C 1 unaffected daughter : 1 colour vision deficient son : 1 unaffected son : 1 carrier daughter

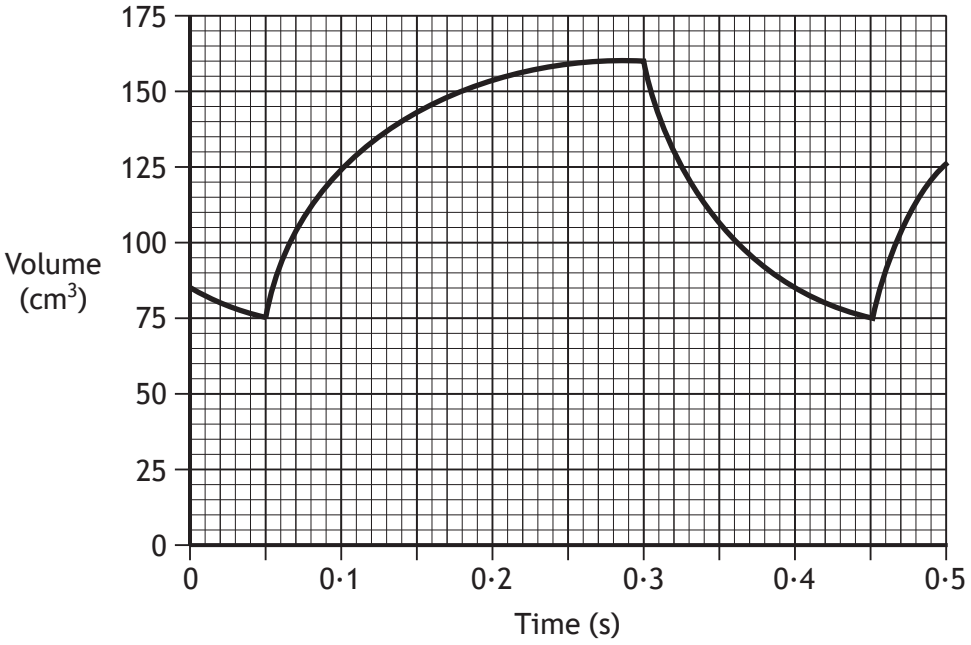
D 1 carrier daughter : 1 colour vision deficient son : 1 unaffected son : 1 colour vision deficient daughter

[Turn over

10. Which row in the table shows how the autonomic nervous system controls an increase in heart rate?

	Branch of autonomic nervous system	*Neurotransmitter*
A	sympathetic	acetylcholine
B	parasympathetic	noradrenaline
C	sympathetic	noradrenaline
D	parasympathetic	acetylcholine

11. The graph shows changes in the volume of blood in the left ventricle of an individual's heart while running.



The cardiac output of this individual is

A 5 100 cm^3/min

B 10 200 cm^3/min

C 12 750 cm^3/min

D 24 000 cm^3/min.

12. During clot formation, thrombin

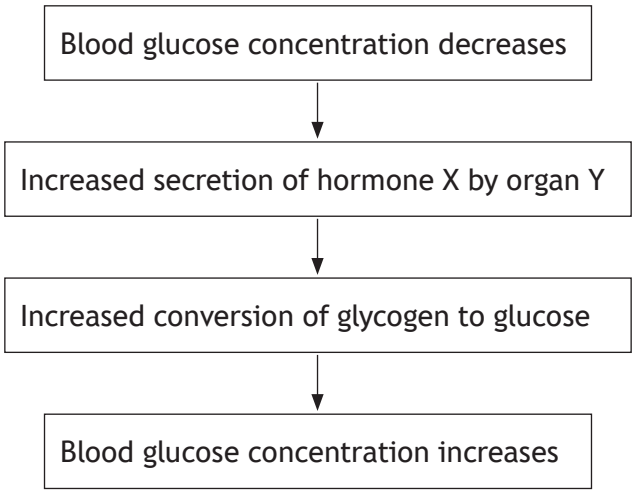
A forms prothrombin

B causes formation of fibrin threads

C causes the release of clotting factors

D forms a meshwork that clots the blood.

13. The flow diagram shows how the concentration of glucose in the blood is controlled during exercise.



Which row in the table identifies hormone X and organ Y?

	Hormone X	*Organ Y*
A	insulin	liver
B	glucagon	liver
C	insulin	pancreas
D	glucagon	pancreas

14. A person is 170 cm tall and weighs 70 kg.

They have a body mass index (BMI) of

A 2·4

B 24·2

C 28·8

D 41·2.

[Turn over

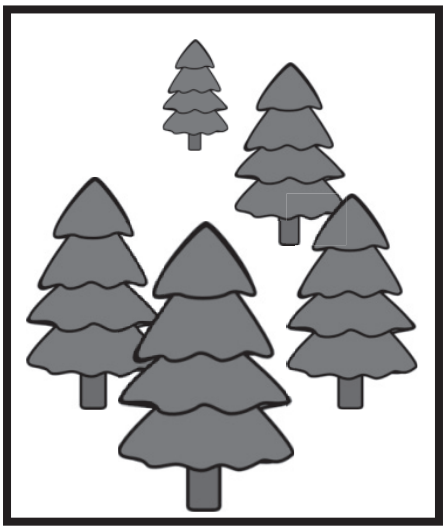
15. The following list shows three areas of the brain.

1 Cortex
2 Limbic system
3 Corpus callosum

Which of these areas are involved in storing spatial memories?

A 1 only
B 2 only
C 1 and 2 only
D 1, 2 and 3

16. The picture shows a scene with trees.



What visual cues are used in the perception of depth in this picture?

1 Relative size
2 Relative height
3 Superimposition
4 Perceptual constancy

A 1 only
B 1 and 2 only
C 1, 2 and 3 only
D 1, 2, 3 and 4

17. Some individuals who suffer head injuries forget the events that happened a few seconds before the injury occurred.

This memory loss is most likely to be due to the injury affecting

A retrieval

B displacement

C long-term memory

D short-term memory.

18. Three groups of students were asked to make paper aeroplanes.

Each student had to make five aeroplanes.

The table shows the conditions under which each group worked.

Group	*Written set of instructions supplied*	*Demonstration given on how to fold the paper*	*Prize awarded to the first student finished*
1	yes	no	no
2	no	yes	no
3	no	no	yes

What behavioural term is used to describe the method of learning used by group 2 only?

A Shaping

B Imitation

C Trial and error

D Reinforcement

[Turn over for next question

19. The table shows the numbers of different types of white blood cells found in blood samples taken from a healthy person and from three different patients.

	White blood cells found in blood sample (cells/mm^3)			
Type of white blood cell	*Healthy person*	*Patient X*	*Patient Y*	*Patient Z*
Phagocyte	7000	7000	8000	7000
Lymphocyte	3000	2000	3000	3500
Mast cell	1000	1000	1000	1500

Use the information above to match each condition to the correct patient.

	Condition		
	Allergic response	*HIV*	*Infected wound*
A	Y	X	Z
B	X	Z	Y
C	Z	Y	X
D	Z	X	Y

20. Which term would be used to describe a global outbreak of an infectious disease?

A Endemic

B Sporadic

C Epidemic

D Pandemic

[END OF SECTION 1. NOW ATTEMPT THE QUESTIONS IN SECTION 2 OF YOUR QUESTION AND ANSWER BOOKLET.]

FOR OFFICIAL USE

National Qualifications 2017

Mark

X740/76/01

Human Biology
Section 1 — Answer Grid and Section 2

TUESDAY, 23 MAY
1:00 PM – 3:30 PM

X7407601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day	Month	Year

Scottish candidate number

Total marks — 100

SECTION 1 — 20 marks

Attempt ALL questions.

Instructions for the completion of Section 1 are given on *Page 02*.

SECTION 2 — 80 marks

Attempt ALL questions.

Question 12 contains a choice.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. You should score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

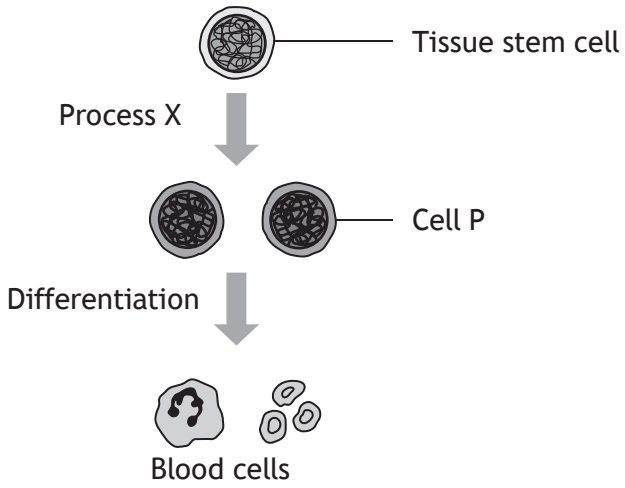
X7407601 01

SECTION 2 — 80 marks

Attempt ALL questions

Question 12 contains a choice

1. The diagram shows some stages in the development of blood cells.



(a) Name process X. **1**

_____

(b) Name the tissue type to which blood cells belong. **1**

_____

(c) Explain why red blood cells contain haemoglobin after differentiation but white blood cells do not. **1**

_____

_____

_____

(d) Describe how a tumour might develop from cell P. **1**

_____

_____

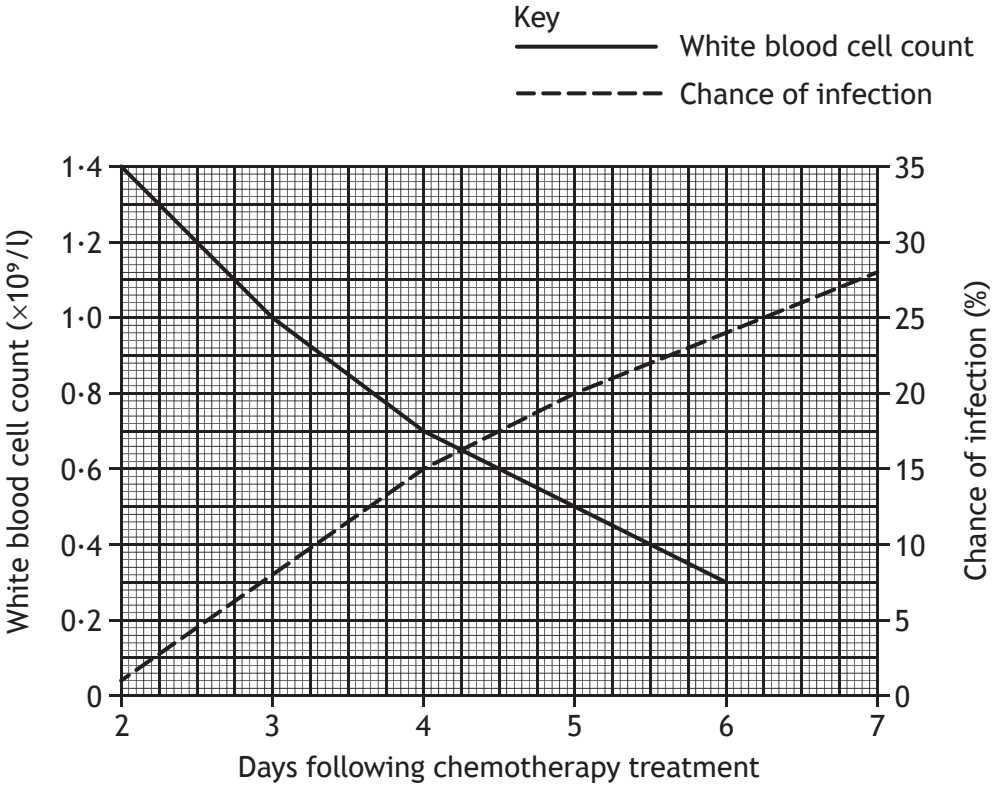
_____

1. (continued)

(e) Cancer patients can be treated using chemotherapy.

This treatment destroys tumour cells but also reduces the number of white blood cells. As a result, patients have a higher chance of infection.

The graph shows the white blood cell count of a cancer patient and their chance of infection in the days following chemotherapy treatment.



(i) State the chance of infection after treatment when the white blood cell count was $0{\cdot}7 \times 10^9$/l. **1**

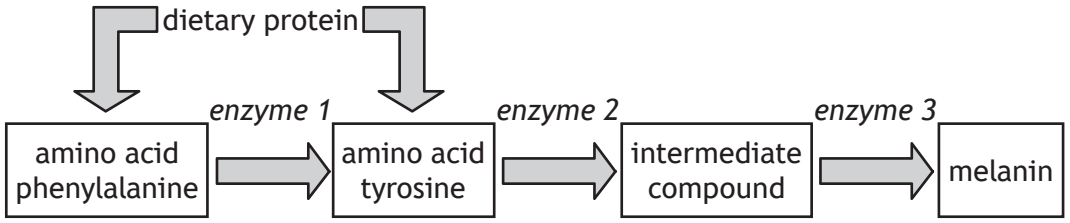
_____ %

(ii) Predict the white blood cell count seven days following chemotherapy treatment. **1**

_____ $\times 10^9$/l

[Turn over

2. Phenylketonuria (PKU) is an example of a genetic disorder which affects the following metabolic pathway.



(a) In PKU enzyme 1 is faulty.

(i) Describe how a substitution mutation would alter the gene coding for enzyme 1. **1**

(ii) Explain how a substitution mutation may cause the synthesis of a faulty enzyme. **1**

(b) Use the metabolic pathway above to suggest

(i) why PKU results in a build-up of phenylalanine; **1**

(ii) why individuals with PKU can still produce melanin. **1**

2. (continued)

(c) Babies born with PKU can develop brain damage from the build-up of phenylalanine and its harmful metabolites.

(i) All babies are tested for PKU immediately after birth.

State the term used to describe this type of diagnostic testing. **1**

_____

(ii) Describe how brain damage can be prevented in babies diagnosed with PKU. **1**

_____

_____

(d) PKU is caused by an autosomal recessive allele.

A couple, who are both unaffected, have a child who has PKU.

Calculate the percentage chance of their next child having this disorder. **1**

Space for calculation

_____%

[Turn over

3. An investigation was carried out into the effect of a competitive inhibitor on the activity of phosphatase at different substrate concentrations.

Phosphatase is an enzyme which catalyses the reaction shown.

$$\underset{\textit{Substrate}}{\text{phenolphthalein phosphate}} \xrightarrow{\text{phosphatase}} \underset{\textit{Products}}{\text{phenolphthalein} + \text{phosphate}}$$

Six test tubes each containing a different concentration of substrate were set up. The inhibitor and then the enzyme were added to each tube.

Figure 1 shows the contents of each tube.

After 30 minutes, 1 cm^3 of alkali was added to each tube.

Phenolphthalein turns pink in the presence of alkali. The more phenolphthalein produced, the more intense the pink colour and the higher the absorbance reading measured by a colorimeter.

Table 1 shows the results of the investigation.

Figure 1

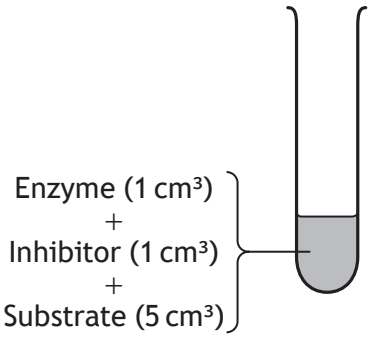


Table 1

Concentration of substrate (M)	*Absorbance* (units)
0·05	0·20
0·10	0·30
0·20	0·48
0·40	0·64
0·60	0·78
0·80	0·90

(a) Suggest why alkali was **not** added to each tube at the start of the investigation. **1**

_____

_____

(b) State **two** variables, other than those shown above, which should be kept constant to make this investigation valid. **2**

1 _____

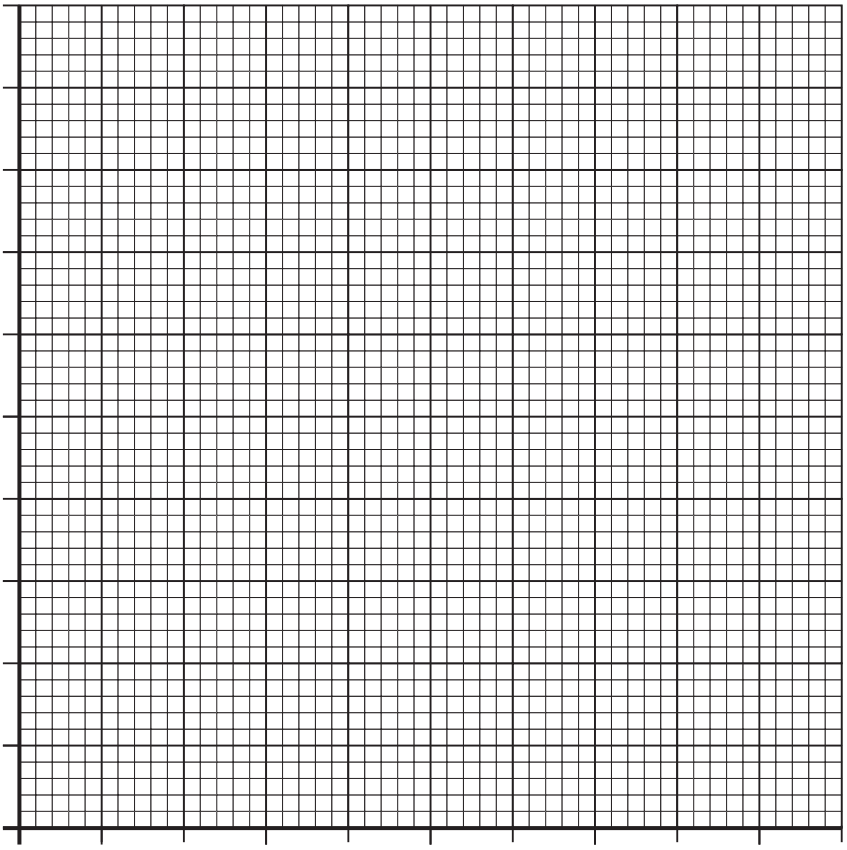
2 _____

MARKS

DO NOT WRITE IN THIS MARGIN

(c) Construct a line graph to show the data in **Table 1**. **2**

(Additional graph paper, if required, can be found on *Page 28*)

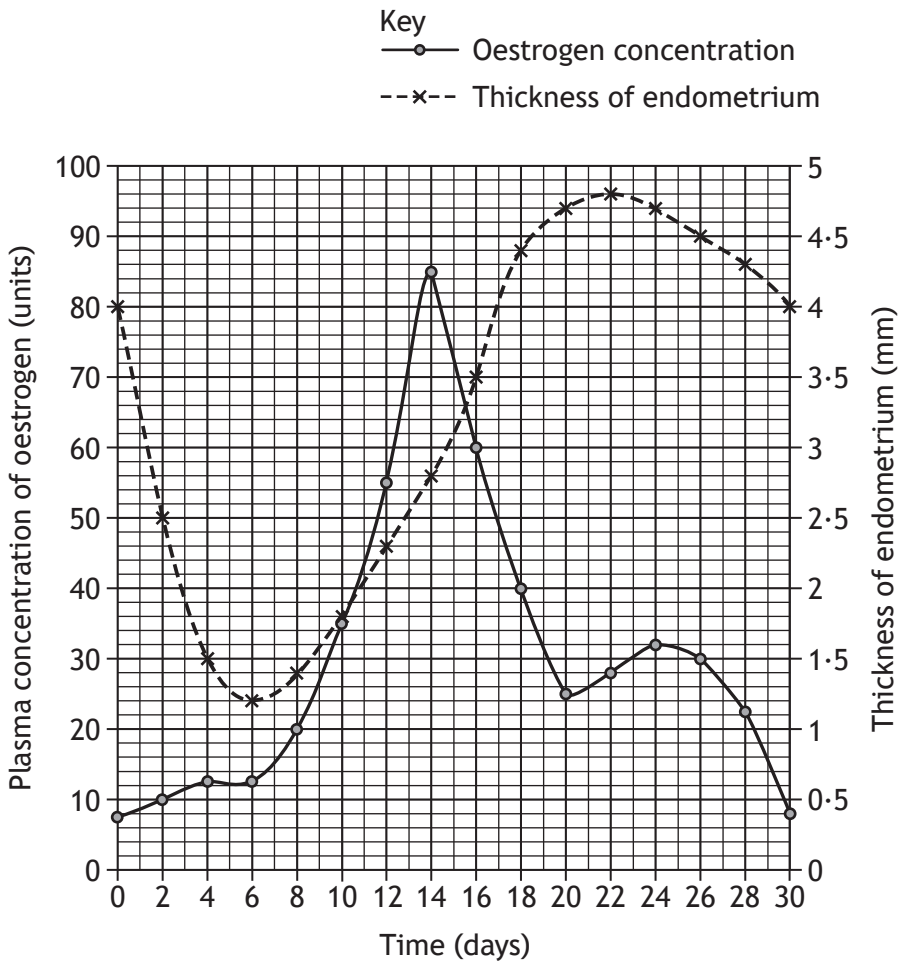


(d) It was concluded that increasing substrate concentration reduces the effect of the competitive inhibitor.

Explain how the results of this investigation support this conclusion. **2**

(e) Suggest how the results of this investigation would be different if a non-competitive inhibitor had been used. **1**

4. The graph shows how the plasma concentration of oestrogen and the thickness of the endometrium vary during a woman's menstrual cycle.



(a) Ovulation occurs on day 15 of this cycle.

(i) Describe the role of oestrogen in triggering this event. **1**

_____

_____

_____

(ii) State the thickness of the endometrium on day 15. **1**

_____ mm

4. (continued)

(b) (i) Express, as a simple whole number ratio, the thickness of the endometrium on day 6 compared to day 22. **1**

Space for calculation

_____ : _____
Day 6 Day 22

(ii) Oestrogen stimulates thickening of the endometrium.

Describe evidence from the graph which indicates that another factor also stimulates thickening of the endometrium. **1**

_____

_____

_____

(c) Suggest **one** way in which the graph for the next menstrual cycle would differ from this one if the woman became pregnant during that cycle. **1**

_____

_____

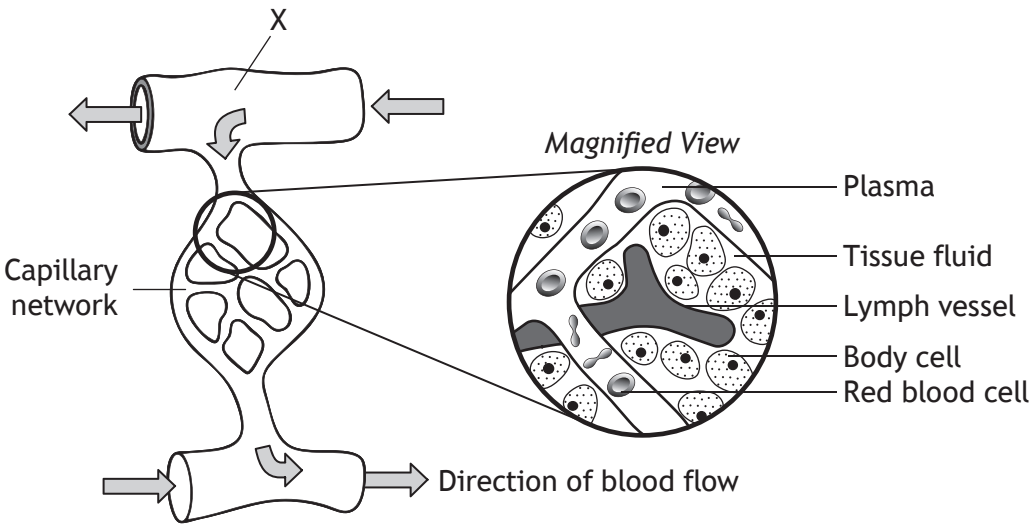
(d) State how fertility drugs stimulate ovulation. **1**

_____

_____

[Turn over

5. The diagram represents a capillary network, associated vessels and cells.



(a) (i) Name the type of blood vessel labelled X. **1**

_____

(ii) State how blood vessel X can reduce blood flow to the capillary network. **1**

_____

_____

(b) (i) Name the layer of cells which forms the wall of a capillary. **1**

_____

(ii) Describe how substances pass from plasma to tissue fluid. **1**

_____

_____

(iii) Name a type of molecule which is present in plasma but absent in tissue fluid. **1**

_____

(c) Describe **one** role of the lymph vessel in the diagram. **1**

_____

_____

6. The picture shows a man having his blood pressure measured.



(a) A blood pressure reading consists of a high systolic value and a lower diastolic value.

Explain the difference between these two values. **1**

(b) Suggest a reason why the pulse in the man's left wrist stops when the cuff is inflated. **1**

(c) The man's blood pressure was measured as 160/100.

(i) Explain how atherosclerosis could have caused this high blood pressure. **2**

(ii) The man's blood HDL to LDL ratio was lower than normal.

Describe how this may have contributed to atherosclerosis. **1**

7. Hormone replacement therapy (HRT) is used by women to relieve symptoms of the menopause, which usually occurs from around the age of 50.

A study was carried out into the effects of HRT on the health of 12 000 women. The women were separated into three equal groups according to their age. Half of each group took a daily HRT tablet while the other half received a placebo. Over the next five years the number of women in each group who developed either a pulmonary embolism or coronary heart disease was recorded.

The results are shown in the table.

	Conditions developed by women in the study			
	Pulmonary Embolism		*Coronary Heart Disease*	
Age group (years)	*Number of cases in women given HRT*	*Number of cases in women given a placebo*	*Number of cases in women given HRT*	*Number of cases in women given a placebo*
50–59	10	5	20	18
60–69	20	12	35	33
70–79	30	15	80	76

(a) Describe **two** trends shown by the results for pulmonary embolism. **2**

1 _____

_____

2 _____

_____

(b) It was concluded that taking HRT has no effect on the risk of coronary heart disease.

Explain why the results of the study support this conclusion. **1**

_____

_____

(c) Suggest a factor, other than HRT, which could have influenced the results of this study. **1**

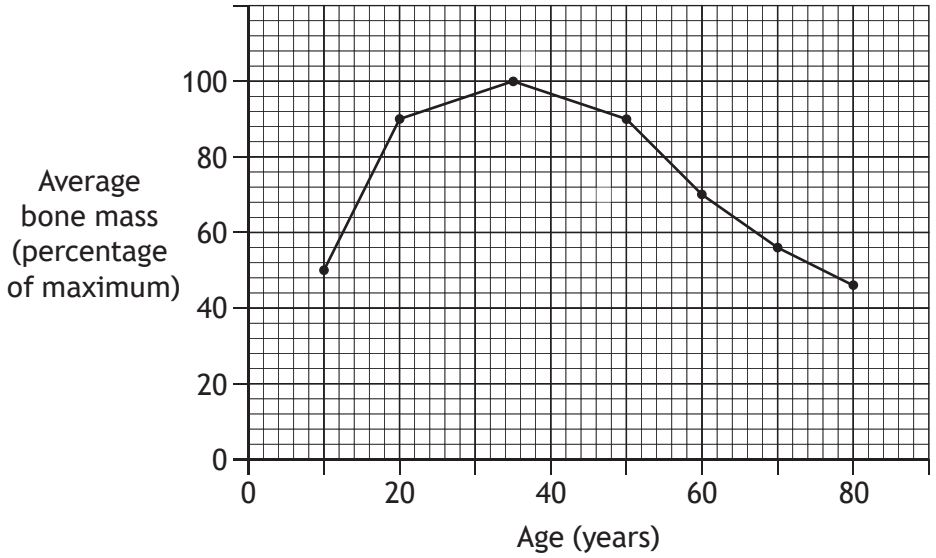
_____

7. (continued)

(d) Describe how the researchers attempted to make sure that the results of this study were reliable. **1**

(e) HRT can be prescribed to treat osteoporosis, a condition which increases the risk of bones thinning and breaking.

The graph shows how the average bone mass of women changes with age.



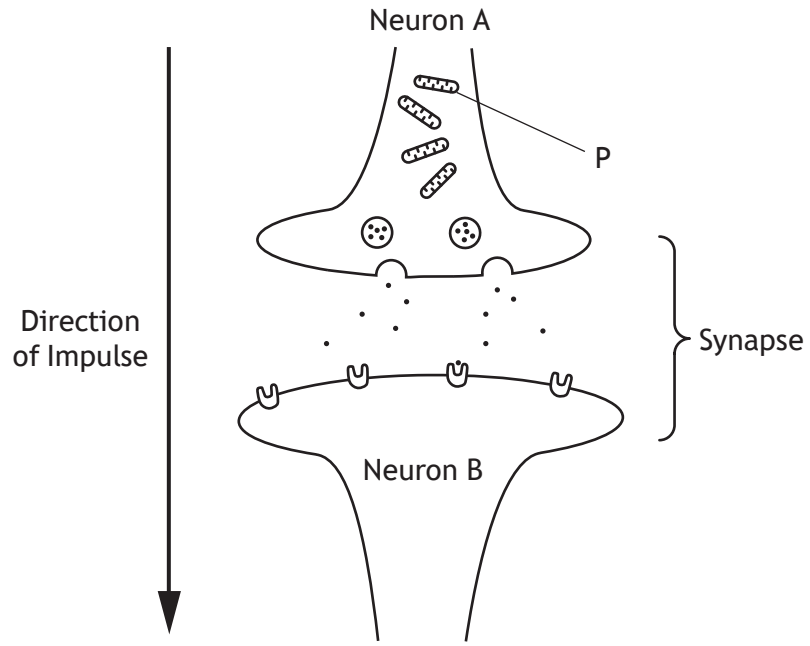
(i) **Use data from the graph** to describe the changes that occur in the average bone mass between the ages of 10 and 80. **2**

(ii) State the number of years the average bone mass of women is at least 80% of the maximum. **1**

_____ years

[Turn over

8. The diagram represents two neurons and the synapse between them.



(a) Name the structure labelled P which generates ATP. **1**

_____

(b) Describe how an impulse is transmitted from Neuron A to Neuron B. **3**

_____

_____

_____

_____

_____

_____

_____

8. (continued)

(c) Many drugs which affect synapses may cause sensitisation over a period of time.

Describe the effect that sensitisation has on the synapse and the consequences for the individual. **2**

Effect _____

_____

Consequences _____

_____

[Turn over

9. Sympathetic and parasympathetic nerves regulate heart rate.

(a) Name the part of the brain that regulates the heart rate. **1**

_____

(b) The sympathetic and parasympathetic nerves work antagonistically.

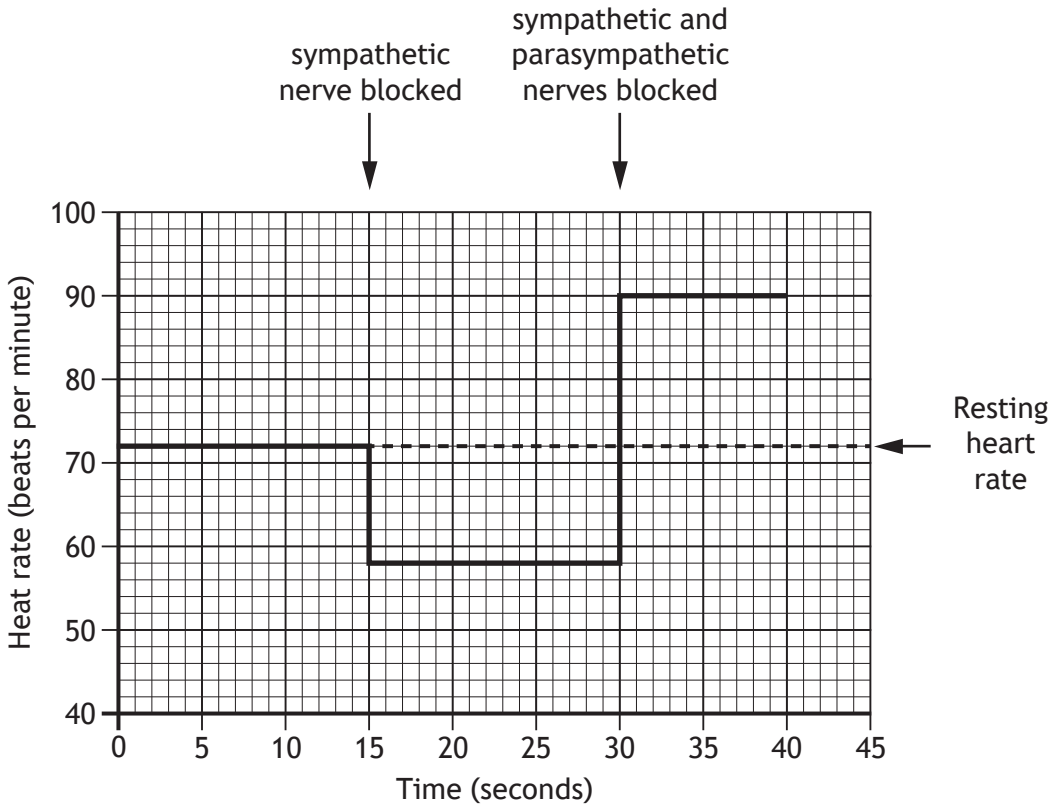
Explain what this statement means. **1**

_____

_____

(c) An investigation was carried out to determine the effects these nerves have on heart rate by firstly blocking the sympathetic nerve and then blocking both nerves.

The graph shows the results of the investigation.



(i) State the heart rate when only the sympathetic nerve is blocked. **1**

_____ beats per minute

9. (c) (continued)

(ii) Calculate the increase in the heart rate which then occurs when the parasympathetic nerve is also blocked. **1**

_____ beats per minute

(d) Explain why the heart continues to contract when both nerves are blocked. **1**

_____

_____

(e) The parasympathetic nerve has a greater effect on the resting heart rate than the sympathetic nerve.

Use information from the graph to justify this statement. **1**

_____

_____

_____

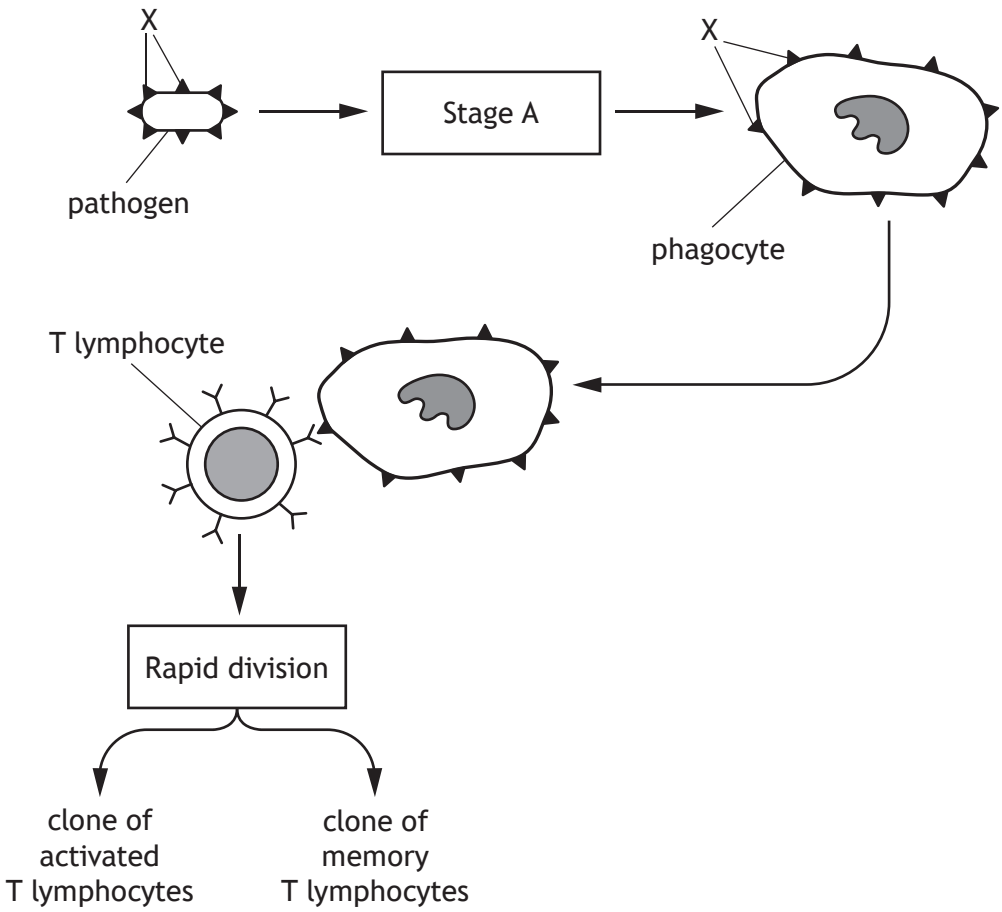
(f) State **one** other effect the sympathetic nervous system has on the body. **1**

_____

_____

[Turn over

10. The diagram shows some of the stages leading to the production of a clone of T lymphocytes by the immune system in response to infection by a pathogen.



(a) Name the structures labelled X. **1**

_____

(b) Describe what happens during Stage A. **2**

_____

_____

_____

_____

(c) Name the chemicals which aid the movement of T lymphocytes to the site of infection. **1**

_____

10. (continued)

(d) The diagram shows how a clone of memory T lymphocytes is produced.

Describe an advantage of having memory cells. **1**

_____

_____

(e) State how a tuberculosis (TB) pathogen avoids immune detection. **1**

_____

_____

[Turn over

11. **Table 1** contains information about the life expectancy of Scottish children between 1861 and 2011.

Life expectancy is the additional number of years a person is expected to survive from a given age.

Table 1

	Average life expectancy of child (years)		
Year	*from birth*	*from age 1*	*from age 15*
1861	42	47	43
1891	46	52	46
1921	55	59	50
1951	66	68	55
1981	72	72	59
2011	78	77	63

(a) (i) Calculate the percentage increase in life expectancy for children born in 2011 compared to children born in 1861. **1**

Space for calculation

_____%

(ii) Give the 30 year period during which the greatest increase in life expectancy from birth occurred. **1**

From _____ to _____

(iii) Suggest **two** reasons for an increase in life expectancy over the 150 year period. **2**

1 _____

_____

2 _____

_____

11. (continued)

(b) **Table 2** contains information about the life span of Scottish children.

Life span is the number of years a person lives for.

Table 2

	Average Life Span (years)		
Year	*from birth*	*from age 1*	*from age 15*
1861	42	48	58
2011			

(i) **Use the information in Table 1 to complete Table 2** to indicate the expected average life span of children in 2011. **1**

(ii) Suggest why the figures for 1861 increase from 42 to 58 years. **1**

_____

_____

[Turn over

12. Answer **either** A **or** B in the space below.

Labelled diagrams may be used where appropriate.

A Describe the structure of DNA and the process of DNA replication. **9**

OR

B Describe the structure of RNA and the process of transcription. **9**

National
Qualifications
2016

X740/76/02

Human Biology
Section 1 — Questions

MONDAY, 9 MAY
1:00 PM – 3:30 PM

Instructions for the completion of Section 1 are given on *Page 02* of your question and answer booklet X740/76/01.

Record your answers on the answer grid on *Page 03* of your question and answer booklet.

Before leaving the examination room you must give your question and answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

SECTION 1 — 20 marks

Attempt ALL questions

1. In a developing embryo, tissues such as muscle and nerve are produced by

A somatic cells dividing by meiosis

B germline cells dividing by meiosis

C somatic cells dividing by mitosis

D germline cells dividing by mitosis.

2. A genetic disorder of the nervous system results from a mutation in which a nucleotide is inserted into a gene.

Which of the following types of mutation causes this genetic disorder?

A nonsense

B missense

C translocation

D frame-shift

3. The following steps occur during the Polymerase Chain Reaction (PCR).

1. Binding of primer
2. Replication of DNA
3. Heating of sample DNA
4. Separation of DNA strands

In which sequence do these steps occur?

A $1 \rightarrow 2 \rightarrow 4 \rightarrow 3$

B $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

C $3 \rightarrow 4 \rightarrow 1 \rightarrow 2$

D $3 \rightarrow 4 \rightarrow 2 \rightarrow 1$

4. The diagrams below represent the shapes of an enzyme molecule and its substrate.

Enzyme molecule *Substrate molecule*



Which row in the table below shows the possible shapes of two types of molecule that could inhibit the enzyme above?

	Competitive Inhibitor	*Non-competitive Inhibitor*
A		
B		
C		
D		

5. During glycolysis, dehydrogenase enzymes catalyse the

A removal of hydrogen ions from $NADH_2$

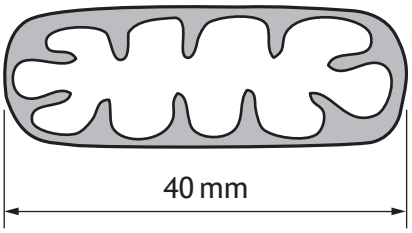
B removal of hydrogen ions from citrate

C transfer of hydrogen ions to glucose

D transfer of hydrogen ions to NAD.

[Turn over

6. The diagram below represents a mitochondrion which has been magnified 10 000 times.



What is the actual length of this mitochondrion?

(1 mm = 1000 micrometres)

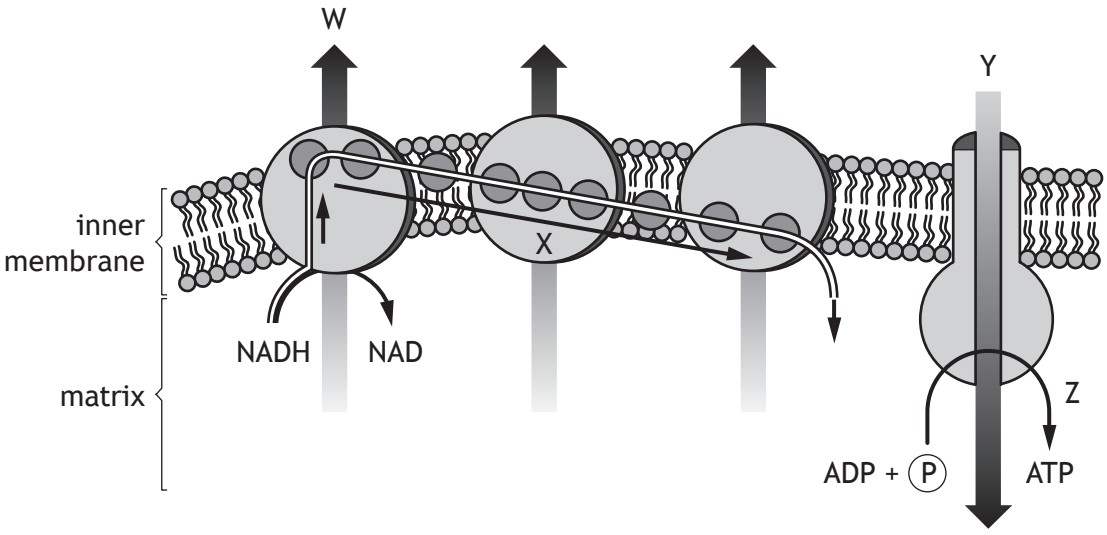
A 0·04 micrometres

B 0·4 micrometres

C 4 micrometres

D 40 micrometres

7. The diagram below represents some of the processes which occur at the inner membrane of a mitochondrion.



Which letter represents the transfer of high energy electrons?

A W

B X

C Y

D Z

8. During cellular respiration, the activity of phosphofructokinase can be inhibited by

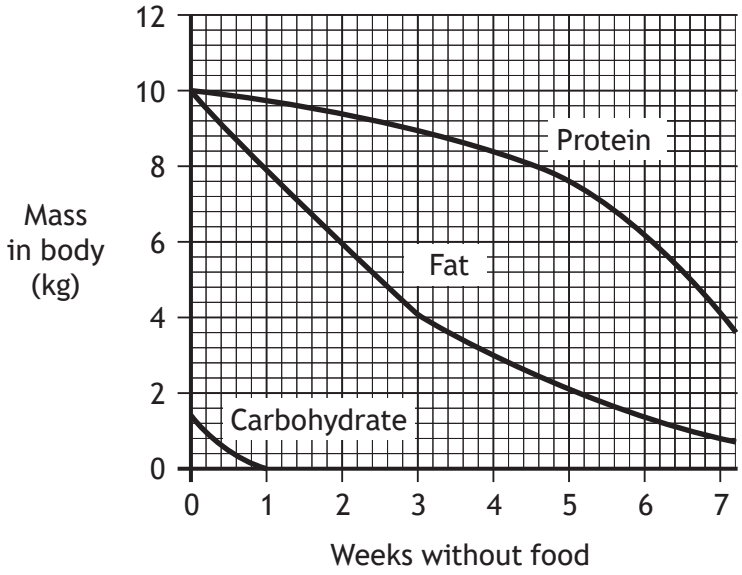
A ATP and citrate

B ADP and citrate

C ATP and lactic acid

D ADP and lactic acid.

9. The graph below shows changes which occur in the masses of protein, fat and carbohydrate in a person's body during seven weeks without food.



The person's starting mass was 60 kg.

Predict their mass after two weeks without food.

A 57 kg

B 54 kg

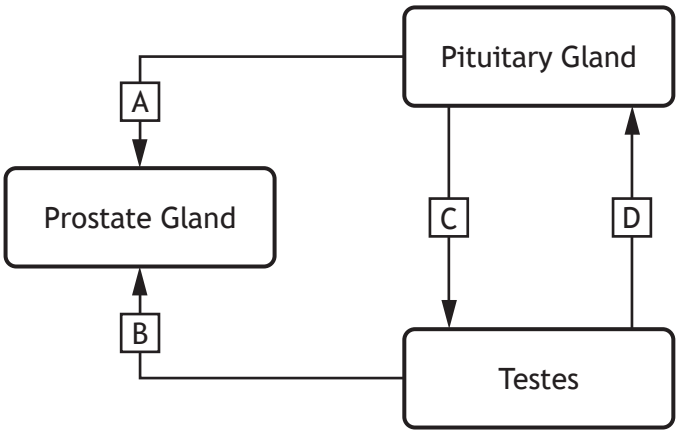
C 50 kg

D 43 kg

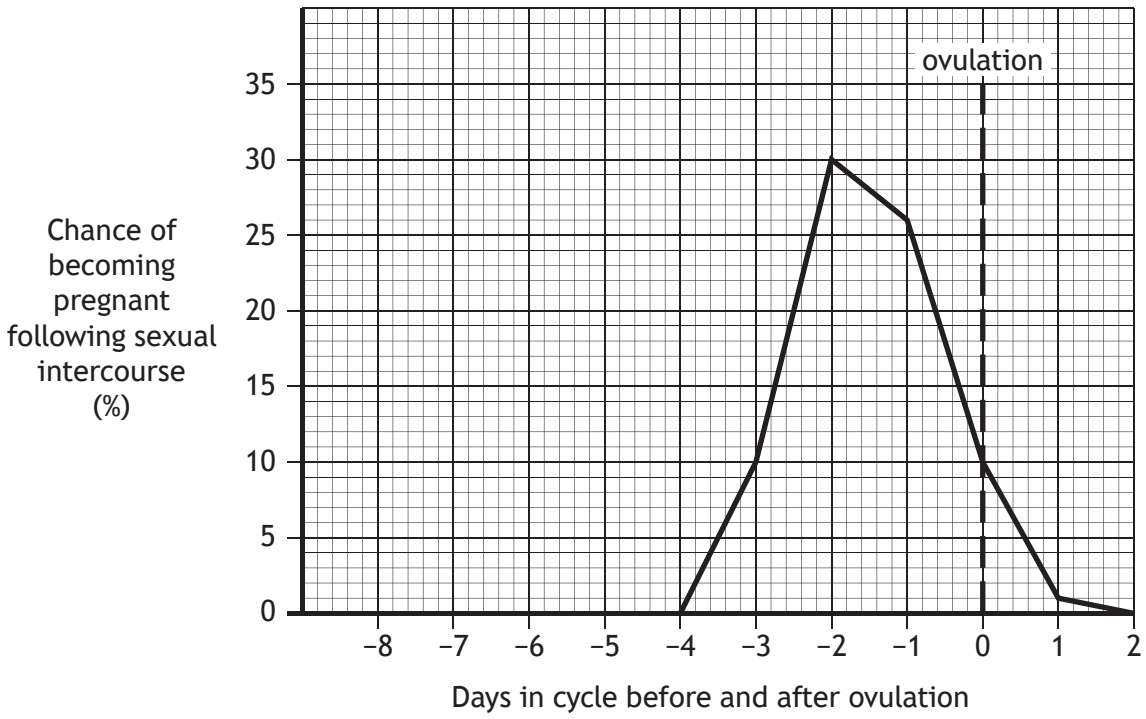
[Turn over

10. The diagram below represents connections between parts of the male reproductive system.

Which arrow in the diagram does **not** represent a male reproductive hormone?



11. The graph below shows the chance of a woman becoming pregnant, following sexual intercourse, on the days before and after ovulation.



This woman has a 28 day menstrual cycle and ovulates on the 3rd of May.

On which day in May would having sexual intercourse give her the best chance of becoming pregnant?

A 3rd May

B 17th May

C 29th May

D 31st May

12. In the treatment of infertility, ovulation can be stimulated by drugs that prevent the negative feedback effect of

A oestrogen on LH secretion

B oestrogen on FSH secretion

C progesterone on LH secretion

D progesterone on FSH secretion.

13. During antenatal care, which **two** techniques can be used to obtain cells for production of a karyotype?

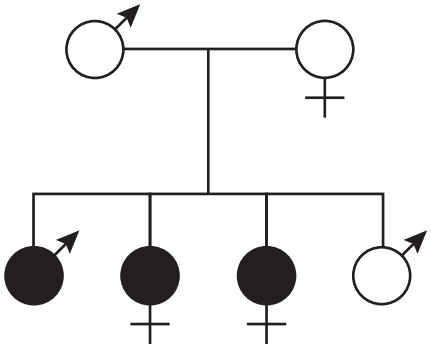
A Chorionic villus sampling (CVS) and amniocentesis

B Ultrasound imaging and chorionic villus sampling (CVS)

C Amniocentesis and pre-implantation genetic diagnosis (PGD)

D Pre-implantation genetic diagnosis (PGD) and ultrasound imaging

14. The inheritance of an allele for deafness is shown in the family tree below.



Key	
♀	Unaffected female
♂	Unaffected male
● ♀	Affected female
● ♂	Affected male

This condition is controlled by an allele which is

A dominant and sex-linked

B recessive and sex-linked

C dominant and not sex-linked

D recessive and not sex-linked.

[Turn over

15. Which of the following memories would be stored in the limbic system only?

A The tune to your favourite song.

B How to keep three balls in the air when juggling.

C The route to your bed across your bedroom in the dark.

D The taste of your favourite food.

16. Playing cards normally have hearts and diamonds in red, and spades and clubs in black.

An investigation showed that the speed and accuracy in recognising the cards decreased when the colours were reversed, for example when hearts appeared black.

This result was most likely to have been caused by the effect of

A a perceptual set

B a binocular disparity

C a segregation into figure and ground

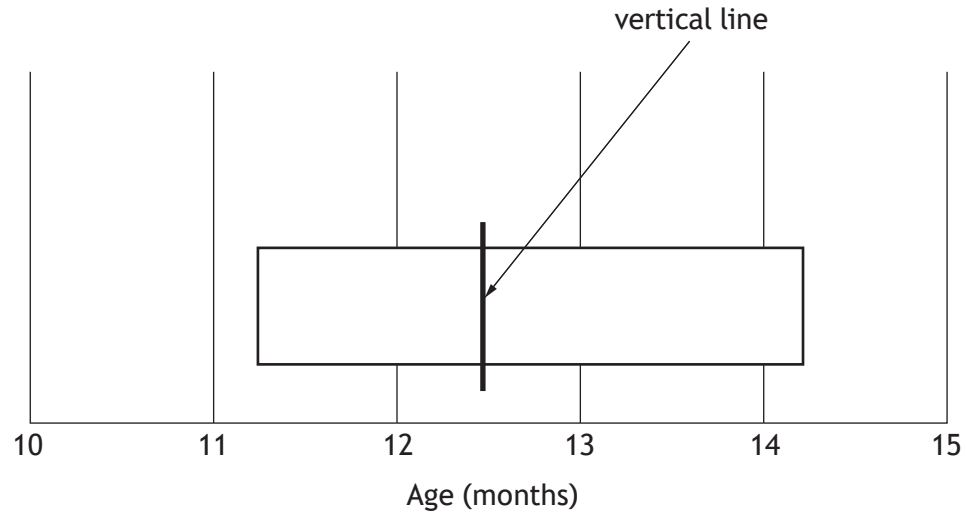
D an organisation into coherent patterns.

17. The diagram below shows the ages at which infants are able to walk unaided.

The left end of the bar shows the age at which 25% of infants can walk unaided.

The right end of the bar shows the age at which 90% of infants can walk unaided.

The vertical line on the bar shows the age at which 50% of infants can walk unaided.



If 24 infants, aged 12 months, were tested, how many would be expected to walk unaided?

A 6

B 10

C 14

D 18

18. The table below contains information about two groups of students who were asked to construct a paper model from a set of instructions.

Group	*Arrangement of students*	*Average time to complete model (s)*
1	all students together in one room	105
2	each student in a separate room	140

The improved performance of the students in group 1 is likely to be due to

A shaping

B discrimination

C deindividuation

D social facilitation.

19. When tissue is damaged, mast cells release histamine which **immediately** results in

A an accumulation of phagocytes

B increased delivery of antimicrobial proteins and clotting elements

C increased localised blood vessel dilation and capillary permeability

D stimulation of a specific immune response by activating lymphocytes.

20. In Scotland cases of influenza are always present but occasionally they rise to unusually high levels. In this case, the disease is said to have changed from being

A epidemic to endemic

B endemic to epidemic

C sporadic to epidemic

D endemic to sporadic.

[END OF SECTION 1. NOW ATTEMPT THE QUESTIONS IN SECTION 2 OF YOUR QUESTION AND ANSWER BOOKLET.]

FOR OFFICIAL USE

National Qualifications 2016

Mark

X740/76/01

Human Biology
Section 1 — Answer Grid and Section 2

MONDAY, 9 MAY
1:00 PM – 3:30 PM

X7407601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day	Month	Year

Scottish candidate number

Total marks — 100

SECTION 1 — 20 marks

Attempt ALL questions.

Instructions for the completion of Section 1 are given on *Page 02*.

SECTION 2 — 80 marks

Attempt ALL questions.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. You should score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

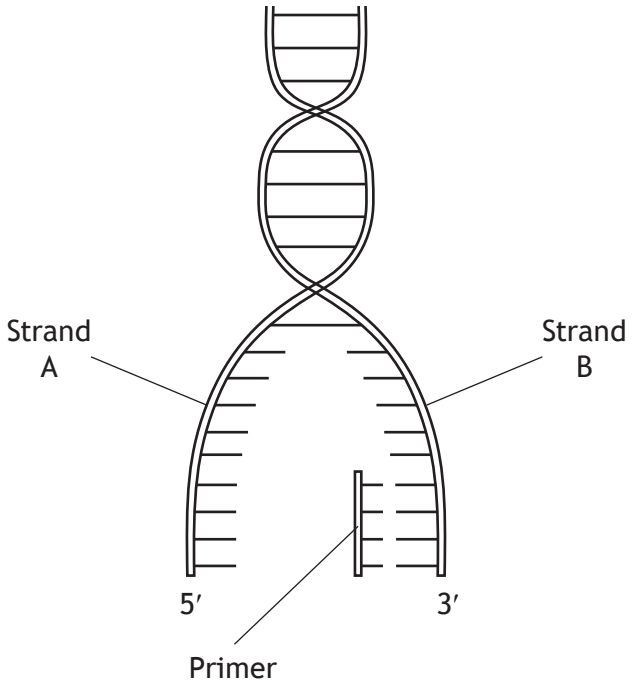
X7407601 01

SECTION 2 — 80 marks

Attempt ALL questions

Note that Question 13 contains a choice.

1. The diagram below represents a stage in the process of DNA replication.



(a) (i) Name the type of bond which links the primer to strand B. **1**

_____

(ii) Name the chemical group found at the 5′ end of a DNA strand. **1**

_____

(b) Strand B is replicated continuously while strand A can only be replicated in fragments.

Explain why the strands are replicated in different ways. **1**

_____

_____

_____

1. (continued)

(c) Describe the role of the following enzymes in DNA replication. **2**

DNA polymerase _____

_____

Ligase _____

_____

2. At the start of polypeptide synthesis in a cell, DNA transcription occurs in the nucleus to form mRNA.

The sequence of bases from a section of a DNA strand is shown below.

..... C A C G A T C G A T A G G A T .....

(a) (i) State the sequence of bases in the primary mRNA transcript formed from this strand of DNA. **1**

(ii) State the term used to describe the coding regions of a primary mRNA transcript. **1**

(iii) Name the process by which the coding regions of a primary mRNA transcript are joined together to produce a mature mRNA transcript. **1**

(iv) The sequence of bases in the mature mRNA transcript, formed from the section of the DNA strand, is shown below.

..... G U G C U A U C C U A .....

Using this mature mRNA transcript, state the order of bases in the intron present in the primary mRNA transcript. **1**

(b) State the location for the translation of a mature mRNA transcript into a polypeptide. **1**

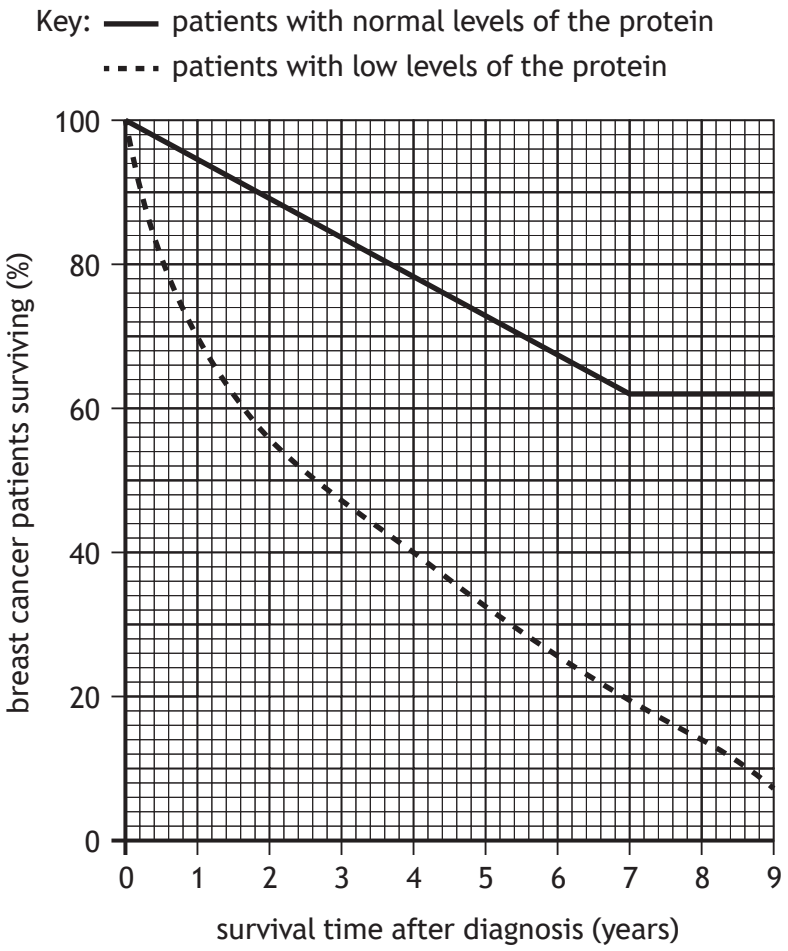
(c) Describe **one** form of post-translational modification of a polypeptide. **1**

[Turn over

3. A naturally occurring cell protein (nm23) has been shown to inhibit the activity of cancer cells.

Individuals produce varying levels of this protein depending on their genetic make-up.

The graph below shows the results of a 9 year study of women diagnosed with breast cancer. The women were divided into two groups according to their production of the protein.



(a) (i) In a city, 1000 women were diagnosed with breast cancer.

Of these women, 900 had normal levels of the protein while 100 had low levels.

Using the results from the study, calculate how many of the 1000 women would be expected to survive for 4 years after diagnosis. **1**

Space for calculation

3. (a) (continued)

(ii) **Use data from the graph** to describe the changes in the percentage of surviving breast cancer patients with normal levels of the protein during the study. **2**

(b) Describe how cancer can develop and spread through the body. **3**

[Turn over

4. A student carried out an investigation into the effect of physical activity on respiration rate.

The rate of respiration of six individuals was measured after carrying out three different activities for five minutes.

Immediately after completing the activity, each individual breathed into a bottle containing a pH indicator solution. This indicator changes colour from blue to yellow in the presence of a high concentration of carbon dioxide.

Figure 1 – Apparatus used

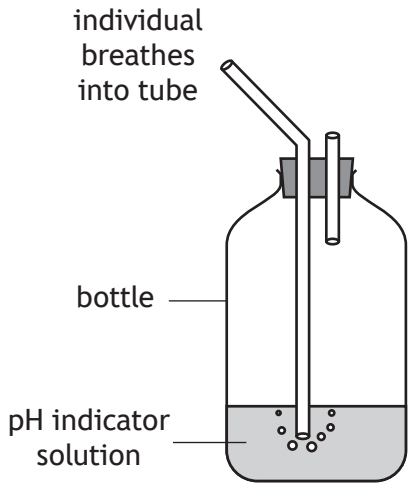


Table 1 – Results of Investigation

Individual	*Time taken for pH indicator to turn yellow* (s)		
	Activity 1 resting	Activity 2 walking	Activity 3 running
1	33	28	22
2	29	14	11
3	22	16	12
4	30	26	20
5	44	35	22
6		31	21
Average time taken (s)	33	25	18

(a) State **two** variables which would have to be kept constant when setting up the apparatus shown in **Figure 1**. **2**

1 _____

2 _____

(b) Calculate the time taken for the indicator to turn yellow after individual 6 had completed Activity 1. **1**

Space for calculation

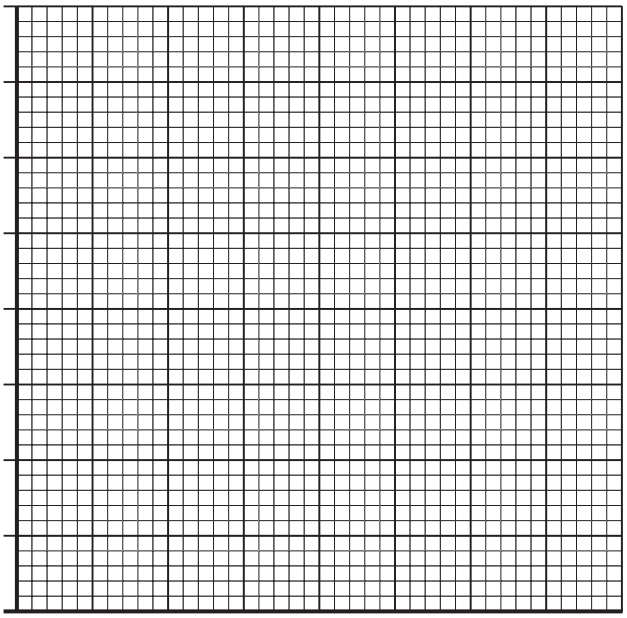
_____ s

4. (continued)

(c) Describe how the student increased the reliability of the results. **1**

(d) Construct a bar graph to show the average results obtained in this investigation. **2**

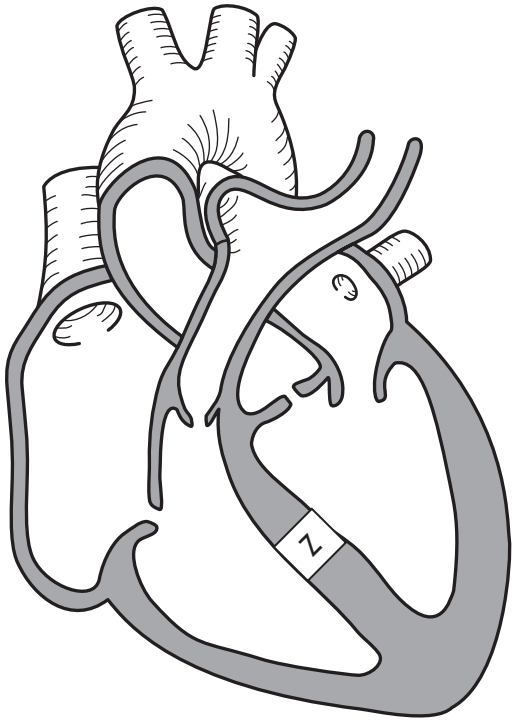
(Additional graph paper, if required, can be found on *Page 31*)



(e) State a conclusion that can be drawn from the results of this investigation. **1**

(f) Suggest an explanation for the results obtained in this investigation. **1**

5. The diagram below represents the structure of the heart and its associated blood vessels.



(a) On the diagram, **label** the pulmonary artery with the letter P. **1**

(b) Sometimes babies can be born with a ventricular septal defect (VSD) in which a "hole" occurs at point Z in the heart.

Explain how the presence of this hole would affect the oxygen concentration of the blood leaving the heart through the aorta. **2**

5. (continued)

(c) Babies with a VSD sometimes have irregular heart rhythms. This can be detected by recording the electrical activity from the heart.

(i) Name the chamber of the heart in which this electrical activity originates. **1**

_____

(ii) Name the type of graph that displays such patterns of electrical activity. **1**

_____

(d) Babies with a VSD often have a lower stroke volume than babies who have a normal heart structure.

Despite this, both groups of babies often have similar cardiac outputs.

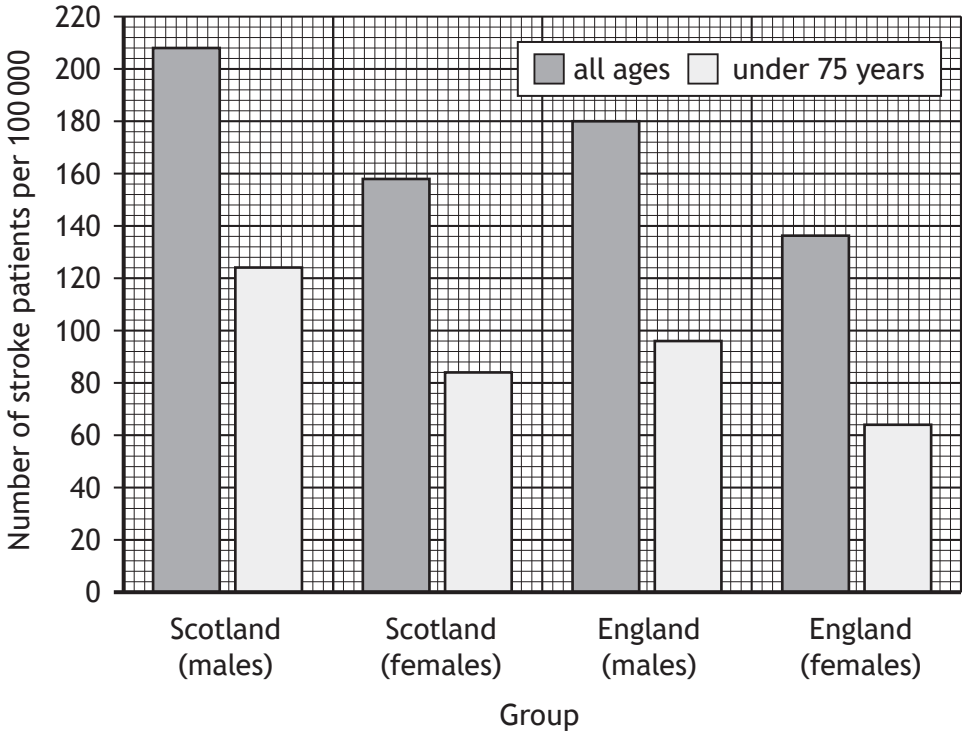
Suggest how babies with a VSD are able to achieve a similar cardiac output to babies with a normal heart structure. **1**

_____

_____

[Turn over

6. The graph below shows the number of stroke patients in different groups from Scotland and England in 2007.



(a) (i) Calculate the difference in the number of male stroke patients of all ages in Scotland and England in 2007. **1**

Space for calculation

_____ per 100 000

(ii) Explain the importance of presenting the data as the number of stroke patients **per 100 000**. **1**

_____

_____

6. (a) (continued)

(iii) Scotland's population was 5·1 million in 2007.

Calculate the number of female stroke patients in Scotland under 75 years of age in this year. **1**

Space for calculation

_____

(iv) Express, as a simple whole number ratio, the number of male stroke patients under 75 years of age compared to female stroke patients under 75 years of age in England in 2007. **1**

Space for calculation

_____ : _____
male patients female patients

(b) Describe what causes a stroke. **1**

_____

_____

(c) Paralysis occurs when voluntary muscle is unable to contract.

Explain how a stroke could lead to muscle paralysis on the left side of the body. **2**

_____

_____

_____

_____

_____

7. The table below contains information about five obese patients who attended a weight loss clinic for 12 weeks.

Patient	*Height* (m)	*Starting weight* (kg)	*Starting BMI*	*Final Weight* (kg)	*Final BMI*
P	1·74	92	30·5	82	27·2
Q	1·68	98	34·8	90	32·1
R	1·81	104	31·8	97	29·7
S	1·89	121	33·9	113	31·4
T	1·90	100	32·3	94	

(a) (i) Calculate the final BMI of patient T. **1**

Space for calculation

Final BMI = _____

(ii) State why patient Q was still classed as obese after 12 weeks. **1**

_____

_____

(b) Explain why all the patients were advised to exercise regularly to increase their weight loss. **1**

_____

_____

_____

7. (continued)

(c) Rugby players may have a BMI which indicates that they are obese.

Suggest why a BMI reading may not be a reliable indicator of obesity in rugby players. **1**

_____

_____

_____

[Turn over

8. Statins are drugs which reduce the production of cholesterol in the liver. A year-long trial was carried out to investigate the effects of taking a newly-developed statin on blood cholesterol levels.

Sixty individuals with raised blood cholesterol levels were selected and divided into two groups of thirty.

Individuals in Group 1 were prescribed a capsule, containing 20 mg of the statin, to take each day. Individuals in Group 2 were the control group. At two-monthly intervals, blood samples were taken from all individuals and their blood cholesterol levels measured.

The results are shown in the table below.

	Average blood cholesterol level (mmol/l)	
Month of trial	Group 1	Group 2
0	6·3	6·3
2	6·3	6·3
4	6·3	6·1
6	6·3	6·3
8	5·6	6·1
10	5·3	6·2
12	5·1	6·1

(a) **Using the results in the table**, give **one** reason why this drug might be **2**

recommended _____

_____

not recommended _____

_____

(b) Suggest what was prescribed to the individuals in Group 2 during this trial. **1**

_____

_____

8. (continued)

(c) Describe the design features which would have been used to ensure that this was both a randomised and a double-blind trial. **2**

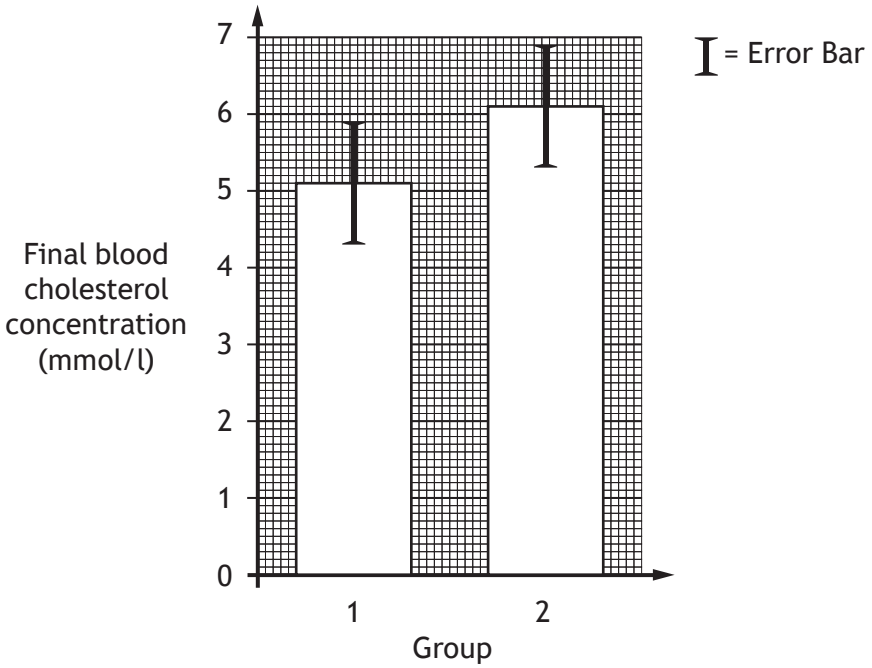
randomised _____

_____

double-blind _____

_____

(d) The bar graph below summarises the data collected in the final month of this trial.



Using evidence from the bar graph, suggest why it was decided that this statin was not worth further development. **1**

_____

_____

(e) Describe **one** function of cholesterol in the human body. **1**

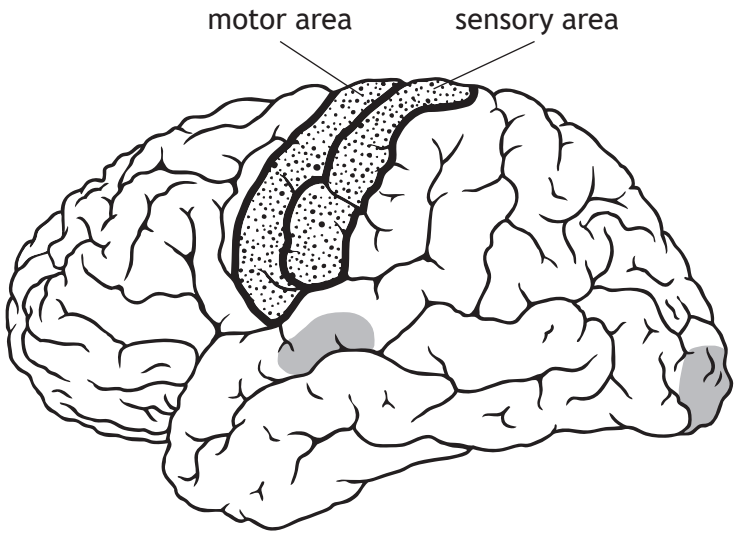
_____

_____

9. The diagram below represents areas of high activity in a part of the brain of an individual as a task is described to them which they then complete.

areas of high activity during the description of the task

areas of high activity during the completion of the task



(a) Name the part of the brain shown in the diagram. **1**

_____

(b) Explain how the diagram supports the suggestion that there is localisation of function in the brain. **1**

_____

_____

(c) Explain the high level of brain activity during the description of the task. **1**

_____

_____

9. (continued)

(d) The task was to fold a piece of paper.

Explain why the diagram shows high levels of activity in the sensory and motor areas. **2**

Sensory area _____

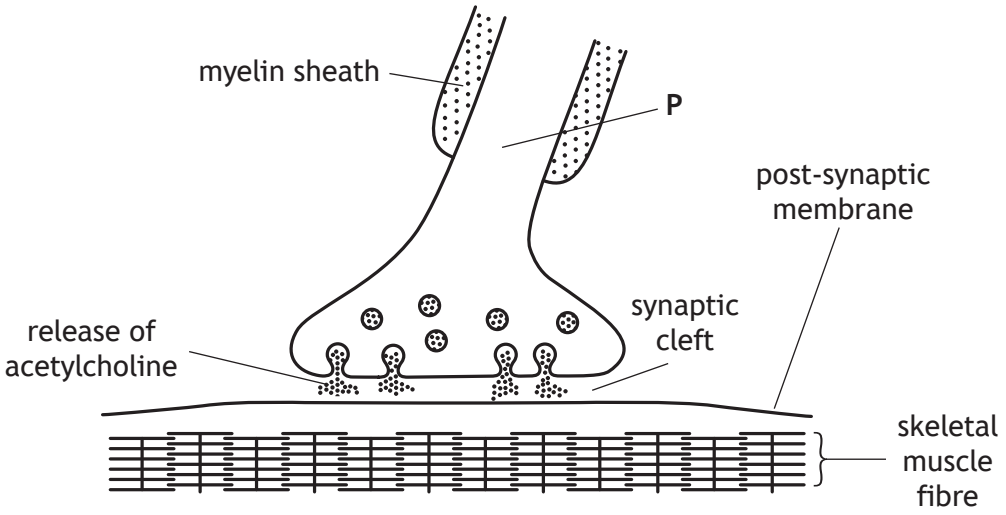
_____

Motor area _____

_____

[Turn over

10. The diagram below shows a synapse in skeletal muscle of a weightlifter.



(a) Name the part of the motor neuron labelled **P**. **1**

_____

(b) Describe what acetylcholine does when it reaches the post-synaptic membrane. **1**

_____

_____

(c) Name the type of skeletal muscle fibres which will be the most common in the arm muscles of a champion weightlifter. **1**

_____

10. (continued)

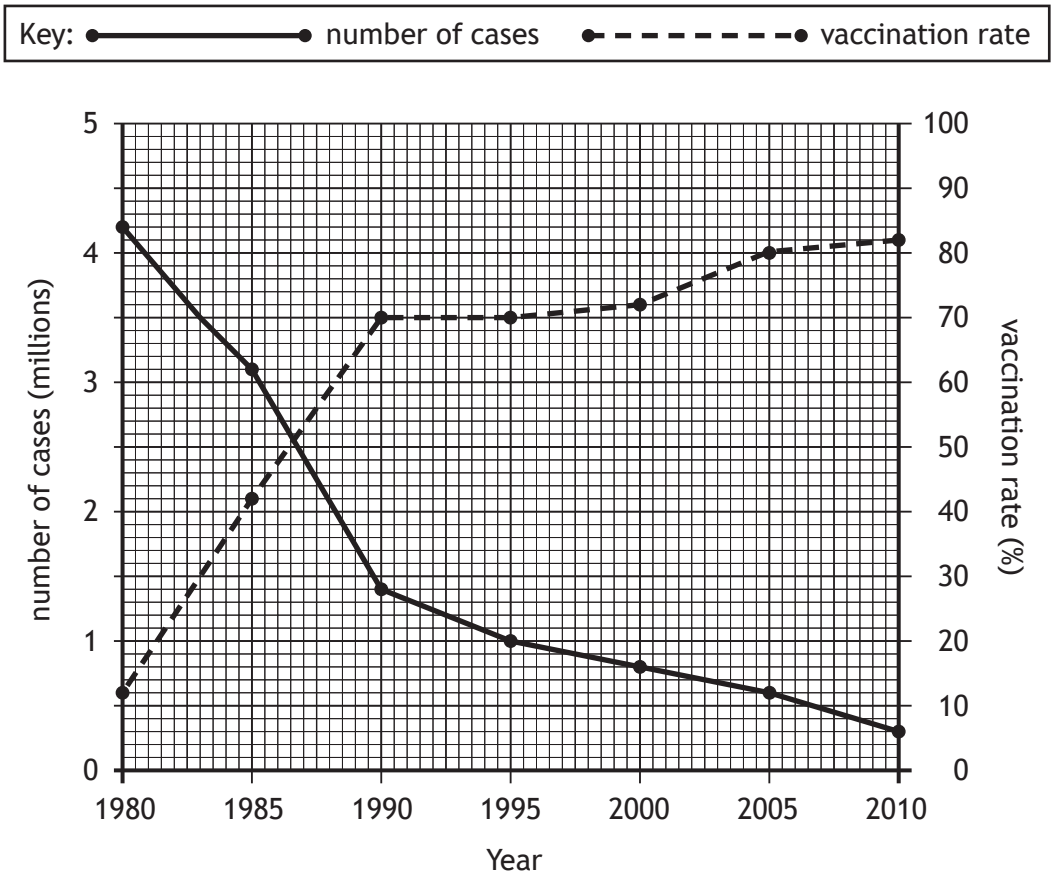
(d) Nicotine is a drug that is an agonist of acetylcholine.

(i) Explain how an agonist works. **1**

(ii) Suggest how nicotine induces feelings of pleasure and so reinforces smoking behaviour. **1**

[Turn over

11. The graph below shows the number of cases of measles that occurred in the world between 1980 and 2010. It also shows the global vaccination rate against measles over the same period.



(a) State how many cases of measles there were in 1985. **1**

_____

(b) State the vaccination rate when there were 3·5 million cases of measles in the world. **1**

_____%

(c) Calculate the percentage decrease in the number of cases of measles between 1995 and 2010. **1**

Space for calculation

_____

11. (continued)

(d) In many countries herd immunity has been established against measles.

(i) Explain why people in these countries who have not been vaccinated are still protected against measles. **1**

(ii) Suggest **one** reason why widespread vaccination programmes against measles are not possible in all countries of the world. **1**

(e) In 2010 the population of the world was 6900 million.

Using the information from the graph, calculate how many people in the world had not been vaccinated against measles in 2010. **1**

Space for calculation

_____ million

(f) The World Health Organisation (WHO) has set a goal of eliminating measles worldwide by 2020.

Explain how the information in the graph indicates that this goal can be achieved. **1**

[Turn over

12. Specific cellular defences are part of the body's immune system and give protection against individual types of pathogen.

(a) (i) Explain how a clonal population of lymphocytes would be formed when a pathogen invades the body. **2**

(ii) Describe the role of phagocytes in the specific immune response. **2**

(b) Failure of the immune system can lead to conditions such as allergy and autoimmune disease.

Choose **one** of these conditions and complete the table below with information about it. **2**

Condition _____

Type of white blood cell involved	*Description of immune system failure*

13. Answer **either** A **or** B in the space below.

Labelled diagrams may be used where appropriate.

A Discuss the causes, development and associated health problems of atherosclerosis. **8**

OR

B Discuss the diagnosis, treatment and role of insulin in Type 1 and Type 2 diabetes. **8**

National
Qualifications
2015

X740/76/02

Human Biology
Section 1 — Questions

WEDNESDAY, 13 MAY
1:00 PM – 3:30 PM

Instructions for the completion of Section 1 are given on *Page two* of your question and answer booklet.

Record your answers on the answer grid on *Page three* of your question and answer booklet.

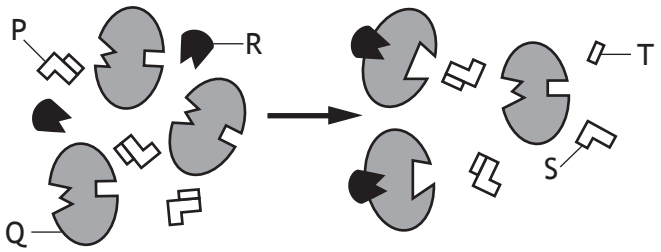
Before leaving the examination room you must give your question and answer booklet to the Invigilator; if you do not you may lose all the marks for this paper.

SQA ©

SECTION 1 — 20 marks

Attempt ALL questions

1. The diagram below shows an enzyme-catalysed reaction taking place in the presence of an inhibitor.



Which line in the table below identifies the molecules in the reaction?

	Inhibitor	*Substrate*	*Product*
A	P	R	S
B	Q	P	S
C	R	P	T
D	R	Q	T

2. A primary transcript is a strand of

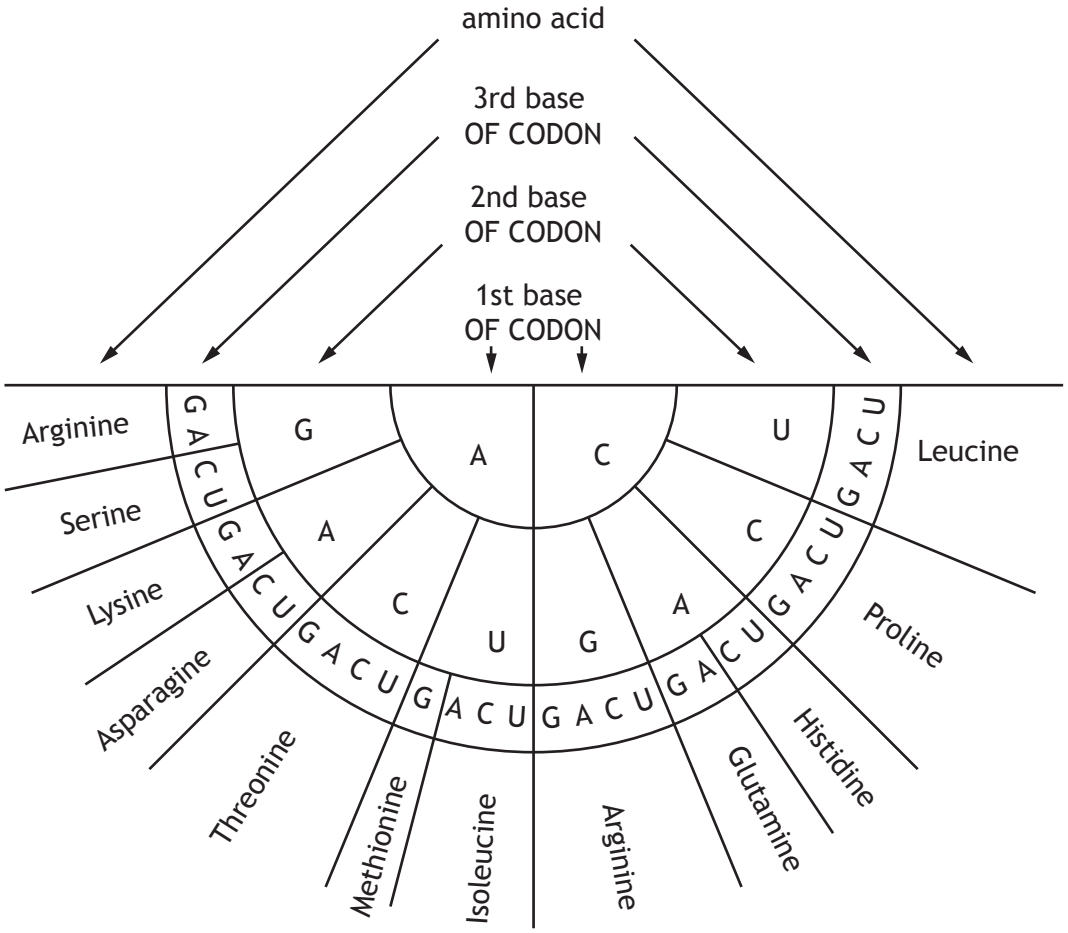
A RNA comprising just exons

B DNA comprising just exons

C RNA comprising introns and exons

D DNA comprising introns and exons.

3. The diagram below can be used to identify amino acids coded for by mRNA codons.



How many different amino acids are coded for by the following mRNA strand?



A 4

B 5

C 6

D 7

[Turn over

4. The following are descriptions of three single gene mutations.

Description 1: exon-intron codons are created or destroyed

Description 2: one amino acid codon is replaced with another

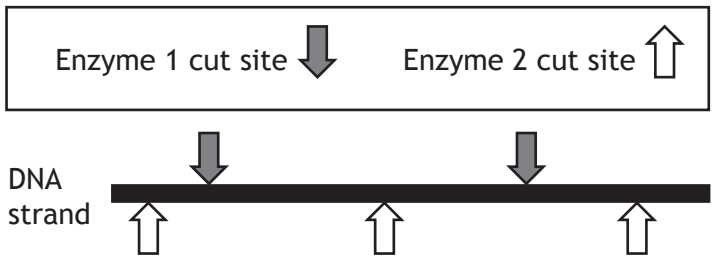
Description 3: one amino acid codon is replaced with a stop codon

Which line in the table below matches the descriptions with the correct gene mutation?

	Gene mutation		
	Missense	*Nonsense*	*Splice site*
A	1	2	3
B	1	3	2
C	2	1	3
D	2	3	1

5. DNA profiling may be used in criminal investigations.

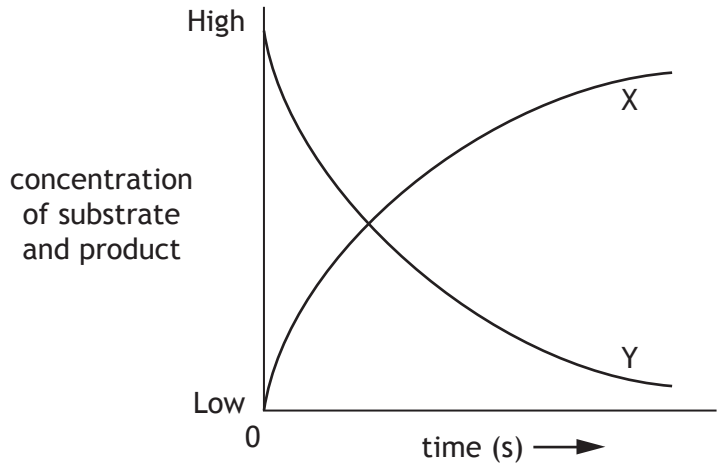
During this procedure DNA is cut into fragments by two different enzymes. Each enzyme cuts DNA at a specific point.



Which line in the table below gives the correct number of DNA fragments produced from this DNA strand?

	Number of fragments produced using		
	enzyme 1 only	*enzyme 2 only*	*enzymes 1 and 2*
A	2	3	5
B	2	3	6
C	3	4	7
D	3	4	6

6. The graph below shows the changes to the concentrations of substrate and product during an enzyme-controlled reaction.

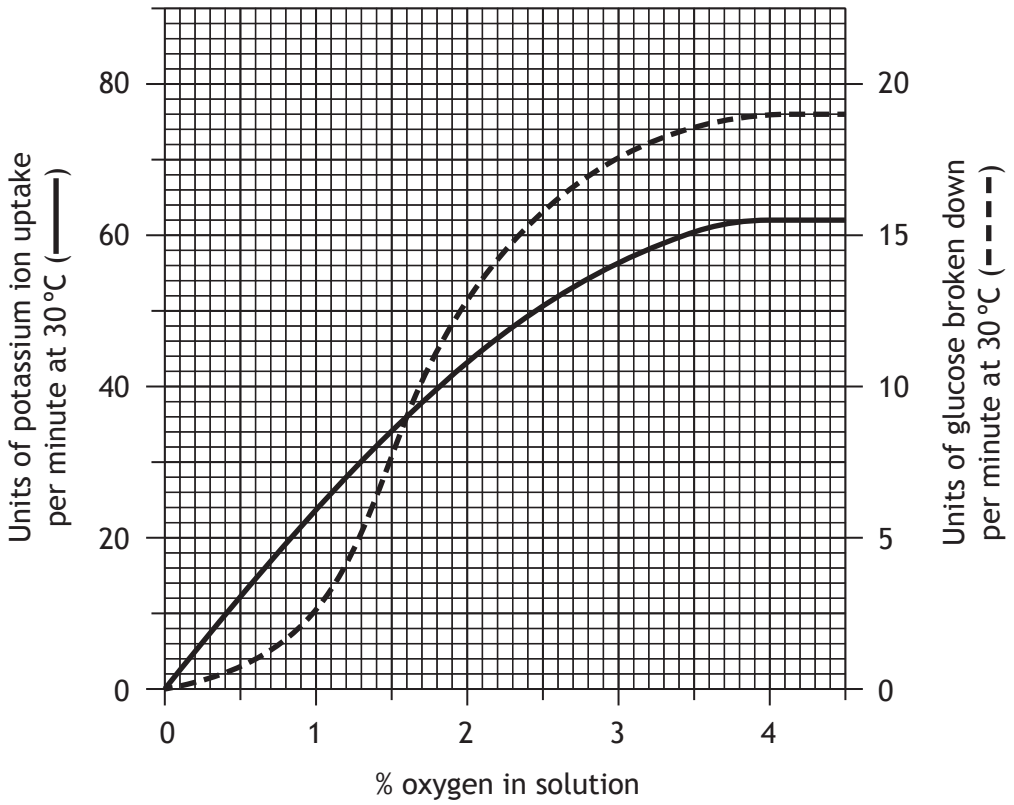


Which line in the table below identifies the substrate, product and the change in the rate of the reaction during the process?

	Substrate	*Product*	*Rate of reaction*
A	X	Y	increasing
B	X	Y	decreasing
C	Y	X	increasing
D	Y	X	decreasing

[Turn over

7. The graph below shows the rate of potassium uptake and glucose breakdown by muscle tissue in solutions of different oxygen concentrations.



How much glucose is broken down per minute when the oxygen concentration is 1%?

A 2·5 units

B 6 units

C 10 units

D 24 units

8. A 40 g serving of a breakfast cereal contains 2 mg of iron. Only 25% of this iron is absorbed into the bloodstream.

If a pregnant woman requires a daily uptake of 6 mg of iron, how much cereal would she have to eat each day to meet this requirement?

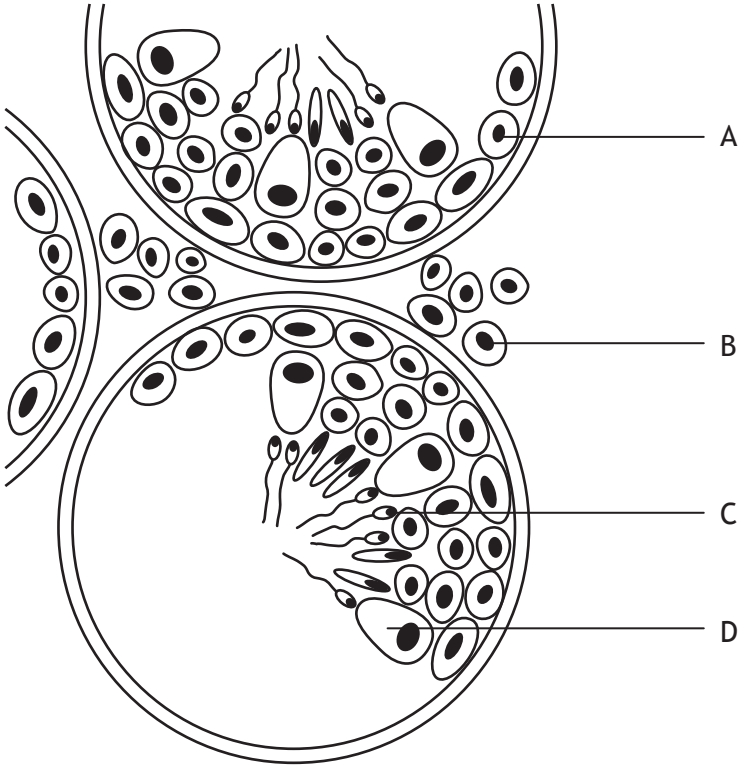
A 60 g

B 120 g

C 240 g

D 480 g

9. The diagram below shows a section through part of the testes.



Which cells produce testosterone?

10. The table below shows some genotypes and phenotypes associated with forms of sickle-cell anaemia.

Genotype	*Phenotype*
AA	unaffected
AS	sickle-cell trait
SS	acute sickle-cell anaemia

A woman with sickle-cell trait and an unaffected man have a child together.

What are the chances that their child will have acute sickle-cell anaemia?

A None

B 1 in 1

C 1 in 2

D 1 in 4

[Turn over

11. The events leading to formation of a blood clot are listed below.

1. Clotting factors are released.
2. An insoluble meshwork forms.
3. Fibrinogen is converted to fibrin.
4. Prothrombin is converted to thrombin.

The correct sequence of these events is

A 4, 2, 3, 1

B 1, 4, 3, 2

C 1, 3, 4, 2

D 4, 3, 1, 2

12. Which of the following statements describes the role of lipoprotein in the transport and elimination of excess cholesterol?

A Low density lipoprotein transports excess cholesterol from the liver to the body cells.

B Low density lipoprotein transports excess cholesterol from the body cells to the liver.

C High density lipoprotein transports excess cholesterol from the liver to the body cells.

D High density lipoprotein transports excess cholesterol from the body cells to the liver.

13. Which of the following describes typical features of Type 1 diabetes?

	Feature of Type 1 diabetes	
A	occurs in childhood	cells unable to produce insulin
B	develops later in life	cells unable to produce insulin
C	occurs in childhood	cells less sensitive to insulin
D	develops later in life	cells less sensitive to insulin

14. The following are types of neural pathways.

1. Diverging
2. Converging
3. Reverberating

Which of these pathways involve nerve impulses being sent back through a circuit of neurons?

A 3 only

B 1 and 2 only

C 1 and 3 only

D 1, 2 and 3

15. After drinking, alcohol is removed from the blood at a constant rate.

The table below shows the average time it takes to remove different alcohol concentrations from the blood.

Blood alcohol concentration ($mg/100\,cm^3$)	*Removal time* (hours)
16	1·0
50	3·125
80	5·0
100	6·25
160	10·0
200	12·5

The legal maximum blood alcohol concentration for driving in some regions of the UK is $80\,mg/100\,cm^3$.

Predict how long it would take before a person with a blood alcohol concentration of $240\,mg/100\,cm^3$ would legally be able to drive in these regions.

A 5 hours

B 10 hours

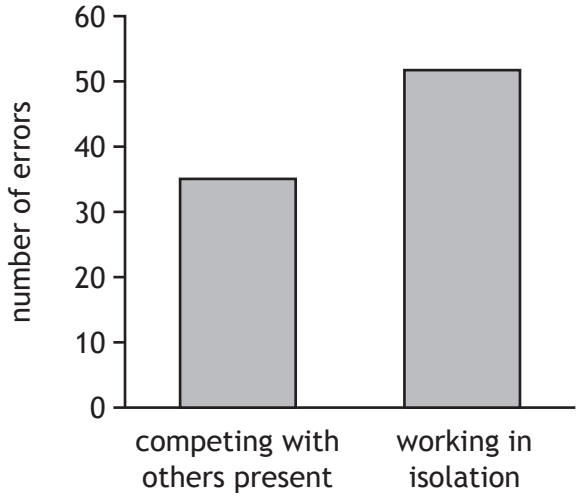
C 15 hours

D 20 hours

[Turn over

16. A number of students were trained to carry out a complex task. Some competed with one another, others worked in isolation.

The graph below shows the number of errors recorded in the training process.



Which process is likely to have caused the difference in the results?

A Deindividuation

B Social facilitation

C Shaping

D Internalisation

17. The pathogen for the disease tuberculosis (TB) evades the specific immune response by

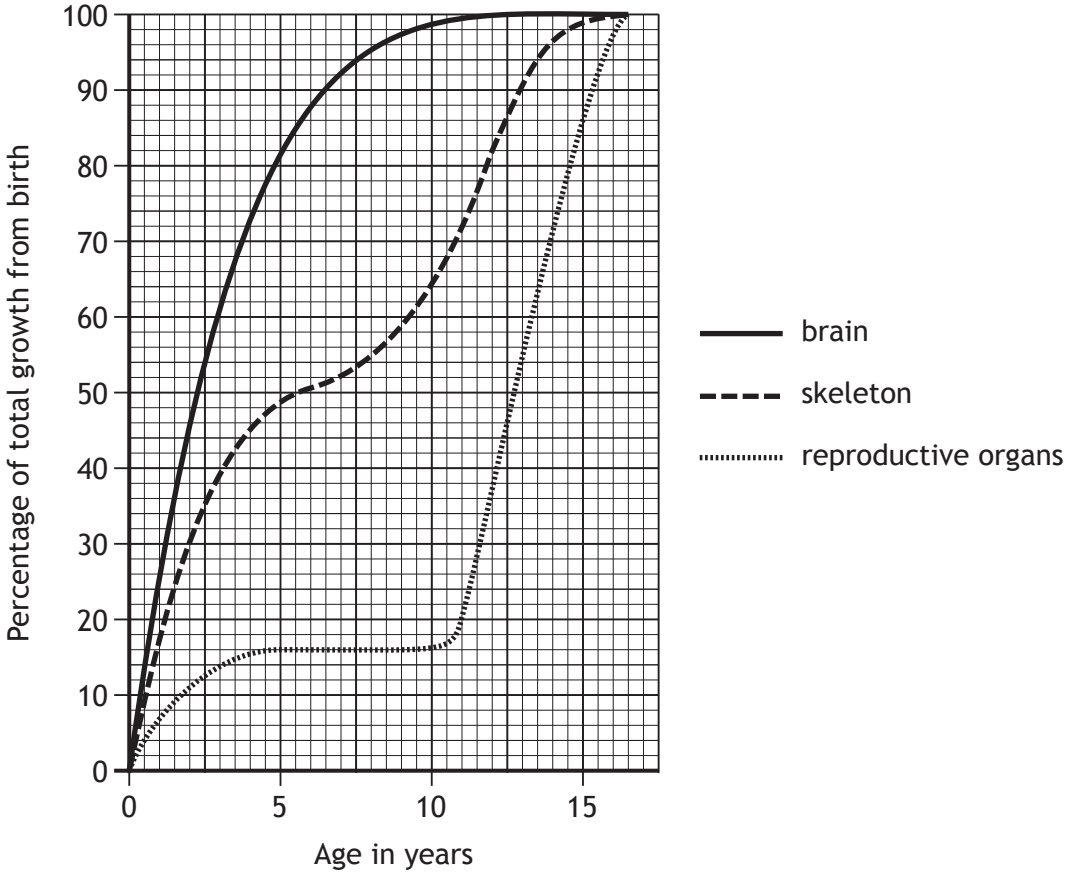
A surviving within phagocytes

B attacking lymphocytes

C attacking phagocytes

D antigenic variation.

18. The graph below shows the average growth rate of body organs in males.



What is the ratio of total growth of brain to skeleton in an 8 year old child?

A 11 : 3

B 3 : 11

C 19 : 11

D 11 : 19

19. Failure in regulation of the immune system leading to an autoimmune disease is caused by a

A B lymphocyte immune response to self antigens.

B T lymphocyte immune response to self antigens.

C B lymphocyte immune response to foreign antigens.

D T lymphocyte immune response to foreign antigens.

[Turn over for Question 20 on *Page twelve*

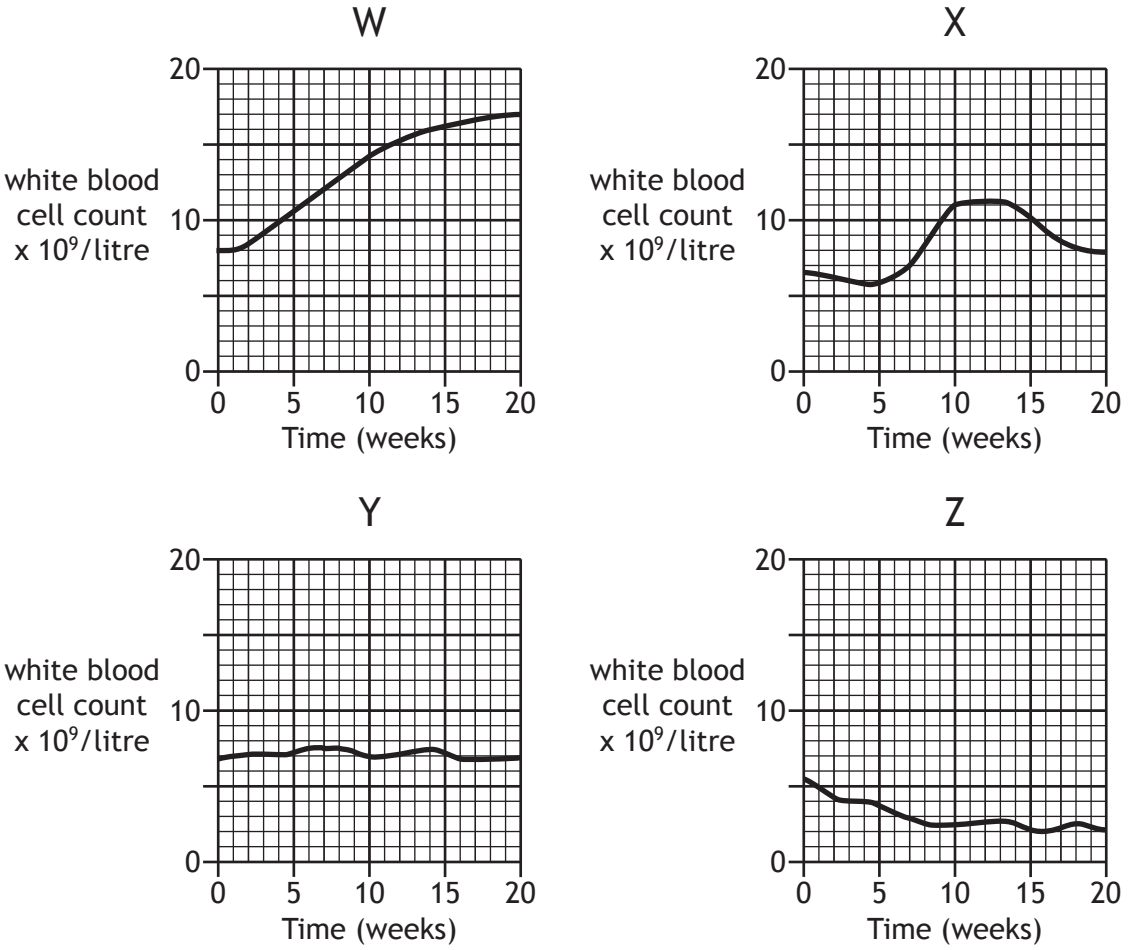
20. Blood tests to measure the number of white blood cells (leucocytes) are often used to indicate infection and/or illness.

Leucopenia, due to starvation or malnutrition, is indicated by white blood cell numbers dropping below 4×10^9/litre.

Leucocytosis, due to fever or tissue damage, is indicated by white blood cell numbers temporarily increasing to 11×10^9/litre.

Leukaemia, due to DNA damage and cell division, is indicated by white blood cell numbers permanently increasing.

The following graphs show the white blood cell count of four patients over 20 weeks.



From the graphs, identify the patients.

	Leukaemia	*Leucocytosis*	*Leucopenia*
A	Y	X	Z
B	Z	W	Y
C	W	X	Z
D	W	Y	X

[END OF SECTION 1. NOW ATTEMPT THE QUESTIONS IN SECTION 2 OF YOUR QUESTION AND ANSWER BOOKLET.]

FOR OFFICIAL USE

National Qualifications 2015

Mark

X740/76/01

Human Biology Section 1 — Answer Grid and Section 2

WEDNESDAY, 13 MAY

1:00 PM – 3:30 PM

X7407601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day | Month | Year

Scottish candidate number

Total marks — 100

SECTION 1 — 20 marks

Attempt ALL questions.

Instructions for completion of Section 1 are given on *Page two*.

SECTION 2 — 80 marks

Attempt ALL questions.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. You should score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

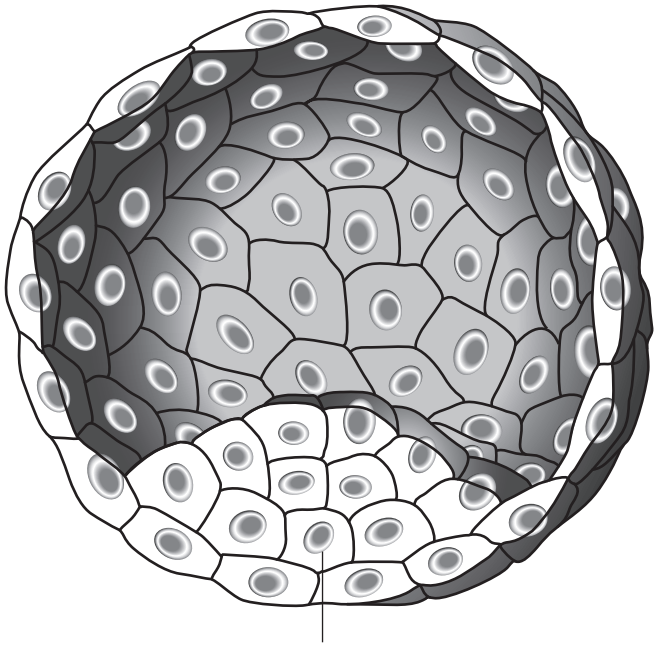
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SECTION 2 — 80 marks

Attempt ALL questions

Note that Question 14 contains a choice

1. The diagram below represents an embryo in the early stages of development and identifies the inner cell mass which is made up of stem cells.



(a) State one feature of stem cells. **1**

(b) Stem cells are also found in tissues throughout the adult body.

Explain how the function of tissue stem cells differs from that of the stem cells found in the inner cell mass of an embryo. **1**

1. (continued)

(c) Stem cells have uses in both therapy and research.

(i) It has been proposed that tissue cells could be used to repair severely damaged muscle tissue.

Suggest how this might be done. **1**

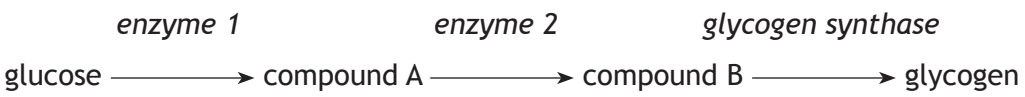
(ii) State how stem cells can be used as model cells in medical research. **1**

[Turn over

MARKS | DO NOT WRITE IN THIS MARGIN

2. Glycogen storage disease is an inherited condition in which the enzyme glycogen synthase does not function.

This enzyme normally catalyses one step in the conversion of glucose to glycogen, for storage, as shown in the diagram below.



(a) State the term which describes a metabolic pathway in which simple molecules are built up into complex molecules. **1**

_____

(b) (i) Describe how the genetic code for glycogen synthase might be altered in an individual with the disease. **1**

_____

_____

(ii) Explain why this altered genetic code fails to produce glycogen synthase. **1**

_____

_____

(c) Suggest why individuals with glycogen storage disease might develop abnormally low blood glucose levels during exercise. **1**

_____

_____

_____

(d) One form of glycogen storage disease is caused by a gene which is recessive and sex-linked.

Describe a pattern of inheritance, shown by a family history, which would indicate that the condition is **2**

recessive _____

_____

sex-linked _____

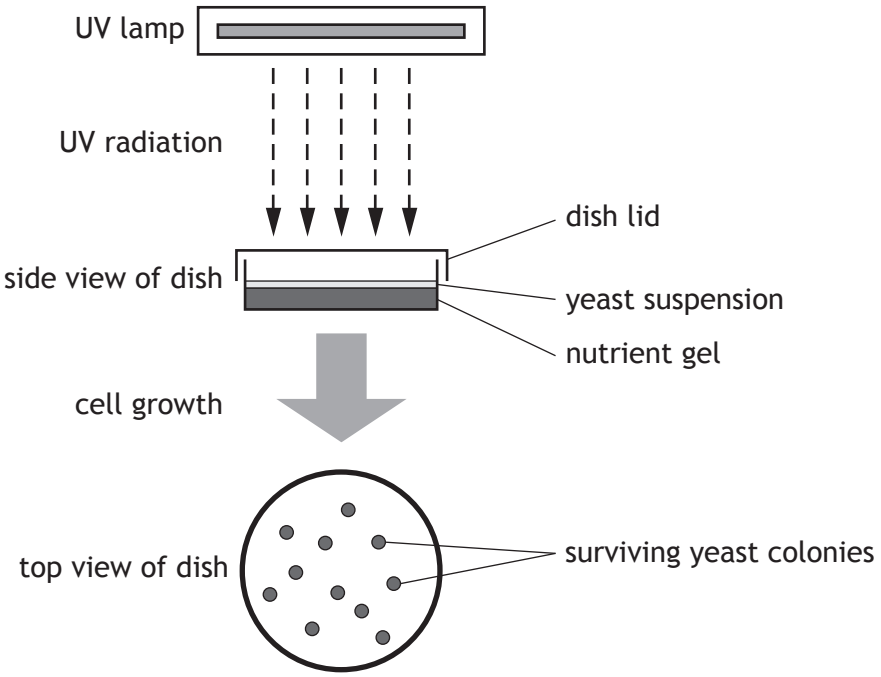
_____

3. Most skin cancers are caused by overexposure to ultraviolet (UV) radiation from the sun or sunbeds. UV radiation damages the DNA in skin cells. Cells normally repair this damage but those which cannot may become cancerous.

A student designed an investigation which used UV-sensitive yeast cells to show the damaging effect of UV radiation. These yeast cells cannot repair DNA damage and die after exposure to UV radiation.

A suspension of UV-sensitive yeast cells was added to dishes which contained a gel that had all the nutrients the yeast needed to grow. The dishes were then exposed to UV radiation for different lengths of time. After exposure, the dishes were placed in an incubator and each of the surviving yeast cells left to grow into a colony on the gel. The number of these colonies was then counted.

The diagram below illustrates this procedure.



(a) List **two** variables which should be kept constant during this investigation. **2**

1 _____

2 _____

[Turn over

3. (continued)

MARKS DO NOT WRITE IN THIS MARGIN

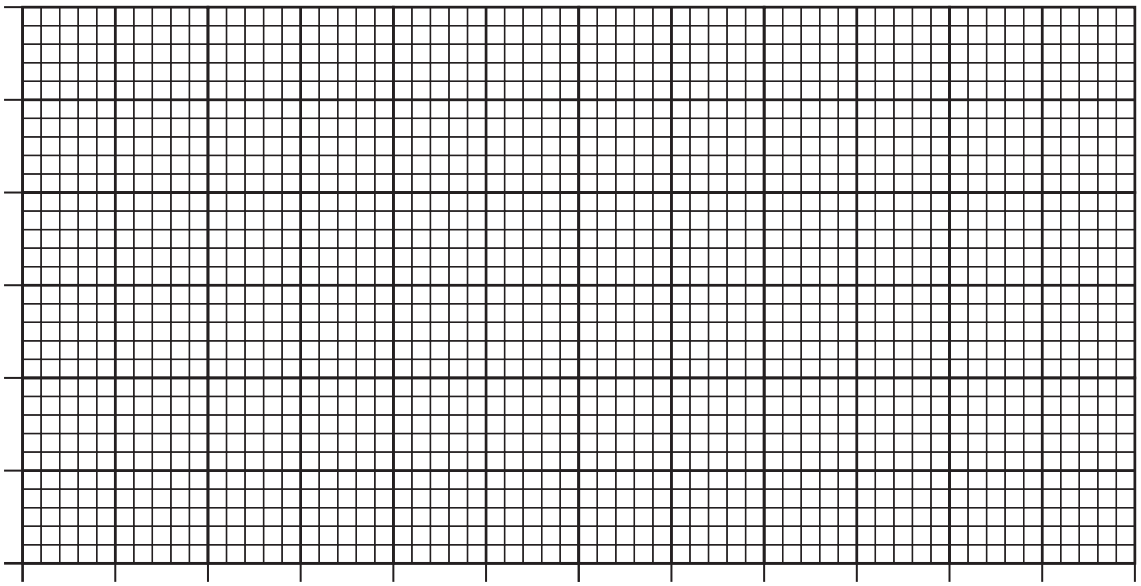
(b) The results of the investigation are shown in **Table 1** below.

Table 1 — Yeast growth after exposure to UV radiation

Length of time of exposure (minutes)	*Number of yeast colonies growing*
10	58
20	32
30	15
40	4
50	1
60	0

(i) Plot a line graph to illustrate the results of the investigation. **2**

(Additional graph paper, if required can be found on *Page thirty-one*)



(ii) State a conclusion that can be drawn from these results. **1**

_____

_____

(iii) State how the reliability of the results in this investigation could be improved. **1**

_____

_____

3. (continued)

MARKS | DO NOT WRITE IN THIS MARGIN

(c) Sunscreens work by blocking UV radiation, preventing it from entering skin cells and causing damage to the DNA, which results in sunburn.

Sunscreens are labelled with a Sun Protection Factor (SPF).

When a sunscreen of SPF 15 is applied to the skin, it will take 15 times longer to burn compared to having no sunscreen applied.

The student carried out a second investigation using UV-sensitive yeast.

The dishes were prepared as before but this time the lids of the dishes were coated with sunscreen of different SPFs. The dishes were then exposed to UV radiation for 30 minutes. After exposure, the dishes were placed in an incubator and the surviving yeast cells left to grow into colonies. The results are shown in **Table 2** below.

Table 2 — Yeast growth after the use of sunscreen protection

Sunscreen used to coat lid (SPF)	*Number of yeast colonies growing*
6	20
15	72
35	74
50	75

(i) Use the information from **Tables 1 and 2** to calculate the percentage increase in yeast cell survival when a sunscreen of SPF 50 is used to coat the lid. **1**

Space for calculation

_____ %

(ii) Official health advice recommends that people should use a sunscreen of SPF 15 when sunbathing for 30 minutes.

State how the results of this investigation support this recommendation. **1**

_____

_____

_____

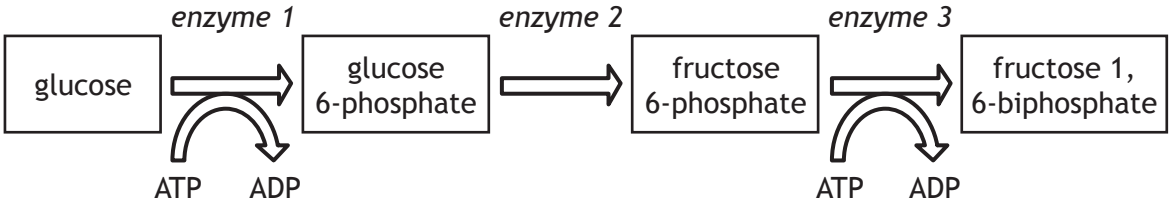
(iii) If skin starts to burn after 10 minutes in strong sunlight, calculate for how long a sunscreen of SPF 35 would protect the skin. **1**

Space for calculation

_____

4. The diagram below represents **three** chemical reactions in the energy investment phase of glycolysis.

MARKS | DO NOT WRITE IN THIS MARGIN



(a) Identify the information, shown **in the diagram**, which confirms that this is the energy investment phase of glycolysis. **1**

(b) Enzyme 1 is activated by the binding of magnesium ions.

Suggest how the binding of these ions leads to an increase in enzyme activity. **1**

(c) Choose an enzyme shown **in the diagram** which is catalysing a phosphorylation reaction.

(Circle) **one** enzyme — Enzyme 1 Enzyme 2 Enzyme 3

Explain what is meant by phosphorylation. **1**

(d) The conversion of glucose 6-phosphate to fructose 6-phosphate is a reversible reaction.

Describe the circumstances under which this reaction would go in the opposite direction to that shown in the diagram. **1**

(e) Following the energy investment phase, glycolysis enters the energy pay off stage, during which ATP is produced.

Enzyme 3 is phosphofructokinase which is inhibited by a build-up of ATP.

Explain how this feedback mechanism conserves the cell's resources. **1**

MARKS | DO NOT WRITE IN THIS MARGIN

5. Muscle cells utilise a variety of energy systems during strenuous activity.

(a) Creatine phosphate is found in muscle cells.

(i) Describe how creatine phosphate supports strenuous muscle activity. **2**

(ii) Explain why this support is not provided to strenuous activities beyond the first 10 seconds. **1**

(b) Name the substance which builds up in muscle cells as they become fatigued. **1**

(c) Choose a sporting activity and decide whether slow twitch or fast twitch muscle fibres would be best suited for the activity.

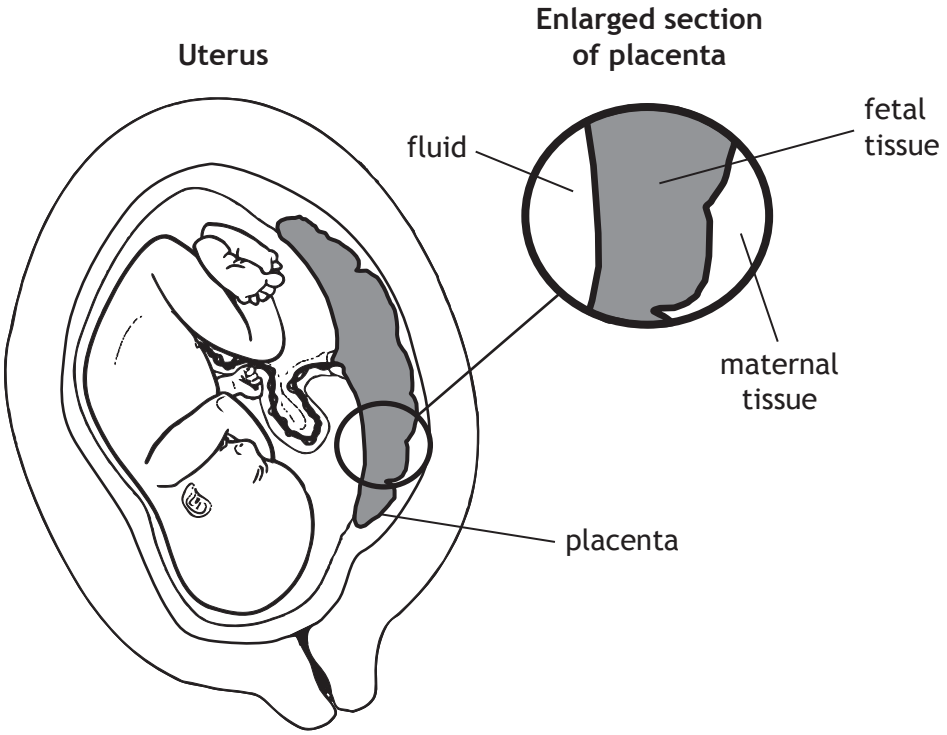
Sporting Activity _____

Slow twitch ☐ Fast twitch ☐

Give reasons to justify your choice of muscle fibre. **3**

6. Chorionic villus sampling (CVS) is a technique which can be used during antenatal screening. The cells obtained from CVS are used to prepare a karyotype.

(a) The diagram below shows the uterus of a pregnant woman with a section of the placenta enlarged.



(i) Place a cross (X) on the diagram of the **enlarged section of placenta** to indicate the area from which cells are removed during CVS. **1**

(ii) Describe the process by which a karyotype is produced from cells removed during CVS. **2**

(iii) Suggest an advantage of using CVS rather than amniocentesis during antenatal screening. **1**

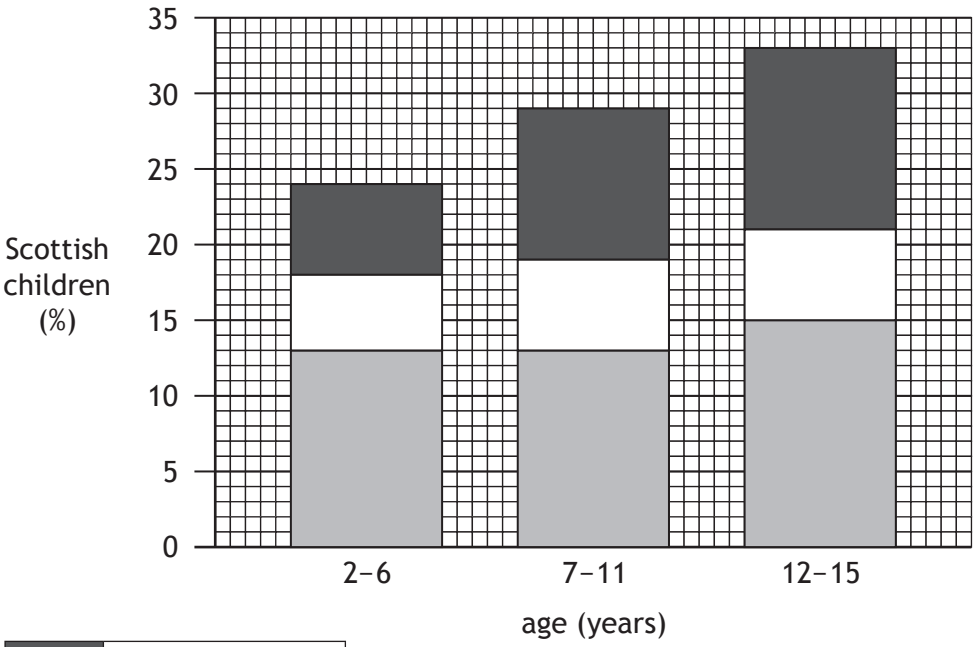
6. (continued)

(b) Name the type of antenatal screening tests which are routinely carried out to monitor the concentration of certain substances, such as protein, in a pregnant woman's blood. **1**

_____

[Turn over

7. The graph below contains information about the body mass index (BMI) of Scottish children in 2009.



	MORBIDLY OBESE BMI above 35
	OBESE BMI 30 to 34·9
	OVERWEIGHT BMI 25 to 29·9

(a) State the percentage of children aged 12 to 15 who had a BMI of more than 30 in 2009. **1**

Space for calculation

_____%

(b) Suggest reasons why the percentage of obese children increased between the ages of 2 and 15. **1**

_____

_____

7. (continued)

(c) Explain how BMI is calculated. **1**

_____

_____

(d) Suggest how children could be encouraged to maintain a healthy BMI by use of the following processes. **2**

Identification _____

_____

Internalisation _____

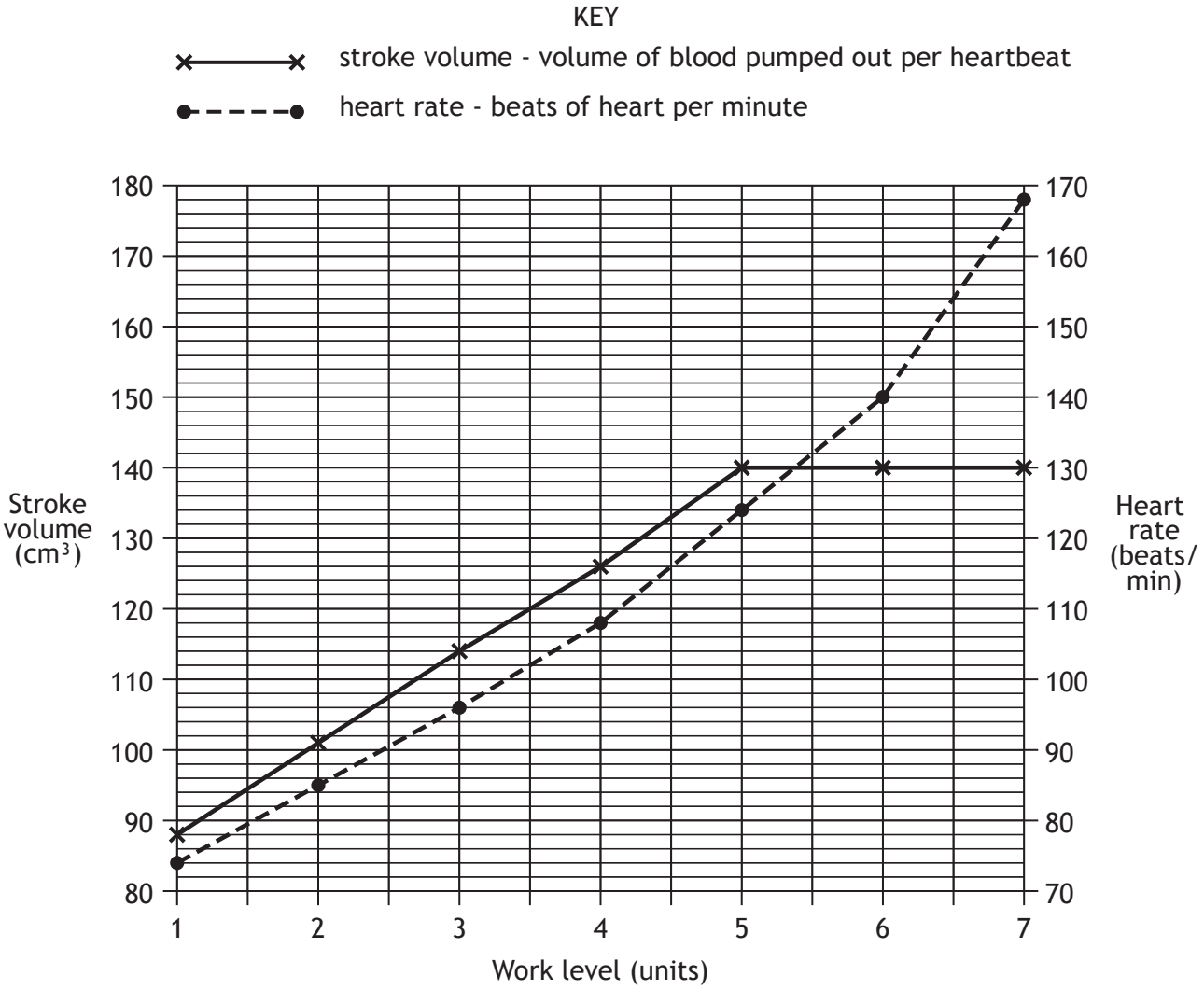
_____

[Turn over

8. The heart rate and stroke volume of a 40 year old cyclist were monitored as he used an exercise bike.

The cyclist was told to pedal at a constant rate as his work level was gradually raised by increasing the resistance to pedalling.

The graph below shows the changes that occurred in the cyclist's heart rate and stroke volume at seven different work levels.



8. (continued)

(a) Use **data from the graph** to describe the changes that occurred in the cyclist's stroke volume when the work level increased from 1 to 7 units. **2**

(b) State what the cyclist's heart rate was when his stroke volume was 120 cm^3. **1**

Space for calculation

_____ beats/min

(c) Cardiac output is the volume of blood leaving the heart in one minute. It is calculated using the formula shown below.

cardiac output = heart rate × stroke volume

Calculate the cyclist's cardiac output when his work level was 6 units. **1**

Space for calculation

_____ cm^3/min

[Turn over

8. (continued)

(d) The table below shows the recommended minimum heart rates that cyclists of different ages should maintain in order to either metabolise fat or improve their fitness.

Age	*Minimum heart rate for metabolising fat* (beats/min)	*Minimum heart rate for improving fitness* (beats/min)
10	136	168
20	130	160
30	123	152
40	116	144
50	110	136
60	104	128

(i) Use information from the **table** and the **graph** to determine the work level that the cyclist should maintain in order to metabolise fat. **1**

_____ units

(ii) Use information from the **table** to predict the minimum heart rate for improving the fitness of a 70 year old. **1**

_____

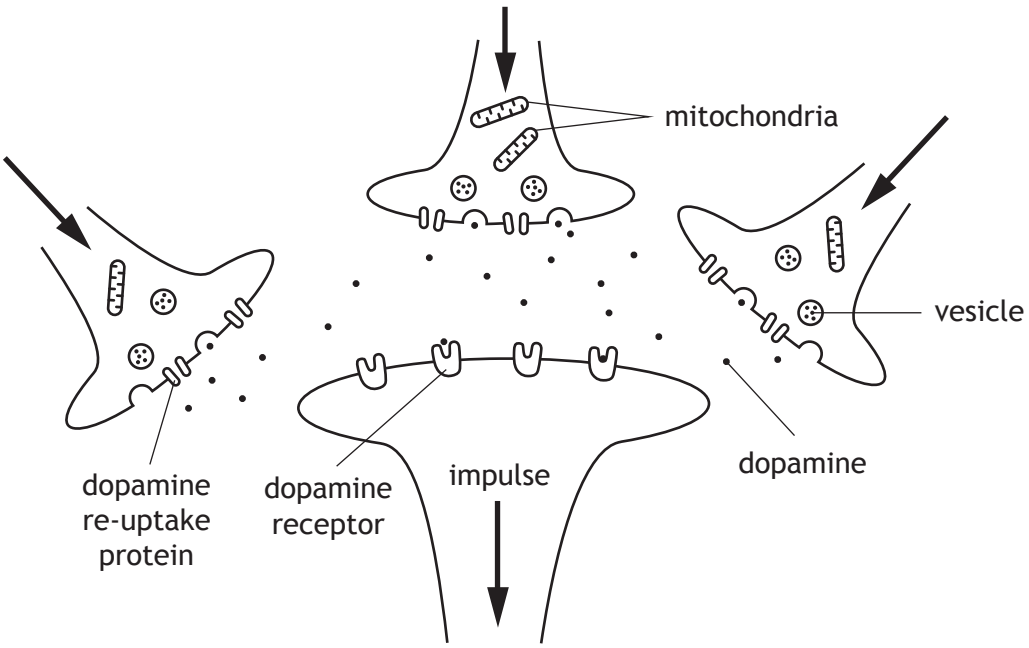
(iii) As an individual gets older, their minimum heart rate for improving fitness decreases.

Use the information from the **table** to calculate the percentage decrease that occurs between the ages of 10 and 60 years. **1**

Space for calculation

_____%

9. The diagram below shows some nerve cells involved in a neural reward pathway.



(a) Using information from the diagram, explain what is meant by the term "summation". **1**

(b) Suggest a function for the mitochondria shown in the diagram. **1**

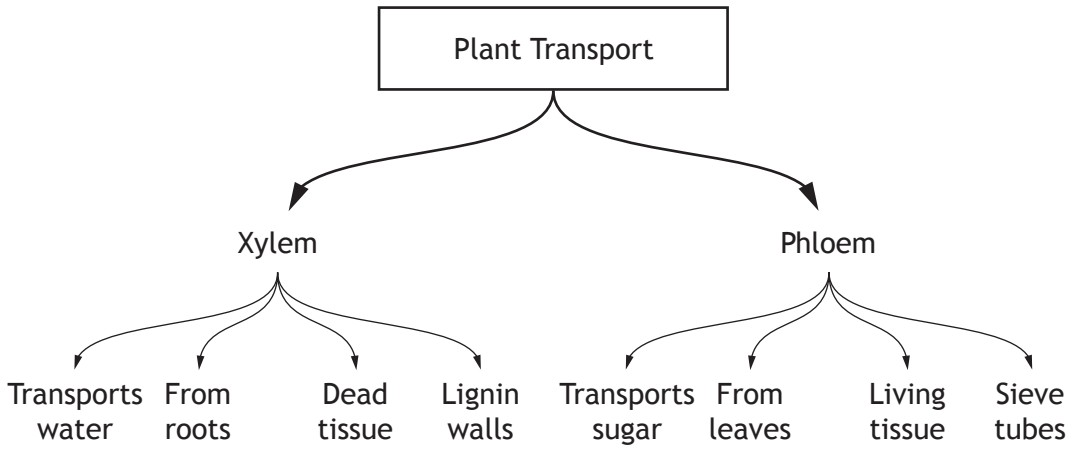
(c) Cocaine is a recreational drug that has an effect at this synapse.

Cocaine binds to the dopamine re-uptake proteins. As a result, the reward pathway is stimulated for longer.

Suggest how cocaine produces this effect. **2**

MARKS | DO NOT WRITE IN THIS MARGIN

10. A biology student produced the following diagram as a memory aid to help her learn about transport in plants.



(a) In producing this diagram the student used various methods to learn the information.

Other than rehearsal, name **two** methods that she used and describe how they helped her transfer the information into her long-term memory. **2**

1 Method _____

Description _____

_____

2 Method _____

Description _____

_____

(b) Any information which is not transferred into long-term memory is displaced. Explain why displacement occurs. **1**

_____

_____

(c) The student is storing a record of facts as she learns this information. State the part of the brain in which such memories are stored. **1**

_____

MARKS | DO NOT WRITE IN THIS MARGIN

11. Various types of white blood cell are involved in the non-specific immune response.

(a) Describe the role of each of the following cells in the non-specific defence of the body.

(i) Mast cells _____ **2**

(ii) Natural killer (NK) cells _____ **1**

(b) Explain how the presence of phagocytes is important in the activation of T lymphocytes. **2**

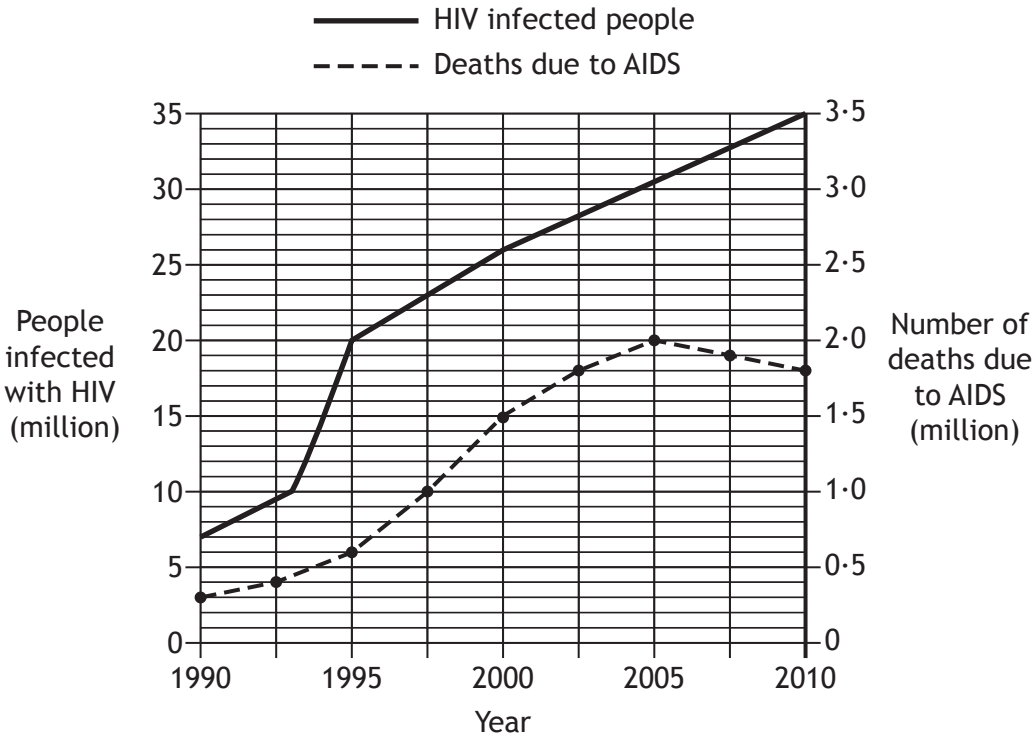
[Turn over

12. HIV is a virus which invades the cells of the immune system.

People infected with HIV may not show symptoms for many years.

AIDS is the condition which may develop from HIV infection, resulting in death.

The graph below shows the number of people in the world infected with HIV, from 1990 to 2010. It also shows the number of people who died from AIDS during this period.



(a) State how many people were infected with HIV in the year 2000. **1**

_____

(b) State how many people died from AIDS when 20 million people in the world were infected with HIV. **1**

_____

(c) Calculate the percentage of HIV-infected people who died from AIDS in 2010. **1**

Space for calculation

_____%

12. (continued)

(d) Describe the evidence from the graph which suggests that the rate of people becoming infected with HIV was greatest between 1993 and 1995. **1**

_____

_____

[Turn over

13. A scientist investigated the effectiveness of four different types of influenza vaccine. A total of 2000 volunteers from a Scottish community were divided into four groups.

Each group was injected with a different vaccine.

The number who developed influenza during the following years was recorded.

The results are shown in the table below.

Type of influenza vaccine	*Developed influenza*	*Did not develop influenza*	*Total*
P	35	495	530
Q	25	455	480
R	24	496	520
S	17		

(a) (i) Suggest **one** way in which the scientist could minimise variation between the four groups of volunteers. **1**

_____

_____

(ii) **Complete the table** for the volunteers who received type S vaccine. **1**

(iii) State which of the vaccines P, Q or R was most effective in this investigation. **1**

_____

(b) Explain why vaccines usually contain an adjuvant. **1**

_____

_____

(c) In 1918 fifty million people died in a global outbreak of influenza.

State the term used to describe such an outbreak. **1**

_____

MARKS | DO NOT WRITE IN THIS MARGIN

14. Answer **either** A **or** B in the space below.

Labelled diagrams may be used where appropriate.

A Describe hormonal control of the menstrual cycle under the following headings:

(i) events leading to ovulation; **6**

(ii) events following ovulation. **4**

OR

B Describe the cardiac cycle under the following headings:

(i) the conducting system of the heart; **5**

(ii) nervous control of the cardiac cycle. **5**

[Turn over

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Higher – Paper 1

Question Paper Finalised Marking Instructions

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Marking instructions for each question

Question	Response	Mark
1.	C	**1**
2.	C	**1**
3.	A	**1**
4.	C	**1**
5.	B	**1**
6.	A	**1**
7.	D	**1**
8.	B	**1**
9.	C	**1**
10.	D	**1**
11.	A	**1**
12.	C	**1**
13.	A	**1**
14.	D	**1**
15.	B	**1**
16.	B	**1**
17.	C	**1**
18.	B	**1**
19.	B	**1**
20.	D	**1**
21.	D	**1**
22.	B	**1**
23.	D	**1**
24.	C	**1**
25.	A	**1**

[END OF MARKING INSTRUCTIONS]

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2026 Human Biology

Higher – Paper 2

Question Paper Finalised Marking Instructions

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Marking instructions for each question

Question			Expected response	Max mark	Additional guidance
1.	(a)	(i)	Differentiation/specialisation	**1**	
		(ii)	Can differentiate/specialise into all/any of the (types of) blood cells. **OR** Can differentiate/specialise into cells within a particular tissue/of the tissue they come from. **OR** Can only differentiate/specialise into a limited range of cells/cell types.	**1**	Do not accept: divide/replicate/ multiply for differentiate
		(iii)	They express different genes.	**1**	
	(b)	(i)	(Cancer cells/they) do not respond to regulatory signals.	**1**	
		(ii)	From (days) 0-12 the white blood cell count decreased from 88 000 cells/microlitre to 4000 cells/microlitre. **OR** From (days) 0-12 the white blood cell count decreased by 84 000 cells/microlitre. **(1)** From (days) 12 – 32 the white blood cell count increased from 4000 cells/microlitre to 80 000 cells/microlitre. **OR** From (days) 12 – 32 the white blood cell count increased by 76 000 cells/microlitre. **(1)**	**2**	Answer must indicate cells/microlitre for at least one figure. Award **one mark** if candidate states that it decreases to (day) 12 and then increases.
		(iii)	90 000	**1**	

Question			Expected response	Max mark	Additional guidance
2.	(a)	(i)	Direction of replication. **OR** Direction nucleotides are being added.	**1**	
		(ii)	DNA polymerase can only add nucleotides to the 3′ end of the primer/new strand. **OR** DNA polymerase can only add nucleotides in the 5′ to 3′ direction on the new strand. **OR** DNA polymerase can only replicate in a 3′ to 5′ direction on the original strand.	**1**	
		(iii)	(DNA) ligase	**1**	
	(b)		Unwinds the helix/DNA/strands. **OR** Breaks the hydrogen bonds (between the bases)/separates the (DNA) strands. **(1)** Adds RNA nucleotides to synthesise mRNA/a primary transcript. **OR** Adds RNA nucleotides using complementary base pairing. **OR** Adds RNA nucleotides that are complementary to the DNA. **(1)**	**2**	

Question			Expected response	Max mark	Additional guidance
2.	(c)		1. (tRNA) folds due to complementary base pairing/hydrogen bonding **or** (tRNA) shape is due to complementary base pairing/ hydrogen bonding 2. (tRNA) is single stranded and is composed of nucleotides 3. (Nucleotides) contain ribose (sugar), phosphate and (one of) the bases cytosine, guanine, adenine and uracil 4. (tRNA) has an anticodon/an exposed triplet of bases (at one end) 5. (tRNA) has an attachment site for a specific amino acid (at the other end) **or** (tRNA) binds to a specific amino acid (at the other end) **Any 3**	**3**	

Question			Expected response	Max mark	Additional guidance
3.	(a)		A C U C G U G A G	**1**	
	(b)	(i)	Substitution	**1**	
		(ii)	One amino acid is replaced by another. **OR** Sequence of amino acids is changed.	**1**	Do not accept: produced/formed/ made for replaced
	(c)		Less oxygen so pyruvate is converted to lactate. **OR** Less oxygen for aerobic respiration/electron transport chain/ATP production. **OR** Less oxygen so can only carry out glycolysis. **OR** Less oxygen so lactate metabolism occurs.	**1**	Accept: ETC for electron transport chain
	(d)		8375	**1**	

Question			Expected response	Max mark	Additional guidance
4.	(a)		Cytoplasm	**1**	
	(b)		(Pyruvate dehydrogenase) removes hydrogen ions and electrons from pyruvate and transfers them to NAD.	**1**	
	(c)		**Substance X** – Oxaloacetate. **(1)** **Role** – Combines with acetyl (to form citrate). **(1)**	**2**	Do not accept: acetyl coenzyme A/ coA
	(d)		**ATP production** - Decreases **(1)** **Lactate production** - Increases **(1)**	**2**	
	(e)	(i)	To select the best/most effective drugs/dosage/medicine. **OR** Use genetic information in choice of drug/dosage/medicine. **OR** Personalised medicine. **OR** To reduce side/harmful effects.	**1**	Do not accept: treatment/ prescription
		(ii)	25%	**1**	

Question			Expected response	Max mark	Additional guidance
5.	(a)	(i)	4:10:13	**1**	
		(ii)	3.5	**1**	
	(b)	(i)	They can contract (relatively) quickly. **OR** Can respire/generate ATP using glycolysis only. **OR** Storage fuel is glycogen.	**1**	
		(ii)	**Function** – (Myoglobin) stores/provides oxygen (to muscle fibres). **OR** (Myoglobin) is an oxygen-storing protein. **(1)** **Explanation** - (Slow twitch muscle fibres) rely on aerobic respiration/need oxygen **AND** to generate ATP/support electron transport chain/for muscle contraction. **(1)**	**2**	Do not accept: carries/transports oxygen
		(iii)	Slow twitch (muscle fibres) have more mitochondria. **OR** Slow twitch (muscles) have larger blood supply/more blood capillaries.	**1**	Accept: converse

Question			Expected response	Max mark	Additional guidance
6.	(a)	(i)	LH/luteinising hormone	**1**	
		(ii)	**Body temperature** – Increases **(1)** **Cervical mucus consistency** – will be thinner. **(1)**	**2**	Accept: converse if answer indicates day 3
	(b)	(i)	**Phase A** – Follicular **(1)** **Phase B** – Luteal **(1)**	**2**	
		(ii)	LH stimulates the follicle to develop into a corpus luteum. **OR** The follicle develops into the corpus luteum following ovulation. **(1)** Corpus luteum/this secretes progesterone. **(1)**	**2**	
		(iii)	Progesterone concentration would remain constant/stay the same/ would not decrease/remain high.	**1**	Accept: oestrogen concentration would remain constant/stay the same/would not decrease/remain high

Question			Expected response	Max mark	Additional guidance
7.	(a)		35	**1**	
	(b)	(i)	The starting lengths are different. **OR** To allow a valid comparison.	**1**	Use of accurate/reliable negates.
		(ii)	Repeat with more blood vessels. **OR** Repeat with more veins and arteries. **OR** Repeat and calculate an average.	**1**	Use of accurate/valid negates.
	(c)	(i)	Arteries have more elastic fibres (than veins). **OR** Veins have fewer elastic fibres (than arteries). **OR** Arteries have higher elasticity/are more elastic (than veins). **OR** Veins have lower elasticity/are less elastic (than arteries).	**1**	Do not accept: veins do not have elastic fibres
		(ii)	To accommodate the surge of the blood/increased blood flow (after each contraction). **OR** To accommodate the increase in blood pressure (after each contraction).	**1**	
	(d)		They (smooth muscle) can vasoconstrict and vasodilate.	**1**	Elastic fibres causing vasodilation/constriction negates.

<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>8.</td><td>(a)</td><td>(i)</td><td>To act as a control.

OR

To provide a valid comparison.

OR

To show the alcohol is reducing sperm count/causing the effect.</td><td>1</td><td>Use of reliable/accurate negates.</td></tr>
<tr><td></td><td></td><td>(ii)</td><td>1. concentration/volume/dosage/ type of alcohol
2. type/breed/age/size/mass/ weight/health/(starting) fertility of mice
3. temperature/conditions mice are kept in
or
activity of mice.
4. type of food/diet of mice.
5. time of day dose is given/semen is collected.
Any 2</td><td>2</td><td>1. accept: volume of water/ solutions.

3. Accept: presence of other mice.

If the candidate lists 3 or more variables and at least 1 is correct award 1 mark for the question.
If all answers are correct award 2 marks.</td></tr>
<tr><td></td><td>(b)</td><td>(i)</td><td>Axes have correct scales and labels. (1)

Points correctly plotted and lines drawn. (1)

<table><tr><th rowspan="2">Day semen sample taken</th><th>Average sperm count (million/cm^3)</th></tr><tr><th>Group 2</th></tr><tr><td>0</td><td>45.0</td></tr><tr><td>30</td><td>42.2</td></tr><tr><td>60</td><td>40.2</td></tr><tr><td>90</td><td>38.2</td></tr><tr><td>120</td><td>36.2</td></tr></table></td><td>2</td><td>Scale breaks are not acceptable.

If the axes are transposed do not award the scale mark.

The line must go through all the points.</td></tr>
<tr><td></td><td></td><td>(ii)</td><td>Regular alcohol consumption reduces the sperm count (of mice).</td><td>1</td><td></td></tr>
<tr><td></td><td></td><td>(iii)</td><td>35.2</td><td>1</td><td></td></tr>
</table>

Question			Expected response	Max mark	Additional guidance
8.	(c)		Several samples of semen/sperm are collected. **OR** A sperm donor can be used. **OR** Semen/sperm is injected directly into the female reproductive tract/uterus.	**1**	

Question			Expected response	Max mark	Additional guidance
9.	(a)	(i)	Component of cell membrane. **OR** Used to produce sex hormones/oestrogen/testosterone/ progesterone.	**1**	
		(ii)	Insulates the axon. **OR** Increases the speed of impulses.	**1**	
	(b)	(i)	**Molecule B** - High density lipoprotein/HDL. **(1)** **Description** - transports excess cholesterol to the liver for elimination. **(1)**	**2**	
		(ii)	Negative feedback **OR** The synthesis of new LDL receptors (in cells) is inhibited.	**1**	
		(iii)	**Structure W** – endothelium. **(1)** **Other substance** – fibrous material/ calcium. **(1)**	**2**	
	(c)	(i)	Hirudin/it binds to the active site of thrombin/the enzyme. **(1)** Fibrinogen would not form fibrin. **OR** Less/no fibrin is produced. **(1)**	**2**	
		(ii)	Increase the concentration of fibrinogen/substrate.	**1**	

Question			Expected response	Max mark	Additional guidance
10.	(a)		0.26	**1**	
	(b)	(i)	Sino atrial node/SAN/SA node	**1**	
		(ii)	(The first sound is) the AV/atrioventricular valves closing (at the end of atrial systole/at the start of ventricular systole). **(1)** (The second sound is) the SL/semi lunar valves closing (at the end of ventricular systole/during diastole). **(1)**	**2**	Award **1 mark** for the (opening and) closing of the valves. Any additional information about systole/diastole must be acceptable to award **2 marks**.
11.	(a)	(i)	Elaboration. **OR** Additional information is provided.	**1**	
		(ii)	Elaboration/it is a deeper form of encoding (than rehearsal). **OR** Elaboration/it provides more contextual cues (than rehearsal).	**1**	Accept: rehearsal is a shallower form of encoding (than elaboration)
	(b)		(The information would) decay/be discarded/be displaced.	**1**	
	(c)		**Name** – Chunking. **(1)** **Suggestion** – This reduces the number of units being encoded. **OR** A description of reduced number of units. **OR** More items/information can be stored (as items are) in units. **(1)**	**2**	Accept: a description of chunking. i.e grouping smaller units of information into larger units e.g. 11 individual units reduced to 3 units (phone number) Do not accept: smaller units

Question			Expected response	Max mark	Additional guidance
12.	(a)	(i)	20	**1**	
		(ii)	44	**1**	
		(iii)	466	**1**	
		(iv)	Increased number of influenza cases. **OR** Fewer people were vaccinated against influenza. **OR** More/increased number of people have rejected the vaccine for influenza. **OR** Wrong strain of influenza in vaccine.	**1**	Accept: an outbreak/epidemic of influenza Accept: influenza vaccine was not as effective (that year)
		(v)	4	**1**	
	(b)	(i)	Inactivated pathogen toxins/dead pathogens/parts of pathogens/ weakened pathogens.	**1**	Do not accept: weakened/inactivated antigens
		(ii)	It makes the vaccine/it more effective. **OR** It enhances the immune response.	**1**	
13.	(a)	(i)	(Individual X) has less dopamine/fewer vesicles containing dopamine.	**1**	
		(ii)	Activates the reward pathway. **OR** (Induces) feelings of pleasure. **OR** Reinforces particular/beneficial behaviour.	**1**	
	(b)		Agonist	**1**	

Question			Expected response	Max mark	Additional guidance
13.	(c)		They diffuse across the cleft/synapse and bind to receptors. **(1)** A minimum number of neurotransmitters must attach (to receptors). **OR** The threshold must be reached. **(1)**	**2**	

Question			Expected response	Max mark	Additional guidance
14.	**A**		1. (Lymphocytes) respond to/recognise specific antigens from pathogens. **or** Lymphocytes have a specific receptor/surface protein for an antigen. 2. Antigen binding leads to lymphocyte division **or** A clonal population is formed. **or** They undergo clonal selection. 3. B lymphocytes release antibodies. 4. Antibodies are Y-shaped proteins that have binding sites/receptors for a specific antigen. **or** Antibodies are Y-shaped proteins that have specific binding sites/receptors for an antigen. 5. An antigen-antibody complex forms. **or** Antibodies inactivate the pathogen. 6. (The antigen-antibody complex/this) can then be destroyed by phagocytosis. 7. T lymphocytes attach onto antigens on infected cells. 8. They release proteins into the infected cells. 9. Causing production of self-destructive enzymes. **or** Cause cell death/apoptosis. 10. The remains of the infected cell are then destroyed by phagocytosis. 11. Some of the (cloned) lymphocytes survive as memory cells. 12. This allows a faster secondary response/antibody production. **or** They destroy the pathogen before the individual develops the same infection/symptoms.	**8**	9. Do not accept cause cell death/apoptosis of a pathogen.

Question			Expected response	Max mark	Additional guidance
14.	**B**		1. Central nervous system/CNS consists of the brain and spinal cord. 2. The peripheral nervous system/PNS consists of the somatic nervous system/SNS and the autonomic nervous system/ANS. 3. The somatic nervous system contains sensory and motor neurons. 4. Sensory neurons take impulses from sense organs/receptors to the CNS. 5. Motor neurons take impulses from the CNS to muscles/glands/ effectors. 6. The sympathetic and parasympathetic nervous systems are antagonistic. 7. The sympathetic system speeds up heart rate/breathing rate **and** The sympathetic system slows down/decreases peristalsis/ (production of) intestinal secretions. 8. A sympathetic nerve/system releases noradrenaline. 9. The parasympathetic system slows down heart rate/breathing rate **and** The parasympathetic system speeds up/increases peristalsis/ (production of) intestinal secretions. 10. A parasympathetic nerve/system releases acetylcholine. 11. The medulla regulates (the rate of) the sino-atrial node/SAN through the autonomic nervous system/ANS **or** The medulla regulates (the rate of) the sino-atrial node/SAN through the sympathetic and parasympathetic nerves/systems.	**8**	

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Human Biology

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.

(c) Do not award half marks.

(d) Where a candidate makes an error in the first part of a question, award marks for subsequent answers that are correct with regard to this original error. Do not penalise candidates more than once for the same error.

(e) Unless a numerical question specifically requires evidence of working to be shown, award full marks for a correct final answer (including units, if appropriate) on its own.

(f) Candidates should not use bulleted lists to answer extended-response questions. They must respond to the 'command' word as appropriate and provide extended answers to communicate fully their knowledge and understanding. Candidate responses in the form of bulleted lists may not be able to access the full range of available marks.

(g) In the detailed marking instructions, if a word is **underlined** then it is essential; if a word is **(bracketed)** then it is not essential.

(h) In the detailed marking instructions, words separated by / are **alternatives**.

(i) A correct response can be negated if the candidate includes:
- an extra, incorrect, response
- additional information that contradicts the correct response

(j) Where the candidate is instructed to choose one question to answer but instead answers two questions, mark both responses and award the higher mark.

(k) Unless otherwise required by the question, the use of abbreviations (for example DNA, ATP) or chemical formulae (for example CO_2, H_20) are acceptable alternatives to naming.

(l) If a numerical answer is required and units are not given in the stem of the question or in the answer space, candidates must supply the units to gain the mark. If units are required on more than one occasion, do not penalise candidates repeatedly.

(m) If incorrect spelling is given:
- If the correct word is recognisable then award the mark.
- If the word can easily be confused with another biological term then **do not** award the mark, for example glucagon and glycogen.

(n) Presentation of data:

- If a candidate provides two graphs, in response to one question, mark both and award the higher mark.
- If a question asks for a particular type of graph/chart and the candidate gives the wrong type, do not award full marks. Candidates cannot achieve the plot mark but **may** be able to achieve the mark for scale and label. If the x and y data are transposed, then do not award the scale and label mark.
- If the graph uses less than 50% of the axes then do not award the scale and label mark.
- If 0 is plotted when no data for this is given, then do not award the plot mark — candidates should only plot the data given.

(o) Only award marks for a valid response to the question asked. For example, in response to questions that ask candidates to:

- **identify, name, give** or **state**, they need only answer or present in brief form
- **describe**, they must provide a statement as opposed to simply one word
- **explain**, they must provide a reason for the information given
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between topics being examined
- **calculate**, they must determine a number from given facts, figures or information
- **predict**, they must indicate what may happen based on available information
- **suggest**, they must apply their knowledge and understanding to a new situation

2026 Higher Human Biology Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	C	9	10	**79**	2	0	0
2	C	6	12	**76**	6	0	0
3	A	**87**	10	2	1	0	0
4	C	11	6	**75**	8	0	0
5	B	14	**64**	16	7	0	0
6	A	**75**	2	8	16	0	0
7	D	3	18	10	**69**	0	0
8	B	13	**51**	18	19	0	0
9	C	7	8	**75**	10	0	0
10	D	9	10	4	**77**	0	0
11	A	**72**	8	11	8	0	0
12	C	2	2	**83**	12	0	0
13	A	**42**	40	10	7	1	0
14	D	5	6	30	**58**	0	0
15	B	12	**73**	7	8	0	0
16	B	8	**89**	2	1	0	0
17	C	3	5	**87**	4	0	0
18	B	4	**92**	3	1	0	0
19	B	8	**81**	9	2	0	0
20	D	16	3	2	**78**	0	0
21	D	11	19	15	**54**	0	0
22	B	3	**84**	1	11	0	0
23	D	15	16	10	**59**	0	0
24	C	6	9	**75**	10	0	0
25	A	**55**	8	5	32	0	0

2026 Higher Human Biology Section 2 Marks Data Analysis

2026 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	1	0.8	83.1%	36	1.8%
Q1aii	1	0.6	57.9%	38	0.7%
Q1aiii	1	0.2	17.8%	35	10.5%
Q1bi	1	0.8	75.4%	46	1.7%
Q1bii	2	1.2	60.5%	43	0.4%
Q1biii	1	0.4	37.9%	16	0.6%
Q2ai	1	0.6	56.3%	39	1.5%
Q2aii	1	0.2	17.6%	35	5.1%
Q2aiii	1	0.9	88.3%	42	0.9%
Q2b	2	0.6	28.1%	55	8.8%
Q2c	3	1.2	39.8%	65	5.4%
Q3a	1	0.7	72.0%	44	1.1%
Q3bi	1	0.6	62.8%	39	2.3%
Q3bii	1	0.4	41.2%	43	4.3%
Q3c	1	0.3	26.1%	46	5.5%
Q3d	1	0.9	92.3%	20	1.3%
Q4a	1	0.8	79.8%	48	1.2%
Q4b	1	0.3	25.0%	44	5.5%
Q4c	2	1	51.4%	57	4.0%
Q4d	2	1.3	65.0%	32	5.7%
Q4ei	1	0.5	54.2%	40	3.3%
Q4eii	1	0.8	76.4%	33	1.8%
Q5ai	1	0.6	60.7%	32	0.4%
Q5aii	1	0.7	66.9%	38	1.4%
Q5bi	1	0.5	45.5%	42	2.3%
Q5bii	2	0.3	15.4%	42	12.5%
Q5biii	1	0.6	60.7%	48	5.7%
Q6ai	1	0.6	62.4%	46	0.5%
Q6aii	2	1.7	83.7%	38	0.7%
Q6bi	2	1.6	79.2%	49	1.6%
Q6bii	2	0.8	40.2%	63	4.5%
Q6biii	1	0.7	67.3%	54	2.4%
Q7a	1	0.7	71.6%	34	3.3%
Q7bi	1	0.3	33.4%	44	3.5%
Q7bii	1	0.7	66.3%	43	0.8%
Q7ci	1	0.2	20.7%	27	1.7%
Q7cii	1	0.3	26.4%	36	1.7%
Q7d	1	0.3	30.4%	37	2.2%
Q8ai	1	0.8	78.3%	32	0.4%
Q8aii	2	1.3	65.3%	28	0.6%
Q8bi	2	1.7	84.4%	31	1.4%
Q8bii	1	0.4	39.1%	39	1.2%
Q8biii	1	0.6	58.4%	26	1.8%
Q8c	1	0.6	56.1%	46	1.9%
Q9ai	1	0.5	48.1%	60	9.9%
Q9aii	1	0.7	71.7%	55	4.2%
Q9bi	2	1	52.2%	65	10.1%
Q9bii	1	0.2	19.5%	44	18.1%
Q9biii	2	1	51.8%	64	7.3%
Q9ci	2	0.5	24.9%	54	7.8%
Q9cii	1	0.6	58.5%	57	12.7%
Q10a	1	0.2	24.6%	49	5.6%
Q10bi	1	0.6	64.4%	54	2.2%
Q10bii	2	0.9	44.2%	41	4.2%
Q11ai	1	0.8	75.6%	50	1.9%
Q11aii	1	0.3	27.4%	44	3.5%
Q11b	1	0.9	91.7%	35	0.7%
Q11c	2	0.9	45.7%	43	4.7%
Q12ai	1	0.3	33.5%	38	4.2%
Q12aii	1	0.6	56.4%	44	1.7%
Q12aiii	1	0.7	68.7%	42	3.3%
Q12aiv	1	0.4	37.1%	25	4.8%
Q12av	1	0.2	15.6%	34	6.9%
Q12bi	1	0.6	55.2%	52	8.2%
Q12bii	1	0.7	66.3%	54	6.5%
Q13ai	1	0.7	74.3%	21	2.8%
Q13aii	1	0.8	76.7%	46	3.8%
Q13b	1	0.7	67.9%	54	5.7%
Q13c	2	0.6	32.0%	60	12.2%
Q14A	8	3.8	47.2%	79	6.6%
Q14B	8	4.7	58.4%	70	0.0%
Paper	95	50.5	53.1%		

2026 Higher Human Biology Section 2 Frequency of Marks Analysis

2026 Item	0	1	2	3	4	5	6	7	8
Q1ai	17	83							
Q1aii	42	58							
Q1aiii	82	18							
Q1bi	25	75							
Q1bii	7	65	28						
Q1biii	62	38							
Q2ai	44	56							
Q2aii	82	18							
Q2aiii	12	88							
Q2b	58	28	14						
Q2c	36	23	27	14					
Q3a	28	72							
Q3bi	37	63							
Q3bii	59	41							
Q3c	74	26							
Q3d	8	92							
Q4a	20	80							
Q4b	75	25							
Q4c	17	63	20						
Q4d	24	22	54						
Q4ei	46	54							
Q4eii	24	76							
Q5ai	39	61							
Q5aii	33	67							
Q5bi	55	45							
Q5bii	75	19	6						
Q5biii	39	61							
Q6ai	38	62							
Q6aii	5	22	73						
Q6bi	16	10	74						
Q6bii	44	31	25						
Q6biii	33	67							
Q7a	28	72							
Q7bi	67	33							
Q7bii	34	66							
Q7ci	79	21							
Q7cii	74	26							
Q7d	70	30							
Q8ai	22	78							
Q8aii	8	53	39						
Q8bi	7	17	76						
Q8bii	61	39							
Q8biii	42	58							
Q8c	44	56							
Q9ai	52	48							
Q9aii	28	72							
Q9bi	36	24	40						
Q9bii	80	20							
Q9biii	29	39	32						
Q9ci	59	32	9						
Q9cii	41	59							
Q10a	75	25							
Q10bi	36	64							
Q10bii	42	28	30						
Q11ai	24	76							
Q11aii	73	27							
Q11b	8	92							
Q11c	24	60	16						
Q12ai	66	34							
Q12aii	44	56							
Q12aiii	31	69							
Q12aiv	63	37							
Q12av	84	16							
Q12bi	45	55							
Q12bii	34	66							
Q13ai	26	74							
Q13aii	23	77							
Q13b	32	68							
Q13c	51	34	15						
Q14A	19	10	10	10	9	10	9	8	15
Q14B	4	10	9	11	13	14	13	12	16

2026 Higher Human Biology Section 3 Assignment Marks Analysis

Item	Max Mark	Average Mark	% Average Mark	Discrimination
1	1	0.9	87.2%	25
2	4	2.4	59.0%	45
3a	1	0.5	52.6%	30
3b	1	1	98.0%	9
3c	1	0.6	55.7%	33
3d	1	0.7	69.6%	26
3e	1	0.5	51.8%	36
4a	1	1	98.6%	15
4b	1	0.8	84.8%	22
4c	1	0.9	91.3%	25
4d	1	0.8	79.7%	26
5	1	0.4	36.2%	43
6	1	0.4	44.8%	33
7	3	1.4	46.8%	50
8	1	0.8	78.9%	18
Paper	20	13.1	65.3%	

Item	Max Mark	0	1	2	3	4
1	1	13	87			
2	4	11	18	23	21	27
3a	1	47	53			
3b	1	2	98			
3c	1	44	56			
3d	1	30	70			
3e	1	48	52			
4a	1	1	99			
4b	1	15	85			
4c	1	9	91			
4d	1	20	80			
5	1	64	36			
6	1	55	45			
7	3	24	30	26	20	
8	1	21	79			
Paper	20					

Item	Max Mark	Item Description
1	1	An aim that describes clearly the purpose of the investigation
2	4	Underlying biology
3a	1	A brief summary of the approach used to collect experimental data.
3b	1	Sufficient raw data from the candidate's experiment.
3c	1	Data, including mean/average values, presented in a correctly produced table(s)
3d	1	Data relevant to the aim from an internet/literature source
3e	1	A citation and reference for a source of the internet/literature data
4a	1	An appropriate format from the options of bar graph or line graph.
4b	1	The axes of the graph have suitable scales.
4c	1	The axes of the graph have suitable labels and units
4d	1	Accurately plotted data points and a line (line graph) or clear bar tops (bar graph)
5	1	A valid comparison or an appropriate calculation linked to the aim of the investigation.
6	1	A valid conclusion that relates to the aim and is supported by all the data in the report.
7	3	Evaluation of the investigation
8	1	A clear and concise report, with an informative title.

National Qualifications 2025

2025 Human Biology

Higher Paper 1

Question Paper Finalised Marking Instructions

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Marking instructions for each question

Question	Response	Mark
1.	A	**1**
2.	B	**1**
3.	A	**1**
4.	D	**1**
5.	B	**1**
6.	C	**1**
7.	C	**1**
8.	D	**1**
9.	B	**1**
10.	B	**1**
11.	A	**1**
12.	D	**1**
13.	C	**1**
14.	C	**1**
15.	D	**1**
16.	B	**1**
17.	D	**1**
18.	B	**1**
19.	C	**1**
20.	A	**1**
21.	B	**1**
22.	D	**1**
23.	A	**1**
24.	C	**1**
25.	A	**1**

[END OF MARKING INSTRUCTIONS]

National
Qualifications
2025

2025 Human Biology

Higher – Paper 2

Question Paper Finalised Marking Instructions

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Marking instructions for each question

Question			Expected response	Max mark	Additional guidance
1.	(a)	(i)	Mitosis	**1**	
		(ii)	Testes/seminiferous tubules/ovaries	**1**	
	(b)		(The cells) switch on/express/ Activate. Specific genes/ genes for that cell/corneal cells. **OR** (The cells) produce/express specific proteins/proteins characteristic for that cell/corneal cells.	**1**	
	(c)	(i)	As age increases (from 0-17) to 70-79, the number of donors increases from 5 to 700/by 695. **(1)** then decreases to 394 (at ages 80-99)/by 306. **(1)**	**2**	Award 1 mark for as age increases to 70-79 the number of donors increases and above 79 it decreases.
		(ii)	There are fewer/not enough donors for the recipients. **OR** The patient would not need to wait to find a donor.	**1**	
	(d)		(Used as model cells) to study disease development. **OR** (Used as model cells) for drug testing. **OR** Provide information on cell processes/cell growth/ differentiation/gene regulation.	**1**	If candidate provides more than one answer, all answers must be correct to award the mark.

Question			Expected response	Max mark	Additional guidance
2.	(a)		Cytosine	**1**	
	(b)		DNA has (two) complementary strands. **OR** DNA has a different order of bases/sequences on each strand. **OR** The strands have a different order of bases/sequences. **(1)** The primers/they bind/attach/are added/are joined/are complementary to each strand/different strands/the 3′ end. **(1)**	**2**	
	(c)		50 – 65	**1**	Accept any temperature within this range.

Question			Expected response	Max mark	Additional guidance
3.	(a)	(i)	Sex linked	**1**	
		(ii)	2100	**1**	
	(b)	(i)	Inversion	**1**	
		(ii)	Coding region (of a gene/DNA/ mRNA). **OR** Codes for amino acids/protein.	**1**	Accept it forms/makes up the mature mRNA/transcript.
		(iii)	(The exons are in the wrong order so) a different sequence/order of amino acids. **(1)** The protein shape/structure is changed. **OR** The protein does not fold correctly. **OR** The bonds do not form (between the amino acids) correctly. **(1)**	**2**	Do not accept peptide bonds.
	(c)		Different mature mRNA transcripts are produced. **OR** Different exons are retained/ removed.	**1**	

Question			Expected response	Max mark	Additional guidance
4.	(a)		It has a large surface area.	**1**	
	(b)		Cytoplasm	**1**	
	(c)		They release/provide energy to pump/move hydrogen (ions) across/through the membrane. **OR** Join with hydrogen (ions) and oxygen to form water.	**1**	Do not accept energy is produced but does not negate second answer.
	(d)		ATP synthase	**1**	

Question			Expected response	Max mark	Additional guidance
5.	(a)		catabolic	**1**	
	(b)	(i)	1. temperature of solutions 2. pH of solutions 3. concentration of catalase/enzyme/copper nitrate/ inhibitor 4. source of catalase 5. size/thickness/type/weight/ mass/diameter of disc 6. same size/diameter/depth of beaker 7. same person timing **OR** same timer/stopwatch **Any 2**	**2**	If the candidate lists 3 or more variables and at least 1 is correct they score 1 mark for the question. If all answers are correct they score 2.
		(ii)	To show/compare the rate of the reaction/catalase activity/results without inhibition. **OR** To show/compare the effect of the inhibitor. **OR** To show copper nitrate is inhibiting catalase.	**1**	

<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>5.</td><td>(c)</td><td></td><td>Axes have correct scales and labels. (1)

Points correctly plotted and line drawn. (1)

<table>
<tr><th rowspan="2">Concentration of hydrogen peroxide (%)</th><th>Time for disc to float to surface (seconds)</th></tr>
<tr><th>Catalase + copper nitrate solution</th></tr>
<tr><td>0.5</td><td>192</td></tr>
<tr><td>1.0</td><td>136</td></tr>
<tr><td>2.0</td><td>72</td></tr>
<tr><td>3.0</td><td>52</td></tr>
<tr><td>5.0</td><td>28</td></tr>
</table>
</td><td>2</td><td>No key required.

Major gridline tick marks should be increments of 0.5 on x-axis and increments of 20 on y-axis.

Ignore any extension beyond 5% hydrogen peroxide.</td></tr>
<tr><td></td><td>(d)</td><td></td><td>As substrate concentration increases, inhibition (of catalase by copper nitrate) decreases.</td><td>1</td><td>Accept hydrogen peroxide for substrate.</td></tr>
<tr><td></td><td>(e)</td><td></td><td>28</td><td>1</td><td></td></tr>
</table>

Question			Expected response	Max mark	Additional guidance
6.	(a)		Low and obese	**1**	
	(b)	(i)	27/26.6/26.64	**1**	
		(ii)	18	**1**	
		(iii)	**Treatment:** Intra-cytoplasmic sperm injection/ICSI. **(1)** **Description:** Head of sperm is (drawn into a needle and) injected/inserted directly into the egg. **(1)** **OR** **Treatment:** Artificial insemination/AI. **(1)** **Description:** Several samples of semen/sperm are collected. **OR** donor semen/sperm is used **and** injected into the uterus/female reproductive tract. **(1)**	**2**	

Question			Expected response	Max mark	Additional guidance
7.	(a)	(i)	Aorta	**1**	
		(ii)	The atrio-ventricular/AV valves are open (and the semi-lunar valves are closed).	**1**	
	(b)		Impulses/they (from the SAN) travel to/are detected by the AVN. **OR** Impulses/they travel from/are produced by the AVN. **(1)** Impulses/they (from the AVN) travel to the fibres/the central wall/the septum/ventricles. **(1)**	**2**	
	(c)		50	**1**	
	(d)		(The medulla) regulates/controls (the rate of) the SAN/pacemaker. **OR** (The medulla) regulates/controls/activates the autonomic nervous system/ANS/ parasympathetic nerves/system. **(1)** The parasympathetic nerves/system release acetylcholine. **(1)**	**2**	

Question			Expected response	Max mark	Additional guidance
8.	(a)	(i)	(A build-up of) fatty material/ cholesterol/fibrous material/ calcium. **(1)** Occurs beneath the endothelium. **(1)**	**2**	
		(ii)	The lumen becomes wider. **OR** The atheroma is pushed back/compressed/held back.	**1**	
	(b)		Thrombus/clot breaks loose/forms an embolus and travels to/blocks the coronary artery. **(1)** This/the embolus/thrombus/clot restricts blood flow to the (heart) muscle/tissue/cells. **OR** This/the embolus/thrombus/clot deprives the (heart) cells/tissue/ muscle of oxygen. **(1)**	**2**	
	(c)	(i)	800	**1**	
		(ii)	To allow a valid/fair comparison between countries/populations. **OR** As the countries/they have different population (sizes/numbers).	**1**	

Question			Expected response	Max mark	Additional guidance
9.	(a)	(i)	Cell membrane component. **OR** (Needed to) make/produce sex hormones/testosterone/oestrogen/ Progesterone.	**1**	
		(ii)	Liver	**1**	
	(b)	(i)	Statins/competitive inhibitors/they bind to/block the active site and prevent the substrate from binding/cholesterol being produced. **OR** Statins/competitive inhibitors/they compete with the substrate for the active site and prevent the substrate/it from binding/cholesterol being produced.	**1**	
		(ii)	1. both sexes are affected equally 2. can occur in each generation/does not skip a generation 3. individuals with FH have at least one affected parent 4. a homozygous dominant parent can only have an affected child 5. two affected/heterozygous parents can have an unaffected child 6. two unaffected/homozygous recessive parents can only have unaffected children **OR** cannot have affected children **Any 2**	**2**	If the candidate lists 3 or more ways and at least 1 is correct they score 1 mark for the question. If all answers are correct, they score 2.

Question			Expected response	Max mark	Additional guidance
10.	(a)		X – peripheral (nervous system)/PNS Y – somatic (nervous system)/SNS	**2**	
	(b)		Motor	**1**	
	(c)		1. glial cells produce the myelin (sheath) 2. myelin (sheath) increases the speed of impulse (transmission)/ insulates the axon 3. myelination/it continues from birth to adolescence/occurs in the early years of life **OR** myelination/it is incomplete at birth/in early years of life/until adolescence 4. myelination/it increases a child's coordination/ability to walk/ ability to respond **OR** coordination improves as myelination occurs/ progresses/continues **Any 3**	**3**	

Question			Expected response	Max mark	Additional guidance
11.	(a)	(i)	Encoding	**1**	
		(ii)	1. rehearsal 2. organisation 3. elaboration **Any 2**	**2**	
	(b)		Cerebral cortex/cerebrum	**1**	
	(c)	(i)	1. words at the start can be rehearsed/repeated and transferred to long-term memory **OR** words at the start have been encoded in long-term memory 2. words in the middle of the list are displaced from short-term memory 3. words at the end of the list are still in/have not been displaced from the short-term memory **Any 2**	**2**	Accept STM/LTM
		(ii)	Repeat with another/a different group (of participants). **OR** Repeat with another/a different list of words. **OR** Repeat the investigation and calculate an average.	**1**	

Question			Expected response	Max mark	Additional guidance
12.	(a)		(Induces) feelings of pleasure. **OR** Reinforces particular behaviour.	**1**	
	(b)		Enzyme degradation **OR** Reuptake	**1**	
	(c)	(i)	(Drug) tolerance **OR** Decrease in the number/sensitivity of receptors. **OR** Desensitisation (of receptors).	**1**	
		(ii)	Double-blind **(1)** Prevents/reduces bias (of interpretation of results). **(1)** **OR** Placebo(-controlled) **(1)** To ensure valid comparisons/results/ trial/investigation. **(1)**	**2**	
		(iii)	Suitable/large group/sample size.	**1**	

Question			Expected response	Max mark	Additional guidance
13.	(a)	(i)	(Antigen/surface/membrane) receptor.	**1**	
		(ii)	(Lymphocyte R) produces/releases antibodies.	**1**	
	(b)		(Phagocytes) engulf and digest/destroy the (antigen and antibody) complex/(inactivated) pathogen/it.	**1**	
	(c)		1. (memory) cells/they rapidly/quickly divide **OR** (memory) cells/they form clones/ identical lymphocytes/a clonal population 2. antibody production is rapid/ greater/faster 3. antibodies destroy/inactivate the pathogen (before the individual shows symptoms) **Any 2**	**2**	
	(d)		They release proteins (into the cell) causing cell death/apoptosis. **OR** They cause the production of self-destructive enzymes causing cell death/apoptosis.	**1**	Use of pathogen rather than infected cell negates.

Question			Expected response	Max mark	Additional guidance
14.	(a)	(i)	75	**1**	
		(ii)	96/96.2/96.23	**1**	
		(iii)	From 1980-1990 vaccination coverage increases from 20% to 75%/by 55%. **(1)** From 1990-2000 it remains constant/at 75%. **(1)**	**2**	Answer must use % for at least one figure. Award one mark if candidate states it increases to 75% and then it levels off/remains constant if no other marks are awarded.
	(b)	(i)	100 : 3	**1**	
		(ii)	1 700 000/1.7 million	**1**	
		(iii)	Poverty/malnutrition . **OR** Rejection/refusal of the vaccine/ lack of education. **OR** Difficulty in population accessing the vaccine/geographical remoteness/ access to doctors.	**1**	

Question			Expected response	Max mark	Additional guidance
15.	**A**		1. follicle stimulating hormone/FSH is released from the pituitary gland AND stimulates the maturation/development of a follicle (in the ovary) 2. the follicle/ovarian tissue secretes oestrogen 3. oestrogen stimulates proliferation/growth/repair/ regeneration/thickening/ development of the endometrium/uterus lining 4. oestrogen affects the consistency of cervical mucus/makes cervical mucus thinner 5. high/peak levels of oestrogen trigger a surge in/release of LH (from the pituitary gland) 6. (surge of) LH causes/triggers ovulation/release of an egg from the follicle 7. this/ovulation occurs midway through in the menstrual cycle/marks the start of the luteal phase/end of the follicular phase 8. the follicle develops/turns into/becomes the corpus luteum which secretes progesterone 9. progesterone promotes/causes development/vascularisation of the endometrium/uterus lining **OR** progesterone maintains/thickens/ increases the endometrium/uterus lining 10. progesterone/oestrogen have a negative feedback effect on the pituitary gland **OR** progesterone/oestrogen inhibit the secretion/release/production of FSH/LH (from the pituitary gland) 11. drop in/lack of FSH prevents further follicles from developing/ maturing 12. drop in/lack of LH leads to degeneration of the corpus luteum and a drop/reduction in progesterone 13. this/low levels/concentrations of progesterone lead to menstruation	**9**	

Question			Expected response	Max mark	Additional guidance
15.	**B**		1. women show cyclical fertility and men show continuous fertility/are continuously fertile 2. women are only fertile for a few days (during each menstrual cycle) 3. men continually produce sperm (in their testes) 4. fertile period is identified by a woman's body temperature rising (by around 0.5°C after ovulation) **OR** fertile period is identified by cervical mucus becoming thin/ watery 5. physical/barrier methods (of contraception) prevent sperm reaching the egg 6. **two** examples of physical/barrier methods 7. chemical/hormonal methods (of contraception) include the oral contraceptive pill, progesterone-only pill/mini pill, and emergency contraceptive/ morning after pill (any two) 8. the oral contraceptive pill contains (synthetic) oestrogen and progesterone 9. the oral contraceptive pill mimics negative feedback effect (of oestrogen and progesterone) and prevents the release of FSH/LH 10. the progesterone only pill/mini pill causes thickening of the cervical mucus 11. emergency contraceptive pills/morning after pill prevent/ delay ovulation 12. emergency contraceptive pill/morning after pill can be taken up to 72-120 hours after (unprotected) sex	**9**	

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Human Biology

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.

(c) Do not award half marks.

(d) Where a candidate makes an error in the first part of a question, award marks for subsequent answers that are correct with regard to this original error. Do not penalise candidates more than once for the same error.

(e) Unless a numerical question specifically requires evidence of working to be shown, award full marks for a correct final answer (including units, if appropriate) on its own.

(f) Candidates should not use bulleted lists to answer extended-response questions. They must respond to the 'command' word as appropriate and provide extended answers to communicate fully their knowledge and understanding. Candidate responses in the form of bulleted lists may not be able to access the full range of available marks.

(g) In the detailed marking instructions, if a word is **underlined** then it is essential; if a word is **(bracketed)** then it is not essential.

(h) In the detailed marking instructions, words separated by / are **alternatives**.

(i) A correct response can be negated if the candidate includes:
- an extra, incorrect, response
- additional information that contradicts the correct response

(j) Where the candidate is instructed to choose one question to answer but instead answers two questions, mark both responses and award the higher mark.

(k) Unless otherwise required by the question, the use of abbreviations (for example DNA, ATP) or chemical formulae (for example CO_2, H_20) are acceptable alternatives to naming.

(l) If a numerical answer is required and units are not given in the stem of the question or in the answer space, candidates must supply the units to gain the mark. If units are required on more than one occasion, do not penalise candidates repeatedly.

(m) If incorrect spelling is given:
- If the correct word is recognisable then award the mark.
- If the word can easily be confused with another biological term then **do not** award the mark, for example glucagon and glycogen.

(n) Presentation of data:

- If a candidate provides two graphs, in response to one question, mark both and award the higher mark.
- If a question asks for a particular type of graph/chart and the candidate gives the wrong type, do not award full marks. Candidates cannot achieve the plot mark but **may** be able to achieve the mark for scale and label. If the x and y data are transposed, then do not award the scale and label mark.
- If the graph uses less than 50% of the axes then do not award the scale and label mark.
- If 0 is plotted when no data for this is given, then do not award the plot mark — candidates should only plot the data given.

(o) Only award marks for a valid response to the question asked. For example, in response to questions that ask candidates to:

- **identify, name, give** or **state**, they need only answer or present in brief form
- **describe**, they must provide a statement as opposed to simply one word
- **explain**, they must provide a reason for the information given
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between topics being examined
- **calculate**, they must determine a number from given facts, figures or information
- **predict**, they must indicate what may happen based on available information
- **suggest**, they must apply their knowledge and understanding to a new situation

2025 Higher Human Biology Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	A	**96**	1	3	1	0	0
2	B	1	**82**	7	10	0	0
3	A	**63**	23	9	5	0	0
4	D	33	15	13	**38**	0	0
5	B	3	**65**	24	8	0	0
6	C	30	9	**45**	16	0	0
7	C	5	2	**85**	8	0	0
8	D	8	18	3	**70**	0	0
9	B	7	**51**	11	30	0	0
10	B	19	**59**	13	9	1	0
11	A	**58**	16	13	13	0	0
12	D	20	2	39	**38**	0	0
13	C	1	19	**31**	49	0	0
14	C	12	14	**67**	6	0	0
15	D	34	11	11	**44**	0	0
16	B	8	**67**	14	11	0	0
17	D	1	14	29	**55**	0	0
18	B	2	**87**	9	2	0	0
19	C	2	1	**95**	2	0	0
20	A	**88**	5	6	1	0	0
21	B	4	**90**	3	2	0	0
22	D	2	1	5	**92**	0	0
23	A	**84**	4	10	1	0	0
24	C	35	4	**59**	1	1	0
25	A	**90**	2	4	5	0	0

2025 Higher Human Biology Section 2 Marks Data Analysis

2025 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	1	0.9	94.7%	27	0.6%
Q1aii	1	0.8	79.1%	35	1.0%
Q1b	1	0.4	38.6%	48	3.6%
Q1ci	2	0.9	46.7%	42	0.4%
Q1cii	1	0.3	27.0%	21	2.9%
Q1d	1	0.6	64.3%	39	1.7%
Q2a	1	0.7	65.2%	27	0.3%
Q2b	2	0.3	16.3%	34	5.7%
Q2c	1	0.8	75.2%	45	0.5%
Q3ai	1	0.6	61.7%	48	4.5%
Q3aii	1	0.7	66.3%	33	1.7%
Q3bi	1	0.6	63.4%	52	2.3%
Q3bii	1	0.6	63.9%	52	6.1%
Q3biii	2	0.3	16.2%	46	7.2%
Q3c	1	0.3	35.0%	53	9.4%
Q4a	1	0.5	51.4%	45	11.9%
Q4b	1	0.7	69.6%	42	2.9%
Q4c	1	0.2	19.6%	45	8.7%
Q4d	1	0.5	47.8%	61	8.2%
Q5a	1	0.7	74.3%	46	6.5%
Q5bi	2	1.4	69.1%	38	0.7%
Q5bii	1	0.3	32.6%	35	3.0%
Q5c	2	1.4	68.2%	42	2.2%
Q5d	1	0.2	19.6%	48	3.4%
Q5e	1	0.4	42.7%	27	5.6%
Q6a	1	0.9	92.9%	19	0.4%
Q6bi	1	0.7	69.5%	51	3.6%
Q6bii	1	0.6	61.6%	47	2.6%
Q6biii	2	1.3	67.4%	45	2.1%
Q7ai	1	0.6	55.9%	36	1.3%
Q7aii	1	0.5	53.3%	54	5.2%
Q7b	2	0.7	33.0%	65	14.5%
Q7c	1	0.4	36.1%	52	8.2%
Q7d	2	0.7	34.5%	64	9.0%
Q8ai	2	1	47.7%	63	7.4%
Q8aii	1	0.5	47.4%	45	3.2%
Q8b	2	0.3	17.4%	54	5.5%
Q8ci	1	0.7	67.4%	35	1.6%
Q8cii	1	0.4	41.5%	50	5.5%
Q9ai	1	0.3	33.0%	56	11.7%
Q9aii	1	0.8	83.9%	39	3.8%
Q9bi	1	0.4	36.0%	44	7.1%
Q9bii	2	0.5	26.9%	49	9.6%
Q10a	2	1.6	79.9%	47	2.1%
Q10b	1	0.5	53.1%	40	5.1%
Q10c	3	1.6	54.7%	70	8.0%
Q11ai	1	0.7	67.4%	51	7.0%
Q11aii	2	1.6	79.6%	53	1.4%
Q11b	1	0.7	73.3%	34	2.1%
Q11ci	2	0.6	28.4%	40	2.2%
Q11cii	1	0.4	43.6%	31	1.8%
Q12a	1	0.6	56.9%	36	6.4%
Q12b	1	0.5	47.4%	58	14.5%
Q12ci	1	0.5	54.6%	52	7.8%
Q12cii	2	1.2	62.0%	54	5.7%
Q12ciii	1	0.3	26.0%	42	11.4%
Q13ai	1	0.6	58.6%	41	5.7%
Q13aii	1	0.7	68.1%	51	5.6%
Q13b	1	0.5	50.3%	37	5.9%
Q13c	2	0.4	18.7%	52	3.7%
Q13d	1	0.2	16.9%	41	8.7%
Q14ai	1	0.7	65.4%	44	1.3%
Q14aii	1	0.6	64.3%	42	2.4%
Q14aiii	2	1	49.2%	51	1.6%
Q14bi	1	0.6	58.0%	32	2.7%
Q14bii	1	0.8	80.3%	37	4.5%
Q14biii	1	0.8	77.2%	32	4.6%
Q15A	9	5.3	58.9%	76	4.1%
Q15B	9	2.9	32.7%	61	0.0%
Paper	95	48.4	51.0%		

2025 Higher Human Biology Section 2 Frequency of Marks Analysis

2025 Item	0	1	2	3	4	5	6	7	8	9
Q1ai	5	95								
Q1aii	21	79								
Q1b	61	39								
Q1ci	32	43	25							
Q1cii	73	27								
Q1d	36	64								
Q2a	35	65								
Q2b	74	19	7							
Q2c	25	75								
Q3ai	38	62								
Q3aii	34	66								
Q3bi	37	63								
Q3bii	36	64								
Q3biii	73	21	6							
Q3c	65	35								
Q4a	49	51								
Q4b	30	70								
Q4c	80	20								
Q4d	52	48								
Q5a	26	74								
Q5bi	13	36	51							
Q5bii	67	33								
Q5c	21	21	58							
Q5d	80	20								
Q5e	57	43								
Q6a	7	93								
Q6bi	30	70								
Q6bii	38	62								
Q6biii	9	48	44							
Q7ai	44	56								
Q7aii	47	53								
Q7b	59	17	25							
Q7c	64	36								
Q7d	52	27	21							
Q8ai	35	36	30							
Q8aii	53	47								
Q8b	71	23	6							
Q8ci	33	67								
Q8cii	58	42								
Q9ai	67	33								
Q9aii	16	84								
Q9bi	64	36								
Q9bii	60	27	13							
Q10a	12	16	72							
Q10b	47	53								
Q10c	24	19	25	32						
Q11ai	33	67								
Q11aii	11	18	70							
Q11b	27	73								
Q11ci	60	23	17							
Q11cii	56	44								
Q12a	43	57								
Q12b	53	47								
Q12ci	45	55								
Q12cii	20	35	44							
Q12ciii	74	26								
Q13ai	41	59								
Q13aii	32	68								
Q13b	50	50								
Q13c	68	28	5							
Q13d	83	17								
Q14ai	35	65								
Q14aii	36	64								
Q14aiii	40	21	39							
Q14bi	42	58								
Q14bii	20	80								
Q14biii	23	77								
Q15A	11	7	7	7	7	8	10	11	10	23
Q15B	15	16	19	14	12	9	7	4	2	3

2025 Higher Human Biology Section 3 Assignment Marks Analysis

Item	Max Mark	Average Mark	% Average Mark	Discrimination
1	1	0.8	85.0%	27
2	4	2.6	63.8%	42
3a	1	0.5	47.1%	30
3b	1	1	97.4%	9
3c	1	0.5	52.6%	35
3d	1	0.6	63.6%	23
3e	1	0.5	49.0%	35
4a	1	1	98.2%	16
4b	1	0.8	84.2%	24
4c	1	0.9	89.5%	25
4d	1	0.8	78.0%	26
5	1	0.3	30.8%	41
6	1	0.4	43.3%	30
7	3	1.3	43.2%	49
8	1	0.8	77.0%	19
Paper	20	12.8	64.0%	

Item	Max Mark	0	1	2	3	4
1	1	15	85			
2	4	9	15	21	22	33
3a	1	53	47			
3b	1	3	97			
3c	1	47	53			
3d	1	36	64			
3e	1	51	49			
4a	1	2	98			
4b	1	16	84			
4c	1	10	90			
4d	1	22	78			
5	1	69	31			
6	1	57	43			
7	3	28	31	26	16	
8	1	23	77			
Paper	20					

Item	Max Mark	Item Description
1	1	An aim that describes clearly the purpose of the investigation
2	4	Underlying biology
3a	1	A brief summary of the approach used to collect experimental data.
3b	1	Sufficient raw data from the candidate's experiment.
3c	1	Data, including mean/average values, presented in a correctly produced table(s)
3d	1	Data relevant to the aim from an internet/literature source
3e	1	A citation and reference for a source of the internet/literature data
4a	1	An appropriate format from the options of bar graph or line graph.
4b	1	The axes of the graph have suitable scales.
4c	1	The axes of the graph have suitable labels and units
4d	1	Accurately plotted data points and a line (line graph) or clear bar tops (bar graph)
5	1	A valid comparison or an appropriate calculation linked to the aim of the investigation.
6	1	A valid conclusion that relates to the aim and is supported by all the data in the report.
7	3	Evaluation of the investigation
8	1	A clear and concise report, with an informative title.

National
Qualifications
2024

2024 Human Biology

Higher – Paper 1

Question Paper Finalised Marking Instructions

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Marking instructions for each question

Question	Response	Mark
1.	A	**1**
2.	C	**1**
3.	D	**1**
4.	C	**1**
5.	C	**1**
6.	A	**1**
7.	B	**1**
8.	D	**1**
9.	A	**1**
10.	D	**1**
11.	D	**1**
12.	D	**1**
13.	C	**1**
14.	B	**1**
15.	C	**1**
16.	B	**1**
17.	C	**1**
18.	A	**1**
19.	A	**1**
20.	C	**1**
21.	A	**1**
22.	B	**1**
23.	B	**1**
24.	B	**1**
25.	D	**1**

[END OF MARKING INSTRUCTIONS]

National
Qualifications
2024

2024 Human Biology

Higher – Paper 2

Question Paper Finalised Marking Instructions

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Marking instructions for each question

Question			Expected response	Max mark	Additional guidance
1.	(a)		A T A G C A	**1**	
	(b)	(i)	Any value from 92 - 98	**1**	Accept any range between 92 - 98
		(ii)	1. allows nucleotides to be added. **OR** allows DNA to be amplified/amplification to start. **OR** provides a starting point for amplification. **(1)** 2. adds/joins (complementary) nucleotides (to a DNA strand/primer). **OR** amplifies DNA. **(1)**	**2**	Accept binds to complementary/target/specific sequence/(section of) DNA. Do not accept bases for nucleotides.
	(c)		1. Solving crimes. 2. Identifying people. 3. Diagnose genetic disorders/mutations. **Any 2 from 3**	**2**	
2.	(a)	(i)	100	**1**	
		(ii)	39/39.3/39.29	**1**	
	(b)	(i)	2679	**1**	
		(ii)	To allow a valid comparison between countries. **OR** (As the countries) have different population sizes/numbers.	**1**	Accept a comparison, e.g. England has a greater/larger population than the other countries.

Question			Expected response	Max mark	Additional guidance
3.	(a)	(i)	1. Concentration/volume of hydrogen peroxide. 2. Volume of copper sulfate/ inhibitor. 3. Concentration/mass/weight of yeast (beads)/type of yeast. **OR** Mass/volume of gel (substance). 4. Size/volume/surface area/ mass/weight/number of beads. 5. Temperature of solution/beads/ (contents of) flask. 6. pH of solutions/beads/(contents of) flask. **Any 2**	**2**	Do not accept enzyme/catalase for yeast. For points 5 & 6 accept yeast/gel (substance)/hydrogen peroxide/ copper sulfate/inhibitor.
		(ii)	The immobilised yeast/beads can be used more than once. **OR** The immobilised yeast/beads can be transferred/removed easily.	**1**	
		(iii)	To allow the beads to absorb the copper sulfate/inhibitor. **OR** To allow the copper sulfate/inhibitor to diffuse into the beads. **OR** To allow for the copper sulfate/inhibitor to react with the catalase.	**1**	

<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>3.</td><td>(b)</td><td></td><td>Axes have correct scale and labels. (1)

Points correctly plotted and line drawn. (1)

<table><tr><th>Concentration of copper sulfate (mmol/l)</th><th>Volume of oxygen collected (cm^3)</th></tr><tr><td>0</td><td>82</td></tr><tr><td>5</td><td>68</td></tr><tr><td>10</td><td>45</td></tr><tr><td>25</td><td>25</td></tr><tr><td>50</td><td>16</td></tr></table></td><td>2</td><td>Candidates cannot access the scale mark if the plotted points use less than half the graph paper.

If candidate draws a bar graph they cannot access the scale mark.</td></tr>
<tr><td></td><td>(c)</td><td></td><td>As the inhibitor/copper sulfate concentration increases, the catalase activity decreases.</td><td>1</td><td></td></tr>
</table>

Question			Expected response	Max mark	Additional guidance
4.	(a)	(i)	Y and Z	**1**	
		(ii)	Pyruvate	**1**	
		(iii)	Citrate	**1**	
	(b)		Inner membrane (in mitochondria from muscle cells) has a greater surface area/more folds/is larger/longer. **(1)** For electron transport chains/ATP synthase/hydrogen ions to be pumped across. **(1)**	**2**	
	(c)		They have many mitochondria. **OR** They have a large blood supply. **OR** They have a high concentration of myoglobin. **OR** Their major storage fuel is fat.	**1**	

Question			Expected response	Max mark	Additional guidance
5.	(a)	(i)	(FSH) stimulates development/maturation/growth of the follicle (in the ovary). **(1)** The (developing) follicle releases oestrogen. **(1)**	**2**	
		(ii)	Oestrogen stimulates proliferation/repair/thickening/ regeneration of the endometrium. **OR** Oestrogen prepares the endometrium for implantation.	**1**	
		(iii)	It stimulates ovulation/release of an egg/ovum (from the follicle/ovary).	**1**	
	(b)	(i)	14	**1**	
		(ii)	Corpus luteum	**1**	
	(c)	(i)	Progesterone (concentration) decreases (after day 25/towards the end of the cycle). **OR** Progesterone (concentration) does not remain high (after day 25/towards the end of the cycle).	**1**	
		(ii)	Treatment – IVF. **(1)** Description – eggs are mixed with sperm/fertilised in a culture dish/outside the body. **OR** Description – eggs are (surgically) removed and mixed with sperm/fertilised. **(1)**	**2**	

<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>6.</td><td>(a)</td><td></td><td>Could lead to a false positive/negative.
OR
Unnecessary (diagnostic) tests could be carried out.</td><td>1</td><td></td></tr>
<tr><td></td><td>(b)</td><td></td><td>(Amniocentesis has) a lower risk/chance of miscarriage.
OR
CVS has a higher risk/chance of miscarriage.</td><td>1</td><td>Accept amniocentesis is safer for the fetus.
OR
CVS is more dangerous for the fetus.
OR
Amniocentesis may be carried out later in the pregnancy.
OR
It was too late in the pregnancy for CVS.</td></tr>
<tr><td></td><td>(c)</td><td>(i)</td><td>Karyotype</td><td>1</td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td>Autosomes</td><td>1</td><td>Accept autosomal</td></tr>
<tr><td>7.</td><td>(a)</td><td>(i)</td><td>300</td><td>1</td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td>2 : 48 : 3</td><td>1</td><td></td></tr>
<tr><td></td><td></td><td>(iii)</td><td><table><tr><th>Part of body</th><th>Change in blood flow rate</th><th>Process controlling blood flow</th></tr><tr><td>Intestines</td><td>decrease</td><td>vasoconstriction</td></tr><tr><td>Skeletal muscles</td><td>increase</td><td>vasodilation</td></tr></table>1 mark for part of body.
1 mark for process.</td><td>2</td><td>Accept description of smooth muscle in artery contracting/relaxing for vasoconstriction/vasodilation.</td></tr>
<tr><td></td><td>(b)</td><td>(i)</td><td>150</td><td>1</td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td>A sympathetic nerve/neuron releases/secretes noradrenaline</td><td>1</td><td></td></tr>
</table>

Question			Expected response	Max mark	Additional guidance
8.	(a)	(i)	Plasma	**1**	
		(ii)	Pressure filtration	**1**	
	(b)		Blood is at a higher pressure (in A). **OR** Vasoconstriction/vasodilation occurs in A but not in D.	**1**	Accept artery/vein for A/D.
	(c)	(i)	Walls are thin/only one cell thick.	**1**	Accept they provide a large surface area (for diffusion). Accept endothelial/endothelium/epithelial/ epithelium for walls.
		(ii)	(Plasma) Protein	**1**	
	(d)	(i)	Absorbs excess tissue fluid. **OR** Carries lymph back to the circulatory system.	**1**	
		(ii)	They prevent the backflow of fluid/lymph. **OR** They make sure that the fluid/lymph only flows in one direction.	**1**	

Question			Expected response	Max mark	Additional guidance
9.	(a)		(An inflatable) cuff stops/restricts blood flow/circulation (in the artery). **OR** (An inflatable) cuff deflates (gradually) so blood flow/circulation (in artery) can resume. **(1)** (Systolic blood) pressure is measured when blood starts to flow/a pulse is detected. **(1)**		
	(b)	(i)	Blood pressure	**1**	
		(ii)	(Consuming) energy drinks increases (diastolic) blood pressure. **OR** (Consuming) energy drinks has no effect on systolic blood pressure.		
		(iii)	97	**1**	
	(c)		Hypertension	**1**	

Question			Expected response	Max mark	Additional guidance
10.	(a)	(i)	Age/gender/ethnicity/other health conditions	**1**	
		(ii)	Type 1 diabetes may cause low vitamin D (concentrations). **OR** Type 1 diabetes may have occurred before/earlier/first. **OR** The results are only from one study/small sample size. **OR** More than half/59% of the control group have low vitamin D concentrations (but not type 1 diabetes).	**1**	
		(iii)	118	**1**	
		(iv)	Repeat the study with larger groups/more individuals in each group. **OR** Repeat the study and calculate an average.	**1**	
	(b)	(i)	From January to May the average hours of sunlight increases/goes from 36 to 184/increases by 148 hours from 36. **(1)** From May to December the average hours of sunlight decreases/goes from 184 to 32 hours/decreases by 152 hours from 184. **(1)**	**2**	Award 1 mark if candidates state it increases to May and then it decreases if no other marks are awarded. Answer must indicate hours for at least one figure.
		(ii)	October to March	**1**	Accept January to March **and** October to December.

Question			Expected response	Max mark	Additional guidance
11.	(a)		Displacement/decay	**1**	
	(b)		Working memory (model)	**1**	
	(c)		STM/it has a limited capacity. **OR** 11 numbers/it exceeds capacity of STM.	**1**	Accept can only hold 5 to 9 numbers/items. Accept can only hold 7 ± 2 numbers/items. Accept any value in the range 5 to 9.
	(d)	(i)	(They could be allocated) by a computer (program)/by selecting names out of a hat.	**1**	Do not accept randomisation only. Accept any other suitable example of random allocation.
		(ii)	(Group 1 used) organisation which allows words to be transferred/ encoded (from STM) to LTM. **OR** Group 2 did not use organisation so fewer words were transferred/ encoded (from STM) to LTM.	**1**	
	(e)		Contextual cues	**1**	

Question			Expected response	Max mark	Additional guidance
12.	(a)		Synapse/synaptic cleft	**1**	
	(b)	(i)	Diverging	**1**	
		(ii)	More than one neuron/muscle/finger is affected/stimulated at the same time. **OR** Impulse/neurotransmitter is sent to many neurons/muscles/fingers giving coordinated movement.	**1**	
	(c)	(i)	Myelin (sheath)	**1**	
		(ii)	It increases the speed of impulses. **OR** It insulates the axon.	**1**	
	(d)	(i)	Fewer receptors/no receptors for the neurotransmitter to bind to. **OR** Fewer receptors/no receptors so impulses/neurotransmitters cannot cause muscle contraction. **OR** The threshold (initiating muscle contraction) cannot be reached.	**1**	Accept neurotransmitters cannot bind to the receptors.
		(ii)	T lymphocytes respond to self-antigens/antigens on own (healthy) cells. **(1)** Attacking/destroying/killing the body's own (healthy) cells. **(1)**	**2**	Accept T lymphocytes fail to recognise self-antigens/antigen on own (healthy) cells. Accept apoptosis of the body's own cells.
		(iii)	10 320	**1**	

Question			Expected response	Max mark	Additional guidance
13.	(a)		1. B lymphocytes have receptors which recognise antigens on pathogen/foreign antigens. 2. (Following recognition) they form clones/a clonal population/copies of themselves. **OR** They undergo clonal selection. 3. B lymphocytes produce antibodies. 4. Antibodies bind to antigens on the pathogen. **OR** The antigen-antibody complex can be destroyed/removed by phagocytosis. **Any 3 from 4**	**3**	
	(b)		Allergy/allergic reaction/ hypersensitive response	**1**	

Question			Expected response	Max mark	Additional guidance
14.	(a)		DNA/RNA/nucleic acid/chromosome	**1**	
	(b)		(Inactivated) pathogen toxins/dead pathogens/weakened pathogens	**1**	
	(c)	(i)	To establish their safety/ effectiveness.	**1**	
		(ii)	Neither the subjects nor the researchers know which group the subjects are in. **OR** Neither the subject nor the researchers know who has been given the placebo/vaccine.	**1**	
	(d)		Antigenic variation occurs. **(1)** Memory cells/memory lymphocytes are not effective against/do not recognise the changed/ mutated/different/new antigen. **OR** Memory cells/memory lymphocytes are not effective against/do not recognise the changed/ mutated/altered pathogen/virus. **(1)**	**2**	

Question			Expected response	Max mark	Additional guidance
15.	**A**		1. A somatic cell is any cell (in the body) other than cells involved in reproduction/germline cells. 2. Germline cells are gametes/ sperm and egg/ova **and** (stem) cells that divide/differentiate to form/produce gametes. 3. Somatic cells divide by mitosis to form (more) somatic cells. 4. Mitosis of germline cells produces (more) germline cells. 5. Meiosis of germline cell produces gametes. 6. Somatic cells/germline stem cells are diploid **and** gametes are haploid. 7. Meiosis has two stages/divisions. 8. First stage/division separates homologous/matching/pairs of chromosomes. 9. Second stage/division separates chromatids. 10. Somatic cells/germline cells divide by mitosis to maintain the (diploid) chromosome number/ complement. 11. Diploid cells have 23 pairs of homologous/matching chromosomes. **OR** Haploid cells/gametes contain 23 single chromosomes/one set of 23 chromosomes. **8 marks from 11**	**8**	3.Accept identical daughter cells for somatic cells. a. Award mark for germline cells can divide by mitosis and meiosis if candidate has **not** been awarded both points 4 and 5.

Question			Expected response	Max mark	Additional guidance
15.	**B**		1. mRNA/primary transcript is transcribed/produced from DNA in the nucleus. 2. RNA polymerase unwinds the DNA **and** breaks the (hydrogen) bonds between the bases/strands. **OR** RNA polymerase unwinds the DNA **and** unzips the strands. 3. RNA polymerase adds complementary RNA nucleotides. **OR** RNA polymerase allows complementary base pairing to occur. 4. RNA contains the (base) uracil instead of thymine. **OR** Uracil in RNA is complementary to/pairs with adenine. 5. The primary mRNA/transcript undergoes (RNA) splicing. 6. Primary mRNA/transcript has introns and exons. 7. Introns are non-coding regions, exons are coding. 8. Introns are removed **and** exons are joined/retained to form mature mRNA/transcript. 9. Different mature mRNA/mature transcripts/proteins can be produced as a result of alternative (RNA) splicing. 10. Different mature mRNA/mature transcripts/proteins can be produced depending on which exons are retained. 11. The order of exons is not changed. **8 marks from 11**	**8**	

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Human Biology

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.

(c) Do not award half marks.

(d) Where a candidate makes an error in the first part of a question, award marks for subsequent answers that are correct with regard to this original error. Do not penalise candidates more than once for the same error.

(e) Unless a numerical question specifically requires evidence of working to be shown, award full marks for a correct final answer (including units, if appropriate) on its own.

(f) Candidates should not use bulleted lists to answer extended-response questions. They must respond to the 'command' word as appropriate and provide extended answers to communicate fully their knowledge and understanding. Candidate responses in the form of bulleted lists may not be able to access the full range of available marks.

(g) In the detailed marking instructions, if a word is **underlined** then it is essential; if a word is **(bracketed)** then it is not essential.

(h) In the detailed marking instructions, words separated by / are **alternatives**.

(i) A correct response can be negated if the candidate includes:
- an extra, incorrect, response
- additional information that contradicts the correct response

(j) Where the candidate is instructed to choose one question to answer but instead answers two questions, mark both responses and award the higher mark.

(k) Unless otherwise required by the question, the use of abbreviations (for example DNA, ATP) or chemical formulae (for example CO_2, H_20) are acceptable alternatives to naming.

(l) If a numerical answer is required and units are not given in the stem of the question or in the answer space, candidates must supply the units to gain the mark. If units are required on more than one occasion, do not penalise candidates repeatedly.

(m) If incorrect spelling is given:
- If the correct word is recognisable then award the mark.
- If the word can easily be confused with another biological term then **do not** award the mark, for example glucagon and glycogen.

(n) Presentation of data:

- If a candidate provides two graphs, in response to one question, mark both and award the higher mark.
- If a question asks for a particular type of graph/chart and the candidate gives the wrong type, do not award full marks. Candidates cannot achieve the plot mark but **may** be able to achieve the mark for scale and label. If the x and y data are transposed, then do not award the scale and label mark.
- If the graph uses less than 50% of the axes then do not award the scale and label mark.
- If 0 is plotted when no data for this is given, then do not award the plot mark — candidates should only plot the data given.

(o) Only award marks for a valid response to the question asked. For example, in response to questions that ask candidates to:

- **identify, name, give** or **state**, they need only answer or present in brief form
- **describe**, they must provide a statement as opposed to simply one word
- **explain**, they must provide a reason for the information given
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between topics being examined
- **calculate**, they must determine a number from given facts, figures or information
- **predict**, they must indicate what may happen based on available information
- **suggest**, they must apply their knowledge and understanding to a new situation

2024 Higher Human Biology Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	A	**78**	13	3	6	0	0
2	C	7	13	**77**	3	0	0
3	D	11	2	7	**80**	0	0
4	C	12	14	**54**	20	0	0
5	C	7	5	**84**	4	0	0
6	A	**75**	5	16	4	0	0
7	B	9	**76**	8	7	0	0
8	D	1	4	1	**94**	0	0
9	A	**52**	14	24	10	0	0
10	D	10	25	14	**50**	0	0
11	D	3	13	14	**70**	0	0
12	D	20	8	39	**33**	0	0
13	C	13	5	**75**	7	0	0
14	B	2	**64**	6	28	0	0
15	C	27	13	**42**	17	0	0
16	B	6	**56**	33	5	0	0
17	C	11	12	**60**	17	0	0
18	A	**82**	9	4	5	0	0
19	A	**87**	5	4	4	0	0
20	C	9	4	**72**	15	0	0
21	A	**33**	6	44	16	0	0
22	B	13	**78**	5	4	0	0
23	B	8	**56**	24	12	0	0
24	B	15	**61**	12	12	0	0
25	D	11	6	7	**76**	0	0

2024 Higher Human Biology Section 2 Marks Data Analysis

2024 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1a	1	0.9	93.8%	17	0.3%
Q1bi	1	0.7	66.8%	40	0.3%
Q1bii	2	0.9	44.0%	58	3.0%
Q1c	2	1.4	70.3%	52	3.9%
Q2ai	1	0.8	84.3%	24	0.1%
Q2aii	1	0.3	34.1%	40	1.1%
Q2bi	1	0.7	73.3%	31	2.0%
Q2bii	1	0.1	14.9%	32	5.3%
Q3ai	2	1.3	67.2%	36	0.3%
Q3aii	1	0.1	6.6%	11	4.7%
Q3aiii	1	0.3	31.3%	26	5.5%
Q3b	2	1.6	80.1%	34	0.8%
Q3c	1	0.3	27.4%	50	1.2%
Q4ai	1	0.4	41.2%	49	0.4%
Q4aii	1	0.6	63.4%	50	3.6%
Q4aiii	1	0.6	64.4%	60	6.4%
Q4b	2	0.7	33.0%	59	4.5%
Q4c	1	0.7	65.6%	50	5.4%
Q5ai	2	0.8	42.2%	66	8.1%
Q5aii	1	0.5	48.1%	51	5.3%
Q5aiii	1	0.6	62.7%	47	7.0%
Q5bi	1	0.7	65.9%	36	3.5%
Q5bii	1	0.7	65.2%	59	4.3%
Q5ci	1	0.6	60.7%	49	2.5%
Q5cii	2	1.2	61.0%	40	4.2%
Q6a	1	0.5	49.2%	45	4.5%
Q6b	1	0.6	60.5%	48	7.1%
Q6ci	1	0.6	56.2%	59	13.6%
Q6cii	1	0.3	32.6%	55	8.8%
Q7ai	1	0.5	51.8%	45	1.2%
Q7aii	1	0.6	64.3%	32	1.7%
Q7aiii	2	1.1	53.2%	50	5.4%
Q7bi	1	0.5	49.8%	49	8.2%
Q7bii	1	0.3	29.8%	55	9.3%
Q8ai	1	0.6	59.2%	29	4.0%
Q8aii	1	0.5	50.0%	64	16.8%
Q8b	1	0.4	38.7%	32	3.2%
Q8ci	1	0.5	46.4%	37	7.1%
Q8cii	1	0.5	52.0%	54	6.7%
Q8di	1	0.3	29.8%	56	14.6%
Q8dii	1	0.2	20.6%	36	5.0%
Q9a	2	0.5	23.7%	47	12.1%
Q9bi	1	0.3	32.9%	36	2.4%
Q9bii	1	0.5	49.8%	28	1.3%
Q9biii	1	0.5	51.0%	50	7.7%
Q9c	1	0.4	39.1%	52	7.6%
Q10ai	1	0.6	57.2%	22	2.3%
Q10aii	1	0.2	19.1%	23	5.7%
Q10aiii	1	0.5	47.6%	36	3.5%
Q10aiv	1	0.4	42.7%	30	1.6%
Q10bi	2	0.8	40.8%	50	1.6%
Q10bii	1	0.8	81.0%	26	1.1%
Q11a	1	0.7	67.3%	48	5.4%
Q11b	1	0.3	26.3%	47	22.8%
Q11c	1	0.6	63.6%	34	1.4%
Q11di	1	0.5	48.7%	33	8.1%
Q11dii	1	0.2	21.8%	40	1.6%
Q11e	1	0.5	47.0%	44	7.0%
Q12a	1	0.3	34.7%	26	2.9%
Q12bi	1	0.7	73.9%	46	5.5%
Q12bii	1	0.1	11.0%	31	11.4%
Q12ci	1	0.8	78.4%	48	5.0%
Q12cii	1	0.5	46.8%	53	8.4%
Q12di	1	0.1	10.8%	29	6.2%
Q12dii	2	0.5	23.5%	54	13.9%
Q12diii	1	0.5	52.7%	35	4.9%
Q13a	3	0.9	28.6%	68	10.6%
Q13b	1	0.6	60.5%	57	10.1%
Q14a	1	0.3	26.7%	27	5.4%
Q14b	1	0.3	34.6%	45	15.3%
Q14ci	1	0.8	85.0%	27	2.4%
Q14cii	1	0.6	58.1%	45	4.9%
Q14d	2	0.6	27.5%	63	4.6%
Q15A	8	2.8	35.0%	60	4.3%
Q15B	8	4.1	51.5%	72	0.0%
Paper	95	44.7	47.1%		

2024 Higher Human Biology Section 2 Distribution of Marks Analysis

2024 Item	0	1	2	3	4	5	6	7	8
Q1a	6	94							
Q1bi	33	67							
Q1bii	40	31	28						
Q1c	18	22	59						
Q2ai	16	84							
Q2aii	66	34							
Q2bi	27	73							
Q2bii	85	15							
Q3ai	14	38	48						
Q3aii	93	7							
Q3aiii	69	31							
Q3b	8	25	68						
Q3c	73	27							
Q4ai	59	41							
Q4aii	37	63							
Q4aiii	36	64							
Q4b	56	21	22						
Q4c	34	66							
Q5ai	46	23	31						
Q5aii	52	48							
Q5aiii	37	63							
Q5bi	34	66							
Q5bii	35	65							
Q5ci	39	61							
Q5cii	25	27	47						
Q6a	51	49							
Q6b	39	61							
Q6ci	44	56							
Q6cii	67	33							
Q7ai	48	52							
Q7aii	36	64							
Q7aiii	19	55	26						
Q7bi	50	50							
Q7bii	70	30							
Q8ai	41	59							
Q8aii	50	50							
Q8b	61	39							
Q8ci	54	46							
Q8cii	48	52							
Q8di	70	30							
Q8dii	79	21							
Q9a	63	26	11						
Q9bi	67	33							
Q9bii	50	50							
Q9biii	49	51							
Q9c	61	39							
Q10ai	43	57							
Q10aii	81	19							
Q10aiii	52	48							
Q10aiv	57	43							
Q10bi	42	35	23						
Q10bii	19	81							
Q11a	33	67							
Q11b	74	26							
Q11c	36	64							
Q11di	51	49							
Q11dii	78	22							
Q11e	53	47							
Q12a	65	35							
Q12bi	26	74							
Q12bii	89	11							
Q12ci	22	78							
Q12cii	53	47							
Q12di	89	11							
Q12dii	63	28	10						
Q12diii	47	53							
Q13a	49	23	22	6					
Q13b	39	61							
Q14a	73	27							
Q14b	65	35							
Q14ci	15	85							
Q14cii	42	58							
Q14d	58	29	13						
Q15A	17	18	17	14	11	8	5	3	5
Q15B	9	8	9	13	16	14	11	10	10

2024 Higher Human Biology Section 3 Assignment Marks Analysis

Item	Max Mark	Average Mark	% Average Mark	Discrimination
1	1	0.8	81.8%	26
2	4	2.3	58.3%	39
3a	1	0.4	41.2%	28
3b	1	1	95.1%	13
3c	1	0.4	41.4%	31
3d	1	0.6	59.3%	20
3e	1	0.4	37.8%	35
4a	1	1	97.2%	16
4b	1	0.8	81.6%	24
4c	1	0.9	85.5%	28
4d	1	0.7	71.7%	30
5	1	0.3	27.9%	41
6	1	0.4	40.8%	32
7	3	1.1	36.0%	49
8	1	0.7	74.8%	17
Paper	20			

Item	Max Mark	0	1	2	3	4
1	1	18	82			
2	4	12	18	22	22	27
3a	1	59	41			
3b	1	5	95			
3c	1	59	41			
3d	1	41	59			
3e	1	62	38			
4a	1	3	97			
4b	1	18	82			
4c	1	14	86			
4d	1	28	72			
5	1	72	28			
6	1	59	41			
7	3	36	31	22	11	
8	1	25	75			
Paper	20					

Item	Max Mark	Item Description
1	1	An aim that describes clearly the purpose of the investigation
2	4	Underlying biology
3a	1	A brief summary of the approach used to collect experimental data.
3b	1	Sufficient raw data from the candidate's experiment.
3c	1	Data, including mean/average values, presented in a correctly produced table(s)
3d	1	Data relevant to the aim from an internet/literature source
3e	1	A citation and reference for a source of the internet/literature data
4a	1	An appropriate format from the options of bar graph or line graph.
4b	1	The axes of the graph have suitable scales.
4c	1	The axes of the graph have suitable labels and units
4d	1	Accurately plotted data points and a line (line graph) or clear bar tops (bar graph)
5	1	A valid comparison or an appropriate calculation linked to the aim of the investigation.
6	1	A valid conclusion that relates to the aim and is supported by all the data in the report.
7	3	Evaluation of the investigation
8	1	A clear and concise report, with an informative title.

National
Qualifications
2023

2023 Human Biology

Higher

Paper 1 – Multiple choice

Finalised Marking Instructions

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Marking instructions for each question

Question	Response	Mark
1.	D	**1**
2.	B	**1**
3.	C	**1**
4.	C	**1**
5.	A	**1**
6.	B	**1**
7.	A	**1**
8.	B	**1**
9.	C	**1**
10.	D	**1**
11.	D	**1**
12.	B	**1**
13.	D	**1**
14.	A	**1**
15.	B	**1**
16.	C	**1**
17.	A	**1**
18.	C	**1**
19.	D	**1**
20.	A	**1**
21.	C	**1**
22.	B	**1**
23.	D	**1**
24.	A	**1**
25.	C	**1**

[END OF MARKING INSTRUCTIONS]

National
Qualifications
2023

2023 Human Biology

Higher – Paper 2

Finalised Marking Instructions

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Marking instructions for each question

Question			Expected response	Max mark	Additional guidance
1.	(a)	(i)	(DNA) Nucleotides	**1**	
		(ii)	Sugar/deoxyribose and phosphate	**1**	
		(iii)	X or Y	**1**	
	(b)	(i)	Transcription	**1**	
		(ii)	Different exons are retained/ removed. **OR** Different mature transcripts are produced.	**1**	Any reference to introns being retained or treated as exons negates.
		(iii)	tRNA/it attaches to a specific amino acid. **OR** tRNA/it carries/brings an amino acid to the ribosome/mRNA. **(1)** Anticodon (on tRNA) aligns/joins/ bonds with codon (on mRNA). **OR** Anticodon is complementary/specific to the codon. **(1)** Results in/ensures the amino acid(s) are in the correct order/position. **OR** tRNA/it detaches/leaves the ribosome and can collect another amino acid/be reused. **(1)** **Any 2**	**2**	Accept anticodon is specific to an amino acid. Suggestion of tRNA has codon/mRNA has anticodon negates this mark.

Question			Expected response	Max mark	Additional guidance
2.	(a)	(i)	A (large) molecule/substrate/ substance is broken down (to smaller molecules). **OR** Two/multiple (smaller) products/ molecules are formed.	**1**	
		(ii)	The active site changes shape. **OR** The active site changes to (better) fit the substrate.	**1**	
		(iii)	The products/they have a low affinity/attraction for the active site. **OR** The products/they no longer fit the active site.	**1**	
	(b)		It is the lowest concentration (of tyrosine/substrate) to produce the maximum/highest concentration/ (~)0.010 (mg/cm^3) of L-dopa/ product. **OR** Raising the concentration of tyrosine above this/5 (mg/cm^3) does not increase the concentration of L-dopa/product. **OR** By raising the concentration of tyrosine above this/5 (mg/cm^3), the concentration of L-dopa/product levels off/is not affected/remains constant/at (~) 0.010 (mg/cm^3).	**1**	Accept: it/5 (mg/cm^3) of tyrosine produces the highest concentration of L-dopa/product before the concentration/it levels off/is not affected/remains constant/at (~) 0.010 (mg/cm^3).

Question			Expected response	Max mark	Additional guidance
2.	(c)		To prevent feedback/end-product inhibition. **OR** A high concentration of end-product inhibits the enzyme/lowers the reaction rate. **OR** Removal of the product drives/ pushes/promotes/causes the reaction to make more product. **OR** It prevents build-up/an increase of product, which would reverse the reaction.	**1**	Accept a description of end-product inhibition.

<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>3.</td><td>(a)</td><td>(i)</td><td>Concentration of enzyme/ dehydrogenase/DCPIP/inhibitor.
OR
Temperature/pH of solution/test tubes.</td><td>1</td><td>Do not accept type of inhibitor.

Accept temperature/pH of contents of tube/named solution.</td></tr>
<tr><td></td><td></td><td>(ii)</td><td>Difficult to identify/measure the (exact) time of decolourisation (of DCPIP)/end point.
OR
Difficult to make sure all drops (of DCPIP) are consistent/the same volume/size.
OR
Adding the DCPIP/it after the reaction starts/enzyme is added.</td><td>1</td><td></td></tr>
<tr><td></td><td></td><td>(iii)</td><td>Repeat (the investigation/ experiment/it) at each (succinate) concentration.
OR
Repeat (the investigation/ experiment/it) and take an average.</td><td>1</td><td></td></tr>
<tr><td></td><td>(b)</td><td></td><td>Axes have correct scales and labels. (1)

Points correctly plotted and lines drawn. (1)

<table><tr><th>Concentration of succinate (M)</th><th>Time to decolourise DCPIP (s)</th></tr><tr><td>0.2</td><td>94</td></tr><tr><td>0.4</td><td>48</td></tr><tr><td>0.6</td><td>24</td></tr><tr><td>0.8</td><td>14</td></tr><tr><td>1.0</td><td>8</td></tr></table></td><td>2</td><td>Any 3 values to establish a linear scale. A number at the origin is not essential.

Data can be plotted outwith the numbered scale.

Scale breaks are not acceptable.
If the axes are transposed do not award the scale mark.

Candidates cannot access the scale mark if the plotted points use less than half the graph paper.

The line must touch all the points.</td></tr>
</table>

Question			Expected response	Max mark	Additional guidance
3.	(c)		As the substrate/succinate concentration increases the time of the reaction/to decolourise decreases. **OR** The maximum rate of reaction is reached. **OR** Increasing substrate concentration reduces/overcomes/reverses inhibition. **OR** The time to decolourise DCPIP with inhibitor reaches the same value as without inhibitor.	**1**	Accept if non-competitive inhibitor is used maximum rate of reaction would not be reached.
	(d)	(i)	Matrix of mitochondria.	**1**	
		(ii)	To remove hydrogen ions)/electrons. **OR** Pass hydrogen (ions)/electrons to NAD (to form NADH).	**1**	Accept H^+/e^- Removal of hydrogen (ions)/ electrons from NADH negates. Reference to dehydrogenase transporting hydrogen (ions)/ electrons to electron transport chain negates.
		(iii)	Oxaloacetate	**1**	

Question			Expected response	Max mark	Additional guidance
4.	(a)	(i)	(One) thymine/T is replaced by adenine/A. **OR** The second/one codon has one base/nucleotide different.	**1**	
		(ii)	The shape/structure of the protein is changed. **OR** The protein does not fold properly. **OR** Different bonding occurs (between the amino acids). **OR** The binding site for oxygen has changed.	**1**	
	(b)		In a missense mutation only one amino acid is changed/replaced/ altered. **(1)** In a frame-shift mutation (the order of) amino acids after the mutation are changed/replaced/altered. **(1)**	**2**	Do not accept production of amino acids.
	(c)	(i)	Both types of haemoglobin are found in their red blood cells. **OR** Individuals/they have both haemoglobin S and normal haemoglobin. **OR** Both (haemoglobin) alleles have an equal chance of being expressed. **OR** Individuals have a mild/less severe form of sickle cell disease.	**1**	Accept both (haemoglobin) alleles are shown in the phenotype.
		(ii)	Their (red) blood cells could stick together/stick to artery walls and forms clots/blockages. **OR** Their (red) blood cells block arteries to the brain.	**1**	
		(iii)	14 700	**1**	

Question			Expected response	Max mark	Additional guidance
5.	(a)	(i)	4	**1**	
		(ii)	Corpus luteum.	**1**	
	(b)	(i)	A lack/decrease in LH/drop in concentration of LH **(1)** (Leads to the) degeneration/break down of the corpus luteum. **(1)**	**2**	Accept a decrease in LH secretion/ production.
		(ii)	Menstruation starts/the endometrium starts to break down/no longer maintained. **OR** Pituitary (gland) secretes FSH. **OR** Inhibition of the pituitary (gland) stops/is removed.	**1**	Accept menstruation. Accept FSH concentration increases.
	(c)		Prevent the negative feedback/ inhibitory effect of oestrogen (on FSH secretion). **OR** Mimic (the action of) FSH/LH.	**1**	Do not accept answers referring to the drugs containing FSH/LH.

Question			Expected response	Max mark	Additional guidance
6.	(a)	(i)	1. As the (number of IVF) cycles increases the birth rate (per embryo transfer) decreases. **(1)** 2. As the (number of IVF) cycles increases the average age of the women (undergoing IVF) increases. **(1)** 3. As the average age of women (undergoing IVF) increases the birth rate (per embryo transfer) decreases. **(1)** **Any 2**	**2**	
		(ii)	35	**1**	
	(b)	(i)	99/99.1/99.06	**1**	
		(ii)	6240	**1**	
		(iii)	As the number of embryos transferred/used is different/ decreases.	**1**	
	(c)	(i)	27	**1**	
		(ii)	The birth rate (per embryo transfer) decreased with age for the women's own eggs but remained constant/did not decrease with age for donor eggs. **OR** The difference in birth rate (per embryo transfer) increases with age.	**1**	
		(iii)	Donor eggs/they may be younger/ come from younger women. **OR** Donor eggs/they may have less mutations/abnormalities.	**1**	Accept donor eggs/they come from women under 35.

Question			Expected response	Max mark	Additional guidance
7.	(a)	(i)	The muscle/connective tissue layer is thicker/larger/wider in X/arteries. **OR** The lumen is smaller/narrower in X/arteries.	**1**	Do not accept arteries have thicker walls/veins have thinner walls Accept converse answers relating to veins.
		(ii)	Endothelium/endothelial (cells)	**1**	Do not accept epithelium/epithelial.
	(b)		1. Pressure filtration (occurs). **OR** Plasma passes through capillary walls. 2. The plasma becomes tissue fluid. **OR** Tissue fluid surrounds the cells. 3. Glucose/oxygen moves/diffuses into/is supplied to the cells. **OR** Carbon dioxide/(metabolic) waste products diffuse/move out of the cells. 4. Tissue fluid moves back into the blood/capillary. **Any 3**	**3**	Accept excess tissue fluid is absorbed/taken in/removed by lymphatic vessels/system.

Question			Expected response	Max mark	Additional guidance
8.	(a)	(i)	Coronary artery.	**1**	
		(ii)	Muscle/cells/tissue are deprived of oxygen/glucose.	**1**	
	(b)		Statins/they reduce (blood) cholesterol (levels/build up). **OR** Prevents synthesis of cholesterol (by liver cells) **(1)** (Statins) reduce atherosclerosis/the chances of a clot/thrombus/ atheroma/embolism. **(1)**	**2**	
	(c)	(i)	Accept any value from 30 to 31	**1**	
		(ii)	115.992/115.99/116	**1**	

Question			Expected response	Max mark	Additional guidance
9.	(a)		Glucose tolerance (test).	**1**	
	(b)		114	**1**	
	(c)		There is a decrease in the number/ sensitivity of insulin receptors. **OR** The cells develop insulin resistance/become less sensitive to insulin. **(1)** This means that liver cells/they can take in less glucose/do not remove glucose from the blood. **OR** (So) less/no glucose is converted to glycogen. **(1)**	**2**	Accept receptors develop insulin resistance.
	(d)		The endothelium/(small) blood vessels/capillaries are damaged. **OR** (Small) blood vessels/capillaries burst/haemorrhage/leak blood.	**1**	Do not accept arteries/veins for blood vessels.

Question			Expected response	Max mark	Additional guidance
10.	(a)	(i)	Axon	**1**	
		(ii)	It increases the speed of impulses.	**1**	Answer must indicate an increase.
		(iii)	Glial	**1**	
	(b)		They have less myelin/myelination is incomplete and impulses are slower. **OR** They have less myelin/myelination is incomplete and they cannot move their muscles as quickly/control their muscles as well.	**1**	Do not accept they have no myelin.
	(c)		Impulse(s)/neurotransmitters can be sent to more than one destination/ neuron. **OR** More than one muscle/effector/ destination/neuron can be stimulated. **OR** (It allows) fine motor control/ coordination of movement.	**1**	Accept examples of fine motor control like writing/threading a needle.

Question			Expected response	Max mark	Additional guidance
11.	(a)		Central Nervous (System)/CNS.	**1**	
	(b)	(i)	Antagonistic	**1**	Do not accept antagonist(s)
		(ii)	Increased heart rate: Increases blood flow to muscles/the body/cells. **OR** Increases glucose/oxygen to muscles/cells. **OR** Increases glucose/oxygen for respiration. **(1)** Vasoconstriction of arteries: Decreases blood flow to small intestine/digestive system/gut. **OR** Increases blood flow to skeletal/ working/exercising muscles. **(1)**	**2**	Accept blood is diverted to the muscles/other areas.

Question			Expected response	Max mark	Additional guidance
12.	(a)		1. (Same) person reading out the words/volume/tone of person (reading words) 2. (Same) gender balance/ratio. 3. Speed of words read out/time between each word/time allowed to write down the words/time between listening and writing. 4. Order of words/the same words. 5. Hearing ability/level of alertness/IQ/literacy level/first language of participant. 6. Volume of music/background noise/distance from music source/speakers **Any 2**	**2**	3. Accept time taken to read the words. If candidate lists three or more variables and at least one is correct award 1 mark. If all answers are correct award 2 marks.
	(b)		1	**1**	
	(c)		No (background/rock) music.	**1**	
	(d)		(Listening to background) rock music decreases the ability to recall/ remember words.	**1**	
	(e)		Words/they may still be in short term memory/STM. **OR** Words/they will not have been displaced/decayed from short term memory/STM.	**1**	

Question			Expected response	Max mark	Additional guidance
13.	(a)	(i)	105	**1**	
		(ii)	125	**1**	
		(iii)	The muscle/cells do not get enough oxygen. **OR** Oxygen debt occurs/the electron transport chain cannot occur. **(1)** Pyruvate is converted to lactate/ lactate is not converted back to pyruvate. **(1)** Lactate production allows the regeneration of NAD/ glycolysis to continue/some ATP to be produced. **(1)** **Any 2**	**2**	
	(b)		Eating/sex.	**1**	Accept laughter. Accept eating any named food. Do not accept injury/stress/pain.
	(c)	(i)	7:10	**1**	
		(ii)	Fitness levels have no effect on endorphin concentrations/levels. **OR** Endorphin concentrations/levels increase to the same extent in both fit and unfit individuals. **OR** Endorphin concentrations/levels in both fit and unfit individuals are the same/similar.	**1**	

Question			Expected response	Max mark	Additional guidance
14.	(a)		Pathogen	**1**	
	(b)		They/phagocytes engulf bacteria/it (into a vacuole). **(1)** Lysosomes release/contain enzymes **(1)** Enzymes destroy/digest/break down the bacteria/it. **(1)** **Any 2**	**2**	Accept lysozyme for enzyme. Do not accept lysosome for enzyme.
	(c)		Attract/signal (more) phagocytes/ lymphocytes/white blood cells (to the wound).	**1**	Accept cause an accumulation for attract.
	(d)		Prevents/reduces the release/ crossing of neurotransmitters. **OR** Prevents/reduces the fusion of vesicles. **OR** Increases the reuptake of the neurotransmitters.	**1**	
	(e)		Tetanus/it is not spread from person to person/from an infected individual.	**1**	Accept tetanus/it is contracted by contact from soil not people.

Question			Expected response	Max mark	Additional guidance
15.	**A**		a. (Recreational drugs can act as) agonists which stimulate receptors. **OR** Agonists mimic (the action of) neurotransmitters. b. (Recreational drugs can act as) antagonists which block receptors. **OR** Antagonists block the action/ binding of neurotransmitters. c. (Recreational drugs) affect neurotransmission/receptors/ synapses in the reward pathway. d. (Some recreational drugs) inhibit reuptake of neurotransmitters/ inhibit the enzymes that degrade neurotransmitters. **Maximum 2 marks** 1. Recreational drugs alter (an individual's) mood/cognition/ perception/behaviour. 2. (Drug) addiction is caused by (repeated/continual use of) antagonists. 3. Antagonists increase the number/sensitivity of receptors. 4. An increase in number/sensitivity of receptors causes sensitisation. 5. (This leads to) addiction as the individual craves (more of) the drug. 6. (Drug) tolerance is caused by (repeated/continual use of) agonists. 7. Agonists decrease the number/ sensitivity of receptors. 8. A decrease in number/sensitivity of receptors causes desensitisation. 9. (This leads to) tolerance where an individual must take more of the drug to get an effect. **Maximum of 6 marks**	**8**	c. Accept (recreational drugs) activate/stimulate the reward pathway 1. Accept description of these terms.

Question			Expected response	Max mark	Additional guidance
15.	**B**		a. Vaccines contain antigens from (infectious) pathogens. b. Vaccines contain inactivated pathogen toxins/dead pathogens/parts of pathogens/ weakened pathogens. c. (These are usually) mixed with/contain an adjuvant. d. An adjuvant makes the vaccine more effective/enhances the immune response. **Maximum of 2 marks**	**8**	Accept bacteria/virus for pathogen.
			1. Clinical trials are used to make sure vaccines/drug are safe. 2. Subjects/individuals are divided/placed into groups in a randomised way. 3. Characteristics/age/gender are controlled/similar. 4. (In a clinical trial) one group/some people receives the vaccine/drug while the second group/some people receive a placebo. 5. Using a placebo ensures valid comparisons/results. 6. In a double-blind trial neither the subjects/participants nor the researchers/doctors know which group subjects are in/treatment is given. 7. Bias is prevented/reduced through randomised allocation/ double-blind (trial) (to increase validity). 8. Groups must be of a suitable/ large size to reduce experimental error/increase reliability. 9. The results are compared to determine any (statistically) significant differences. **Maximum of 6 marks**		Use of accuracy does not negate. 2. Accept an example of randomisation (eg names out of a hat/computer generated). 5. Use of reliability negates. 6. Accept a description of the groups/treatment (eg which group/who receives vaccine/ placebo). 7. Accept a description of protocol(s). 7. Use of reliability negates if not already penalised in point 5. 8. Use of validity negates.

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Human Biology

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.

(c) Do not award half marks.

(d) Where a candidate makes an error in the first part of a question, award marks for subsequent answers that are correct with regard to this original error. Do not penalise candidates more than once for the same error.

(e) Unless a numerical question specifically requires evidence of working to be shown, award full marks for a correct final answer (including units, if appropriate) on its own.

(f) Candidates should not use bulleted lists to answer extended-response questions. They must respond to the 'command' word as appropriate and provide extended answers to communicate fully their knowledge and understanding. Candidate responses in the form of bulleted lists may not be able to access the full range of available marks.

(g) In the detailed marking instructions, if a word is **underlined** then it is essential; if a word is **(bracketed)** then it is not essential.

(h) In the detailed marking instructions, words separated by / are **alternatives**.

(i) A correct response can be negated if the candidate includes:
- an extra, incorrect, response
- additional information that contradicts the correct response

(j) Where the candidate is instructed to choose one question to answer but instead answers two questions, mark both responses and award the higher mark.

(k) Unless otherwise required by the question, the use of abbreviations (for example DNA, ATP) or chemical formulae (for example CO_2, H_20) are acceptable alternatives to naming.

(l) If a numerical answer is required and units are not given in the stem of the question or in the answer space, candidates must supply the units to gain the mark. If units are required on more than one occasion, do not penalise candidates repeatedly.

(m) If incorrect spelling is given:
- If the correct word is recognisable then award the mark.
- If the word can easily be confused with another biological term then **do not** award the mark, for example glucagon and glycogen.

(n) Presentation of data:

- If a candidate provides two graphs, in response to one question, mark both and award the higher mark.
- If a question asks for a particular type of graph/chart and the candidate gives the wrong type, do not award full marks. Candidates cannot achieve the plot mark but **may** be able to achieve the mark for scale and label. If the x and y data are transposed, then do not award the scale and label mark.
- If the graph uses less than 50% of the axes then do not award the scale and label mark.
- If 0 is plotted when no data for this is given, then do not award the plot mark — candidates should only plot the data given.

(o) Only award marks for a valid response to the question asked. For example, in response to questions that ask candidates to:

- **identify, name, give** or **state**, they need only answer or present in brief form
- **describe**, they must provide a statement as opposed to simply one word
- **explain**, they must provide a reason for the information given
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between topics being examined
- **calculate**, they must determine a number from given facts, figures or information
- **predict**, they must indicate what may happen based on available information
- **suggest**, they must apply their knowledge and understanding to a new situation

2023 Higher Human Biology Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	D	7	8	6	**79**	0	0
2	B	10	**79**	7	4	0	0
3	C	4	9	**30**	57	0	0
4	C	4	40	**51**	6	0	0
5	A	**77**	7	12	4	0	0
6	B	14	**43**	26	16	0	0
7	A	**81**	14	2	3	0	0
8	B	11	**35**	42	12	0	0
9	C	7	4	**78**	10	0	0
10	D	12	14	14	**60**	0	0
11	D	8	14	8	**69**	0	0
12	B	23	**65**	8	4	0	0
13	D	11	8	4	**77**	0	0
14	A	**46**	13	26	15	0	0
15	B	7	**78**	6	9	0	0
16	C	21	20	**54**	5	0	0
17	A	**64**	10	14	13	0	0
18	C	0	7	**90**	2	0	0
19	D	6	6	4	**83**	0	0
20	A	**82**	5	7	6	0	0
21	C	3	23	**69**	4	1	0
22	B	3	**75**	6	15	0	0
23	D	15	8	42	**34**	0	0
24	A	**60**	13	20	6	0	0
25	C	1	4	**48**	47	0	0

2023 Higher Human Biology Section 2 Marks Data Analysis

2023 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	1	0.7	74.5%	47	0.9%
Q1aii	1	0.8	77.2%	43	1.4%
Q1aiii	1	0.8	77.4%	32	0.1%
Q1bi	1	0.7	72.4%	53	3.0%
Q1bii	1	0.2	24.1%	46	7.5%
Q1biii	2	0.6	30.6%	62	7.7%
Q2ai	1	0.8	81.5%	45	2.7%
Q2aii	1	0.7	69.8%	49	2.6%
Q2aiii	1	0.4	35.8%	52	4.3%
Q2b	1	0.3	29.8%	31	3.4%
Q2c	1	0.3	27.0%	48	8.1%
Q3ai	1	0.5	52.8%	38	0.9%
Q3aii	1	0.4	39.1%	18	5.4%
Q3aiii	1	0.6	60.3%	45	1.1%
Q3b	2	1.6	80.1%	36	0.8%
Q3c	1	0.4	39.0%	37	4.9%
Q3di	1	0.7	70.1%	46	1.2%
Q3dii	1	0.5	50.2%	45	7.6%
Q3diii	1	0.6	58.1%	50	3.6%
Q4ai	1	0.6	62.1%	16	1.3%
Q4aii	1	0.2	21.2%	37	4.0%
Q4b	2	0.8	38.5%	60	8.4%
Q4ci	1	0.1	12.2%	23	9.5%
Q4cii	1	0.1	13.7%	22	3.3%
Q4ciii	1	0.9	89.5%	20	1.6%
Q5ai	1	0.9	88.9%	30	0.2%
Q5aii	1	0.7	74.1%	51	2.4%
Q5bi	2	0.6	32.5%	61	8.0%
Q5bii	1	0.4	40.2%	49	5.6%
Q5c	1	0.4	41.0%	49	3.0%
Q6ai	2	1.2	59.7%	45	0.9%
Q6aii	1	0.8	81.4%	40	0.6%
Q6bi	1	0.7	70.0%	42	2.1%
Q6bii	1	0.6	57.0%	42	4.7%
Q6biii	1	0.1	9.7%	32	7.9%
Q6ci	1	0.7	67.3%	33	1.4%
Q6cii	1	0.3	34.1%	39	2.0%
Q6ciii	1	0.6	58.8%	27	2.4%
Q7ai	1	0.6	63.8%	43	0.7%
Q7aii	1	0.6	62.2%	48	3.6%
Q7b	3	1.1	36.9%	69	11.7%
Q8ai	1	0.3	34.3%	51	2.9%
Q8aii	1	0.3	28.7%	46	5.6%
Q8b	2	0.9	47.4%	63	8.0%
Q8ci	1	0.6	64.9%	44	3.1%
Q8cii	1	0.5	53.0%	56	4.8%
Q9a	1	0.5	47.1%	47	9.3%
Q9b	1	0.8	81.2%	36	3.3%
Q9c	2	0.7	34.7%	64	7.6%
Q9d	1	0.1	12.3%	36	22.5%
Q10ai	1	0.9	87.0%	45	3.2%
Q10aii	1	0.5	46.3%	49	4.3%
Q10aiii	1	0.8	81.0%	53	5.5%
Q10b	1	0.3	25.5%	41	5.2%
Q10c	1	0.5	51.9%	46	6.9%
Q11a	1	0.9	89.1%	32	0.9%
Q11bi	1	0.6	63.3%	52	8.8%
Q11bii	2	0.7	37.4%	38	4.1%
Q12a	2	1.3	65.5%	24	0.9%
Q12b	1	0.1	14.2%	16	3.0%
Q12c	1	0.6	64.1%	38	4.5%
Q12d	1	0.3	28.6%	40	1.1%
Q12e	1	0.5	50.2%	32	1.4%
Q13ai	1	0.8	81.8%	35	0.9%
Q13aii	1	0.5	49.1%	43	2.9%
Q13aiii	2	0.7	35.3%	58	5.3%
Q13b	1	0.8	79.7%	10	1.5%
Q13ci	1	0.8	83.7%	33	1.7%
Q13cii	1	0.6	57.3%	23	2.8%
Q14a	1	0.7	73.9%	54	7.0%
Q14b	2	1.3	63.1%	60	2.7%
Q14c	1	0.5	52.1%	59	5.4%
Q14d	1	0.3	31.7%	49	11.9%
Q14e	1	0.5	46.2%	29	4.1%
Q15A	8	4.6	58.0%	79	5.6%
Q15B	8	4.4	54.4%	67	0.0%
Paper	95	49.7	52.3%		

2023 Higher Human Biology Section 2 Distribution of Marks Analysis

2023 Item	0	1	2	3	4	5	6	7	8
Q1ai	26	74							
Q1aii	23	77							
Q1aiii	23	77							
Q1bi	28	72							
Q1bii	76	24							
Q1biii	56	26	18						
Q2ai	18	82							
Q2aii	30	70							
Q2aiii	64	36							
Q2b	70	30							
Q2c	73	27							
Q3ai	47	53							
Q3aii	61	39							
Q3aiii	40	60							
Q3b	7	26	67						
Q3c	61	39							
Q3di	30	70							
Q3dii	50	50							
Q3diii	42	58							
Q4ai	38	62							
Q4aii	79	21							
Q4b	49	24	27						
Q4ci	88	12							
Q4cii	86	14							
Q4ciii	10	90							
Q5ai	11	89							
Q5aii	26	74							
Q5bi	52	32	17						
Q5bii	60	40							
Q5c	59	41							
Q6ai	27	27	46						
Q6aii	19	81							
Q6bi	30	70							
Q6bii	43	57							
Q6biii	90	10							
Q6ci	33	67							
Q6cii	66	34							
Q6ciii	41	59							
Q7ai	36	64							
Q7aii	38	62							
Q7b	42	24	17	18					
Q8ai	66	34							
Q8aii	71	29							
Q8b	37	31	32						
Q8ci	35	65							
Q8cii	47	53							
Q9a	53	47							
Q9b	19	81							
Q9c	50	31	19						
Q9d	88	12							
Q10ai	13	87							
Q10aii	54	46							
Q10aiii	19	81							
Q10b	75	25							
Q10c	48	52							
Q11a	11	89							
Q11bi	37	63							
Q11bii	40	46	14						
Q12a	12	45	43						
Q12b	86	14							
Q12c	36	64							
Q12d	71	29							
Q12e	50	50							
Q13ai	18	82							
Q13aii	51	49							
Q13aiii	47	35	18						
Q13b	20	80							
Q13ci	16	84							
Q13cii	43	57							
Q14a	26	74							
Q14b	15	43	41						
Q14c	48	52							
Q14d	68	32							
Q14e	54	46							
Q15A	17	8	7	6	6	6	9	9	32
Q15B	5	8	11	12	15	16	15	10	9

National
Qualifications
2022

2022 Human Biology

Higher

Paper 1 – Multiple choice

Finalised marking instructions

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Marking instructions for each question

Question	Response	Mark
1.	C	**1**
2.	D	**1**
3.	C	**1**
4.	C	**1**
5.	A	**1**
6.	A	**1**
7.	D	**1**
8.	A	**1**
9.	B	**1**
10.	C	**1**
11.	B	**1**
12.	B	**1**
13.	C	**1**
14.	B	**1**
15.	D	**1**
16.	D	**1**
17.	D	**1**
18.	A	**1**
19.	A	**1**
20.	C	**1**
21.	D	**1**
22.	B	**1**
23.	A	**1**
24.	C	**1**
25.	B	**1**

[END OF MARKING INSTRUCTIONS]

National
Qualifications
2022

2022 Human Biology

Higher – Paper 2

Finalised marking instructions

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Marking instructions for each question

Question			Expected response	Max mark	Additional guidance
1.	(a)	(i)	Germline (stem cell)	**1**	Accept gamete mother cell.
		(ii)	Mitosis	**1**	Do not accept meiosis.
		(iii)	They can differentiate into/form all cell types/any cell type **OR** All the genes can be switched on	**1**	Accept specialise/becomes for differentiate. Do not accept divide for differentiate. Accept all genes are switched on/ none of the genes are switched off.
	(b)		(The tissue stem cell) switches on/ expresses/activates/triggers specific genes. **OR** (The tissue stem cell) produces/ expresses specific proteins/ proteins characteristic for that cell/ red blood cells.	**1**	Accept certain/particular for specific. Accept different genes are expressed. Addition of switching off genes does not negate.
	(c)		(Patient's own stem) cells will not be attacked/destroyed by the immune system/white blood cells. **OR** (Stem) cells/tissue will not be rejected. **OR** Patient does not need a donor/can be treated faster.	**1**	Do not accept answers about ethical benefits. Accept allergic reaction for rejection. Accept no need to take immune-suppressing drugs.

Question			Expected response	Max mark	Additional guidance
2.	(a)		The temperature/it decreases to 50 to 65ºC. **OR** The temperature/it decreases from 92 to 98º C to 50 to 65 º C. **OR** The temperature/it decreases by 27 to 48ºC.	**1**	Accept any value within the ranges. Units must be given correctly at least once. If any value is given for step 1 or step 2, it must be correct. If all PCR temperature changes are given, candidate must highlight step 2.
	(b)	(i)	DNA polymerase.	**1**	Accept Taq polymerase. Do not accept polymerase on its own.
		(ii)	It will not be denatured (when heated/by high temperatures). **OR** It will allow multiple cycles (without the enzyme needing replaced). **OR** The temperature range of PCR is greater than the temperature range of DNA polymerase.	**1**	Accept the optimum temperature is higher.
	(c)		1024	**1**	

Question			Expected response	Max mark	Additional guidance
3.	(a)		1. Volume/thickness/area/mass/ weight/concentration/pH/type of (nutrient) gel. 2. Size/diameter of dish/thickness of dish lid/type of dish. 3. Temperature of incubator/dishes/ apparatus/suspension/water. 4. Intensity/wavelength of radiation/use the same lamp. **OR** Distance/strength of lamp. **Any 2**	**2**	Only penalise the use of amount once. Do not accept concentration/type/pH of yeast. Do not accept temperature of lamp/UV/room/environment. Accept level/sources of background light/keep in darkness. If candidate lists three or more variables and at least one is correct they score one mark for the question. If all answers are correct they score two marks.
	(b)		To prevent too many cells/ colonies growing (in the dishes). **OR** To spread out/separate the cells/ colonies. **OR** So the (number of) cells/colonies growing could be easily counted/ seen/(accurately) measured.	**1**	Accept to allow (all) cells/colonies to access space/nutrients/resources. Accept to allow (all) cells/colonies to be exposed to the UV light. Accept yeast for cells/colonies.

<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>3.</td><td>(c)</td><td>(i)</td><td>Axes have correct scales and labels. (1)
Points correctly plotted and lines drawn. (1)

<table><tr><th>Length of UV exposure time (min)</th><th>Number of yeast cell colonies</th></tr><tr><td>1</td><td>70</td></tr><tr><td>3</td><td>66</td></tr><tr><td>5</td><td>58</td></tr><tr><td>7</td><td>46</td></tr><tr><td>9</td><td>30</td></tr></table></td><td>2</td><td>Any 3 values to establish a linear scale. Zero at the origin is not essential.

Data can be plotted outwith the numbered scale.

Scale breaks are not acceptable.

If the axes are transposed do not award the scale mark.

Candidate cannot access the scale mark if the plotted points use less than half the graph paper.

The line must touch all points.</td></tr>
<tr><td></td><td></td><td>(ii)</td><td>As the length of UV exposure time increases fewer yeast cells/colonies survive/live.
OR
As the length of UV exposure time increases more yeast cells/colonies die/do not survive.</td><td>1</td><td>Do not accept as the length of UV exposure time increases the number of yeast cells/colonies decreases.

Answer must not only refer to number of yeast cells growing.</td></tr>
<tr><td></td><td></td><td>(iii)</td><td>10</td><td>1</td><td></td></tr>
</table>

Question			Expected response	Max mark	Additional guidance
4.	(a)		Exons	**1**	
	(b)		848	**1**	
	(c)	(i)	Alternative (RNA) splicing.	**1**	Do not accept alternative DNA splicing.
		(ii)	Different mature transcripts are formed. **OR** Different exons/coding regions remain/are retained/expressed/ translated.	**1**	Accept different exons are left out/excluded.
	(d)	(i)	An/one/the <u>intron</u> has been retained. **OR** Not all <u>introns</u> are removed.	**1**	Do not accept an intron being inserted/substituted/added.
		(ii)	The protein will not fold properly. **OR** The protein shape/length will change. **OR** The sequence/number of amino acids will have changed.	**1**	Answer must refer to structure. Accept longer or shorter for length will change. Accept answers linking stop codon to making the protein shorter. However, any mention of other named mutations negates.

Question			Expected response	Max mark	Additional guidance
5.	(a)		The concentration/solubility/ mass/weight/size of each tablet/ painkiller/inhibitory substance may have been different.	**1**	Accept strength/dosage for concentration. Accept they/them for painkillers.
	(b)		(15 cm^3 of) water and (5 cm^3 of) pepsin. **OR** Keep everything the same but replace painkiller solution with water/placebo.	**1**	Incorrect volumes negate. Mention of egg white does not negate.
	(c)		Original mass/2 g minus final mass equals mass lost (of egg white). **OR** Calculate the difference/change/ decrease in mass (of egg white).	**1**	Answer must describe how mass of egg white broken down was calculated. Accept weight for mass.
	(d)		Aspirin	**1**	
	(e)		Repeat the investigation with each painkiller. **OR** Repeat the investigation and calculate averages/an average.	**1**	Accept experiment for investigation. Accurate/valid negates.

Question			Expected response	Max mark	Additional guidance
6.	(a)	(i)	Matrix	**1**	
		(ii)	Electron transport chain	**1**	Do not accept E.T.C.
		(iii)	(It contains) ribosomes which are the site of protein synthesis/translation. **OR** (It contains) DNA which can be transcribed/copied into mRNA (to help form proteins).	**1**	Ribosome answer must refer to location of protein synthesis.
	(b)	(i)	Muscle (tissue) contains many/ more mitochondria. **OR** Muscle (tissue) requires more/lots of ATP/energy. **OR** Muscle (tissue) has a higher rate of respiration.	**1**	
		(ii)	Slow twitch.	**1**	
	(b)	(iii)	Their mitochondria do not produce enough ATP. **OR** There is a lack/low supply of ATP/ energy available (to the muscle). **OR** They/their muscles cannot carry out aerobic respiration (efficiently)/can only carry out glycolysis/fatigue quickly. **OR** They/their muscles produce (too much) lactate/have a buildup of lactate. **OR** Endurance activities/long distance running requires (a high proportion of) slow twitch fibres.	**1**	Do not accept slow twitch muscles.
	(c)		It converts lactate to pyruvate. **OR** It converts pyruvate into acetyl/ acetyl CoA.	**1**	

Question			Expected response	Max mark	Additional guidance
7.	(a)	(i)	FSH – (stimulates) development/ maturation/growth of the follicle. **OR** FSH – stimulates/causes the production of oestrogen (by the follicle). **(1)** Oestrogen – (stimulates) proliferation/repair/thickening/ regeneration of the endometrium. **OR** Oestrogen – prepares the endometrium for implantation. **OR** Oestrogen – makes cervical mucus thinner/less viscous/easier for sperm to swim through. **OR** Oestrogen – stimulates/causes (a surge in) the secretion of LH (by the pituitary gland). **OR** Oestrogen – (high concentrations) inhibit FSH/LH production/pituitary gland or causes negative feedback. **(1)**	**2**	Do not accept formation/production of follicle. Accept lining of the uterus for endometrium. Accept alters/changes the viscosity/consistency of cervical mucus.
		(ii)	Corpus luteum	**1**	
	(b)	(i)	The pill/it/oestrogen/progesterone has a negative feedback/inhibitory effect on the pituitary (gland). **(1)** FSH (concentrations) decrease/ remain low/do not increase (throughout the menstrual cycle). **(1)**	**2**	Answer must refer to negative feedback/inhibition, not a general description.
		(ii)	Low/decreased concentrations of progesterone/oestrogen/hormones (cause the breakdown of the endometrium/menstruation). **OR** The inactive pills/they do not contain progesterone/oestrogen/ hormones.	**1**	Accept levels for concentration. Accept less/a drop in progesterone/ oestrogen/hormones.
	(c)		It prevents/stops/delays ovulation.	**1**	Any reference to implantation negates.

Question			Expected response	Max mark	Additional guidance
8.	(a)		37.2	**1**	
	(b)		9	**1**	
	(c)	(i)	0.59	**1**	
		(ii)	Ovulation/release of an ovum/egg	**1**	
	(d)	(i)	Contraction of the ventricles/ ventricular systole.	**1**	Do not accept systole on its own. Do not accept a heartbeat. Any reference to atrial systole or diastole negates.
		(ii)	3	**1**	
	(e)		The <u>cervical/cervix</u> mucus is thin(ner)/watery.	**1**	Accept thickness/viscosity/ consistency of cervical mucus.

Question			Expected response	Max mark	Additional guidance
9.	(a)		X – Glucagon Y - Liver	**2**	
	(b)		1. (The individual) fasts/does not eat for a period of time. 2. (The individual) then drinks a glucose solution. 3. Blood glucose concentrations/ levels are then measured regularly/for at least two hours. **Any 2 from 3** 4. A diabetic's blood glucose concentration/level is higher/ starts higher. **OR** A diabetic's blood glucose concentration/level increases to a much higher level. **OR** A diabetic's blood glucose concentration/level takes longer to return to the starting concentration/does not return to initial value. **(1)**	**3**	Accept sugar for glucose. Accept monitored for measured regularly. Point 4. must be comparative.

Question			Expected response	Max mark	Additional guidance
10.	(a)		X – cell body Y – axon	**2**	Accept cytoplasm for X
	(b)	(i)	Broken down by enzymes/enzyme degradation. **OR** Reuptake/reabsorption **OR** Taken back into presynaptic neuron.	**1**	Do not accept diffusion.
		(ii)	To prevent continuous/prolonged stimulation. **OR** To prevent continuous/repeated impulses/impulse transmission. **OR** To allow another/a new/the next impulse to be transmitted.	**1**	
	(c)	(i)	Insulates the axon. **OR** Increases the speed of impulses (conduction).	**1**	
		(ii)	Glial (cells)	**1**	
	(d)		The (number of) cases in males peaks at/decreases from/increases to (age) 55-64 while females peaks at/decreases from/increases to (age) 45-54. **OR** From (age) 0-14 to 45-54 the (number of) cases rise faster in females than in males. **(1)** The (number of) cases in females is always higher (than in males). **(1)**	**2**	Answer must refer to (number of) cases. Accept highest for peaks. Do not accept single age group comparisons. Do not accept changes which only consider age 45-54 to 55-64. Do not accept age 0-54. Accept the (number of) cases in males is always lower (than in females).

Question			Expected response	Max mark	Additional guidance
11.	(a)		They take impulses/signals from (sensory) receptors/sense organs to the CNS/brain/spinal cord/inter neurons.	**1**	Accept examples of sense organs.
	(b)	(i)	Several/four (rod) cells/neurons (link) to one neuron/cell.	**1**	Answer must refer to cells/neurons rather than impulses only. An incorrect number of neurons negates.
		(ii)	Each rod cell releases a small number of neurotransmitters (when stimulated). **OR** Summation of weak stimuli occurs. **(1)** Enough neurotransmitter molecules are released (to trigger an impulse). **OR** The threshold is reached (to trigger an impulse). **(1)**	**2**	
	(c)		Autosomes/autosomal chromosomes. **OR** Any chromosome that is not a sex chromosome.	**1**	Accept any chromosome number between 1 and 22. Do not accept chromosomes on its own.

Question			Expected response	Max mark	Additional guidance
12.	(a)		(Skiing) increases levels/production/ it.	**1**	
	(b)		Endorphins/they reduce the intensity of pain/act as a (natural) painkiller.	**1**	
	(c)	(i)	(Morphine binds to and) stimulates endorphin receptors. **OR** (Morphine) mimics (the action of) endorphins.	**1**	
		(ii)	It decreases the number/sensitivity of receptors. **OR** It causes desensitisation of receptors.	**1**	Do not accept the amount of/a decrease in receptors. Accept less/fewer receptors for a decrease in number. Any reference to addiction does not negate.

Question			Expected response	Max mark	Additional guidance
13.	(a)		(As age increases) from 20-29 to 30-39 the number of cases/ it increases and then it decreases from 30-39 to 90-99. **(1)** (It increases) from 24 to 38 per 100 000/increases by 14 per 100 000. **OR** (It decreases) from 38 to 8 per 100 000/decreases by 30 per 100 000. **(1)**	**2**	Accept it increases to 30-39 and then it decreases. Must use age ranges from the graph rather than single years (eg 20-29 to 30-39 not 20-39). Must mention per 100 000 once.
	(b)	(i)	4.5	**1**	
		(ii)	Many females with cervical cancer/ it have died by/before this age. **OR** There are less/decreased cases/ low number of females with it.	**1**	Do not accept general answers that refer to dying from old age.
	(c)		19:2	**1**	
	(d)		(Cancer) cells fail to attach to each other/to the tumour. **OR** (Cancer) cells spread through the body/in the blood.	**1**	Answers must include cells.
	(e)		There are no cases below 20/in teenage years/between 0-19/ 10-19. **OR** From (age) 20-29/after teenage/in adult years there are cases (of cervical cancer). **OR** Cervical cancer is common in (age) 20-29/30-39/40-49.	**1**	Accept 20s/30s/40s

Question			Expected response	Max mark	Additional guidance
14.	(a)	(i)	To prevent the virus/cell/it from reproducing/replicating/producing proteins (and so causing influenza). **OR** To prevent it causing influenza/flu/ disease/illness/symptoms (in the vaccinated individual).	**1**	Do not accept to stop you getting the virus.
		(ii)	The surface proteins/they are/ act as antigens. **OR** The surface proteins/they are recognised as foreign/non-self/by lymphocytes. **(1)** Formation of memory cells/ production of antibodies/ lymphocytes occurs **OR** They trigger/cause the immune response **(1)**	**2**	Accept lymphocyte receptors binding to the surface proteins/them as recognition by lymphocytes.
		(iii)	To enhance/improve the immune response. **OR** To make the vaccine/it more effective/efficient.	**1**	Do not accept enhances the vaccine.

Question			Expected response	Max mark	Additional guidance
14.	(b)	(i)	(Antibody concentration/it) rises faster. **OR** (Antibody concentration/it) reaches its maximum/peak in a shorter time. **OR** (Antibody concentration/it) takes less time to start increasing/there is no delay in the increase in antibody concentration. **OR** (Antibody concentration/it) remains high for longer/decreases more slowly. **OR** (Antibody concentration/it) doesn't return to low levels. **Any 2 from 5**	**2**	
		(ii)	Memory cells/they produce B lymphocytes/specific lymphocytes/ clones of lymphocytes. **OR** Memory cells/they divide (rapidly) to form a clonal population.	**1**	Do not accept lymphocytes on its own. T lymphocytes negates.
	(c)		Phagocytes engulf the pathogen/ virus/antibody complex/bacteria. **(1)** Lysosomes release/contain enzymes. **(1)** Enzymes destroy/digest/break down the pathogen/virus/antibody complex/bacteria. **(1)** **Any 2 from 3**	**2**	

Question			Expected response	Max mark	Additional guidance
15.	A		**Formation** 1. The formation of a thrombus/clot is referred to as thrombosis. 2. (Atheromas may) rupture/damage the endothelium/artery wall. 3. Clotting factors are released/ produced. 4. Clotting factors activate a cascade/series of reactions. 5. (The enzyme) prothrombin is converted to thrombin. 6. Thrombin causes fibrinogen to form fibrin. 7. Fibrin forms a mesh(work)/ net(work) (of threads) that clots the blood/seal the wound. **OR** Fibrin provides a scaffold for the formation of scar tissue. **Effects** a. A thrombus/clot may break loose/detaches forming an embolus and travels through the blood(stream). b. A clot/thrombus/thrombosis embolus/embolism in a <u>coronary artery</u> may lead to a heart attack/ myocardial infarction/MI. c. A clot/thrombus/thrombosis/ embolus/embolism in an <u>artery</u> in the brain may lead to a stroke. d. Cells/tissues are deprived of oxygen leading to their death. e. A deep vein thrombosis/DVT can occur (most commonly) in the leg/a peripheral vein. f. The clot/thrombus from DVT can (break off and) cause a <u>pulmonary</u> embolism (in the lungs).	**9**	5. Use of inactive/active must be correct. 6. Use of soluble/insoluble must be correct. a. Accept blocking blood flow as travels through the blood. e. Do not accept peripheral artery.

Question			Expected response	Max mark	Additional guidance
15.	B		**Production** a. Cholesterol is produced/ synthesised by (all) cells/in the liver. b. A diet high in saturated fats causes an increase in cholesterol (levels). c. Statins reduce cholesterol (levels)/inhibit cholesterol production. **Transport** 1. HDL/High density lipoproteins transport (excess) cholesterol to the liver (for elimination). 2. This prevents accumulation/ build-up of cholesterol in the blood. 3. (Regular) physical activity/ exercise raises HDL/HDL:LDL/ lowers cholesterol levels (in the blood). 4. LDL/Low density lipoproteins transport cholesterol to (body) cells. 5. LDL receptors then take cholesterol/LDL into the cell (where cholesterol is released). 6. Negative feedback inhibits the synthesis of new (LDL) receptors/ prevents the uptake of (more) cholesterol/LDL. 7. LDL can deposit (excess) cholesterol in the arteries/under the endothelium. 8. (Excess/increased) cholesterol can form atheromas/plaques/ lead to atherosclerosis. **Maximum 7 marks** **Role** d. Cholesterol is found in/forms part of the cell membrane. e. Cholesterol is used to make sex hormones/testosterone/ oestrogen/progesterone.	**9**	**To achieve full marks must have at least one mark from each section.** 6. Accept once the cell has enough cholesterol as negative feedback. 8. High HDL:LDL/higher HDL can decrease (the chance of) an atheroma/plaque (size)/ atherosclerosis.

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Human Biology

Always apply these general principles. Use them in conjunction with the marking instructions for each question, which identify the key features required in candidates' responses.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.

(c) Do not award half marks.

(d) Where a candidate makes an error in the first part of a question, award marks for subsequent answers that are correct with regard to this original error. Do not penalise candidates more than once for the same error.

(e) Unless a numerical question specifically requires evidence of working to be shown, award full marks for a correct final answer (including units, if appropriate) on its own.

(f) Candidates should not use bulleted lists to answer extended-response questions. They must respond to the 'command' word as appropriate and provide extended answers to communicate fully their knowledge and understanding. Candidate responses in the form of bulleted lists may not be able to access the full range of available marks.

(g) In the detailed marking instructions, if a word is **underlined** then it is essential; if a word is **(bracketed)** then it is not essential.

(h) In the detailed marking instructions, words separated by / are **alternatives**.

(i) A correct response can be negated if the candidate includes:
- an extra, incorrect, response
- additional information that contradicts the correct response

(j) Where the candidate is instructed to choose one question to answer but instead answers two questions, mark both responses and award the higher mark.

(k) Unless otherwise required by the question, the use of abbreviations (for example DNA, ATP) or chemical formulae (for example CO_2, H_20) are acceptable alternatives to naming.

(l) If a numerical answer is required and units are not given in the stem of the question or in the answer space, candidates must supply the units to gain the mark. If units are required on more than one occasion, do not penalise candidates repeatedly.

(m) If incorrect spelling is given:
- If the correct word is recognisable then award the mark.
- If the word can easily be confused with another biological term, then **do not** award the mark, for example glucagon and glycogen.

(n) Presentation of data:

- If a candidate provides two graphs, in response to one question, mark both and award the higher mark.
- If a question asks for a particular type of graph/chart and the candidate gives the wrong type, do not award full marks. Candidates cannot achieve the plot mark but **may** be able to achieve the mark for scale and label. If the x and y data are transposed, then do not award the scale and label mark.
- If the graph uses less than 50% of the axes then do not award the scale and label mark.
- If 0 is plotted when no data for this is given, then do not award the plot mark — candidates should only plot the data given.

(o) Only award marks for a valid response to the question asked. For example, in response to questions that ask candidates to:

- **identify, name, give** or **state**, they need only answer or present in brief form
- **describe**, they must provide a statement as opposed to simply one word
- **explain**, they must provide a reason for the information given
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between topics being examined
- **calculate**, they must determine a number from given facts, figures or information
- **predict**, they must indicate what may happen based on available information
- **suggest**, they must apply their knowledge and understanding to a new situation

2022 Higher Human Biology Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	**C**	4	2	**83**	11	0	0
2	**D**	2	9	51	**38**	0	0
3	**C**	8	16	**64**	11	1	0
4	**C**	1	2	**62**	35	0	0
5	**A**	**65**	26	1	8	0	0
6	**A**	**71**	9	15	5	0	0
7	**D**	9	7	6	**78**	0	0
8	**A**	**58**	13	9	20	0	0
9	**B**	12	**82**	4	2	0	0
10	**C**	9	5	**40**	45	0	0
11	**B**	43	**44**	10	4	0	0
12	**B**	11	**83**	4	3	0	0
13	**C**	22	38	**39**	1	0	0
14	**B**	15	**47**	14	24	1	0
15	**D**	19	15	10	**55**	0	0
16	**D**	5	1	8	**86**	0	0
17	**D**	1	5	49	**45**	0	0
18	**A**	**78**	6	9	7	0	0
19	**A**	**80**	3	4	12	0	0
20	**C**	6	8	**56**	30	0	0
21	**D**	17	8	9	**66**	0	0
22	**B**	7	**78**	6	9	0	0
23	**A**	**82**	3	2	13	0	0
24	**C**	15	16	**56**	13	0	0
25	**B**	21	**41**	25	13	0	0

2022 Higher Human Biology Section 2 Marks Data Analysis

2022 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	1	0.69	69%	44	1%
Q1aii	1	0.58	58%	24	2%
Q1aiii	1	0.71	71%	43	2%
Q1b	1	0.36	36%	43	8%
Q1c	1	0.26	26%	17	3%
Q2a	1	0.41	41%	38	3%
Q2bi	1	0.66	66%	53	4%
Q2bii	1	0.58	58%	37	5%
Q2c	1	0.66	66%	43	3%
Q3a	2	1.25	62%	33	1%
Q3b	1	0.13	13%	22	11%
Q3ci	2	1.71	85%	28	0%
Q3cii	1	0.27	27%	35	1%
Q3ciii	1	0.54	54%	21	1%
Q4a	1	0.76	76%	50	3%
Q4b	1	0.38	38%	49	4%
Q4ci	1	0.29	29%	53	7%
Q4cii	1	0.16	16%	45	16%
Q4di	1	0.30	30%	40	6%
Q4dii	1	0.35	35%	30	7%
Q5a	1	0.52	52%	27	3%
Q5b	1	0.23	23%	36	7%
Q5c	1	0.38	38%	40	6%
Q5d	1	0.62	62%	36	1%
Q5e	1	0.51	51%	39	1%
Q6ai	1	0.60	60%	46	5%
Q6aii	1	0.55	55%	53	6%
Q6aiii	1	0.29	29%	41	15%
Q6bi	1	0.30	30%	29	8%
Q6bii	1	0.63	63%	49	11%
Q6biii	1	0.37	37%	34	5%
Q6c	1	0.14	14%	34	14%
Q7ai	2	1.03	52%	65	5%
Q7aii	1	0.60	60%	58	6%
Q7bi	2	0.76	38%	49	7%
Q7bii	1	0.26	26%	30	6%
Q7c	1	0.14	14%	18	6%
Q8a	1	0.58	58%	37	0%
Q8b	1	0.79	79%	31	1%
Q8ci	1	0.55	55%	40	2%
Q8cii	1	0.67	67%	37	3%
Q8di	1	0.33	33%	54	5%
Q8dii	1	0.34	34%	43	9%
Q8e	1	0.52	52%	48	4%
Q9a	2	1.37	69%	60	1%
Q9b	3	2.14	71%	54	4%
Q10a	2	1.29	65%	56	3%
Q10bi	1	0.46	46%	60	16%
Q10bii	1	0.39	39%	56	17%
Q10ci	1	0.54	54%	57	6%
Q10cii	1	0.72	72%	56	11%
Q10d	2	0.83	41%	35	1%
Q11a	1	0.22	22%	43	5%
Q11bi	1	0.41	41%	36	6%
Q11bii	2	0.16	8%	39	15%
Q11c	1	0.21	21%	48	15%
Q12a	1	0.61	61%	38	8%
Q12b	1	0.81	81%	34	3%
Q12ci	1	0.33	33%	51	10%
Q12cii	1	0.42	42%	58	5%
Q13a	2	0.75	37%	51	1%
Q13bi	1	0.55	55%	37	2%
Q13bii	1	0.23	23%	7	3%
Q13c	1	0.57	57%	39	2%
Q13d	1	0.68	68%	39	2%
Q13e	1	0.47	47%	27	2%
Q14ai	1	0.27	27%	30	6%
Q14aii	2	0.56	28%	55	10%
Q14aiii	1	0.56	56%	51	12%
Q14bi	2	0.99	49%	48	6%
Q14bii	1	0.09	9%	34	8%
Q14c	2	1.02	51%	62	11%
Q15A	9	6.13	68%	63	4%
Q15B	9	5.84	65%	60	0%
Paper	95	46.47	49%		

2022 Higher Human Biology Section 2 Distribution of Marks Analysis

2022 Item	0	1	2	3	4	5	6	7	8	9
Q1ai	31	69								
Q1aii	42	58								
Q1aiii	29	71								
Q1b	64	36								
Q1c	74	26								
Q2a	59	41								
Q2bi	34	66								
Q2bii	42	58								
Q2c	34	66								
Q3a	13	49	38							
Q3b	87	13								
Q3ci	4	20	75							
Q3cii	73	27								
Q3ciii	46	54								
Q4a	24	76								
Q4b	62	38								
Q4ci	71	29								
Q4cii	84	16								
Q4di	70	30								
Q4dii	65	35								
Q5a	48	52								
Q5b	77	23								
Q5c	62	38								
Q5d	38	62								
Q5e	49	51								
Q6ai	40	60								
Q6aii	45	55								
Q6aiii	71	29								
Q6bi	70	30								
Q6bii	37	63								
Q6biii	63	37								
Q6c	86	14								
Q7ai	34	29	37							
Q7aii	40	60								
Q7bi	38	47	14							
Q7bii	74	26								
Q7c	86	14								
Q8a	42	58								
Q8b	21	79								
Q8ci	45	55								
Q8cii	33	67								
Q8di	67	33								
Q8dii	66	34								
Q8e	48	52								
Q9a	11	41	48							
Q9b	12	11	29	48						
Q10a	23	25	52							
Q10bi	54	46								
Q10bii	61	39								
Q10ci	46	54								
Q10cii	28	72								
Q10d	35	47	18							
Q11a	78	22								
Q11bi	59	41								
Q11bii	87	9	3							
Q11c	79	21								
Q12a	39	61								
Q12b	19	81								
Q12ci	67	33								
Q12cii	58	42								
Q13a	46	34	20							
Q13bi	45	55								
Q13bii	77	23								
Q13c	43	57								
Q13d	32	68								
Q13e	53	47								
Q14ai	73	27								
Q14aii	55	34	11							
Q14aiii	44	56								
Q14bi	33	35	32							
Q14bii	91	9								
Q14c	29	39	31							
Q15A	10	4	4	5	6	7	8	10	10	37
Q15B	7	5	7	7	7	7	9	11	10	30

National
Qualifications

X840/76/12

Human Biology
Paper 1 — Multiple choice

Marking Instructions

Please note that these marking instructions have not been standardised based on candidate responses. You may therefore need to agree within your centre how to consistently mark an item if a candidate response is not covered by the marking instructions.

©

Marking instructions for each question

Question	Answer	Mark
1.	C	**1**
2.	A	**1**
3.	C	**1**
4.	C	**1**
5.	B	**1**
6.	A	**1**
7.	D	**1**
8.	C	**1**
9.	C	**1**
10.	A	**1**
11.	C	**1**
12.	B	**1**
13.	B	**1**
14.	D	**1**
15.	D	**1**
16.	B	**1**
17.	B	**1**
18.	D	**1**
19.	B	**1**
20.	C	**1**
21.	A	**1**
22.	A	**1**
23.	D	**1**
24.	B	**1**
25.	D	**1**

[END OF MARKING INSTRUCTIONS]

Marking instructions for each question

Question			Expected response	Max mark	Additional guidance
1.	(a)		Bone marrow	**1**	
	(b)		They only differentiate into blood cells. **OR** They only differentiate into cells of their tissue.	**1**	
	(c)		<u>Therapeutic</u> Corneal repair/regeneration of damaged skin. **(1)** <u>Research</u> (Used as model cells) to study how diseases develop. **OR** (Used as model cells) for drug testing. **OR** Used to study cell processes/cell growth/cell differentiation/gene regulation. **(1)**	**2**	

Question			Expected response	Max mark	Additional guidance
2.	(a)	(i)	Stomach	**1**	
		(ii)	1032	**1**	
		(iii)	20	**1**	
		(iv)	Male deaths from breast cancer may/will be included in (the figures for) other types of cancer.	**1**	
		(v)	More men (compared to women) have stopped smoking (since 2007).	**1**	
	(b)		More cases will be detected/treated earlier. **OR** Treatment is more likely to be effective when detected sooner/at an early stage.	**1**	
	(c)		Cells within the tumour fail to attach to each other (and spread throughout the body).	**1**	

Question			Expected response	Max mark	Additional guidance
3.	(a)	(i)	(Primers) bind/attach to DNA/ strands/3′ end (of the DNA) **OR** (Primers) bind/attach to a complementary section/target sequence/complimentary nucleotides. **(1)** (Primers) allow DNA polymerase to attach/start replicating/start copying/add nucleotides (to DNA). **(1)**	**2**	Accept Taq for DNA Answer must indicate link between the primer and DNA polymerase. Accept: DNA polymerase adds nucleotides to primers/them.
		(ii)	DNA polymerase can only add DNA nucleotides in one direction/to the 3′/deoxyribose end of the new strand (which is forming).	**1**	
		(iii)	R	**1**	
		(iv)	(DNA) ligase	**1**	
	(b)	(i)	25	**1**	
		(ii)	To separate the (DNA) strands. **OR** To break the hydrogen bonds/bonds between the bases/strands (in DNA).	**1**	Accept: to denature the DNA.
		(iii)	Allows primers to bind (to DNA/target sequences/ complementary nucleotides).	**1**	Accept: anneal for bind

Question			Expected response	Max mark	Additional guidance
4.	(a)		Substrate and enzyme (with no inhibitor present).	**1**	Accept: any named substrate with its correct enzyme
	(b)		Inhibitor B Increasing the substrate concentration does not reverse inhibition. **OR** Concentrations of above 0·4 M /high substrate concentrations have no effect on/do not increase the rate of reaction.	**1**	
	(c)		36	**1**	
	(d)		50	**1**	
	(e)	(i)	Type of reaction – Anabolic **(1)** Reason - The reaction involves the build-up of molecules/substrates. **OR** Small molecules are built-up/ combined into a large molecule. **(1)**	**2**	Accept synthesis Only accept synthesis in place of build-up if anabolic is given as the type of reaction.
		(ii)	(When the substrate binds) the shape of the active site changes. **(1)** This brings substrate molecules closer together/closer to the active site. **OR** This weakens/breaks chemical bonds. **OR** This lowers the activation energy/ energy needed to start the reaction. **(1)**	**2**	

Question			Expected response	Max mark	Additional guidance
5.	(a)		Drug/inhibitor binds to the (proton) pump/protein **AND** changes the shape of the (proton) pump/protein/ proton attachment site/active site. **(1)** (This means that) protons cannot attach to the (proton) pump/protein **OR** Protons cannot be pumped into the stomach/across the membrane. **(1)**	**2**	
	(b)	(i)	They would be allocated/assigned randomly. **OR** Example describing random allocation.	**1**	Examples could be drawing names out of a hat or using a computer program to allocate them.
		(ii)	Between 0 and 4 hours acid production decreased and then it increased from 4 to 8 hours. **(1)** It decreased from 190 to 60 cm^3/ by 130 cm^3. **OR** It increased from 60 to 110 cm^3/by 50 cm^3. **(1)**	**2**	Accept it decreases for 4 hours and then increases. Must state hours at least once. Must state cm^3 at least once.
		(iii)	The lowest acid volume/reading might have been at a different time between 2 and 6 hours/not been exactly at 4 hours. **OR** The acidity is only checked every 2 hours (so the time when the drug is most effective may be missed).	**1**	
	(c)		(Hydrogen ions) are pumped across the (inner) mitochondrial membrane. **(1)** They flow (back) through ATP synthase (generating ATP). **(1)**	**2**	

Question			Expected response	Max mark	Additional guidance
6.	(a)		Hypothalamus	**1**	
	(b)		Stimulates the development of a follicle. **OR** Stimulates the production of oestrogen (by the follicle).	**1**	
	(c)		63	**1**	
	(d)		Luteal	**1**	
	(e)		(Lack of LH) causes degeneration/ break down of the corpus luteum. **(1)** (This leads to) a decrease in progesterone levels (triggering menstruation). **(1)**	**2**	
	(f)		Cycle starts to repeat/repeats between days 28 and 36. **OR** After 28 days the cycle starts to repeat/repeats. **OR** The changes in days 28 to 36 are the same as days 0 to 8.	**1**	
	(g)		They prevent the negative feedback of oestrogen on FSH secretion. **OR** They mimic the action of FSH/LH.	**1**	

Question			Expected response	Max mark	Additional guidance
7.	(a)	(i)	Autosome	**1**	
		(ii)	P/the grandfather has passed the condition to R/his son.	**1**	
		(iii)	75%	**1**	
		(iv)	Twins	**1**	
	(b)		Cholesterol is deposited in the arteries (by LDL) **OR** Cholesterol cannot be deposited in the cells (by LDL). **(1)** Atherosclerosis occurs/atheromas form/plaques form (increasing blood pressure). **OR** Artery lumen (diameter) narrows/arteries narrow/arteries lose elasticity (increasing blood pressure). **(1)**	**2**	
	(c)	(i)	Statins	**1**	
		(ii)	Regular physical activity/it increases HDL levels/increases the ratio of HDL to LDL. **OR** Regular physical activity/it decreases the ratio of LDL to HDL. **(1)** This lowers (blood) cholesterol levels/concentrations. **(1)**	**2**	Not acceptable: regular physical activity lowers LDL levels.

<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>8.</td><td>(a)</td><td>(i)</td><td>Speed of treadmill.
OR
Length of time of exercise (at each level).
OR
Time between exercising at each level.</td><td>1</td><td>Accept temperature of room.

Accept resting heart rate before exercising at each level.</td></tr>
<tr><td></td><td></td><td>(ii)</td><td>Repeat the investigation at all gradients.
OR
Repeat the investigation with more individuals.
OR
Repeat the investigation and calculate an average.</td><td>1</td><td></td></tr>
<tr><td></td><td>(b)</td><td>(i)</td><td>95</td><td>1</td><td>Accept 95 only.</td></tr>
<tr><td></td><td></td><td>(ii)</td><td>Axes have correct scale and labels. (1)

Points correctly plotted and line drawn. (1)

<table><tr><th>Treadmill gradient</th><th>Cardiac output (litres/min)</th></tr><tr><td>0</td><td>8·6</td></tr><tr><td>2</td><td>9·8</td></tr><tr><td>4</td><td>12·4</td></tr><tr><td>6</td><td>16·6</td></tr><tr><td>8</td><td>17·4</td></tr><tr><td>10</td><td>17·6</td></tr></table></td><td>2</td><td>Scales using one large box for gradients of 2 on the X-axis are acceptable.

Any 3 values to establish a linear scale. A number at the origin is not essential.
Data can be plotted outwith the numbered scale.
Scale breaks are not acceptable.

If the axes are transposed do not award the scale mark.

Gradient alone is acceptable for the X-axis label.

The line must go through all points.</td></tr>
<tr><td></td><td></td><td>(iii)</td><td>As intensity of exercise increases cardiac output increases.</td><td>1</td><td>Not acceptable: as treadmill gradient increases cardiac output increases.</td></tr>
<tr><td></td><td>(c)</td><td></td><td>The time for blood filling the heart/ ventricles decreases.
OR
Less/a smaller volume of blood is able to enter the heart/ventricles.
OR
The heart rate was too high to allow the heart/ventricles to fully fill with blood.</td><td>1</td><td></td></tr>
</table>

Question			Expected response	Max mark	Additional guidance
9.	(a)		A – Dendrite **(1)** B – Synapse/synaptic cleft/ neuromuscular junction **(1)**	**2**	
	(b)		The myelin sheath/myelin/glial cells are attacked/damaged (by the immune system). **(1)** T lymphocytes respond to/attack self–antigens/the body's own cells. **(1)**	**2**	
	(c)		There is a decrease in the speed of impulse transmissions. **OR** Impulses are conducted at a slower rate.	**1**	

Question			Expected response	Max mark	Additional guidance
10.	(a)		X – Peripheral **(1)** Y – Autonomic **(1)**	**2**	
	(b)		They take impulses from sense organs/receptors to the central nervous system/CNS/interneurons.	**1**	Accept brain/spinal cord for CNS.
	(c)		Breathing - It decreases the rate/depth of breathing. **(1)** Digestive system - It increases peristalsis/the production of intestinal secretions/digestive enzymes. **(1)**	**2**	Accept – it increases blood flow to the digestive system.

Question			Expected response	Max mark	Additional guidance
11.	(a)	(i)	Sensory	**1**	
		(ii)	The diagram/it may trigger memories/remind them of when they coloured in/labelled the heart/diagram.	**1**	Not acceptable: general answers that do not relate to this activity.
	(b)		Rehearse/repeat the information (a few times). **(1)** Elaborate on/add more detail/add meaning to the information. **(1)** Organise/sort the information into groups/categories. **(1)** Elaboration is (an example of) a deep form of encoding. **OR** Rehearsal is (an example of) a shallow form of encoding. **(1)** **Any 3 from 4**	**3**	Candidates can be awarded one mark for use of the three terms rehearsal, elaboration and organisation where there are no marks awarded for descriptions.

Question			Expected response	Max mark	Additional guidance
12.	(a)		4797	**1**	
	(b)	(i)	Similarity - both increase (from 2013/14) to 2014/15. **OR** Both decrease from 2014/15 to 2015/16. **OR** Both peak/highest in 2014/15. **(1)** Difference - England increases between 2015/16 and 2016/17 while Scotland decreases. **OR** There are always more deaths in England/fewer deaths in Scotland. **(1)**	**2**	Not acceptable: they are both high in 2014/15.
		(ii)	Express the data as death rates. **OR** Produce figures per same number of people in each country. **OR** Relate the number of deaths to the size of populations in each country. **OR** Produce figures by dividing the number of deaths by (the size of) the populations of each country.	**1**	Accept examples of numbers; eg per 100 000.
		(iii)	9 : 100	**1**	
	(c)		9750	**1**	
	(d)		It alters/changes its antigens/ surface proteins. **OR** It undergoes antigenic variation.	**1**	
	(e)		Epithelial	**1**	

Question			Expected response	Max mark	Additional guidance
13.	(a)	(i)	Reduced uptake of the vaccine/ vaccine rejection. **OR** Herd immunity threshold for the disease not reached in these years. **OR** Reduced effectiveness of the vaccine. **OR** Antigenic variation of the bacteria. **OR** Mutation of the bacteria.	**1**	
		(ii)	104·76/104·8/105	**1**	
	(b)		The vaccine/it contains inactivated/ killed/weakened pathogens. **OR** The vaccine/it contains inactivated toxins from the pathogens.	**1**	Accept virus or bacteria for pathogens. Accept contains attenuated pathogens.
	(c)		Adjuvant	**1**	

Question			Expected response	Max mark	Additional guidance
14.	A		1. Antenatal screening identifies the risk of a disorder (so that further tests/a prenatal diagnosis can be offered). 2. Ultrasound imaging is offered between 8 and 14 weeks and between 18 and 20 weeks. 3. The dating/8 to 14 weeks scan determines pregnancy stage and due date. 4. The anomaly/18 to 20 weeks scan may detect physical abnormalities in the fetus. 5. Marker chemicals (which vary normally during pregnancy) are tested for. 6. Blood/urine tests are carried out regularly during pregnancy. 7. Measuring a chemical at the wrong time could lead to a false positive result. 8. An atypical/abnormal chemical concentration can lead to diagnostic testing. 9. Amniocentesis/chorionic villus sampling (CVS) (used to diagnose a range of genetic conditions). 10. CVS can be carried out earlier in pregnancy than amniocentesis, although it has a higher risk of miscarriage. **OR** Amniocentesis is carried out later in pregnancy but has a lower risk of miscarriage than CVS. 11. Cells (from samples) can be cultured/grown to produce a karyotype. 12. A karyotype shows an individual's chromosomes arranged as homologous pairs. 13. There are risks/decisions to consider before proceeding with diagnostic tests/these tests. **Any 9 from 13**	**9**	For points 2, 3 and 4, accept any number between 8 and 14 for the dating scan and any number between 18 and 20 for the anomaly scan. Point 8: Accept a specific example of a substance/chemical tested for. Point 13: Accept specific examples of risks/decisions.

Question			Expected response	Max mark	Additional guidance
14.	B		1. Arteries/veins/capillaries/blood vessels have a central lumen lined with endothelial cells/endothelium. 2. Arteries/veins have a middle layer containing smooth muscle and elastic fibres. 3. Arteries/veins have an outer layer of connective tissue containing elastic fibres. 4. Veins have a thinner muscle layer/muscular wall than arteries. **OR** Arteries have a thicker muscle layer/muscular wall than veins. 5. Veins contain valves to prevent the backflow of blood/to allow blood to flow in one direction. 6. The elastic walls of the arteries (stretch/recoil) to accommodate/ tolerate the surge/pulse of blood from the heart. 7. Artery (smooth) muscle contracts causing vasoconstriction/relaxes causing vasodilation. 8. Vasoconstriction/vasodilation controls blood flow through the artery. 9. Blood circulates/travels from the heart through the arteries to the capillaries then to the veins and back to the heart. 10. There is a decrease in blood pressure as blood moves away from the heart. **Maximum 8 marks from points 1 to10** 11. Capillaries allow exchange of substances with tissues/(body) cells through their thin walls. 12. Pressure filtration causes plasma to pass through capillary walls into the tissue fluid (surrounding the cells). 13. The plasma (leaving the capillary)/tissue fluid can exchange oxygen/glucose/ carbon dioxide/ (metabolic) waste. **To achieve full marks must have one from points 11 to 13**	**9**	Point 10: Accept: Blood pressure is higher in arteries compared to veins **OR** Blood pressure is lower in veins compared to arteries

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Human Biology

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.

(c) Do not award half marks.

(d) Where a candidate makes an error in the first part of a question, award marks for subsequent answers that are correct with regard to this original error. Do not penalise candidates more than once for the same error.

(e) Unless a numerical question specifically requires evidence of working to be shown, award full marks for a correct final answer (including units, if appropriate) on its own.

(f) Candidates should not use bulleted lists to answer extended-response questions. They must respond to the 'command' word as appropriate and provide extended answers to communicate fully their knowledge and understanding. Candidate responses in the form of bulleted lists may not be able to access the full range of available marks.

(g) In the detailed marking instructions, if a word is **underlined** then it is essential; if a word is **(bracketed)** then it is not essential.

(h) In the detailed marking instructions, words separated by/are **alternatives**.

(i) A correct response can be negated if the candidate includes:
- an extra, incorrect, response
- additional information that contradicts the correct response

(j) Where the candidate is instructed to choose one question to answer but instead answers two questions, mark both responses and award the higher mark.

(k) Unless otherwise required by the question, the use of abbreviations (eg DNA, ATP) or chemical formulae (for example CO_2, H_20) are acceptable alternatives to naming.

(l) If a numerical answer is required and units are not given in the stem of the question or in the answer space, candidates must supply the units to gain the mark. If units are required on more than one occasion, do not penalise candidates repeatedly.

(m) If incorrect spelling is given, sound out the words.
- If the correct word is recognisable then award the mark.
- If the word can easily be confused with another biological term then **do not** award the mark, for example glucagon and glycogen.

(n) Presentation of data:

- If a candidate provides two graphs, in response to one question, mark both and award the higher mark.
- If a question asks for a particular type of graph/chart and the candidate gives the wrong type, do not award full marks. Candidates cannot achieve the plot mark but **may** be able to achieve the mark for scale and label. If the x and y data are transposed, then do not award the scale and label mark.
- If the graph uses less than 50% of the axes then do not award the scale and label mark.
- If 0 is plotted when no data for this is given, then do not award the plot mark – candidates should only plot the data given.

(o) Only award marks for a valid response to the question asked. For example, in response to questions that ask candidates to:

- **identify, name, give** or **state**, they need only answer or present in brief form
- **describe**, they must provide a statement as opposed to simply one word
- **explain**, they must provide a reason for the information given
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between topics being examined
- **calculate**, they must determine a number from given facts, figures or information
- **predict**, they must indicate what may happen based on available information
- **suggest**, they must apply their knowledge and understanding to a new situation

National
Qualifications
2019

2019 Human Biology

Higher – Paper 1 – Multiple Choice

Finalised Marking Instructions

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Marking instructions for each question

Question	Answer	Mark
1.	D	**1**
2.	C	**1**
3.	A	**1**
4.	D	**1**
5.	C	**1**
6.	B	**1**
7.	B	**1**
8.	A	**1**
9.	C	**1**
10.	D	**1**
11.	A	**1**
12.	A	**1**
13.	B	**1**
14.	B	**1**
15.	C	**1**
16.	A	**1**
17.	C	**1**
18.	B	**1**
19.	D	**1**
20.	C	**1**
21.	B	**1**
22.	D	**1**
23.	D	**1**
24.	A	**1**
25.	B	**1**

[END OF MARKING INSTRUCTIONS]

National
Qualifications
2019

2019 Human Biology

Higher – Paper 2

Finalised Marking Instructions

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Marking instructions for each question

Question			Expected response	Max mark	Additional guidance
1.	(a)	(i)	Pluripotent	**1**	
		(ii)	All genes are switched on/can be expressed/can be switched on **OR** No genes are switched off **OR** All proteins can be made by embryonic cells	**1**	Do not negate for all genes can be switched on or off. Do not accept all genes can be switched off on its own. Do not accept all genes <u>are</u> expressed.
	(b)		23	**1**	
	(c)		To study diseases/(genetic) disorders/drug testing. **OR** To study cell processes/cell growth/cell differentiation/gene regulation. **OR** As model cells.	**1**	Do not accept therapeutic uses such as for corneal repair or for growing skin.
	(d)		Embryos do not need to be used/ destroyed. **OR** It means a (potential) life does not need to be ended/destroyed.	**1**	

Question			Expected response	Max mark	Additional guidance
2.	(a)		1. To separate/split the DNA/ strands. **OR** To break <u>hydrogen</u> bonds. **OR** To break bonds <u>between the bases</u>. **(1)** 3. Any temperature between 70 and 80. **(1)**	**2**	Accept to denature the DNA. Unzipping/unwinding the DNA negates.
	(b)		33	**1**	
	(c)		Stage 2 - (Primers) bind/attach to DNA/3'end/strands. **OR** Stage 2 – (Primers) bind/anneal/ attach to a complementary section/ target sequence/complementary nucleotides. **(1)** Stage 3 – (Primers) allow <u>DNA/Taq polymerase</u> to attach/start replicating/start copying/add nucleotides (to DNA). **(1)**	**2**	Answer must indicate link between the primer and DNA polymerase. Accept DNA/Taq polymerase adds nucleotides to <u>primers</u>.
	(d)		<u>DNA</u> can be amplified/copied/ replicated/multiplied.	**1**	

Question			Expected response	Max mark	Additional guidance
3.	(a)		Acetyl group (is produced/present/ made). **(1)** Acetyl group combines with oxaloacetate to form citrate. **OR** Acetyl group enters the citric acid cycle. **OR** Acetyl group combines with coenzyme A to form acetyl coenzyme A. **(1)**	**2**	Acetyl coenzyme A entering the citric acid cycle/combining with oxaloacetate does not negate.
	(b)	(i)	It contains a different/wrong <u>amino acid</u> (in the enzyme). **OR** The shape of the enzyme/active site is changed.	**1**	Accept that the sequence of <u>amino acids</u> has changed/been altered. Do not accept that the sequence of <u>amino acids</u> after the mutation has changed/been altered. Do not accept the amino acids are altered.
		(ii)	Acetaldehyde/the toxic compound builds up/cannot be broken down (quickly). **OR** They are less efficient/slower at breaking down acetaldehyde/the toxic compound. **OR** They are less efficient/slower at forming/cannot form the non-toxic compounds.	**1**	Do not accept that they cannot fully break down alcohol on its own.
	(c)		It binds to/blocks/competes for the active site. **OR** It prevents the substrate/ acetaldehyde binding to the enzyme.	**1**	

Question			Expected response	Max mark	Additional guidance
4.	(a)		1. Concentration/volume/mass/ dosage of supplement/placebo taken. 2. Time of day for exercise/taking the supplement/taking the placebo/testing. 3. Brand/type of supplement. 4. Other/additional (types of) exercise/training. 5. Number of times weight is lifted/ number of repetitions/time spent on training/intensity of training. 6. Food/fluid/other supplements/ medicines/drugs consumed. 7. Student health issues/smoking status/ethnicity **Any 2**	**2**	Only penalise the use of amount once. If candidate lists three or more variables and at least one is correct they can be awarded (**1 mark)**. If all answers are correct award **(2 marks).**
	(b)	(i)	Axes have correct scales and labels. **(1)** Points correctly plotted and lines drawn. **(1)** Labels or key added to distinguish between group A/protein supplement and group B/placebo. **(1)**	**3**	Scales using one large box for 2 weeks on the X-axis and one large box for 10kgs on the Y axis are acceptable. Lines must go through all points. Only the X-axis scale must start at zero. If the axes are transposed the candidate loses the scale mark.

	Average maximum weight raised (kg)					
Time(weeks) / Group	0	2	4	6	8	10
A	52	57	64	72	86	95
B	50	55	60	68	74	80

Question			Expected response	Max mark	Additional guidance
		(ii)	The protein supplement improved the ability (of leg muscles) to raise weights. **OR** The protein supplement allowed students to lift heavier weights/ more weight.	**1**	Do not accept the protein supplement improved performance. Answer must not simply describe the results.
		(iii)	Training/repeated use of apparatus/ use of weights.	**1**	Accept - the placebo effect/they believed they were taking the supplement.
	(c)		Group A/group taking the supplement had a greater increase in body mass compared to group B. **OR** Group A/group taking the supplement lost more fat compared to group B. **(1)** The change/increase in body mass must be due to muscle. **OR** The supplement/protein is used to synthesise/make muscle. **OR** This suggests body fat is being replaced by/converted to muscle. **(1)**	**2**	Accept group A/group taking the supplement gained 1·8kg/3% in body mass while group B gained 0·8kg/ 1·3% Accept group A/group taking the supplement lost 4·5% fat while group B lost 2·1%.

Question			Expected response	Max mark	Additional guidance
5.	(a)		ATP is produced/released/given out/gained (by the cell)/removed (from the pathway)/lost (from the pathway).	**1**	Do not accept that there is a <u>net</u> gain of ATP.
	(b)		They remove/release hydrogen (ions)/electrons. **OR** They pass hydrogen (ions)/electrons to NAD.	**1**	Do not accept collects hydrogen (ions)/electrons. Accept H^+ for hydrogen ions. Accept H/H_2 for hydrogen. Accept e^- for electrons.
	(c)		Glucagon	**1**	Accept - adrenaline/epinephrine
	(d)		(No/less intermediate Q so) no/less nucleic acids/nucleotides/lipids. **(1)** Nucleic acids/nucleotides are needed for cell division/making chromosomes/DNA replication/ protein synthesis. **OR** Lipids are needed for membrane formation. **(1)**	**2**	

Question			Expected response	Max mark	Additional guidance
6.	(a)	(i)	Any interstitial cell between the seminiferous tubules labelled with a T.	**1**	If no line is drawn, the letter T must be near or on the interstitial cells.
		(ii)	1. It stimulates/promotes sperm/ gamete production. **(1)** 2. It activates/triggers the prostate gland/seminal vesicles (to produce secretions). **(1)** 3. It inhibits/stops the production of FSH/ICSH/the pituitary gland. **(1)** **Any 2 from 3**	**2**	Accept – it triggers the development of (secondary) sexual characteristics. Accept specific examples such as growth of facial hair and deepening of voice. Do not accept – it triggers puberty. Do not accept negative feedback of the pituitary on its own. Do not accept LH in place of ICSH but additional use does not negate.
		(iii)	(Low levels of) testosterone fails to inhibit the <u>pituitary</u> gland. **(1)** (An increased concentration of) <u>ICSH</u> is released (by the pituitary gland). **(1)** <u>Interstitial cells</u> produce testosterone. **(1)** **Any 2 from 3**	**2**	Accept low levels of testosterone stimulate the <u>pituitary</u> gland. Use of FSH in addition to ICSH does not negate. Do not accept LH in place of ICSH.
	(b)		AI - several samples of <u>semen</u> are collected (over a period of time). **OR** Sperm/semen is injected directly into the (female) reproductive tract/vagina/uterus/womb. **(1)** ICSI – The (head of the) sperm is injected directly into the egg (to achieve fertilisation). **(1)**	**2**	Answer must indicate insertion of sperm/semen by artificial means.

Question			Expected response	Max mark	Additional guidance
7.	(a)	(i)	<u>Similarity</u> Both increase up to 28 years. **OR** Both decrease from 28 years. **OR** Both peak at 28 years. **(1)** <u>Difference</u> The pregnancy rate is (always) higher (than the birth rate). **OR** The birth rate is (always) lower (than the pregnancy rate). **OR** The pregnancy rate increases faster than the birth rate up to 28 years. **OR** The birth rate decreases to zero (at 46 years) while the pregnancy rate does not. **(1)**	**2**	Answers must relate to trends and not single points. Accept same age for 28 years.
		(ii)	350	**1**	
	(b)	(i)	36	**1**	
		(ii)	The women who donated the eggs will be younger. **OR** <u>The</u> (donated) <u>eggs</u>/<u>they</u> will be younger.	**1**	Answer must indicate a comparison. Accept – <u>their</u> eggs are older. Do not accept their eggs are newer.

Question			Expected response	Max mark	Additional guidance
	(c)		PGD identifies genetic disorders/ chromosome abnormalities. **OR** A parent/family member has a genetic disorder/chromosome abnormality. **OR** Biological tests indicate a genetic disorder/chromosome abnormality. **OR** To ensure that the embryo/fetus/ baby/child does not have the genetic disorder/chromosome abnormality.	**1**	Accept genetic diseases/inherited diseases/genetic mutations/ genetic conditions for genetic disorders. Accept gene mutations/ chromosome mutations/genetic abnormalities for chromosome abnormalities. Do not accept answers that suggest PGD identifies risk/likelihood of having the genetic disorder.

Question			Expected response	Max mark	Additional guidance
8.	(a)		An autosomal disorder/gene/allele/ mutation is not located on a sex chromosome while a sex-linked disorder/gene/allele/mutation is (located on a sex/X chromosome). **OR** An autosomal disorder/gene/allele/ mutation is located on autosomes/ chromosomes 1 to 22 while a sex-linked disorder/gene/allele/ mutation is located on the sex chromosomes/X chromosome. **OR** Autosomal disorders affect males and females (equally) while sex-linked disorders affect more males (than females).	**1**	Mention of the Y chromosome does not negate. Accept chromosome 23 for X chromosome.
	(b)	(i)	Aa	**1**	Only accept answers using 'a'.
		(ii)	You cannot tell the genotype (of Q and R). **OR** You cannot tell if they/Q and R are carriers/heterozygous/homozygous/ Aa/AA.	**1**	Accept – they can both have two dominant alleles or one recessive allele. Do not negate if the letters are wrong eg Tt
	(c)		Stem cells/bone marrow can start producing/produce/differentiate into lymphocytes/white blood cells.	**1**	Do not accept – bone marrow contains lymphocytes/white blood cells.
	(d)		Advantage – (amniocentesis has a) lower risk of miscarriage. **(1)** Disadvantage – (amniocentesis is carried out) later in the pregnancy. **(1)**	**2**	Answers must indicate a comparison. Accept – amniocentesis takes longer to get the results.

Question			Expected response	Max mark	Additional guidance
9.	(a)		300	**1**	
	(b)		120	**1**	
	(c)		12 500	**1**	
	(d)	(i)	Cells are less sensitive to insulin/ resistant to insulin. **OR** Cells/they have fewer insulin receptors/have less sensitive insulin receptors/less effective insulin receptors. **(1)** They cannot convert/convert less glucose to glycogen. **OR** Less glucose is taken into the cells. **(1)**	**2**	Accept insulin receptors are desensitised. Answer must indicate the receptors are linked to insulin. Accept that it will take longer to convert glucose into glycogen.
		(ii)	Glucose/it is lost in urine/removed by the kidneys. **OR** Glucose/it is used in respiration/ in glycolysis/cell activities/to produce energy.	**1**	
	(e)		The man is able to produce insulin/ insulin is present. **OR** The (blood) insulin concentration increases/is high.	**1**	
	(f)		Exercise. **OR** Lose weight/fat.	**1**	Answers indicating insulin negate.

Question			Expected response	Max mark	Additional guidance
10.	(a)	(i)	Men in England/English men	**1**	
		(ii)	Death rate is (always) higher in men/ lower in women. **OR** Death rate is (always) higher in Scotland/lower in England. **OR** English women have the lowest death rate/Scottish men have the highest death rate. **OR** Death rates decreased more in England/less in Scotland.	**1**	Answers must indicate a comparison and must refer to death rate. Do not accept answers that relate to chance/risk of death rather than death rate. Accept more die/more deaths in place of death rate is higher. Accept faster for more and slower for less.
		(iii)	9:11	**1**	
		(iv)	The death rate/it is <u>per 100 000</u>.	**1**	Answers that indicate the populations are the same size negate.
	(b)	(i)	Neither the participants/men nor the researchers/doctors know which group each man is placed into/who is getting the drug/placebo.	**1**	Candidate must indicate person giving drug and person receiving the drug don't know.
		(ii)	Large numbers/3000 men were used **OR** There were large numbers/1500 men in each group. **OR** A large sample size was used.	**1**	
		(iii)	Less cholesterol is deposited in arteries/the blood. **(1)** Less chance of atherosclerosis/the formation of an atheroma/the formation of plaque occurring (which causes CVD). **(1)**	**2**	Both answers must indicate less but answers that do not indicate less which are otherwise correct should be awarded (1 mark). eg LDL deposits cholesterol in the blood which leads to atheromas being formed (1 mark).

Question			Expected response	Max mark	Additional guidance
11.	(a)	(i)	So water flea gets used to conditions/reacts to conditions/ becomes acclimatised. **OR** Allow the heart rate to stabilise/ adjust/settle/return to its resting rate. **OR** Allow time for caffeine to take effect/enter the flea.	**1**	Do not accept to allow the reaction to take place. Incorrect use of accuracy or reliability does not negate an otherwise correct answer.
		(ii)	So you can play it back. **OR** To slow it down. **OR** To get a more accurate reading. **OR** To minimise counting error/make it easier to count.	**1**	Use of reliable negates. Accept – as it is difficult to count (the heart beats).
		(iii)	As the caffeine concentration increases from 0 to 0·8 g/l the heart rate increases and then it levels off/ remains constant from 0·8 g/l to 1 g/l. **(1)** It increases from 135 to 270 bpm/by 135 bpm. **OR** It levels off/remains constant at 270 bpm. **(1)**	**2**	Accept it increases to 0·8 g/l then it levels off/remains constant. Answer must indicate g/l for at least one figure. Answer must indicate bpm for at least one figure.
		(iv)	Repeat the investigation/experiment at each (caffeine) concentration. **OR** Repeat the investigation/experiment with more water fleas. **OR** Repeat the investigation/experiment and take an average.	**1**	Do not accept use additional (caffeine) concentrations. Accept with another water flea. Do not accept using different model organisms.

Question			Expected response	Max mark	Additional guidance
	(b)		1. The <u>medulla</u> controls the autonomic nervous system/ sympathetic nervous system/the sympathetic nerve/sino-atrial node/SAN. 2. The sino-atrial node/SAN sends impulses/signals through the heart/across the atria/to the atrio-ventricular node/AVN. **OR** The sino-atrial node/SAN is located in the wall of the right atrium (of the heart). 3. A sympathetic neuron/nerve/ nervous system releases noradrenaline/norepinephrine. 4. Noradrenaline/norepinephrine/ the sympathetic nervous system/ increases the heart rate. **Any 3 from 4**	**3**	

Question			Expected response	Max mark	Additional guidance
12.	(a)	(i)	(Prolonged) exercise/eating/sex/ laughter.	**1**	Do not accept stress/injury but these do not negate a correct answer. Accept any named sporting activity for exercise. Accept eating any named food. Accept taking (recreational) drugs or named drug eg cocaine, alcohol, nicotine.
		(ii)	They reduce (the intensity of) pain **OR** They induce feelings of pleasure/ euphoria/happiness.	**1**	Accept act as natural painkillers. Accept that they inhibit pain receptors/signals. Activating the reward pathway negates.
	(b)		Acts as a dopamine agonist/mimics the action of dopamine. **OR** Prevents/slows the breakdown of dopamine. **OR** Prevents/slows the reuptake of dopamine.	**1**	Do not accept – take synthetic dopamine/L-dopa unless qualified with a correct mode of action. All answers must relate to a drug treatment.
	(c)	(i)	It has reduced the number of/there are less receptors.	**1**	Mention of sensitivity of receptors does not negate. Do not accept amount of receptors. Use of antagonist does not negate.
		(ii)	The individual must take more of the drug (to get an effect).	**1**	Do not accept it leads to increased tolerance of the drug but this does not negate. Use of addiction does not negate.

Question			Expected response	Max mark	Additional guidance
13.	(a)	(i)	They form a (physical) barrier **OR** They are closely packed.	**1**	Accept that they produce a waterproof barrier. Accept that they produce sweat/ sebum/chemical secretions. Do not accept stomach acid, saliva, mucus and tears. Do not accept that they form a chemical barrier.
		(ii)	(Blood) capillary	**1**	
	(b)	(i)	Mast (cells)	**1**	
		(ii)	Vasodilation/increased blood flow **OR** Increased permeability of capillaries.	**1**	Accept from a buildup of tissue fluid. Do not accept vasodilation of capillaries but this does not negate if answer includes correct reference to capillary permeability. Accept more porous for increased permeability.
	(c)	(i)	It recognises an <u>antigen</u> **OR** <u>Antigens</u> on the pathogen. **OR** It has <u>specific/antigen</u> receptors/ binding sites (to the pathogen).	**1**	Do not accept the word antigen on its own.
		(ii)	They cause the cells to produce self-destructive enzymes **OR** They induce/cause apoptosis/ (programmed) cell death.	**1**	Do not accept T-lymphocytes inject self-destructive proteins/enzymes into infected cells. Accept cell suicide for cell death. Answer must indicate the role of T-lymphocytes so apoptosis on its own is incorrect.
	(d)	(i)	The immune response/lymphocytes/ T cells have destroyed most of the viruses/HIV/infected cells.	**1**	Accept the immune response/ lymphocytes/T cells are attacking/ removing the virus/HIV/the infection. Do not accept the immune response/lymphocytes/T cells are <u>fighting</u>/<u>combating</u> the virus/HIV/ the infection.
		(ii)	The HIV virus is attacking/ destroying T lymphocytes/T cells/ them.	**1**	Accept that the HIV virus kills T lymphocytes.

Question			Expected response	Max mark	Additional guidance
14.	A		**Encoding** 1. Encoding is transferring information into memory/short-term memory (STM)/ long-term memory (LTM). 2. Transfer occurs from sensory memory to STM to LTM. 3. Only selected images/sounds are encoded/transferred into STM/transferred from sensory memory. 4. Information is transferred from STM to long-term memory (LTM) by rehearsal/ organisation/elaboration. **Any 2 for the mark** 5. Rehearsal is the repeating of information/organisation involves putting information into related groups/ elaboration involves adding meaning to information. 6. Rehearsal is a shallow form of encoding/elaboration is a deeper form of encoding (information into LTM). **Storage** 7. Sensory memory retains/holds information for a few seconds. 8. STM has a limited (storage) capacity/ holds information for a short time. 9. The number of items held in STM can be increased by <u>chunking</u>. 10. Information is retained in STM by rehearsal. 11. Information is lost from STM by displacement/decay. 12. LTM has an unlimited (storage) capacity/holds information for a long time. **Retrieval** a. Retrieval of information (from LTM) is aided/helped by the use of <u>contextual cues</u>. b. Contextual cues relate to the time/ place where the information was (firstly) encoded/memory was created.	**9**	5. Candidates cannot explain rehearsal/organisation and elaboration by only using the terms as their explanation. 8. Accept any number between 5 to 9 items for limited STM capacity. 8. Accept 30 seconds or less for short time. 9. Accept an example that illustrates <u>chunking</u> increases the number of items held in STM. 9. The use of increased memory span does not negate answers relating to <u>chunking</u>. a. trigger a memory from LTM is the same as retrieval of information from LTM.

Question			Expected response	Max mark	Additional guidance
14.	B		**Vaccination** 1. Vaccination develops/provides immunity against a pathogen/disease. **OR** Vaccination produces memory cells against a pathogen. 2. Antigens from pathogens are used to create memory cells/immunological memory. 3. If a vaccinated individual becomes infected/exposed again, antibody production/the immune/the secondary response is greater/more rapid/ prevents symptoms developing. 4. Vaccines can contain inactivated (pathogen) toxins/dead pathogens/parts of pathogens/weakened pathogens. 5. Antigens are usually mixed with an adjuvant **OR** Vaccines usually contain/are mixed with an adjuvant. 6. An adjuvant makes the vaccine more effective/enhances the immune response. **Herd Immunity** a. Mass vaccination programmes are designed to establish herd immunity (to a disease). b. Herd immunity occurs when a large percentage/high numbers of a population are immune/immunised/ vaccinated. c. Herd immunity reduces the spread of diseases. d. Herd immunity protects non-immune individuals as they have a lower probability/less chance of coming into contact with infected individuals. e. Diseases have a threshold for herd immunity. f. The threshold depends on the type of disease/the effectiveness of the vaccine/the density of the population. **Any 2 for the mark** g. Herd immunity/vaccination may not be possible due to poverty. h. Herd immunity/vaccination may not occur due to vaccine rejection.	**9**	Accept viruses or bacteria for pathogens The terms vaccine/ vaccination/immunisation are interchangeable. d. Answer must indicate coming into contact with an individual who has the disease. g. Accept lack of access to medical facilities/doctors as poverty. h. Accept beliefs/allergy as rejection.

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Human Biology

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.

(c) Do not award half marks.

(d) Where a candidate makes an error in the first part of a question, award marks for subsequent answers that are correct with regard to this original error. Do not penalise candidates more than once for the same error.

(e) Unless a numerical question specifically requires evidence of working to be shown, award full marks for a correct final answer (including units, if appropriate) on its own.

(f) Candidates should not use bulleted lists to answer extended-response questions. They must respond to the 'command' word as appropriate and provide extended answers to communicate fully their knowledge and understanding. Candidate responses in the form of bulleted lists may not be able to access the full range of available marks.

(g) In the detailed marking instructions, if a word is **underlined** then it is essential; if a word is **(bracketed)** then it is not essential.

(h) In the detailed marking instructions, words separated by/are **alternatives**.

(i) A correct response can be negated if the candidate includes:
- an extra, incorrect, response
- additional information that contradicts the correct response

(j) Where the candidate is instructed to choose one question to answer but instead answers two questions, mark both responses and award the higher mark.

(k) Unless otherwise required by the question, the use of abbreviations (for example DNA, ATP) or chemical formulae (for example CO_2, H_20) are acceptable alternatives to naming.

(l) If a numerical answer is required and units are not given in the stem of the question or in the answer space, candidates must supply the units to gain the mark. If units are required on more than one occasion, do not penalise candidates repeatedly.

(m) If incorrect spelling is given, sound out the words.
- If the correct word is recognisable then award the mark.
- If the word can easily be confused with another biological term then **do not** award the mark, for example glucagon and glycogen.

(n) Presentation of data:

- If a candidate provides two graphs, in response to one question, mark both and award the higher mark.
- If a question asks for a particular type of graph/chart and the candidate gives the wrong type, do not award full marks. Candidates cannot achieve the plot mark but **may** be able to achieve the mark for scale and label. If the x and y data are transposed, then do not award the scale and label mark.
- If the graph uses less than 50% of the axes then do not award the scale and label mark.
- If 0 is plotted when no data for this is given, then do not award the plot mark — candidates should only plot the data given.

(o) Only award marks for a valid response to the question asked. For example, in response to questions that ask candidates to:

- **identify, name, give** or **state**, they need only answer or present in brief form
- **describe**, they must provide a statement as opposed to simply one word
- **explain**, they must provide a reason for the information given
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between topics being examined
- **calculate**, they must determine a number from given facts, figures or information
- **predict**, they must indicate what may happen based on available information
- **suggest**, they must apply their knowledge and understanding to a new situation

2019 Higher Human Biology Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	**D**	2	3	1	**92**	0	2
2	**C**	4	4	**72**	12	0	8
3	**A**	**49**	4	36	1	0	9
4	**D**	9	8	10	**65**	0	8
5	**C**	1	4	**73**	13	0	9
6	**B**	12	**79**	1	4	0	4
7	**B**	3	**43**	37	7	0	10
8	**A**	**75**	2	6	12	0	5
9	**C**	22	20	**43**	7	0	8
10	**D**	12	6	9	**67**	0	7
11	**A**	**67**	13	10	5	0	5
12	**A**	**28**	46	7	9	0	10
13	**B**	3	**59**	30	2	0	7
14	**B**	34	**41**	12	2	0	11
15	**C**	3	13	**74**	7	0	3
16	**A**	**62**	11	1	21	0	6
17	**C**	11	3	**79**	2	0	4
18	**B**	22	**67**	1	4	0	6
19	**D**	1	1	1	**94**	0	3
20	**C**	41	5	**44**	1	0	8
21	**B**	3	**84**	6	3	0	4
22	**D**	5	5	35	**48**	0	8
23	**D**	4	14	10	**64**	0	7
24	**A**	**70**	15	2	8	0	4
25	**B**	4	**77**	8	6	0	5

National
Qualifications
2018

2018 Human Biology

Higher

Finalised Marking Instructions

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Marking Instructions for each question

Section 1

Question	Answer	Mark
1.	B	**1**
2.	A	**1**
3.	B	**1**
4.	D	**1**
5.	B	**1**
6.	D	**1**
7.	C	**1**
8.	A	**1**
9.	A	**1**
10.	C	**1**
11.	D	**1**
12.	D	**1**
13.	C	**1**
14.	B	**1**
15.	C	**1**
16.	A	**1**
17.	B	**1**
18.	D	**1**
19.	D	**1**
20.	C	**1**

Section 2

Question			Expected response	Max mark	Additional guidance
1.	(a)	(i)	Cytoplasm	**1**	
		(ii)	(Two) ATP (molecules) are used. **OR** ATP provides phosphate (to phosphorylate glucose/intermediates). **OR** Phosphorylation of glucose/ intermediates occur.	**1**	Do not accept ATP molecules are invested. Accept ATP is converted/ broken down to ADP and Pi. Any mention of net gain of ATP / production of ATP negates.
	(b)	(i)	NADH /$NADH^+$	**1**	Accept $NADH_2$
		(ii)	NAD allows ATP production (during glycolysis). **OR** NAD allows glycolysis to continue. **OR** NAD accepts / collects / carries hydrogen / hydrogen ions. **OR** NAD allows dehydrogenase enzymes to function.	**1**	Do not accept so NAD can be reused. Do not accept NAD removes hydrogen / hydrogen ions. Accept NAD is reduced (to NADH) Any mention of the electron transport chain/citric acid cycle/next stage negates.
		(iii)	Not enough / no oxygen (reaches the muscle cells).	**1**	Accept conditions become anaerobic / an oxygen debt has built up.

<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>2.</td><td>(a)</td><td></td><td>Any two from:

• mass/weight/surface area/size/ type of liver;
• volume of inhibitor / alcohol / solution;
• <u>initial</u> alcohol concentration;
• pH of <u>solution</u>;
• type of alcohol / inhibitor.</td><td>2</td><td>If candidate lists three or more variables and at least one is correct they can be awarded 1 mark.

Do not accept size of test tube.

Do not accept ‘amount’ but do not penalise twice if used in both responses.

Do not accept ‘alcohol concentration’.</td></tr>
<tr><td></td><td>(b)</td><td></td><td>It binds elsewhere (on the enzyme) / the allosteric site / not to the active site and changes the shape of the <u>active site</u>.</td><td>1</td><td>Use of denatures negates.</td></tr>
<tr><td></td><td>(c)</td><td>(i)</td><td>Axes have correct scales and labels. (1)

Points correctly plotted and line drawn. (1)

<table><tr><th>Inhibitor concentration (mM)</th><th>Final alcohol concentration (% of initial concentration)</th></tr><tr><td>0·5</td><td>20</td></tr><tr><td>1·5</td><td>28</td></tr><tr><td>2·5</td><td>60</td></tr><tr><td>3·5</td><td>96</td></tr><tr><td>4·5</td><td>100</td></tr><tr><td>5·5</td><td>100</td></tr></table></td><td>2</td><td>Candidate cannot access the scale mark if the plotted points use less than half of the graph paper.

If the scales start at zero there must be a zero entered. A common zero is acceptable if both scales start at zero. The scales do not have to start at zero but final number must be equal to or above 5·5mM and equal to 100%.

If the axes are transposed the candidate loses the scale mark.

Line must go through all points.

If all points correctly plotted, assume horizontal line between 4·5 and 5·5 is correct.</td></tr>
<tr><td></td><td></td><td>(ii)</td><td>As inhibitor concentration increases (enzyme) <u>activity</u> decreases. (1)

Beyond 4·5 (mM) there is no (enzyme) <u>activity</u>/<u>activity</u> stops/<u>activity</u> levels off/ <u>activity</u> remains constant. (1)</td><td>2</td><td>Answer must indicate the change happens from 4·5 mM onwards rather than only at 4·5 mM.</td></tr>
</table>

Question			Expected response	Max mark	Additional guidance
2.	(d)		The (final) alcohol concentration will not change / will not decrease / will stay at 100% / will remain high. **OR** (Increasing the initial) alcohol/ substrate concentration will have no effect (on the results).	**1**	Candidates are not expected to state whether they are discussing initial or final alcohol concentration. Do not accept answers in terms of enzyme activity.

Question			Expected response	Max mark	Additional guidance
3.	(a)	(i)	Deletion	**1**	
		(ii)	The order/sequence of <u>amino acids</u> would change **OR** <u>Amino acids</u> after the mutation would change **OR** The protein is shorter due to the presence of a <u>stop codon</u>.	**1**	Do not accept one / wrong amino acid is changed. Do not accept stop codon is produced (without reference to shorter protein).
	(b)		Systematics	**1**	
	(c)	(i)	15 : 5 : 1	**1**	
		(ii)	The more an animal has the better is its sense of smell. **OR** Rats have the best / a better sense of smell because they have most (functional genes/nasal receptors). **OR** Chickens have the worst / a poorer sense of smell because they have fewest (functional genes/nasal receptors).	**1**	Answer must link to number of genes.
	(d)		1. <u>Deletion</u> is the loss of part of a chromosome / genes. 2. <u>Duplication</u> is the repeat of a part of a chromosome / genes. OR <u>Duplication</u> – the breaking off of part of a chromosome / genes and it reattaching to its homologous partner. 3. <u>Translocation</u> the moving/swapping/ transfer of parts of chromosomes / genes to another chromosome / non-homologous chromosome. **Any 2 from 3** 4. These mutations/changes can be lethal/be fatal to/kill the individual/life threatening. **OR** An example named such as cri-du-chat for deletion, myeloid leukaemia or <u>familial</u> Down's syndrome for translocation. **(1)**	**3**	Give 1 mark for naming two types of mutation but providing no / incorrect descriptions. Cannot use the name of the mutation in its description. E.g. deletion is the deletion of genes. Example must be correctly linked to the chromosome mutation.

Question			Expected response	Max mark	Additional guidance
4.	(a)	(i)	5·6 per 100 000	**1**	
		(ii)	37·5	**1**	
		(iii)	46	**1**	
		(iv)	Between 2007 and 2009 there is an increase in breastfeeding while infant mortality continues to increase/does not decrease.	**1**	Accept 2007 and 2008 **or** 2008 and 2009 as an alternative to between 2007 and 2009. Ignore any Y-axis figures.
	(b)	(i)	6	**1**	
		(ii)	Bottles/water/milk can be contaminated with bacteria/viruses/ pathogens/ microbes. **OR** Breastfed babies can get antibodies/ immunity from their mother.	**1**	Do not accept germs.
5.	(a)		Medulla **(1)** SAN/right atrium (wall)/pacemaker **(1)**	**2**	
	(b)		Sympathetic nervous system increases heart rate while parasympathetic system decreases it. **(1)** Sympathetic system releases/produces noradrenaline/norepinephrine while the parasympathetic system releases/produces acetylcholine. **(1)**	**2**	
	(c)	(i)	60	**1**	
		(ii)	Ventricles are contracting/pumping blood/ventricular systole. **OR** Impulses are spreading through the ventricles.	**1**	Mention of atria contracting negates.

Question			Expected response	Max mark	Additional guidance
6.	(a)		15-44	**1**	
	(b)	(i)	Statins	**1**	
		(ii)	2007/2008	**1**	
	(c)		Component of the cell membrane. **OR** Used to synthesise steroids/sex hormones/testosterone/oestrogen/ progesterone.	**1**	
7.	(a)	(i)	30·68/30·7/31	**1**	Ignore units.
		(ii)	The office worker has/ they have less muscle/more fat. **OR** The swimmer has more muscle/less fat.	**1**	A comparison between individuals must be made. Do not accept 'more muscles'. Accept 'bigger muscles' or more muscular.
	(b)		Cells/receptors have become less sensitive/(more) resistant/desensitised to insulin. **OR** Cells have less insulin receptors. **(1)** This means that glucose cannot enter the cells. **OR** This means that glucose cannot be converted to / stored as glycogen. **OR** This means less glucose is converted to / stored as glycogen **(1)**	**2**	If the cell is specified accept liver, muscle and fat, but not pancreas. Accept this means that glucose channels do not open.

Question			Expected response	Max mark	Additional guidance
8.	(a)	(i)	Encoding	**1**	
		(ii)	Short-term memory has a limited capacity/span /stores only 5-9 pieces of information.	**1**	Accept any number between 5 and 9.
		(iii)	Information can put into categories/related groups. **OR** Information can have detail/meaning added to it.	**1**	Candidate must indicate the information in the groups is related / linked. Accept a correct description of either organisation or elaboration. If these terms are used they must be correctly described. Use of any incorrect term with correct description (of organisation / elaboration) negates. Do not accept chunking.
		(iv)	These trigger memories relating to the circumstances/conditions/ environment/stimuli present when the memory was formed. **OR** Suitable example described.	**1**	e.g. they help you remember as that scent was present when the memory was made.
	(b)		Cerebrum/cerebral cortex	**1**	Only accept these two terms.

Question			Expected response	Max mark	Additional guidance
9.	(a)	(i)	It/cars/objects which are further away look/appear/are smaller. **OR** It/cars/objects which are nearer look/appear/are bigger.	**1**	
		(ii)	The view of cars / objects which are further away is (partially) blocked / screened / obstructed /covered by nearer cars / objects.	**1**	Answer must convey 'overlapping'.
	(b)		Binocular disparity	**1**	
	(c)	(i)	16	**1**	Accept if correct answer is in the table and there is no answer in the answer space.
		(ii)	To prevent them learning the position of the coin. **OR** To prevent them remembering where the coin was. **OR** To make sure that the experiment / results were valid.	**1**	The use of reliable or accurate negates any answer. Accept so student did not know where the coin would be / where to place their finger Accept to prevent them creating a motor pathway of where the coin is.
		(iii)	The (student's) left eye is better/more accurate at judging distance. **OR** The (student's) right eye is worse/less accurate at judging distance.	**1**	

Question			Expected response	Max mark	Additional guidance
10.	(a)	(i)	1. Less/a lower number/a lower percentage of females smoke than males in all age groups. **OR** More/a higher number/ a higher percentage of males smoke than females in all age groups. 2. The percentage / number of female smokers decreases with age. 3. The percentage / number of male smokers increases to 25-34 years and then decreases. **Any 2 from 3**	**2**	Accept there are always less/ a lower number/a lower percentage of female smokers than males. Accept there are always more/a higher number/ a higher percentage of male smokers than females. Do not accept the number /percentage of male smokers decreases from 25-34 years.
		(ii)	Many older smokers have died (from smoking related illnesses).	**1**	Accept 65+ as older.
	(b)	(i)	Nicotine binds to acetylcholine receptors. **OR** Nicotine mimics the action of acetylcholine.	**1**	Do not accept answers that state neurotransmitter instead of acetylcholine.
		(ii)	Dopamine induces feelings of pleasure/euphoria/makes them feel good. **OR** Dopamine stimulates / activates / triggers the reward pathway.	**1**	Answer must indicate that the reward pathway is activated. Do not accept affects / links to / reinforces the reward pathway on their own.
		(iii)	It causes a decrease in the number/ sensitivity of receptors.	**1**	Amount is acceptable for number.
	(c)		Individuals identify with the celebrity and stop smoking/change their behaviour/change their beliefs to be like them/as they admire them.	**1**	Answer must contain identification / identify. Role model/ idol indicates admiration of celebrity.
	(d)		43 000	**1**	

Question			Expected response	Max mark	Additional guidance
11.	(a)	(i)	There were a high/higher number of cases in 1991 and 1992 / two years.	**1**	
		(ii)	Endemic	**1**	
		(iii)	The cases would not occur every year / regularly / constantly. **OR** The cases would occur occasionally / randomly / irregularly / less frequently.	**1**	Do not accept there would be fewer cases, unless qualified.
	(b)	(i)	Individuals who had not been immunised were unlikely to come into contact with infected individuals.	**1**	Answer must indicate coming into contact with an individual who has the disease and not simply the disease. Do not accept contact with unvaccinated individuals instead of infected individuals.
		(ii)	Due to malnutrition/poverty/rejection by some of the population/lack of education/lack of access to medical resources or vaccines/geographical remoteness.	**1**	Rejection can include for beliefs and because individuals are allergic to vaccines / immunosuppressed / have a weakened immune system. Accept vaccination may not result in immunity / work. Do not accept health problems.

Question			Expected response	Max mark	Additional guidance
12.	(a)	(i)	Epithelial / epithelium.	**1**	
		(ii)	They produce secretions / chemicals / mucus / sweat / stomach acid / tears / sebum.	**1**	pH must be linked to a secretion. Do not accept lowers pH of skin. Any answer must indicate production / secretion.
	(b)		Chemical: histamine **(1)** Explanation: causes vasodilation/ increases capillary permeability **(1)**	**2**	 Vasodilation of capillaries negates.
	(c)	(i)	Phagocyte: engulfs it/ ingests it / takes it in <u>and</u> digests it / breaks it down. **(1)** NK cell: induces/triggers apoptosis of (infected) <u>cell</u> **OR** induces/triggers production of self-destructive enzymes/self-destruction/programmed <u>cell</u> death (in the infected <u>cell</u>). **(1)**	**2**	Do not accept destroys instead of digests. Answer must relate to the cell and not the pathogen. If candidate mentions cell assume it is infected e.g. target cell.
		(ii)	They release / produce <u>cytokines</u>.	**1**	

Question			Expected response	Max mark	Additional guidance
13.	A		1. At puberty a (releaser) hormone is produced by the hypothalamus. 2. The pituitary gland releases follicle stimulating hormone/FSH <u>and</u> interstitial cell stimulating hormone /ICSH. 3. FSH/ICSH acts on/targets the testes. 4. FSH stimulates sperm production in the seminiferous tubules. 5. ICSH stimulates the production of testosterone in the interstitial cells. 6. Testosterone stimulates sperm production. 7. Sperm/gametes are produced from germline cells by meiosis. 8. Testosterone activates the prostate gland/seminal vesicles. 9. Secretions affect/maintain sperm mobility/viability **or** named example 10. Negative feedback control / feedback inhibition of sperm / testosterone production exists. 11. High levels of testosterone inhibits/reduces/controls FSH/ICSH production. **8 marks from 11**	**8**	Accept LH in place of ICSH for point 2 only. Point 11 must refer to the production of FSH / ICSH

Question			Expected response	Max mark	Additional guidance
13.	B		1. Drug treatment / fertility drugs are used to stimulate/trigger ovulation/super ovulation. 2. (Some) drugs prevent the negative feedback of oestrogen on FSH production. 3. Other drugs mimic the effect of FSH/LH. 4. Artificial Insemination is used if a man has a low sperm count. 5. (Several) sperm/semen samples are collected. **OR** Sperm can be provided by a donor. 6. Sperm/ semen is inserted into the female reproductive tract by means other than sexual intercourse / syringe. 7. Intra-cytoplasmic sperm injection/ ICSI is used when man has a low sperm count/when sperm are defective. 8. The <u>head</u> of the sperm is (directly) injected/inserted into the egg. 9. During In vitro fertilisation/IVF eggs are fertilised outside the body. 10. Zygotes/fertilised eggs are incubated. **OR** Blastocysts/ balls of cells/ embryos are transferred into the uterus/womb. 11. Pre-implantation genetic diagnosis / PGD identifies genetic disorders/ chromosome abnormalities. **8 marks from 11**	**8**	Accept that ovulatory drugs indicate stimulation of ovulation. Accept medication for drugs. Point 5 Accept sperm is collected over a period of time. Accept vagina / uterus / womb for female reproductive tract Accept descriptions of defective sperm eg mobility problems. Accept zygotes/fertilised eggs divide before being placed in the uterus.

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Human Biology

This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this paper. These principles must be read in conjunction with the detailed marking instructions, which identify the key features required in candidate responses.

(a) Marks for each candidate response must **always** be assigned in line with these general marking principles and the detailed marking instructions for this assessment.

(b) Marking should always be positive. Marks should be awarded for what is correct and not deducted for errors or omissions.

(c) If a specific candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you should seek guidance from your team leader.

(d) There are no half marks awarded.

(e) Where a candidate makes an error in the first part of a question, credit should normally be given for subsequent answers that are correct with regard to this original error. Candidates should not be penalised more than once for the same error.

(f) Unless a numerical question specifically requires evidence of working to be shown, full marks should be awarded for a correct final answer (including units, if appropriate) on its own.

(g) Bulleted lists should not be used for extended response questions. Candidates must respond to the "command" word as appropriate and write extended answers in order to communicate fully their knowledge and understanding. Candidate responses in the form of bulleted lists may not be able to access the full range of available marks.

(h) In the detailed marking instructions, if a word is **underlined** then it is essential; if a word is **(bracketed)** then it is not essential.

(i) In the detailed marking instructions, words separated by / are **alternatives**.

(j) A correct answer can be negated if:
- an extra, incorrect, response is given
- additional information that contradicts the correct response is included.

(k) Where the candidate is instructed to choose one question to answer but instead answers both questions, both responses should be marked and the better mark awarded.

(l) Unless otherwise required by the question, use of abbreviations (eg DNA, ATP) or chemical formulae (eg CO_2, H_20) are acceptable alternatives to naming.

(m) If a numerical answer is required and units are not given in the stem of the question or in the answer space, candidates must supply the units to gain the mark. If units are required on more than one occasion, candidates should not be penalised repeatedly.

(n) Incorrect spelling is given. Sound out the word(s).
- If the correct word is recognisable then give the mark.
- If the word can easily be confused with another biological term then **do not** give the mark eg glucagon and glycogen.

(o) Presentation of data:

- If a candidate provides two graphs, in response to one question, mark both and give the higher score.
- If a question asks for a particular type of graph/chart and the wrong type is given, then full marks cannot be awarded. Candidates cannot achieve the plot mark but **may** be able to achieve the mark for scale and label. If the x and y data are transposed, then do not give the scale and label mark.
- If the graph uses less than 50% of the axes then do not give the scale and label mark.
- If 0 is plotted when no data for this is given, then do not give the plot mark (ie candidates should only plot the data given).

(p) Marks are awarded only for a valid response to the question asked. For example, in response to questions that ask candidate to:

- **identify, name, give** or **state**, they need only answer or present in brief form;
- **describe**, they must provide a statement as opposed to simply one word;
- **explain**, they must provide a reason for the information given;
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between topics being examined;
- **calculate**, they must determine a number from given facts, figures or information;
- **predict**, they must indicate what may happen based on available information;
- **suggest**, they must apply their knowledge and understanding to a new situation.

2018 Higher Human Biology Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	**B**	8	**86**	3	2	1	0
2	**A**	**70**	4	15	11	1	0
3	**B**	16	**39**	36	8	1	0
4	**D**	5	15	18	**61**	1	0
5	**B**	12	**68**	13	6	2	0
6	**D**	4	2	7	**86**	1	0
7	**C**	3	5	**68**	23	1	0
8	**A**	**53**	9	20	17	1	0
9	**A**	62	13	11	13	1	0
10	**C**	21	14	**54**	10	1	0
11	**D**	2	3	2	**92**	1	0
12	**D**	41	12	12	**33**	1	0
13	**C**	3	10	**84**	1	1	0
14	**B**	19	**54**	13	13	1	0
15	**C**	12	18	**59**	10	1	0
16	**A**	**70**	2	3	24	1	0
17	**B**	13	**82**	3	1	1	0
18	**D**	13	11	4	**71**	1	0
19	**D**	6	10	23	**60**	1	0
20	**C**	22	29	**34**	15	1	0

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2017 Human Biology

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Finalised Marking Instructions

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Marking Instructions for each question

Section 1

Question	Answer	Mark
1.	C	**1**
2.	D	**1**
3.	A	**1**
4.	C	**1**
5.	A	**1**
6.	B	**1**
7.	C	**1**
8.	A	**1**
9.	A	**1**
10.	C	**1**
11.	C	**1**
12.	B	**1**
13.	D	**1**
14.	B	**1**
15.	B	**1**
16.	C	**1**
17.	D	**1**
18.	B	**1**
19.	D	**1**
20.	D	**1**

Section 2

Question			Expected answer(s)	Max mark	Additional guidance
1.	(a)		Mitosis.	**1**	Do not accept cell division.
	(b)		Connective.	**1**	
	(c)		The gene(s) coding for haemoglobin is switched on/expressed in the red blood cell but is switched off/ not expressed in the white blood cell.	**1**	Comparison must be made between red and white blood cells. Accept 'it' for haemoglobin.
	(d)		Cell divides excessively/unregulated division occurs. **OR** Cell does not respond to regulatory signals.	**1**	Accept replicates/multiplies but not reproduces for division. Accept uncontrollably for excessively. Accept mitosis for cell division.
	(e)	(i)	15	**1**	
		(ii)	0·1	**1**	

Question			Expected answer(s)	Max mark	Additional guidance
2.	(a)	(i)	One nucleotide/base is replaced with another.	**1**	Accept substituted/changed/ swapped for replaced. Do not accept references to mRNA. Do not accept it changes the nucleotide/base sequence/codon.
		(ii)	One amino acid is changed/the sequence of amino acids is changed. **OR** The enzyme is shorter due to the presence of a stop codon.	**1**	Accept the protein/enzyme will contain a different amino acid. Do not accept the active site is changed unless qualified by reference to amino acids. Accept that the wrong/a different amino acid will be formed/ produced/made.
	(b)	(i)	Phenylalanine cannot be broken down/converted (to tyrosine). **OR** Enzyme 1 does not break down phenylalanine/lead to the production of tyrosine.	**1**	Accept 'it' for phenylalanine. Do not accept 'transferred' for broken down. Accept 'it' for phenylalanine.
		(ii)	They get tyrosine from their diet.	**1**	Answer must indicate that the dietary protein contains tyrosine. Accept – the intermediate compound could be produced from a different metabolic pathway.
	(c)	(i)	Post-natal (screening).	**1**	
		(ii)	Restricted diet/low phenylalanine diet.	**1**	Accept altering/controlling their diet.
	(d)		25	**1**	

Question			Expected answer(s)	Max mark	Additional guidance
3.	(a)		To allow (time for) the reaction to take place/phenolphthalein to be produced/substrate to be broken down. **OR** So the enzyme/phosphatase is not denatured **OR** So the enzyme/phosphatase can work at its optimum pH.	**1**	Accept the alkali could stop/ inhibit/change/affect the reaction. Accept to allow the enzyme and inhibitor/substrate to react.
	(b)		1.Concentration of inhibitor / alkali / enzyme / phosphatase. 2. Type of alkali. 3. Temperature of solutions/test tubes. 4. pH of solutions/test tubes/alkali. (**Any 2 from 4**)	**2**	Do not accept 'amount' in place of concentration. Do not accept use the same colorimeter. Do not accept temperature of room/environment. Accept enzyme/inhibitor/substrate for solutions. *If candidate lists three or more variables and one is wrong they can only score 1 mark.*
	(c)		Axes have correct scales and labels. **(1)** Points correctly plotted and line drawn. **(1)** *Concentration of substrate* (M) / *Absorbance* (units) 0·05 / 0·20 0·10 / 0·30 0·20 / 0·48 0·40 / 0·64 0·60 / 0·78 0·80 / 0·90	**2**	Candidate cannot access the scale mark if they use less than half of the graph paper. If the scales start at zero there must be a zero entered. A common zero is acceptable if both scales start at zero. The scales do not have to start at zero but final number must be equal to or above 0.8M and 0.9 Units. If the axes are transposed the candidate loses the scale mark. Line must go through all points.

Question			Expected answer(s)	Max mark	Additional guidance
	(d)		As the substrate concentration increases there is an increase in absorbance/intensity of colour. **(1)** This indicates that more phenolphthalein/product has been produced **OR** This indicates that there is greater enzyme activity. **(1)**	**2**	
	(e)		The absorbance levels will remain low/lower/not increase/increase less (at each substrate concentration).	**1**	Do not accept that substrate concentration will have no effect on absorbance/will decrease absorbance.

Question			Expected answer(s)	Max mark	Additional guidance
4.	(a)	(i)	Oestrogen stimulates/causes the production/triggers the release of LH.	**1**	
		(ii)	3·1	**1**	
	(b)	(i)	1:4	**1**	
		(ii)	Endometrium thickness continues to increase after oestrogen concentration decreases (between day 14 and day 20).	**1**	Answer should focus on trends and not compare individual points.
	(c)		Endometrium thickness would not fall/not decrease/remain high/stay above 4·5. **OR** Oestrogen concentration would remain constant/stay the same/ stay around 30 units/would not decrease/remain high.	**1**	
	(d)		Fertility drugs prevent the negative feedback effect of oestrogen. **OR** Fertility drugs mimic the action of FSH/LH. **OR** Fertility drugs stimulate the production of FSH/LH.	**1**	

Question			Expected answer(s)	Max mark	Additional guidance
5.	(a)	(i)	Artery/arteriole.	**1**	
		(ii)	Vasoconstriction/contraction/ constriction. **OR** Narrowing of the lumen/cavity/ channel.	**1**	
	(b)	(i)	Endothelium/epithelium/ endothelial/epithelial.	**1**	
		(ii)	Pressure forces substances out/ through the capillary wall. **OR** By pressure filtration.	**1**	Do not accept diffusion on its own but it does not negate if pressure description is also given. Accept pushed/forced as alternatives for pressure. Do not accept squeezed.
		(iii)	Proteins.	**1**	Do not accept red blood cells/ haemoglobin. Accept enzyme/antibody/ hormone.
	(c)		Absorbs (excess) tissue fluid. **OR** Returns lymph/tissue fluid to the blood/circulatory system.	**1**	

Question			Expected answer(s)	Max mark	Additional guidance
6.	(a)		Systolic is when the heart beats/contracts/constricts and diastolic is when the heart relaxes/rests/does not beat.	**1**	Accept atria and ventricles in place of heart but do not accept atria on its own.
	(b)		(It/the cuff/pressure) squeezes/ closes/squashes/compresses the artery/vessel. **OR** (It/the cuff/ pressure) cuts off/ restricts/prevents blood flow/ circulation.	**1**	
	(c)	(i)	Fatty material/cholesterol/fibrous material/calcium builds up. **OR** An atheroma/plaque forms. **(1)** Diameter of lumen/cavity/channel of artery reduced. **OR** Loss of elasticity in arterial wall. **(1)**	**2**	Accept hardening of the arteries.
		(ii)	The cholesterol level/ concentration increases in the arteries/vessels/blood/ endothelium. **OR** Less cholesterol is removed from the arteries/vessels/blood/is taken to the liver.	**1**	Answers mentioning from the body cells do not negate but must be linked to the liver.

Question			Expected answer(s)	Max mark	Additional guidance
7.	(a)		The number of cases (of pulmonary embolism) increases with age. **(1)** The number of cases is higher in women taking HRT compared to those taking the placebo/not taking HRT. **(1)**	**2**	Do not accept answers that focus on one specific age group. Candidates must compare taking HRT to taking the placebo.
	(b)		The difference in cases between taking HRT or the placebo is not large/may not be significant. **OR** There are only a small number of cases compared to the large sample size/from 12 000 women.	**1**	Accept the difference between the two groups of women is small/ not significant. Answer must indicate that the difference is between two groups.
	(c)		(Differences in) diet/ smoking/degree of exercise/level of cholesterol/blood pressure/ genes/weight/drugs/medication/ alcohol/diabetes/high blood pressure/obesity.	**1**	Do not accept lifestyle/age/ gender/ill health/underlying medical conditions Accept ethnicity.
	(d)		They used a large sample size/ 12 000 women/large groups/ groups of 4 000 women.	**1**	Do not accept answers linked to validity such as a wide range of women/different age groups but these answers do not negate correct answers. Do not accept used a suitable number of women/repeated the experiment.
	(e)	(i)	Between the ages of 10 and 35 average bone mass increased and then it decreased between 35 and 80. **(1)** It increased from 50% to 100%/by 50% **OR** It decreased from 100% to 46%/by 54%. **(1)**	**2**	Accept it increases up to the age of 35 and then decreases. Must indicate % for at least one figure.
		(ii)	Any value from 37 to 38.	**1**	

Question			Expected answer(s)	Max mark	Additional guidance
8.	(a)		Mitochondrion/mitochondria.	**1**	
	(b)		1. Vesicle fuses with membrane/ releases neurotransmitter. 2. Neurotransmitter diffuses/moves across the gap /synaptic cleft/synapse 3. Neurotransmitter binds to/joins with the receptor. 4. Sufficient/a minimum amount of neurotransmitter is required (to transmit impulse). **(Any 3 from 4)**	**3**	Answer must indicate attachment to receptor. Accept – threshold must be reached.
	(c)		Effect - Increased number of/sensitivity of receptors. Consequence - Leads to addiction.	**2**	

Question			Expected answer(s)	Max mark	Additional guidance
9.	(a)		Medulla.	**1**	
	(b)		They cause opposite effects/work against each other. **OR** The sympathetic speeds up heart (rate) and the parasympathetic slows down heart (rate).	**1**	Accept they do opposite things. Accept one speeds up the heart and one slows it down.
	(c)	(i)	58	**1**	
		(ii)	32	**1**	
	(d)		The SAN controls the heart (rate/beat)/acts as a pacemaker/ still sends impulses (to the AVN)/is auto rhythmic.	**1**	
	(e)		There is a greater change (from the resting rate) when the parasympathetic nerve is blocked (compared to when the sympathetic is blocked) **OR** Blockage of the parasympathetic nerve raises heart rate by 18/32 (beats) while blockage of the sympathetic nerve lowers it by 14 (beats).	**1**	Answer must indicate a comparison. Answer must refer to nerve blockage. Units not essential.
	(f)		It increases breathing rate. **OR** It decreases digestive processes/ digestion/peristalsis.	**1**	*Also accept any of the following.* It stimulates the adrenal gland. It increases the release of adrenaline. It diverts blood to the skeletal muscles. It diverts blood away from the digestive system. It increases sweating/perspiration. It increases the volume of air entering the lungs. It dilates pupils in the eye. Do not accept prepares body for 'fight or flight' response.

Question			Expected answer(s)	Max mark	Additional guidance
10.	(a)		Antigens.	**1**	
	(b)		Phagocytosis is carried out on the pathogen. **OR** Pathogen is engulfed/captured by phagocyte and broken down. **(1)** Phagocyte displays (fragments of) antigen/X (on the surface/ membrane). **OR** Phagocyte becomes an antigen-presenting cell. **(1)**	**2**	Phagocytosis must be described.
	(c)		Cytokines.	**1**	
	(d)		Allows a faster/greater response to the same pathogen/antigen/ infection. **OR** Individual does not develop the same infection again.	**1**	Accept that second/secondary exposure/response indicates the same pathogen is involved. Accept rapid cell division for faster response but do not accept rapid reproduction of cells. Accept it is easier to fight the same pathogen/infection again.
	(e)		It survives/hides within phagocytic cells/phagocytes.	**1**	Accept antigen presenting cell for phagocyte.

Question			Expected answer(s)	Max mark	Additional guidance
11.	(a)	(i)	85·71/85·7/86	**1**	
		(ii)	1921 - 1951	**1**	
		(iii)	1. Development or availability of vaccines/immunisation/ establishment of herd immunity 2. Improvement in water supply/chlorination of drinking water 3. Improved sanitation/waste disposal systems/hygiene/ housing/living conditions 4. Improved storage/handling/ availability/production of food/diet 5. Improved health care/ medical facilities/medical treatments/medicines/ drug therapy/antisepsis/ sexual health. **(Any 2 from 5)**	**2**	**Note** – if no direction is given in both answers give one mark. e.g. health care and sanitation with no indication of improvement. **Note** candidates may access two marks within one sentence. This is acceptable here.
	(b)	(i)	78, 78, 78	**1**	
		(ii)	(In 1861) a large number of deaths occurred in childhood. **OR** (In 1861) there was large childhood/infant mortality. **OR** Individuals have survived childhood. **OR** Individuals have survived/ avoided illnesses/diseases (in childhood).	**1**	Do not accept answers that relate to immunity.

Question			Expected answer(s)	Max mark	Additional guidance
12.	A		**Structure** 1 DNA is composed of nucleotides containing deoxyribose (sugar), phosphate and base. 2 There is a sugar phosphate backbone. 3 The four bases are adenine, thymine, cytosine and guanine. 4 Cytosine bonds/pairs with Guanine and Adenine bonds with Thymine. 5 Bases/two strands are joined by hydrogen bonds. 6 Strands are anti-parallel/run in opposite directions/3' to 5' and 5' to 3'.j 7 DNA/two strands form a double helix (shape). **Replication** a DNA/double helix is unwound /unzipped. b A primer is needed at the start of replication/attaches to DNA strand. c DNA polymerase adds nucleotides to the 3' end (of a new/growing strand/primer). d One strand is replicated continuously and the other strand is replicated in fragments/discontinuously. e The strand replicated in fragments is joined together by ligase.	**9**	Accept A, T, C and G for point 4. Accept double-stranded for two strands. Award mark if candidate clearly indicates when primer is needed.

Question			Expected answer(s)	Max mark	Additional guidance
12.	B		**Structure** 1 RNA is single stranded. 2 It is composed of nucleotides containing ribose (sugar), phosphate and base. 3 The four bases are uracil, adenine, cytosine and guanine. 4 Groups of three bases/nucleotides form codons in mRNA/anticodons in tRNA. 5 Start/stop codons exist. 6 tRNA folds due to base pairing/has an attachment site for a specific amino acid. 7 rRNA (and protein) forms a ribosome. **Transcription** a Transcription occurs in the nucleus. b RNA polymerase forms mRNA/ unwinds and unzips DNA. c (Complementary) base pairing occurs - adenine with uracil, guanine with cytosine. d The primary transcript/mRNA contains introns and exons. e Introns are removed/ exons remain after (RNA) splicing. f (Splicing)forms the mature transcript/mRNA.	**9**	Accept helicase unwinds and unzips DNA for point b. Accept A, U, C and G for point c.

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Human Biology / Biology

This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this paper. These principles must be read in conjunction with the detailed marking instructions, which identify the key features required in candidate responses.

(a) Marks for each candidate response must <u>always</u> be assigned in line with these general marking principles and the Detailed Marking Instructions for this assessment.

(b) Marking should always be positive. Marks should be awarded for what is correct and not deducted for errors or omissions.

(c) If a specific candidate response does not seem to be covered by either the principles or detailed Marking Instructions, and you are uncertain how to assess it, you should seek guidance from your Team Leader.

(d) There are no half marks awarded.

(e) Where a candidate makes an error in the first part of a question, credit should normally be given for subsequent answers that are correct with regard to this original error. Candidates should not be penalised more than once for the same error.

(f) Unless a numerical question specifically requires evidence of working to be shown, full marks should be awarded for a correct final answer (including units, if appropriate) on its own.

(g) Bulleted lists should not be used for extended response questions. Candidates must respond to the "command" word as appropriate and write extended answers in order to communicate fully their knowledge and understanding. Candidate responses in the form of bulleted lists may not be able to access the full range of available marks.

(h) In the detailed marking instructions, if a word is **<u>underlined</u>** then it is essential; if a word is **(bracketed)** then it is not essential.

(i) In the detailed marking instructions, words separated by / are **alternatives**.

(j) A correct answer can be negated if:

- an extra, incorrect, response is given
- additional information that contradicts the correct response is included.

(k) Where the candidate is instructed to choose one question to answer but instead answers both questions, both responses should be marked and the better mark awarded.

(l) Unless otherwise required by the question, use of abbreviations (e.g. DNA, ATP) or chemical formulae (e.g. CO_2, H_20) are acceptable alternatives to naming.

(m) If a numerical answer is required and units are not given in the stem of the question or in the answer space, candidates must supply the units to gain the mark. If units are required on more than one occasion, candidates should not be penalised repeatedly.

(n) Incorrect spelling is given. Sound out the word(s).

- If the correct word is recognisable then give the mark.
- If the word can easily be confused with another biological term then **do not** give the mark e.g. glucagon and glycogen.

(o) Presentation of data:

- If a candidate provides two graphs, in response to one question, mark both and give the higher score.
- If a question asks for a particular type of graph/chart and the wrong type is given, then full marks cannot be awarded. Candidates cannot achieve the plot mark but **may** be able to achieve the mark for scale and label. If the x and y data are transposed, then do not give the scale and label mark.
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- **calculate**, they must determine a number from given facts, figures or information;
- **predict**, they must indicate what may happen based on available information;
- **suggest**, they must apply their knowledge and understanding to a new situation.

2017 Higher Human Biology Section 1 Multiple Choice Analysis

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2	**D**	7	5	37	**51**	0	0
3	**A**	**84**	13	2	2	0	0
4	**C**	8	20	**59**	14	0	0
5	**A**	**35**	34	21	10	0	0
6	**B**	17	**48**	14	21	0	0
7	**C**	32	30	**21**	16	0	0
8	**A**	**48**	12	12	28	0	0
9	**A**	**43**	10	30	17	0	0
10	**C**	18	11	**64**	7	0	0
11	**C**	12	27	**34**	25	2	0
12	**B**	19	**50**	17	14	0	0
13	**D**	5	29	25	**41**	0	0
14	**B**	27	**55**	8	10	0	0
15	**B**	18	**52**	27	3	0	0
16	**C**	5	18	**48**	29	0	0
17	**D**	20	8	3	**69**	0	0
18	**B**	4	**90**	4	2	0	0
19	**D**	32	7	16	**45**	1	0
20	**D**	1	1	8	**89**	0	0

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Finalised Marking Instructions

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Marking Instructions for each question

Section 1

Question	Answer	Mark
1.	C	**1**
2.	D	**1**
3.	C	**1**
4.	B	**1**
5.	D	**1**
6.	C	**1**
7.	B	**1**
8.	A	**1**
9.	B	**1**
10.	A	**1**
11.	C	**1**
12.	B	**1**
13.	A	**1**
14.	D	**1**
15.	C	**1**
16.	A	**1**
17.	B	**1**
18.	D	**1**
19.	C	**1**
20.	B	**1**

Section 2

Question			Expected Answer(s)	Max Mark	Additional Guidance
1.	(a)	(i)	Hydrogen	**1**	Do not accept H.
		(ii)	Phosphate	**1**	Do not accept P/Pi.
	(b)		Nucleotides can only be added to the 3′/deoxyribose end (of a new strand/primer). **OR** DNA/strands can only be replicated from 5′ to 3′.	**1**	Do not accept bases instead of nucleotides.
	(c)		DNA polymerase adds nucleotides (to the new strand/primer) **(1)** Ligase joins fragments (of DNA/ lagging strand) **(1)**	**2**	Do not accept bases instead of nucleotides. Accept alternatives to 'adds' as long as it is clear that the nucleotide is being added to the new strand/primer/sugar-phosphate backbone.
2.	(a)	(i)	GUG CUA GCU AUC CUA	**1**	
		(ii)	exons	**1**	
		(iii)	(RNA/alternative) splicing.	**1**	Do not accept DNA splicing
		(iv)	G C U A	**1**	
	(b)		ribosome	**1**	
	(c)		Polypeptide chains can be cut/ cleaved (and recombined). **OR** Phosphate/carbohydrate groups may be added (to the polypeptide).	**1**	

Question			Expected Answer(s)	Max Mark	Additional Guidance
3.	(a)	(i)	742	**1**	
		(ii)	Between 0 and 7 years after diagnosis the percentage of surviving patients decreased and then it remained constant between 7 and 9 years. **(1)** It decreased from 100% to 62% **OR** It decreased by 38%. **OR** It remained constant at 62%. **(1)**	**2**	Accept – the percentage surviving decreases to/for 7 years and then levels out.
	(b)		1. (Cancer) cells divide excessively and this leads to a mass of abnormal cells/tumour. 2. These cells don't respond to regulatory signals. 3. Cells fail to attach to each other/the tumour. **OR** Cells detach from each other/ the tumour. 4. They/cells spread to form secondary tumours/undergo metastasis. **(Any 3)**	**3**	1. Accept replicate/multiply for division but do not accept reproduce. Accept rapidly/uncontrollably – answer must indicate more than normal.

Question			Expected Answer(s)	Max Mark	Additional Guidance
4.	(a)		1. Volume of solution. 2. Concentration of solution. 3. Initial pH of solution. 4. Diameter of tube/length of tube/size of tube/position of tube in indicator/size of bottle. 5. Temperature of air/bottle/ solution. **(Any 2)**	**2**	
	(b)		40	**1**	Accept if 40 is written in Table 1
	(c)		They repeated the investigation/ experiment and took an average/ averages. **OR** They repeated the investigation/ experiment at each activity.	**1**	Accept they used six/several individuals and took an average/ averages. Accept they used six/several individuals for each activity.
	(d)		Correct scale on vertical axis and correct labels on both axes. **(1)** Bars are correctly drawn. (33, 25 and 18) **(1)**	**2**	Candidate cannot access the scale mark if they use less than half of the graph paper. Minimum label for Y axis is 'Average time taken (s)'. Label for X axis must contain Activity 1 resting, Activity 2 walking, Activity 3 running, unless some of this information is given in a key. If the vertical scale starts at zero there must be a zero entered at the bottom. A common zero is incorrect. The scale does not have to start at zero. There must be a number equal to or above 33 on the scale.
	(e)		Increasing physical activity increases the respiration rate. **OR** Running produces the highest respiration rate.	**1**	Any correct conclusion must mention respiration rate.

Question			Expected Answer(s)	Max Mark	Additional Guidance
	(f)		**Increased respiration/activity produces more carbon dioxide.** **OR** Increased exercise/physical activity uses more ATP/energy. **OR** Running produces the most carbon dioxide/uses the most ATP (or opposite for resting).	**1**	Do not accept answers that refer only to breathing.
5.	(a)		Label correctly showing pulmonary artery.	**1**	Label can be anywhere on the artery above the semilunar valve. Label can be just above the end of the artery. Label can contain the full name instead of P.
	(b)		The blood would contain a lower concentration of oxygen/less oxygen. **(1)** Deoxygenated blood enters the left ventricle/side of the heart. **(1)**	**2**	Do not accept less oxygen would leave the heart.
	(c)	(i)	Right atrium/atria.	**1**	
		(ii)	Electrocardiogram/ECG.	**1**	
	(d)		They have a higher heart rate/ pulse rate.	**1**	Do not accept increased heartbeat. Do accept increased heartbeat per minute.

Question			Expected Answer(s)	Max Mark	Additional Guidance
6.	(a)	(i)	28	**1**	
		(ii)	It allows groups <u>of different sizes</u> to be compared. **OR** It allows <u>different populations</u> to be compared.	**1**	
		(iii)	4284	**1**	
		(iv)	3:2	**1**	
	(b)		A blockage/clot/embolism in an artery/blood vessel leading to/in the brain.	**1**	Mentioning carotid artery indicates candidate knows it is the brain that is affected.
	(c)		Lack of oxygen kills (brain) <u>cells/ tissues.</u> Stroke/damage/lack of oxygen occurs in the <u>right side</u>/<u>hemisphere</u> of the brain/motorcortex/cerebrum. Impulses/signals are not transmitted/ sent to the muscles (on the left side of the body). **OR** Impulses/signals are not transmitted/ sent so preventing movement (on the left side of the body). **(Any 2 from 3)**	**2**	Accept description of stroke – blood clot in an artery.

Question			Expected Answer(s)	Max Mark	Additional Guidance
7.	(a)	(i)	26 or 26·0	**1**	Do not accept answers to more than one decimal place.
		(ii)	Their BMI is greater than 30.	**1**	
	(b)		Exercise increases energy expenditure/increases respiration rate/uses up (stored) fats.	**1**	Accept exercise burns fat. Accept exercise increases metabolism. Accept exercise increases HDLs/ HDL:LDL ratio and so removes fat. Do not accept exercise increases their muscle to fat ratio.
	(c)		They have a (relatively) high muscle mass.	**1**	Accept they have more muscle. Do not accept that they have more muscles. Accept muscle is heavier more dense than fat. Accept that lean tissue is muscle tissue.

Question			Expected Answer(s)	Max Mark	Additional Guidance
8.	(a)		**Recommended** – It lowers cholesterol levels. **(1)** **Not recommended** – It takes a long time to work. **(1)**	**2**	Accept any figures that specify six months or above.
	(b)		A placebo **OR** A capsule containing no statin/ no drug.	**1**	Do not accept a sugar pill/dummy pill/fake pill. Answers like this do not negate if the term placebo is given.
	(c)		**Randomised –** All individuals have an equal chance of being in either group. **OR** Example describing this. **(1)** **Double-blind –** Neither the participants/patients or the researchers/doctors should know which group participants are placed into/who is getting the drug. **(1)**	**2**	Do not accept that individuals are <u>randomly</u> placed in each group unless qualified by an example. Examples could be drawing numbers out of a hat or using a computer program to allocate them. Must mention <u>both</u> groups of individuals.
	(d)		The error bars overlap. **OR** There is no <u>significant</u> difference between the group results.	**1**	
	(e)		Found in cell membranes. **OR** Forms hormones/forms (other) steroids/is a precursor for steroids (being synthesised).	**1**	

Question			Expected Answer(s)	Max Mark	Additional Guidance
9.	(a)		Cerebrum/cerebral hemisphere(s)/cerebral cortex.	**1**	
	(b)		Different/some areas of the brain are used/active during different aspects/parts of the task.	**1**	Accept – different parts of the brain are involved in description and completion of the task.
	(c)		These areas are receiving signals/ impulses from eyes/ears. **OR** These are the hearing/auditory/ visual areas. **OR** These are areas where sounds/ language/images are processed.	**1**	
	(d)		**Sensory area** – individual was touching/feeling (the paper). **(1)** **Motor area** – individual was using (muscles in) hands/fingers (to fold the paper). **(1)**	**2**	Accept – individual uses muscles to fold the paper.
10.	(a)		Axon	**1**	
	(b)		It attaches to a <u>receptor</u>/ diffuses into the <u>receptor</u> (on the postsynaptic membrane).	**1**	Additional information does not negate if receptor is mentioned.
	(c)		Fast twitch/Type 2.	**1**	
	(d)	(i)	Agonists (bind to and) stimulate (neurotransmitter) <u>receptors</u>. **OR** Agonists mimic (the action of) <u>neurotransmitters</u>.	**1**	Accept nicotine in place of agonist and acetylcholine in place of neurotransmitter.
		(ii)	Nicotine triggers/causes the (increased) release of/activates <u>dopamine/endorphins</u>. **OR** Nicotine acts as an agonist of/ mimics <u>dopamine</u>. **OR** Nicotine stimulates/reinforces the <u>reward</u> pathway/circuit. **OR** Nicotine blocks/prevents/inhibits the reuptake of dopamine.	**1**	Do not accept nicotine releases dopamine.

Question			Expected Answer(s)	Max Mark	Additional Guidance
11.	(a)		3·1 million/3 100 000	**1**	Do not accept 3·1
	(b)		30	**1**	
	(c)		70	**1**	Accept − 70
	(d)	(i)	They have a very low chance of coming into contact with/being exposed to someone who has measles/the disease/is carrying the pathogen.	**1**	Answer must indicate coming into contact with an individual who has the disease.
		(ii)	Malnutrition/poverty/rejection by some of the population/lack of education/lack of access to medical resources or vaccines/ geographical remoteness.	**1**	
	(e)		1242	**1**	
	(f)		If the rate of decrease in the number of measles cases remains the same there will be no cases of measles (by 2020). **OR** Between 2005 and 2010 the number of cases decreased by 300 000. This suggests that measles will be eliminated (by 2015/2020).	**1**	Use of alternative correct figures is acceptable eg a decrease of 500 000 occurred between 2000 and 2010.

Question			Expected Answer(s)	Max Mark	Additional Guidance
12.	(a)	(i)	The receptors on the lymphocyte bind to the antigen (on the pathogen). **(1)** This leads to (repeated) division (of the lymphocyte to form a clone). **(1)**	**2**	 Accept mitosis for division. Accept lymphocyte creates many copies of itself for division.
		(ii)	Phagocytes capture/engulf pathogens/bacteria/viruses and display antigens/become antigen-presenting cells. **(1)** These activate/stimulate T-lymphocytes. **OR** These cause the production of T-lymphocytes. **(1)**	**2**	
	(b)		*Allergy or example eg hayfever* B lymphocyte. **(1)** Attack/respond to a harmless antigen. **(1)** **OR** *Autoimmune disease or example eg arthritis* T lymphocyte. **(1)** Attack/respond to self-antigens. **(1)**	**2**	If both answers are correct but no condition indicated give 1 mark. Do not accept mast cells. Accept recognises own antigens as foreign and attacks them.

Question			Expected Answer(s)	Max Mark	Additional Guidance
13.	A		<u>Causes</u> Too much fat/cholesterol in the diet/blood. **(1)** High LDL levels/low HDL levels **or** High LDL:HDL or Low HDL:LDL. **(1)** Lack of exercise/inactive lifestyle. **(1)** Genetic condition/familial hypercholesterolaemia/FH. **(1)** Diabetes/high blood glucose levels. **(1)** <u>Development</u> There is an accumulation of fatty/ fibrous material/ cholesterol/calcium**.** **(1)** The atheroma/plaque forms beneath the <u>endothelium</u> of an artery. **(1)** Artery (wall) thickens/lumen narrows. **(1)** Blood flow is reduced/ restricted/prevented. **(1)** Loss of elasticity in artery (wall)/hardening of the arteries occurs. **(1)** <u>Health Problems</u> Raises blood pressure/causes hypertension. **(1)** Causes CHD/angina/ heart attack/stroke/PVD **(any 2)**. **(1)** Description of CHD/angina/ heart attack/stroke/PVD. **(1)**	**8**	*In order to score 8 marks candidate must mention at least one point from each of the three areas.* Do not give marks for any points that indicate prevention eg reduce fat in the diet/increase exercise. Do not accept in/on the endothelium. Do not accept arteries narrow. Do not accept CVD unless two specific examples are given. Do not accept heart disease for CHD.

Question			Expected Answer(s)	Max Mark	Additional Guidance
13.	B		<u>Diagnosis</u> Glucose presence in urine suggests diabetes. **(1)** (Diagnosis made by carrying out a) <u>glucose tolerance test</u>. **(1)** Individual fasts/does not eat prior to the test. **(1)** Individual drinks a glucose solution/drink. **(1)** Blood glucose concentration that remains high indicates diabetes. **(1)** Type 1 diabetes tends to be diagnosed in children <u>while</u> type 2 diabetes tends to be diagnosed in adults/later in life. **(1)** <u>Treatment</u> Type 1 diabetes is treated with regular doses/ injections of insulin. **(1)** Type 2 diabetes is treated/controlled by lifestyle changes/weight loss/exercise/ dietary changes. **(1)** <u>Role of Insulin</u> Insulin is produced in the <u>pancreas</u>. **(1)** Type 1 diabetics are unable to produce insulin. **(1)** Insulin converts <u>glucose into glycogen</u>. **(1)** Type 2 diabetics can produce insulin <u>but</u> cells are less sensitive/ resistant to it. **(1)** In Type 2 diabetics there are <u>less insulin receptors</u> on cells. **(1)**	**8**	*In order to score 8 marks candidate must mention at least one point from each of the three areas.* Ignore any times for fasting. Ignore any values for the concentration/volume/of the glucose solution. Accept glucose cannot be converted to glycogen (due to a lack of insulin). Glycogen must be spelt correctly. Accept <u>insulin receptors</u> are desensitised/less sensitive to insulin.

[END OF MARKING INSTRUCTIONS]

General Marking Principles for Higher Human Biology

This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this paper. These principles must be read in conjunction with the detailed marking instructions, which identify the key features required in candidate responses.

(a) Marks for each candidate response must always be assigned in line with these General Marking Principles and the Detailed Marking Instructions for this assessment.

(b) Marking should always be positive. This means that, for each candidate response, marks are accumulated for the demonstration of relevant skills, knowledge and understanding: they are not deducted from a maximum on the basis of errors or omissions.

(c) If a specific candidate response does not seem to be covered by either the principles or detailed Marking Instructions, and you are uncertain how to assess it, you must seek guidance from your Team Leader.

(d) There are no half marks awarded.

(e) Where a candidate makes an error at an early stage in a multi stage calculation, credit should normally be given for correct follow on working subsequent stages, unless the error significantly reduces the complexity of the remaining stages. The same principle should be applied in questions which require several stages of non-mathematical reasoning.

(f) Unless a numerical question specifically requires evidence of working to be shown, full marks should be awarded for a correct final answer (including units) on its own.

(g) Bulleted lists should not be used for extended response questions. Candidates must respond to the "command" word as appropriate and write extended answers in order to communicate fully their knowledge and understanding. Candidate responses in the form of bulleted lists may not be able to access the full range of available marks.

(h) In the mark scheme, if a word is **underlined** then it is essential; if a word is **(bracketed)** then it is not essential.

(i) In the mark scheme, words separated by/are **alternatives**.

(j) If two answers are given that contradict one another the first answer should be taken. However, there are occasions where the second answer negates the first and no marks are given. There is no hard and fast rule here, and professional judgement must be applied.

(k) Where the candidate is instructed to choose one question to answer but instead answers both questions, both responses should be marked and the better mark awarded.

(l) The assessment is of skills, knowledge and understanding in Human Biology, so marks should be awarded for a valid response, even if the response is not presented in the format expected. For example, if the response is correct but is not presented in the table as requested, or if it is circled rather than underlined as requested, give the mark.

(m) Unless otherwise required by the question, use of abbreviations (eg DNA, ATP) or chemical formulae (eg CO_2, H_20) are acceptable alternatives to naming.

(n) Content that is outwith the course assessment specification should be given credit if used appropriately eg metaphase of meiosis.

(o) If a numerical answer is required and units are not given in the stem of the question or in the answer space, candidates must supply the units to gain the mark. If units are required on more than one occasion, candidates should not be penalised repeatedly.

(p) Incorrect **spelling** is used:

- if the term is recognisable then give the mark
- if the term can easily be confused with another biological term then **do not** give the mark eg ureter and urethra
- if the term is a mixture of other biological terms then **do not** give the mark, eg mellum, melebrum, amniosynthesis

(q) Presentation of data:

- if a candidate provides two graphs or bar charts, in response to one question (eg one in the question and another at the end of the booklet), mark both and give the higher score
- for marking purposes no distinction is made between bar charts (used to show discrete features, have descriptions on the x-axis and have separate columns) and histograms (used to show continuous features, have ranges of numbers on the x-axis and have contiguous columns)
- other than in the case of bar charts/histograms, if the question asks for a particular type of graph or chart and the wrong type is given, then do not give the mark(s) for this. Where provided, marks may still be awarded for correctly labelling the axes, plotting the points, joining the points either with straight lines or curves (best fit rarely used), etc.
- the relevant mark should not be awarded if the graph uses less than 50% of the axes; if the x and y data are transposed; if 0 is plotted when no data for this is given (ie candidates should only plot the data given)

(r) Marks are awarded only for a valid response to the question asked. For example, in response to questions that ask candidate to:

- **identify, name, give** or **state**, they need only name or present in brief form;
- **describe**, they must provide a statement or structure of characteristics and/or features;
- **explain**, they must relate cause and effect and/or make relationships between things clear;
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between things;
- **calculate**, they must determine a number from given facts, figures or information;
- **predict**, they must suggest what may happen based on available information;
- **evaluate**, they must make a judgement based on criteria;
- **suggest**, they must apply their knowledge and understanding of Human Biology to a new situation. Marks will be awarded for any suggestions that are supported by knowledge and understanding of Human Biology

National
Qualifications
2015

2015 Human Biology

New Higher

Finalised Marking Instructions

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Detailed Marking Instructions for each question

Section 1

Question	Answer	Mark
1.	C	**1**
2.	C	**1**
3.	B	**1**
4.	D	**1**
5.	D	**1**
6.	D	**1**
7.	A	**1**
8.	D	**1**
9.	B	**1**
10.	A	**1**
11.	B	**1**
12.	D	**1**
13.	A	**1**
14.	A	**1**
15.	B	**1**
16.	B	**1**
17.	A	**1**
18.	C	**1**
19.	B	**1**
20.	C	**1**

Section 2

Question			Expected Answer(s)	Max Mark	Additional Guidance
1.	(a)		(Relatively) unspecialised (cells) **or** Capable of (repeated) division **or** Can differentiate (into specialised cells) **or** Are totipotent.	**1**	Accept undifferentiated (cells) Accept can divide by mitosis Accept can develop/specialise into / become any cell.
	(b)		Embryonic stem cells/inner cell mass cells can form all cells types /are totipotent or pluripotent. while tissue/adult stem cells can only form a limited range of cell types /are multipotent.	**1**	Answer must give a comparison between the two types of cells. Accept – tissue stem cells can only form cells from the tissues they are found in.
	(c)	(i)	Tissue / stem cells are cultured/ grown (in laboratory/outside body). **or** Tissue/stem cells are transplanted/placed into the muscle/tissue/damaged area.	**1**	Do not accept skin grafts. Do not accept 'into the body'.
		(ii)	Stem cells can be used to study diseases/cancer. **or** Stem cells can be used for drug/ medicine testing/treatment.	**1**	

Question			Expected Answer(s)	Max Mark	Additional Guidance
2.	(a)		Anabolic/synthetic/biosynthetic /synthesis	**1**	
	(b)	(i)	It will contain a different nucleotide/base. **or** It will contain a different codon /stop codon.	**1**	Accept a specific description of a gene mutation. eg insertion of a nucleotide/base. Accept a description of missense, nonsense or splice-site mutations e.g. substitution of a nucleotide/ base.
		(ii)	The protein/enzyme/glycogen synthase contains a different amino acid(s).	**1**	Accept 'shorter protein/ polypeptide produced' if linked to stop codon (in bi).
	(c)		Glucose is used up in respiration/to provide energy/ ATP and they have no glycogen stores to provide more glucose.	**1**	Do not accept glucose is used up in exercise. Accept glycogen is not converted to glucose.
	(d)		*Recessive* Disease skips generations/does not appear in every generation. **or** Two unaffected/heterozygous/ carrier parents can have an affected child. *1 mark* *Sex-linked* **More males** will be affected than females. **or** **Affected males** do not pass the allele/condition to their sons. **or** **Affected males** can only pass the allele/condition to their daughters. **or** **Unaffected males** cannot pass the allele/condition to their daughters. **or** Only **affected/carrier females** pass the condition to their sons. *1 mark*	**2**	Accept homozygous recessive for affected child. Do not accept 'only males are affected'. Accept all males that have the allele will be affected.

Question			Expected Answer(s)	Max Mark	Additional Guidance
3.	(a)		Volume of yeast suspension/ solution/cells. Concentration of yeast suspension/solution/number of yeast cells/mass of yeast. Type/age/source of yeast cells. Area/size/diameter/volume/ thickness/type of gel or dish. Concentration of nutrients in gel/pH of gel. Strength or intensity of lamp/use same lamp/ distance of lamp. Temperature of incubator/ dishes. Time for yeast to grow/dishes left in incubator. *Any two*	**2**	Do not accept amount but do not penalise twice if used in both answers. Do not accept quantity. Accept mass of gel. Do not accept temperature on its own.
	(b)	(i)	Axes have correct scales and labels – *1 mark* Points correctly plotted and line drawn (touching each point). – *1 mark*	**2**	Remove 1 mark if less than half the graph paper is used. Remove 1 mark if axes are transposed. Remove 1 mark if no zero at the origin. One common zero is acceptable.
		(ii)	Increasing the exposure (to UV radiation) <u>increases</u> the number of yeast cells/colonies that <u>die</u>/are <u>damaged</u>. **Or** Increasing the exposure (to UV radiation) <u>decreases</u> the number of yeast cells/colonies that <u>survive</u>.	**1**	Answer referring to the number of yeast cells <u>growing</u> is wrong <u>unless</u> there is a further correct reference to cells that die/are damaged/survive. Accept DNA damage as cell damage.
		(iii)	Repeat the investigation <u>at each exposure</u> (time). **Or** Repeat the investigation and calculate <u>averages</u>.	**1**	Do not accept repeat the results/ measurements.
	(c)	(i)	400	**1**	
		(ii)	The number of yeast cells/ colonies at SPF 15 is almost as much as with higher SPF values. **Or** There are more yeast cells/ colonies using SPF 15 compared to when using no sunscreen.	**1**	Do not accept more yeast cells/ colonies using SPF 15 compared to SPF 6. Accept figures for more (72) and none (15).
		(iii)	350 minutes/5 hours 50 minutes	**1**	Units are essential.

Question			Expected Answer(s)	Max Mark	Additional Guidance
4.	(a)		ATP is broken down/used up/ converted to ADP. **Or** ATP is put into the reaction.	**1**	Accept phosphorylation of intermediates/glucose/fructose 6 phosphate. Do not accept phosphorylation occurs on its own. Do not accept ATP is invested.
	(b)		It changes the shape/form of the active site (to suit the substrate molecule). **Or** It induces a better fit with the substrate. **Or** It lowers the activation energy.	**1**	Do not accept references to inhibitors. Accept it increases the enzyme's affinity for the substrate.
	(c)		Enzyme 1 **or** Enzyme 3 Explanation: The transfer of phosphate/ addition of phosphate (from ATP)	**1**	Accept underlining/ticking.
	(d)		When there has been a build-up/ too much/an increased concentration of fructose-6-phosphate.	**1**	
	(e)		This ensures the cell only produces ATP when required **or** This ensures that glucose is only used when it is required/ conserved.	**1**	Do not accept 'conserves energy/ ATP'. Accept excess ATP not produced.
5.	(a)	(i)	Releases/ supplies energy (rapidly / at a fast rate). *1 mark* Phosphate (released) is used to convert ADP to ATP/to create ATP. *1 mark*	**2**	This can refer to creatine phosphate or ATP. Do not accept creatine phosphate for phosphate. Accept creatine phosphate is used to phosphorylate ADP.
		(ii)	The creatine phosphate supply runs out (after 10 seconds).	**1**	Accept there is a limited supply of creatine phosphate.
	(b)		Lactic acid/lactate.	**1**	

Question			Expected Answer(s)	Max Mark	Additional Guidance
5.	(c)		Muscle fibre: Slow twitch Sport: any suitable endurance sport Reasons: • they contract (relatively) slowly • can contract over a (relatively) long period • have many mitochondria • have a large blood supply • rely on aerobic respiration (to generate ATP) • have a high concentration of myoglobin • stores/energy source is mainly fat. Muscle fibre: Fast twitch Sport: any sport requiring bursts of energy Reasons – • they contract (relatively) quickly • can contract over a (relatively) short period • have few mitochondria • have a low blood supply • rely on glycolysis (to generate ATP) • have a low concentration of myoglobin • stores/main energy source is glycogen/creatine phosphate. *Any 3 reasons for 3 marks.*	**3**	Annotate answer with ticks. *Remove 1 mark if wrong type of muscle fibre is picked for chosen sport.* Accept <u>any</u> comparison with fast twitch e.g. contains more mitochondria. Accept <u>any</u> comparison with slow twitch eg they have a lower blood supply.

Question			Expected Answer(s)	Max Mark	Additional Guidance
6.	(a)	(i)	Cross (centre) is placed in the fetal tissue area.	**1**	Accept if cross is placed on the placenta in the main diagram.
		(ii)	Cells are cultured/allowed to divide (to obtain sufficient cells) *1 mark* (Karyotypes then show) the chromosomes (from the cells). *1 mark*	**2**	Candidates can get these marks if discussing amniocentesis. Do not accept cells are amplified.
		(iii)	CVS can be carried out earlier (in pregnancy than amniocentesis).	**1**	Answer must show a length of time comparison.
	(b)		Biochemical	**1**	
7.	(a)		18	**1**	
	(b)		(As children get older) they eat less 'healthy' food/have a higher fat diet/have more sugar in their diet. **Or** (As children get older) they choose/control what they eat. **Or** (As children get older) they exercise less/carry out less physical activities.	**1**	Do not accept have a worse diet. Do not accept children eat more.
	(c)		Weight divided by height squared.	**1**	
	(d)		Identification – use celebrities/ role models/someone they admire to promote a healthy lifestyle/healthy diet / exercise. 1 mark Internalisation – Use adverts/ media/parents/reasoned arguments/persuasion to get children to adopt a healthy lifestyle/healthy diet/exercise. 1 mark	**2**	If the idea of a) use celebrities/role models/someone they admire and b) Use adverts media/ parents/reasoned arguments/persuasion are both given correctly but there is no mention of lifestyle/ eating habits/exercise - *1 mark* Do not accept to promote/ encourage a healthy BMI.

Question			Expected Answer(s)	Max Mark	Additional Guidance
8.	(a)		Between work levels 1 to 5 the stroke volume increased and then it remained constant between levels 5 and 7. *1 mark* It increased from 88 cm^3 to 140 cm^3 **or** it remained constant at 140 cm^3. *1 mark*	**2**	Accept – stroke volume increases up to work level 5 and then levels out. Correct units needed at least once.
	(b)		102	**1**	
	(c)		19 600	**1**	
	(d)	(i)	4·5	**1**	
		(ii)	120 beats/min	**1**	Units are essential. bpm is acceptable.
		(iii)	23·81/23·8/24	**1**	
9.	(a)		Enough / increased/more/a higher concentration of dopamine /neurotransmitter (is released) to trigger an impulse/reach threshold. **or** Many/a series of weak stimuli trigger an impulse/reach threshold.	**1**	Do not accept weak impulses instead of stimuli.
	(b)		Provides energy/ATP to make/ release neurotransmitter or dopamine/form vesicles/allow vesicles to move/allow vesicles to fuse with the membrane/to reuptake neurotransmitter.	**1**	
	(c)		(Cocaine) blocks/inhibits/acts as an antagonist to the re-uptake proteins. **or** (Cocaine) prevents the proteins from reabsorbing/taking up/ removing dopamine. *1 mark* Dopamine/neurotransmitter remains in the synapse/at the receptors. **or** Dopamine/neurotransmitter continues to stimulate receptors/fire impulses. *1 mark*	**2**	Do not accept cocaine binds to/ attaches to the re-uptake proteins. Do not accept agonist. Do not accept dopamine continues to stimulate the reward pathway.

Question			Expected Answer(s)	Max Mark	Additional Guidance
10.	(a)		Organisation Related information is grouped together. **Or** Information is put into categories /headings. *1 mark* Elaboration Additional information is given (about each term). **Or** Meaningful information is given (about each term). *1 mark*	**2**	Names of both methods are correct but incorrect descriptions are given – award 1 mark. Do not accept - put the information into groups. Do not accept put information into a story / into context. If elaboration of meaning is given as the method, the description must still indicate meaning to gain the mark.
	(b)		Short-term memory/STM has a limited capacity/span/only holds around 7 items of information.	**1**	Accept anything between 5 and 9 items. Mention of time (30s) does not negate.
	(c)		Cerebrum/cortex	**1**	
11.	(a)	(i)	Release histamine *1 mark* This causes vasodilation/ increased capillary permeability *1 mark* **Or** Release cytokines *1 mark* This leads to an accumulation of phagocytes/the delivery of antimicrobial proteins/clotting elements *1 mark*	**2**	Accept a description of vasodilation e.g. blood vessels widen. Do not accept vasodilation of capillaries. Accept this attracts phagocytes.
		(ii)	NK cells induce/cause the infected/invaded cell to self-destruct/undergo apoptosis/ undergo programmed cell death. *1 mark*	**1**	
	(b)		Phagocytes engulf/capture/ digest the bacteria/pathogen. *1 mark* They display the bacteria's/ pathogen's antigens on their surface (activating T-lymphocytes). *1 mark*	**2**	Do not accept phagocytes engulf antigens/invaders/foreign cells. Do not accept phagocytes destroy pathogens. Accept phagocytes destroy pathogens by phagocytosis. Accept – they become antigen presenting cells if it is linked to engulfing the pathogen.

Question			Expected Answer(s)	Max Mark	Additional Guidance
12.	(a)		26 million/26 000 000	**1**	
	(b)		0·6 million/600 000	**1**	
	(c)		5·14/5·1/5	**1**	
	(d)		The steepest/steeper part of the (HIV infected) graph/line was between 1993 and 1995 **Or** The graph/line sharply increases between 1993 and 1995 while the rest of the graph/line more steadily increases.	**1**	Do not accept the greatest increase occurs between 1993 and 1995. The answer must refer to the gradient of the line, so greatest slope is acceptable. The answer must indicate a comparison to other areas. So it rises steeply between 1993 and 1995 is not correct.
13.	(a)	(i)	Each group should contain individuals of similar ages/a similar age range. **or** Each group should contain the same number of males and females/same gender mix. **or** Individuals in groups should have no recent history of influenza. **or** Individuals should not be allowed to travel abroad during the study **or** Individuals should all be in good general health.	**1**	
		(ii)	Did not develop influenza = 453 and Total = 470	**1**	
		(iii)	R	**1**	
	(b)		It enhances/improves the immune response/antibody production. **or** It improves the effectiveness of the vaccine.	**1**	Accept it makes the immune response work better/last longer. Accept it makes the vaccine work better/longer. Do not accept - it enhances the vaccine/speeds up the immune system.
	(c)		A pandemic	**1**	

Question			Expected Answer(s)	Max Mark	Additional Guidance
14.	(a)	(i)	1. Pituitary gland secretes/ produces FSH/LH. 2. FSH stimulates growth of <u>follicle</u> (in the ovary). 3. Follicle/ovary produces oestrogen. 4. Oestrogen stimulates growth/ repair/proliferation/ thickening of endometrium/ uterus lining. 5. Oestrogen stimulates production of LH. 6. LH (surge) brings about ovulation/release of the egg. 7. <u>Rising/high levels</u> of oestrogen inhibit FSH production. 8. This is negative feedback	**6**	Any 6 from 8 Do not accept oestrogen causes vascularisation.
		(ii)	a. The follicle develops into the corpus luteum. b. Corpus luteum secretes progesterone (and oestrogen). c. Progesterone maintains/ increases/thickens the endometrium/uterus lining. d. Progesterone inhibits <u>FSH/ LH</u> production. e. Progesterone/oestrogen levels decrease (towards the end of the cycle). f. This/corpus luteum degeneration triggers menstruation/breakdown of the endometrium.	**4**	Any 4 from 6 Do not accept progesterone <u>repairs</u> the endometrium. Accept – progesterone causes vascularisation of the endometrium/uterus lining. Note – point 8 may be awarded here if negative feedback is linked to point d.

Question			Expected Answer(s)	Max Mark	Additional Guidance
14	(b)	(i)	1. Pacemaker/SAN contains autorhythmic cells/is where the heart beat originates/is found in the right atrium. 2. Impulse/wave of excitation spreads across the atria/cause the atria to contract/cause atrial systole. 3. (Impulses) reach/stimulate the atrioventricular node/ AVN. 4. AVN found at junction of atria and ventricles/at base of atria. 5. Impulses from AVN spread through ventricles. 6. (Cause) contraction of ventricles/ventricular systole. 7. (This is followed by) relaxation/resting/diastolic phase/diastole.	**5**	Any 5 from 7 Check blood flow answers carefully as candidates can get points 6 and 7.
		(ii)	a. Medulla controls the cardiac cycle/regulates the SAN. b. Autonomic nervous system (carries impulses to heart). c. Sympathetic nerve speeds up the heart rate. d. Sympathetic nerve releases noradrenaline/norepinephrine. e. Parasympathetic nerve slows down the heart rate. f. Parasympathetic nerve releases acetylcholine. g. Sympathetic and parasympathetic systems are antagonistic to each other	**5**	Any 5 from 7 Give 1 mark (h) if candidate mentions sympathetic and parasympathetic but no/wrong description of speeds up and slows down. Give 1 mark (i) if candidate mentions noradrenaline and acetylcholine but no/wrong link to sympathetic and parasympathetic nerves.

[END OF MARKING INSTRUCTIONS]

General Marking Principles for Higher Human Biology

This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this paper. These principles must be read in conjunction with the detailed marking instructions, which identify the key features required in candidate responses.

(a) Marks for each candidate response must always be assigned in line with these General Marking Principles and the Detailed Marking Instructions for this assessment.

(b) Marking should always be positive. This means that, for each candidate response, marks are accumulated for the demonstration of relevant skills, knowledge and understanding: they are not deducted from a maximum on the basis of errors or omissions.

(c) If a specific candidate response does not seem to be covered by either the principles or detailed Marking Instructions, and you are uncertain how to assess it, you must seek guidance from your Team Leader.

(d) There are no half marks awarded.

(e) Where a candidate makes an error at an early stage in a multi stage calculation, credit should normally be given for correct follow on working subsequent stages, unless the error significantly reduces the complexity of the remaining stages. The same principle should be applied in questions which require several stages of non-mathematical reasoning.

(f) Unless a numerical question specifically requires evidence of working to be shown, full marks should be awarded for a correct final answer (including units) on its own.

(g) Larger mark allocations may be fully accessed whether responses are provided in continuous prose, linked statements or a series of discrete developed points.

(h) In the mark scheme, if a word is **underlined** then it is essential; if a word is **(bracketed)** then it is not essential.

(i) In the mark scheme, words separated by / are **alternatives**.

(j) If two answers are given that contradict one another the first answer should be taken. However, there are occasions where the second answer negates the first and no marks are given. There is no hard and fast rule here, and professional judgement must be applied.

(k) Where the candidate is instructed to choose one question to answer but instead answers both questions, both responses should be marked and the better mark awarded.

(l) The assessment is of skills, knowledge and understanding in Human Biology, so marks should be awarded for a valid response, even if the response is not presented in the format expected. For example, if the response is correct but is not presented in the table as requested, or if it is circled rather than underlined as requested, give the mark.

(m) Unless otherwise required by the question, use of abbreviations (eg DNA, ATP) or chemical formulae (eg CO_2, H_20) are acceptable alternatives to naming.

(n) Content that is outwith the course assessment specification should be given credit if used appropriately eg metaphase of meiosis.

(o) If a numerical answer is required and units are not given in the stem of the question or in the answer space, candidates must supply the units to gain the mark. If units are required on more than one occasion, candidates should not be penalised repeatedly.

(p) Incorrect **spelling** is given: sound out the word(s):

- if the correct item is recognisable then give the mark
- if the word can easily be confused with another biological term then **do not** give the mark eg ureter and urethra
- if the term is a mixture of other biological terms then **do not** give the mark, eg mellum, melebrum, amniosynthesis

(q) Presentation of data:

- if a candidate provides two graphs or bar charts, in response to one question (eg one in the question and another at the end of the booklet), mark both and give the higher score
- for marking purposes no distinction is made between bar charts (used to show discontinuous features, have descriptions on the x-axis and have separate columns) and histograms (used to show continuous features, have ranges of numbers on the x-axis and have contiguous columns)
- other than in the case of bar charts/histograms, if the question asks for a particular type of graph or chart and the wrong type is given, then do not give the mark(s) for this. Where provided, marks may still be awarded for correctly labelling the axes, plotting the points, joining the points either with straight lines or curves (best fit rarely used), etc.
- the relevant mark should not be awarded if the graph uses less than 50% of the axes; if the x and y data are transposed; if 0 is plotted when no data for this is given (ie candidates should only plot the data given)

(r) Marks are awarded only for a valid response to the question asked. For example, in response to questions that ask candidate to:

- **identify, name, give** or **state**, they need only name or present in brief form;
- **describe**, they must provide a statement or structure of characteristics and/or features;
- **explain**, they must relate cause and effect and/or make relationships between things clear;
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between things;
- **calculate**, they must determine a number from given facts, figures or information;
- **predict**, they must suggest what may happen based on available information;
- **evaluate**, they must make a judgement based on criteria;
- **suggest**, they must apply their knowledge and understanding of Human Biology to a new situation. Marks will be awarded for any suggestions that are supported by knowledge and understanding of Human Biology